BEHAVIORAL MEDICINE NOTES

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UNIT- I

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POST GRADUATE INSTITUE OF BEHAVIORAL AND MEDICIAL SCIENCES, RAIPUR, C.G.

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A BRIEF HISTORY OF BEHAVIOURAL MEDICINE

Initially formulated as a reaction to psychoanalytic psychosomatic medicine and buttressed by the more empirically grounded behaviourist movement (Pomerleau & Brady, 1979), behavioural medicine was developed in the 1970s to address psycho-behavioural risk factors in health and disease (Mc Kegney & Schwartz, 1986). At the time, the prevalent view of health and disease was biomedical, with a reductionistic focus on organs and organ systems. By the 1970s, a more systematic and empirical evaluation of the interaction between behaviour and illness generated a movement to integrate behavioural techniques and conferred credibility on the mind-body relationship. This movement evolved into a discipline now referred to as "behavioral medicine." As behavioural medicine expanded during the late 1970s to explore the interaction between behaviour and illness more systematically, diverse and innovative behavioural medicine techniques were effectively applied to various medical populations.

Over 30 years, the emergence of several overlapping disciplines may have created "drift" in the definition of behavioural medicine. The medical specialty of psychiatry adopted the term, often using it as a synonym for its traditional focus on psychopathology. An increase in the promotion and use of therapeutic approaches in the realm of Complementary and Alternative Medicine (CAM) typically predicated on the interaction of biological, psychological, social, and spiritual influences on health—also involves behavioural approaches and expanded the boundaries of behavioural medicine.

DEFINITION

In 1977, behavioral medicine was defined as “the interdisciplinary field concerned with the development and integration of behavioural and biomedical science knowledge and techniques relevant to health and illness and the application of this knowledge and these techniques to prevention, diagnosis, treatment, and rehabilitation” (Schwartz & Weiss, 1978). A broader, and more current, definition comes from the charter of the **International Society of Behavioural Medicine (2008)**, which states that behavioural medicine is “the interdisciplinary field concerned with the development and integration of socio-cultural, psychosocial, behavioural, and biomedical knowledge relevant to health and illness and the application of this knowledge to disease prevention, health promotion, etiology, diagnosis, treatment, and rehabilitation.”

Behavioral medicine is an interdisciplinary field combining both medicine and psychology and is concerned with the integration of knowledge in the biological, behavioral, psychological, and social sciences relevant to health and illness. Thses science include epidemiology, anthropology,sociology,psychology,physiology,pharmacology,neuroanatomy,endocrinology,and immunology. The practice of behavioral medicine encompasses health psychology, but also includes applied psycho-physiological therapies such as biofeedback, hypnosis, and bio-behavioral therapy of physical disorders, aspects of occupational therapy, rehabilitation medicine, and physiatry, as well as preventive medicine. Behavioral medicine emphasizes remediation and healing of illness. Practitioners of behavioural medicine include appropriately qualified nurses, socialworkers, psychologists, and physicians (including medical students and residents), and these professionals often act as behavioural change agents, even in their medical roles.

HEALTH

Health is the level of functional or metabolic efficiency of a living organism. In humans, it is the general condition of a person's mind and body, usually meaning to be free from illness, injury or pain (as in "*good health*" or "*healthy*"). The **World Health Organization (WHO)** defined health in its broader sense in **1946** as "a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity."

The **WHO**'s **1986** ***Ottawa Charter for Health Promotion*** further stated that health is not just a state, but also "a resource for everyday life, not the objective of living. Health is a positive concept emphasizing social and personal resources, as well as physical capacities."

ILLNESS

Illness is a condition of being unhealthy in our body or mind. It may be a specific condition that prevents our body or mind from working normally: a sickness or disease. *Illness* and *sickness* are generally used as synonyms for *disease*. However, this term is occasionally used to refer specifically to the patient's personal experience of his or her disease (Emson & Mc Whinney, 1987). In this model, it is possible for a person to have a disease without being ill (to have an objectively definable, but asymptomatic, medical condition), and to be *ill* without being *diseased* (such as when a person perceives a normal experience as a medical condition, or medicalizes a non-disease situation in his or her life). Illness is often not due to infection, but a collection of evolved responses—sickness behavior by the body—that helps clear infection. Such aspects of illness can include lethargy, depression, anorexia, sleepiness, hyperalgesia, and inability to concentrate (Hart, 1988; Johnson, 2002 & Kelley et al., 2003).

Recent advances in psychological, medical, and physiological research have led to a new way of thinking about health and illness. This conceptualization, which has been labeled the bio-psychosocial model, views health and illness as the product of a combination of factors including biological characteristics (e.g., genetic predisposition), behavioral factors (e.g., lifestyle, stress, health beliefs), and social conditions (e.g., cultural influences, family relationships, social support).

BOUNDARY

Boundaries are the lines we draw between and around things. They can be physical, defining where one country begins and another one ends; biological, differentiating cells, organs, and organisms from one another; or social; delineating norms or rules for what is appropriate or relevant in a particular relationship, conversational exchange, or individual behaviour. Boundaries are necessary, but must be flexible enough to adapt to different situations, different cultures, and other factors. Boundaries allow for relationships to form, but if too rigid, can be inhibiting. Likewise, where boundaries are too porous and unclear, they can make relationships unsafe and conflicted. Confusion arises when boundaries are unclear, participants disagree on what they should be, multiple relationships with the same person (e.g., physician and friend) exist simultaneously and require different boundaries, or when one or both participants do not feel entitled to, or do not know how to create and maintain healthy boundaries.

Learning to create and maintain appropriate boundaries, while developing meaningful and caring doctor-patient relationships is a significant challenge for many health care trainees. The medical student or intern who becomes over involved with his or her suffering, needy (and sometimes manipulative) patient, and becomes overwhelmed, is an archetypal story in medicine. So too are stories about physicians who cope by erecting rigid walls between themselves and their patients. These physicians find themselves cut off from some of the major rewards of doctoring. They often feel a lack of work satisfaction and are vulnerable to early burnout. In addition, they may find themselves dealing with uncooperative, dissatisfied, and possibly litigious patients. Some of the more challenging boundary issues in medical practice are “The limits of caring,”“Sexuality in the provider-patient relationship,” and “Medical advice for family members”.

Boundary Challenges 1:The Limits of Caring

An important function of boundaries in the physicianpatient relationship is to create a sense of “safe space” (partly physical and partly felt or conceptual), within which each person feels protected, autonomous, and comfortable. When this space is mutually understood and respected, it is possible for interaction to be trusting and reciprocal. When relationship boundaries are violated, most people go into defensive mode, putting their efforts toward strengthening and protecting their own personal boundaries rather than connecting with others.

Boundary Challenges 2: Sexuality & Professionalism

Sexuality and sexual feelings are a normal part of being human. Sexual feelings between a doctor and patient are not abnormal, but they can sometimes be confusing and troublesome. When acted upon, they can be dangerous to the relationship and potentially exploitative. Very infrequently, the feelings led to a boundary crossing where physician and patient mutually disclosed their feelings of attraction to one another. Much more frequently, the physician took steps to avoid the feelings by fleeing the encounter or skipping portions of the physical examination.

Some Tips for dealing with Boundary threats are discussed below:

1. Take a breather: excuse yourself, get out of the room, and take a few minutes to clear your head and think about what's happening.
2. Get immediate advice: find a nurse, or a colleague or supervisor you trust, tell them what's happening and get another, hopefully clearer, perspective.
3. Bring a third person (supervisor, nurse, patient family member) into the room to diffuse the situation (the resident in our story brought in a nurse, which was very helpful).
4. Consider changing your physical position in relation to the patient (e.g., put more distance between you or put a barrier like a desk between you).
5. Try to disentangle your own assumptions and feelings (sexual or otherwise) from what you actually observe in a patient.
6. Finally, re-establishing (in one's own mind and perhaps with an appropriate question or statement) the interaction as a doctor-patient encounter may help. Similarly, try reminding yourself that the patient to whom you are attracted (or toward whom you have other confused feelings) is actually a person in need who has sought your professional help.

Boundary Challenges 3: Medical Advice For Family Members

Boundary confusion often results when two different relationships exist simultaneously. Often when boundaries become unclear, there is a temptation to go against one's better judgment and ignore that little voice that whispers "maybe that's not such a good idea. . . . " Somehow it's easier to justify questionable beliefs and actions to oneself than to risk offending or making the other person uncomfortable.

Guidelines For Practitioners

There is no set formula for setting and working with boundary issues. Different practitioners find different boundaries issues difficult based on past history, personality, and so on. Similarly, different types of patient and physician personalities create different physician-patient relationships. However, some guidelines to help physicians develop an individualized approach to dealing with boundaries in practice are:

1. Learn to recognize the warning signs of boundary confusion, including:

a. Difficulty thinking as clearly as usual, judgment feels clouded.
b. A request or demand by the patient that doesn't feel quite right.
c. Changes in patient behavior, such as increasing numbers of calls, no-shows, demands for care, asking personal questions, or being sexually provocative which may indicate a shift or confusion in boundaries.
d. A change in the practitioners' usual behavior may also indicate impending Boundary confusion.
e. Fear of hurting or angering the patient, or not wanting to be "the bad guy."

2. Be aware of your own feelings and needs, and take care of them as much as possible; remembering that:

• Creating safe and healthy boundaries is your right as well as your responsibility to yourself and your patient.

• Provider feelings and needs which are unmet, or unrecognized, are much more likely to spill over into the doctor-patient encounter and cause boundary confusion. For example, a physician who is single and perhaps feeling lonely may be more likely to develop sexual or romantic feelings toward a patient.

• Honesty with oneself and others (where appropriate) helps ensure healthy boundaries.

3. Get together with colleagues and/or teachers to share concerns in a safe, supportive, nonjudgmental atmosphere. Balint or personal awareness groups are ideal for this activity.

4. Practice the skills involved; setting clear and appropriate boundaries requires practice. Workshops and courses devoted to boundary setting can be found on the American Academy for Communication in Healthcare (AACH) web site; cases for discussion can be found on the American Board of Internal Medicine (ABIM), Association of American Medical Colleges (AAMC), and American Medical Association (AMA) web sites.

5. Consider professional counselling. Brief counselling or psychotherapy may be a useful tool for defining personal boundaries in particularly challenging situation.

THE BIO-PSYCHOSOCIAL MODEL

Perhaps the best way to understand the bio-psychosocial model is to contrast it with the biomedical model. The biomedical model, which governed the thinking of most health practitioners for the past 300 years, maintains that all illness can be explained on the basis of aberrant somatic processes, such as biochemical imbalances or neuropsychological abnormalities. The biomedical model assumes that psychological processes are largely independent of the disease process.

The biomedical model implicitly assumes a mind-body dualism, maintaining the mind and body are separate entities. Finally, the biomedical model clearly emphasizes illness over health. That is, it focuses on aberrations that lead to illness rather than on the conditions that might promote health (Engel, 1977).

The bio-psychosocial model maintains that biological, psychological and social factors are all-important determinants of health and illness. As such, both macro level processes (such as the existence of social support, the presence of depression) and micro level processes (such as cellular disorders or chemical imbalances) interact to produce a state of health or illness.

The bio-psychosocial model maintains that health and illness are caused by multiple factors and produce multiple effects. The model further maintains that the mind and body cannot be distinguished in matters of health and illness because both so clearly influence an individual's state of health. The bio- psychosocial model emphasizes health and illness both rather than regarding illness as a deviation from some steady state. From this viewpoint, health becomes something that one achieves through attention to biological, psychological and social needs rather than something that is taken for granted (WHO, 1948). Researchers have adopted a **systems theory** approach to health and illness. Systems theory maintains that all levels of organization in any entity are linked to each other hierarchically and that change in any one level will affect change in all the other levels. This means that the micro level processes (such as cellular changes) are nested within the macro level processes (such as societal values) and that changes on the micro level can have macro level effects (and vice versa).

Consequently, health, illness and medical care are all interrelated processes involving interacting changes both within the individual and on these various levels. There are several implications of the bio-psychosocial model for clinical practice with patients. **First**, the model maintains that the process of diagnosis should always consider the interacting role of biological, psychological and social factors in assessing an individual's health or illness (Oken, 2000). Therefore, an interdisciplinary team approach may be the best way to make a diagnosis (Schwartz, 1982). **Second**, this model maintains that recommendations for treatment must also examine all three sets of factors. By doing this, it should be possible to target therapy uniquely to a particular individual, consider a person's health status in total, and make treatment recommendations that can deal with more than one problem simultaneously. Again, a team approach may be most appropriate (Schwartz, 1982). **Third**, the model makes explicit the significance of the relationship between patient and practitioner. An effective patient-practitioner relationship can improve a patient's use of services as well as the efficacy of treatment and the rapidity with which illness is resolved (Belar, 1977).

STRESS

Stress is a negative emotional experience accompanied by predictable biochemical, physiological, cognitive, and behavioural changes that are directed either toward altering the stressful event or accommodating to its effects (Baum, 1990). The condition of stress has two components: *physical*, involving direct material or bodily challenge, and *psychological*, involving how individuals perceive circumstances in their lives (Lovallo, 2005). These components can be examined in three ways (Dougall & Baum, 2001). One approach focuses on the environment: stress is seen as a *stimulus*, as when we have a demanding job or experience severe pain from arthritis or a death in the family. Physically or psychologically challenging events or circumstances are called **stressors**. The second approach treats stress as a *response*, focusing on people's reactions to stressors. We see an example of this approach when people use the word *stress* to refer to their state of tension. Our responses can be psychological, such as your thought patterns and emotions when you ''feel nervous,'' and physiological, as when your heart pounds, your mouth goes dry, and you perspire. The psychological and physiological response to a stressor is called **strain**. The third approach describes stress as a *process* that includes stressors and strains, but adds an important dimension: the relationship between the person and environment (Lazarus, 1999;
Lazarus & Folkman, 1984). This process involves continuous interactions and adjustments—called **transactions** with the person and environment each affecting and being affected by the other. According to this view, stress is not just a stimulus or a response, but rather a process in which the person is an active agent who can influence the impact of a stressor through behavioral, cognitive, and emotional strategies.

NEURO-ENDOCRINE RESPONSES TO STRESS

A stressor can be defined as a certain stimulus of the external or internal receptor. The stressors are usually divided into macroscopic threats (e.g. fight with enemy, fear, pain) and microscopic threats (targeting at epithelial or endothelial barriers e.g. infection or tissue damage). This neuro-endocrine – immunologic interrelations are also vital in the clinical situations.

- **Cortisol**

The stress response is the complex process that can be initiated by immune or central nervous system. The central nervous system reacts against macroscopic threats and controls whole body response. The hypothalamus – pituitary – adrenal axis is activated and vasopressin, prolactin and growth hormone are released. In clinical settings corticotropin realizing hormone and vasopressin (both stimulated by adreno-cortical signals e.g. pain, fear, hypovolemia or immunologic stimuli e.g. interleukins, TNF, cytokines) synergistically increase adrenocorticotropin (ACTH) secretion. ACTH induces conversion of cholesterol to cortisol which cooperates with sympathetic nervous system to prepare a body for response by mobilization of energetic substrates, increase of intravascular volume and blood pressure enhancement. The immune system reacts against microscopic threats infringing endothelial or epithelial barriers. The initial signal is amplified by cascade of lymphokines and activated cells and stimulates central stress response which eventually terminates system overstimulation.

Immune response and tissue damage contribute to systemic inflammatory response syndrome (SIRS) development. These inflammatory signals are transferred to the central nervous system by vagus nerve and activate HPA axis. After acute stress response when ACTH, prolactin, growth hormone, and thyroid hormone are elevated, the pulsatile, more physiologic pattern of neurohormones concentration appears. Although normal limits of plasma neurohormones levels in stress response are not known, inadequate concentrations can lead to acute failure and shock.

- **Sympathetic Nervous System** –

Sympathetic nervous system stimulation is a part of central regulatory mechanism. It exerts many effects on the cardiovascular system by nor-epinephrine and epinephrine. Afferent baroreceptor signaling to the brain signals low cardiac output and efferent sympathetic pathways are activated. The main results of it are vasoconstriction (increased after-load, decreased renal perfusion), increased heart rate and contractility (increased cardiac output and wall stress), activation of RAAS.

- **Hypothalamic-pituitary-adrenal axis (HPA Axis)** –

Hypothalamic – pitutitary – adrenal system is the central stress response system linking neural regulation to neuro-hormonal and humoral control. In response to cortical signals e.g. fear, pain, deep emotions or immune derived factors like TNF α, Il-6 corticotropin realizing hormone, vasopressin, prolactin and growth hormone are released. Corticotropin releasing hormone stimulates sympathetic system and ACTH secretion. It reaches the adrenal cortex and stimulates cortisol production from cholesterol. Cortisol cooperates with sympathetic activation to prepare metabolism for stress response. These mechanisms inhibit all growth and developmental functions; prepare metabolic substrates (glucose, fatty acids, aminoacids), increase blood pressure and intravascular volume.

- **Vasopressin system** –

Vasopressin (ADH) is released by the hypothalamus as a result of baroreceptor, osmotic, and neuro-hormonal stimuli. It normally maintains body fluid balance, vascular tone, and regulates contractility. The important mechanism of vasopressin action in stress states is its potentiating effect on ACTH secretion leading to cortisol release. Although vasopressin is a powerful vasoconstrictor it dilates the pulmonary, cerebral, and myocardial circulations helping to preserve vital organ blood flow.

Chronically high levels of catecholamines and corticosteroids, such as cortisol, can contribute the development and progression of atherosclerosis (Lundberg, 1999; Matthews et al., 2006). But social support may help: people with high levels of social support tend to exhibit lower endocrine reactivity than with those with lower levels (Seeman & McEwen, 1996). Stress also seems to contribute to health through endocrine system pathways that involve fat stored in the abdominal cavity. The *metabolic syndrome* (Kyrou & Tsigos, 2009) is a set of risk factors including high levels of cholesterol and other blood fats; elevated blood pressure; high levels of insulin in the blood or impairments in the ability of insulin to facilitate transportation of glucose out of the blood stream; and larger fat deposits in the abdomen. The metabolic syndrome seems to be made worse by exposure to stressors and the related physiological stress responses, especially heightened neuro-endocrine activity.

NEUROTRANSMITTER AND STRESS

Neurotransmitters send and receive messages between brain cells. There are two kinds of messengers: "happy" messengers (cheerful and enthusiastic messages) and "sad" messengers (cheerless and silencing messages). Too much stress causes the happy messengers to eventually begin to fail. Sad mesages overtake happy ones, causing a chemical imbalance. This chemical imbalance is overstress. Everyone experiences short durations of overstress. The three "happy" messengers are: Serotonin, Noradrenalin, and Dopamine.

Serotonin

Our body clock is located in a supply of Serotonin in the Pineal Gland in the brain. Our body clock coordinates our body functions to the same rhythm. For example, it sets our physiology for sleeping and waking-up. Our body clock also controls the secretion of the chief stress fighting hormone Cortisol. Disruption of our Cortisol cycle makes sleeping more difficult. Serotonin is often the first happy messenger to malfunction under stress. Thus, lack of restful sleep is usally the first symptom of overstress.

Noradrenalin

Noradrenalin sets our energy levels, and makes us feel energized. Failure of this happy messenger causes overstressed people to feel as though they don't have the energy to do much of anything.

Dopamine

Dopamine is located adjacent to where Endorphin is released in the brain. And so there is a link, so that when Dopamine function declines, so does Endorphin function. Endorphins regulate pain. And so, pain increases when stress causes the Dopamine function to fail. Dopamine also operates our Pleasure Center. Thus, when stress incurs on our Dopamine function, it can also result in a loss of pleasure in normally pleasureful affairs.

Gamma-Aminobutyric Acid (GABA)

GABA is a major neurotransmitter widely distributed throughout the CNS. GABA – the most important inhibitory neurotransmitter in the brain provides this inhibition, acting like a “brake” during times of runaway stress.

L-Tyrosine

Findings from several studies suggest supplementation with tyrosine might, under circumstances characterized by psychosocial and physical stress, reduce the acute effects of stress and fatigue on task performance. In humans, tyrosine supplementation alleviates stress-induced decline in nervous system norepinephrine and subsequently enhancing performance under a variety of circumstances, including sleep deprivation, combat training, cold exposure, and unpleasant background noise.

NEURO-IMMUNE RESPONSES TO STRESS

The immune system is regulated by neural input from sensory, sympathetic and parasympathetic nerves, as well as by circulating hormones, of which the glucocorticoids are among the most prominent. Long regarded as inhibitors of immune function, adrenal steroids have now been recognized as having biphasic effects upon immune function. Under acute stress, energy reserves are mobilized, vegetative processes and reproduction are suppressed, and the body is made ready for fight or flight, which may involve wounding. Thus the immune defense system should acutely gear up to protect the organism from infections and accelerate wound healing.

A primary underlying mechanism for these effects is the translocation or 'trafficking' of immune cells between the blood and different primary, secondary and tertiary immune tissues. Elevations of stress hormones, both glucocorticoids and catecholamines, direct the movement of various cell types of the immune system. Lymphocytes, monocytes and NK cells are all reduced in number in blood and increased in number in tissues, such as the skin, as a result of acute stress or acute glucocorticoid administration. Once immune cells have marginated and begun to enter the tissue, other factors become involved as local mediators of further activation of immune function.

The release of catecholamines and corticosteroids during arousal affects health in another way: these stress responses alter the functioning of the immune system (Kemeny, 2007; Segerstrom & Miller, 2004). Brief stressors typically activate some components of the immune system, especially non-specific immunity, while suppressing specific immunity. Chronic stressors, in contrast, more generally suppress both non-specific and specific immune functions. Chronic stressors also increase *inflammation*, an important process that disrupts immune function when it occurs on a long-term basis (Kemeny, 2007; Segerstrom & Miller, 2004). So, rather than a simple ''up or down'' effect of stress on this vital system, stress dysregulates or disrupts it. Research has shown that high levels of stress reduce the production of these enzymes and the repair of damaged DNA (Glaser et al., 1985; Kiecolt-Glaser & Glaser, 1986). Given that the immune system has far-reaching protective effects, if stress disrupts the immune system it can affect a great variety of health conditions from the common cold to herpes virus infections (Chida & Mao, 2009) to cancer.

NEGATIVE AFFECTIVITY

Certain people are predisposed by their personalities to experience stressful events as especially stressful, which may, in turn, affect their psychological distress, their physical symptoms, and/or their rates of illness. This line of research has focused on a psychological state called **negative affectivity** (Watson & Clark, 1984), a pervasive negative mood marked by anxiety, depression and hostility.

Individuals high in negative affectivity express distress, discomfort, and dissatisfaction across a wide range of situations (Gunthert, Cohen, & Armeli, 1999). People who are high in negative affectivity are more prone to drink heavily (Frances, Franklin & Flavin, 1986), to be depressed (Francis, Fyer & Clarkin, 1986), and to engage in suicidal gestures or even suicide (Cross & Hirschfeld, 1986).

Negativity is related to poor health. In a review of literature relating personality factors to five diseases – asthma, arthritis, ulcers, headaches, and coronary artery disease – Friedman & Booth-Kewley (1987) found weak but consistent evidence of a relationship between these disorders and negative emotions. They suggested that psychological distress involving depression, anger, hostility, and anxiety may constitute the basis of a "disease-prone" personality that predisposes people to these disorders. Negative affectivity can be associated with elevated cortisol secretion, and this increased adrenocortical activity may provide a possible bio-psychosocial pathway linking negative affectivity to adverse health outcomes (van Eck, Berkhof, Nicolson, & Sulon, 1996). Negative affectivity can also affect adjustment to treatment.

People who are high in negative affectivity report higher levels of distressing physical symptoms, such as headaches, stomachaches, and other pains, especially under stress (Watson & Pennebaker, 1989), but in many cases, there is no evidence of an underlying physical disorder (Diefenbach, Leventhal, Leventhal, & Patrick-Miller, 1996). People high in negative affectivity also often appear more vulnerable to illness because they are more likely to use health services during stressful times than are people low in negative affectivity (Cohen & Williamson, 1991). Thus, individuals who are chronically high in negative affect may be more likely to get sick, but they also show distress, physical symptoms, and illness behaviour even when they are not getting sick.

BEHAVIOURAL PATTERNS & COPING STYLES

Defining Behavior Patterns

The **Type A behavior pattern** consists of four characteristics (Chesney, Frautschi, & Rosenman, 1985; Friedman & Rosenman, 1974):

1. *Competitive achievement orientation*. Type A individuals strive toward goals with a sense of being in competition—or even opposition—with others, and not feeling a sense of joy in their efforts or accomplishments.
2. *Time urgency*. Type A people seem to be in a constant struggle against the clock. Often, they quickly become impatient with delays and unproductive time, schedule commitments too tightly, and try to do more than one thing at a time, such as reading while eating or watching TV.
3. *Anger/hostility*. Type A individuals tend to be easily aroused to anger or hostility, which they may or may not express overtly.
4. *Vigorous Vocal Style*. Type A people speak loudly, rapidly, and emphatically, often ‘‘taking over’’ and generally controlling the conversation. In contrast, the **Type B behavior pattern** consists of low levels of competitiveness, time urgency, and hostility. People with the Type B pattern tend to be more

easygoing and ‘‘philosophical’’ about life— they are more likely to ‘‘stop and smell the roses.’’ In conversations, their speech is slower, softer, and reflects a more relaxed ‘‘give and take.’’

Behaviour Patterns and Stress

Individuals who exhibit the Type A behavior pattern react differently to stressors from those with the Type B pattern. Type A individuals respond more quickly and strongly to stressors, often interpreting them as threats to their personal control (Glass, 1977). Type A individuals also often choose more demanding or pressured activities at work and in their leisure times, and they often evoke angry and competitive behavior from others (Smith & Anderson, 1986). Hence, they have greater exposure to stressors, too.

Type A Behaviour and Health

How are people’s health and behavior patterns related? Researchers have studied this issue in two ways. First, studies have examined whether Type A individuals are at greater risk than Type Bs for becoming sick with any of a variety of illnesses, such as asthma and indigestion, but the associations appear to be weak and inconsistent (Orfutt & Lacroix, 1988; Suls & Sanders, 1988).

Second, studies have focused on the Type A pattern as a risk factor for **coronary heart disease (CHD)**—illnesses involving the narrowing of the coronary arteries, which supply blood to the heart muscle. This narrowing is called atherosclerosis, and causes several manifestations of CHD.

Anger / hostility is the main aspect of Type A behavior in the link with CHD (Everson-Rose & Lewis, 2005; Smith & Gallo, 2001). Anger / hostility seems to be Type A’s deadly emotion: people who are chronically hostile have an increased risk of developing CHD. Further, among people who already have CHD, anger and hostility are associated with increased risk of poor medical outcomes, such as additional heart attacks or death from CHD. Angry and hostile people experience more conflict with others at home and work (Smith et al., 2004), indicating greater stress exposure. The suspicious and mistrusting style of hostile persons is likely tomake them cold and argumentative during interactions with others, sometimes even with friends and family members. The resulting conflict and reduced social support may, in turn, contribute to the maintenance or even worsening of their hostile behavior toward others in a vicious cycle or self-fulfilling prophecy (Smith et al., 2004). Further, in difficult interpersonal situations in general, and at work and with family members in particular, they show greater physiological reactivity or strain (Brondolo et al., 2009; Chida & Hamer, 2008; Smith & Gallo, 1999). Further, unlike non-hostile people, hostile people do not respond to social support with reduced physiological reactivity during stressful situations (Holt-

Lunstad et al., 2008; Vella et al., 2008), perhaps because they are too distrusting or worry that support providers will evaluate them negatively. After a stressful situation, hostile people show delayed or incomplete recovery of their physiological stress responses, perhaps because they are more likely to brood or ruminate about upsetting events (Neuman et al., 2004). Also, their sleep quality is more likely to suffer during stressful periods (Brissette & Cohen, 2002). Combined, these stress processes can produce a lot of wear and tear on the cardiovascular system.

Anger might not be the only unhealthy Type A behavior (Houston et al., 1992, 1997). Social dominance—the tendency or motive to exert power, control, or influence over other people—is also associated with coronary atherosclerosis and CHD (Siegman et al., 2000; Smith et al., 2008). Further, this personality trait is associated with greater physiological reactivity or strain during challenging interpersonal tasks and situations, like arguments or debates, and efforts to influence other people also evoke larger increases in blood pressure and stress hormones (Newton, 2009; Smith et al., 2000).

Coping is the process by which people try to *manage the perceived discrepancy* between the demands and resources they appraise in a stressful situation. It indicates that coping efforts can be quite varied and do not necessarily lead to a solution of the problem. Although coping efforts can be aimed at correcting or mastering the problem, they may also simply help the person alter his or her perception of a discrepancy, tolerate or accept the harm or threat, or escape or avoid the situation (Lazarus & Folkman, 1984; Carver & Connor-Smith, 2010).

We cope with stress through our cognitive and behavioral transactions with the environment. The coping process is not a single event. Because coping involves continuous transactions with the environment, the process is best viewed as a dynamic series of appraisals and reappraisals that adjust to shifts in person–environment relationships. And so, in coping with the threat of serious illness, people who try to change their lifestyles may receive encouragement and better relationships with their physician and family. But individuals who ignore the problem are likely to experience worse and worse health and relations with these people. Each shift in one direction or the other is affected by the transactions that preceded it and affects subsequent transactions (Lazarus & Folkman, 1984).

There are several **components** of the coping process. **First**, appraisals of the harm or loss posed by the stressor (Lazarus, 1981) are thought to be important determinants of coping. **Second**, appraisal of the degree of controllability of the stressor is a determinant of coping strategies selected. A **third** component is the person’s evaluation of the outcome of their coping efforts and their expectations for future success in coping with the stressor. These evaluative judgments lead to changes in the types of coping employed.

Functions of Coping

According to Richard Lazarus and his colleagues, coping can serve two main functions (Lazarus, 1999; Lazarus & Folkman, 1984). It can alter the *problem* causing the stress or it can regulate the *emotional* response to the problem.

Emotion-focused coping is aimed at controlling the emotional response to the stressful situation. People can regulate their emotional responses through *behavioral* and *cognitive* approaches. Examples of *behavioral* approaches include using alcohol or drugs, seeking emotional social support from friends or relatives, and engaging in activities, such as sports or watching TV, which distract attention from the problem. *Cognitive* approaches involve how people think about the stressful situation. In one cognitive approach, people *redefine* the situation to put a good face on it, such as by noting that things could be worse, making comparisons with individuals who are less well off, or seeing something good growing out of the problem.

Other emotion-focused cognitive processes include strategies Freud called ‘‘defense mechanisms,’’ which involve distorting memory or reality in someway (Cramer, 2000). For instance, when something is too painful to face, the person may deny that it exists. This defense mechanism is called *denial*. In medical situations, individuals who are diagnosed with terminal diseases often use this strategy and refuse to believe they are really ill. This is one way by which people cope by using *avoidance* strategies.

People tend to use emotion-focused approaches when they believe they can do little to change the stressful conditions (Lazarus & Folkman, 1984). Coping methods that focus on emotions are important because they sometimes interfere with getting medical treatment or involve unhealthful behaviors, such as using cigarettes, alcohol, and drugs to reduce tension. People often use these substances in their efforts toward emotion-focused coping (Wills, 1986).

Problem-focused coping is aimed at reducing the demands of a stressful situation or expanding the resources to deal with it. Everyday life provides many examples of problem-focused coping, including quitting a stressful job, negotiating an extension for paying some bills, devising a new schedule for studying (and sticking to it), choosing a different career to pursue, seeking medical or psychological treatment, and learning new skills. People tend to use problem-focused approaches when they believe their resources or the demands of the situation are changeable (Lazarus & Folkman, 1984). For example, caregivers of terminally ill patients use problem-focused coping more in the months prior to the death than during bereavement

(Moskowitz et al., 1996).

Another very important distinction among coping responses is between **approach and avoidance, or engagement and disengagement responses** (Roth and Cohen, 1986; Skinner et al, 2003). Approach coping strategies are efforts to deal with the stressor or related emotions. Avoidance strategies are attempts to escape from having to deal with the stressor.

Disengagement coping is often emotionfocused, because it involves an attempt to escape from feelings of distress. Sometimes disengagement coping is almost literally an effort to act as though the stressor does not exist, so that it does not have to be reacted to. Wishful thinking and fantasy distance the person from the stressor, at least temporarily, and denial creates a boundary between reality and the person's experience.

Avoidance coping can be useful in the short term, but it is generally ineffective when confronting a stressor that poses a real threat, that is, something that will have to be dealt with eventually. If you go out partying to avoid a stressor, it is likely to still be there the next day. Indeed, for many stressors, the longer you avoid dealing with it, the more difficult and urgent the problem becomes.

Personality: The Styles of Coping

Coping style is typically the term used to refer to characteristic methods individuals use to deal with threatening situations. Cohen and Lazarus (1977) described what they termed *anticipatory coping*, that is, the characteristic ways in which the individual not only adapts to but also shapes his or her environment. To them, coping represents more than just a response to confrontations or stress. It is an ongoing "personality" process, a habitual style of relating to events and structuring life.

Introversive Style

According to Millon *et al.* (1982), these patients are rather colorless and emotionally flat, tending to be quiet and untalkative. Often unconcerned about their problems, they typically are vague and difficult to pin down concerning symptoms and may be passive with regard to taking care of themselves. Lipowski (1970) describes patients such as these as employing a cognitive coping style termed *minimization,* characterized by a tendency to ignore, deny, or rationalize the personal significance of information input.

Inhibited Style

These personalities are characteristically shy and ill at ease; they expect to be hurt, are disposed to feel rejected, and are overly concerned about whether others will think well or ill of them (Millon *et al.*, 1982). Fearing that they will be taken advantage of, they are inclined to keep their problems to themselves. Lipowski refers to personalities similar to these as seeing illness as "punishment." They typically interpret this form of punishment as both expected and just; as a consequence, they are likely to offer little resistance to what they see as the inevitable and often adopt a rather fatalistic attitude toward illness.

Cooperative Style

Millon *et al.* (1982) note the eagerness with which these personalities seek to attach themselves to supportive persons and their willingness to follow advice religiously as long as they need assume little or no responsibility for themselves. According to Lipowski (1970), patients such as these may be disposed to see illness as "relief," that is, a welcome respite from the demands and responsibilities of being well. Leigh and Reiser (1980) saw these individuals as "dependent, demanding" patients. Their underlying striving is a "regressive wish to be cared for as if by an idealized, nurturant mother." This need for reassurance and care is unusually exaggerated among these people and the sick role is quickly adopted as an opportunity to return to a state of infantile dependency.

Sociable Style

Described by Millon *et al.* (1982) as outgoing, talkative and charming, these individuals are often undependable, highly changeable in their likes, more concerned with appearances than substance, and disinclined to deal with serious matters or personal problems. To Lipowski (1970), illness is seen by such patients as a "strategy" to secure attention, support, and compliance from others. Leigh and

Reiser (1980) label these individuals "dramatic, emotional" types. Their major goal is the wish to be attractive and desirable, often leading them to be concerned more with their masculinity or femininity than with their health.

Confident Style

In a brief description of this type, Millon *et al.* (1982) note their characteristic calm and somewhat supercilious manner. Despite their narcissistic airs of superiority, they fear bodily harm and are often highly motivated to regain a state of well-being. It is typical of these personalities both to seek and to expect to be given special treatment; they are also likely to take unjust advantage of others. Lipowski (1970), referring to an "avoiding" behavioral style, notes that it is observed most often among individuals for whom acceptance of the sick role signifies a severe threat to the self-image as independent, masculine, and invulnerable.

Forceful Style

Similar in certain respects to the foregoing type, these individuals are more overtly aggressive and hostile in their behavior (Millon *et al.,* 1982). Acting in a domineering and tough-minded fashion, they often go out of their way to be intimidating and to undermine the efforts of others. Their behavioral coping style, according to Lipowski (l970), might be termed *tackling*, that is, an unwillingness to accept the sick role and a disposition to go on the attack in dealing with the challenges and limitations posed by a disability. This coping attitude would be seen most dramatically in tendencies to fight illness at any cost, such as insisting on using a traumatized body part as if it were intact.

Respectful Style

These individuals are overly responsible and conforming, usually going out of their way to impress others with their self-control, discipline, and seriousmindedness (Millon *et al.,* 1982). To Lipowski (1970), a major element in the behavior of these types is their inclination to see illness as weakness, that is, as a failure on their part and a shameful loss of personal control. Also characteristic is their strong desire to deny or conceal any problem that might prove publicly humiliating. When such denial is impossible, they become "model" patients.

Sensitive Style

These are unpredictable and moody types who are often displeased or dissatisfied with much in their life (Millon *et al.,* 1982). Lipowski (1970) describes patients similar to these as seeing illness as possessing "value" in the sense that "sickness makes health pleasant." Referred to as "long-suffering, self-sacrificing" persons by Leigh and Reiser (1980), they can often be diagnosed simply by the tone of their first words. Frequently speaking in a wailing and complaining voice, they report a history replete with medical misdiagnoses and complicated surgical procedures. They act as if they were "born to suffer" and, in fact, many have suffered.

PSYCHOPHYSIOLOGICAL MODELS OF DISEASE

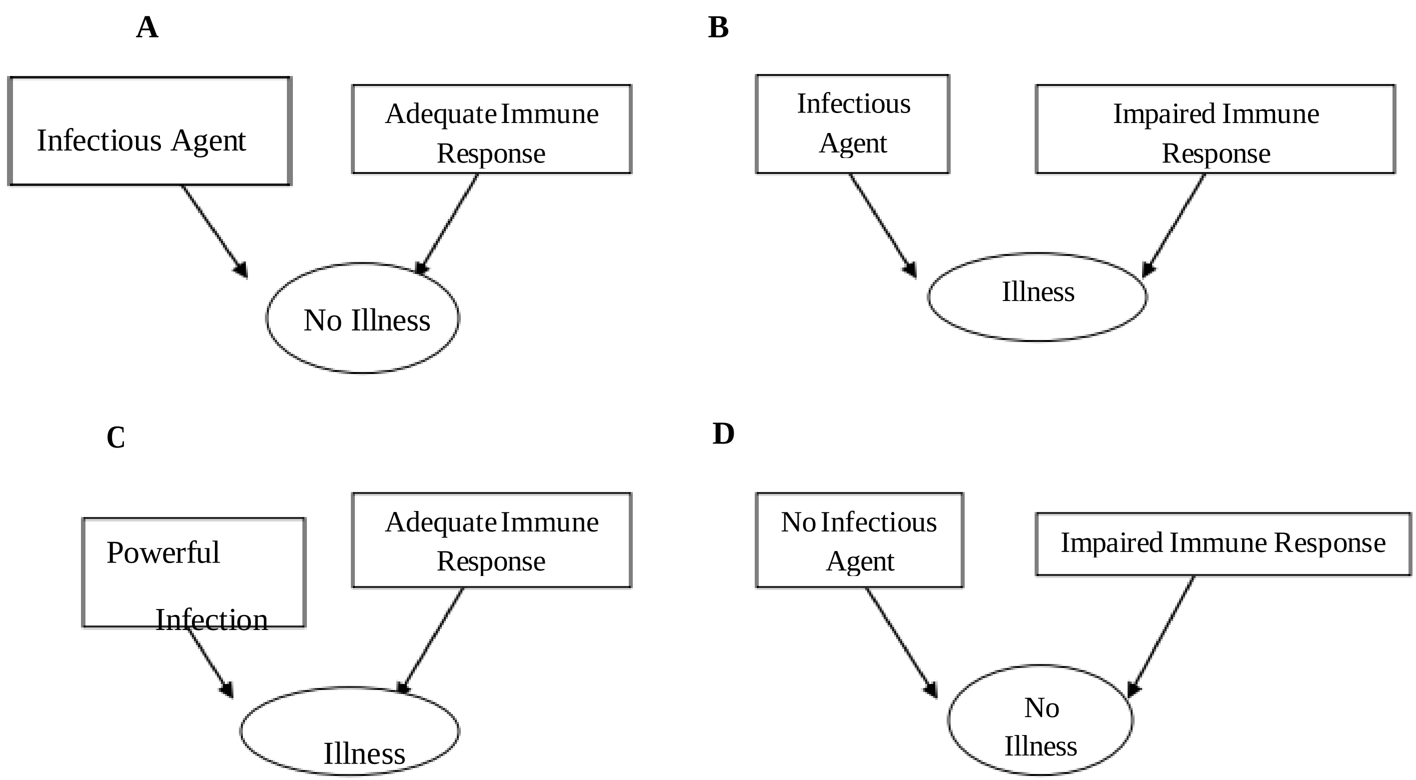
It is important to recognize the differences between several types of mechanism, since the role played by psycho-physiological factors in disease pathology is somewhat varied.

Physiological Reactivity as a Causal Factor:

The first possibility is that psycho-physiological reactivity is directly responsible for disease. Particular individuals show heightened reactivity in specific physiological parameters such as blood pressure, gastrointestinal motility, or muscle tension in the head and neck. Through regular or sustained exposure to psychosocial demands that over-tax resources, these physiological responses may be repeatedly elicited and in due course may gravitate from acute reactions to sustained pathology. The type of study needed to evaluate this mechanism is a longitudinal investigation in which psycho-biological predispositions and reactivity patterns are measured at the outset, and exposure to the stressors that trigger appropriate physiological responses is tracked. It would then be predicted that people with heightened reactivity in a particular physiological response system and who also experience psychosocial demands that elicit these a sustained period, will be at increased risk for developing the disease. More readily available in humans is evidence that people with a variety of disorders tend to react to cognitive and emotional challenges with heightened responses in the physiological systems relevant to their disorder. Thus people with hypertension typically show larger blood pressure responses to mental stress tests than do those with normal blood pressure.

Physiological Reactivity as an Inhibitor of Host Resistance and Defense:

The second mechanism through which psychosocially-induced physiological reactions may influence disease is by altering physical vulnerability in ways that render the person more susceptible to invasive organisms.



The mechanism is presented diagrammatically showing four possible scenarios linking exposure to pathogens and bodily defenses. Panel A represents the situation when the organism mounts an adequate immune response that tackles the infection effectively and prevents disease. In panel B, exposure takes place in the presence of a stress-induced impairment of immune response, allowing the infection to be acquired and illness to develop. This is the situation that pertains when physiological reactivity alters host resistance. Panel C describes a situation when the infectious agent to which the individual is exposed is particularly virulent. Under these circumstances, illness may occur even though immune responses are intact. Another possibility is that bodily defenses are disrupted, but since there is no simultaneous exposure to a pathogen, no illness results (Panel D). The ideal type of study of this mechanism is a longitudinal investigation in which psychosocial factors are monitored together with measures of bodily defenses, exposure to infectious agents, and illness.

In studies of patho-physiology, there is a new understanding that infectious processes are not confined to traditional self-limiting disorders, but may contribute to conditions such as peptic ulcer, gastric cancer, dementia, vasculitis, and insulin-dependent diabetes (Lorber, 1996; O'Connor, Buckley, & O'Morain, 1996).

Physiological Reactivity as a Disruptive Factor:

The role of physiological reactivity as a disruptive factor differs from its involvement in the processes which have discussed before. This mechanism is operative in people already suffering from disease, although the pathology may not necessarily have reached the stage of clinical diagnosis. It is particularly likely to be active in chronic or episodic conditions such as pain syndromes, bronchial asthma, and diabetes. The influence of psycho-physiological responses may be manifest at the level of day-to-day clinical status, and consequently relatively minor daily hassles may be influential. The physiological reactions may either be directly involved in pathology (such as disturbances of bronchoconstriction in asthma) or may disrupt host resistance, as in the case of autoimmune conditions. There may be some problems such as insulin-dependent diabetes where effects could be mediated either directly through disturbance of insulin metabolism or indirectly through failures in resistance to enteroviruses.

Physiological Reactivity as a Trigger of Acute Clinical Events:

The fourth distinct psycho-physiological mechanism in disease concerns the elicitation of acute clinical events. It is possible that psycho-physiological responses can be so intense as to trigger episodes of serious illness, and possibly even death. Such an effect is unlikely in the absence of severe underlying disease rendering the victim especially vulnerable. The mechanism is distinctive in that a single occurrence may have serious clinical consequences. Probably the best illustration of the impact of psycho-physiological processes on acute clinical events is in the cardiovascular field, with the triggering of myocardial infarction and sudden cardiac death. Recent studies of myocardial infarction victims suggest that the disturbances of cardiac rhythm and coronary blood flow may precipitate clinical events in susceptible individuals.

The psycho-physiology of disease involves the integration of several disciplines including neuroscience, pathophysiology, and health psychology, and in each of these areas new discoveries are constantly changing our levels of understanding. The complexity of links between the brain, peripheral physiological function, and disease risk is formidable, and linear models are of limited value.

HEALTH BEHAVIOUR

Interest in health behaviours is derived from two assumptions; that a substantial proportion of the mortality from the leading causes of death is attributable to the behavior of individuals, and that the behavior is modifiable (Stroebe & Stroebe, 1995). The health behaviours examined have been many and varied; from health enhancing behaviours such as exercise and healthy eating, on the one hand, to avoidance of health harming behaviours such as smoking and excessive alcohol consumption, on the other. Each of these behaviours has immediate or long-term effects upon the individual's health and is to varying extents within the individual's control. We might define health behavior as any activity taken for the purpose of preventing or detecting disease or for improving general well-being (Conner & Norman, 1996). The behaviours within this definition include medical service usage (e.g., physician visits, vaccination, screening), compliance with medical regimens (e.g., dietary, diabetic, antihypertensive regimens), and self- directed health behaviours (e.g., diet, exercise, smoking, alcoholconsumption). A clearer understanding of why individuals perform health behaviours might assist in the development of interventions to help individuals gain the benefits of improved health and well-being. A variety of factors have been found to account for individual differences in the performance of various health behaviours, including demographic factors, social factors, emotional factors, perceived symptoms, factors relating to access to medical care, personality factors and cognitive factors (Adler & Mathews, 1994; Rosenstock, 1974; Taylor, 1991).

THEORETICAL MODELS OF HEALTH BEHAVIOUR

Two types of Social Cognition Models (SCMs) have been applied in health psychology, predominantly to explain health-related behaviours and response to treatment (Conner, 1993). The first type focuses on individual's understanding of the causes of health-related events and are best typified by attribution models (e.g., King, 1982). The second types are more diverse in nature and attempt to predict future health-related behaviours and outcomes. These include the Health Belief Model (HBM), Health Locus of Control (HLOC), Protection Motivation Theory (PMT), Theory of Reasoned Action / Theory of Planned Behaviour (TRA / TPB) and Self-Efficacy (SE). Other models include self-regulation theory, the trans-theoretical model of change, the precaution-adoption process and the model of goal achievement. To the prediction of health behaviours at present, the most widely used models are HBM, TPB, HLOC, PMT and SE.

Health Belief Model (HBM):

This model was originally developed by US public health researchers attempting to develop models upon which to base health education programs (Hochbaum, 1958; Rosenstock, 1966). The model attempts to conceptualize the health beliefs which make a behavior more or less attractive. In particular, the key health beliefs were seen to be the likelihood of experiencing a health problem, the severity of the consequences of the health problem and the perceived costs and benefits of the health behavior. Thus, the HBM employs two aspects of individual's representations of health behavior in response to threat of illness: perceptions of the threat of illness and evaluation of the effectiveness of behaviours to counteract this threat. Threat perceptions depend upon two beliefs: the perceived susceptibility to the illness and the perceived severity of the consequences of the illness. Together these two variables determine the likelihood of the individual following a health-related action, although their effect is modified by individual differences in demographic variables, social pressure and personality. The particular action taken is determined by evaluation of the possible alternatives. This behavioural evaluation depends upon beliefs concerning the benefits or efficacy of the health behavior and the perceived costs or barriers to performing the behavior. Hence, individuals are likely to follow a particular health action if they believe themselves to be susceptible to a particular condition or illness which they consider to be serious, and believe the benefits of the action taken to counteract the condition or illness outweigh the costs.

Cues to action and health motivation are two other variables commonly included in the model. Cues to action include a diverse range of triggers to the individual taking action and are commonly divided into factors which are internal (e.g., physical symptom) or external (e.g., mass media campaign, advice from others such as physicians) to the individual (Janz & Becker, 1984). Becker (1974) has argued that the HBM should also contain a measure of health motivation (readiness to be concerned about health matters) because certain individuals may be predisposed to respond to cues to action because of the value they place on their health. Other influences upon the performance of health behaviours, such as demographic factors or psychological characteristics (e.g., personality, peer pressure, perceived control over behavior), are assumed to exert their effect via changes in the six components of the HBM.

Theory of Planned Behaviour (TPB):

The TPB was developed by social psychologists and has been widely applied to the understanding of a variety of behaviours including health behaviours. The TPB details how the influences upon an individual determine the individual's decision to follow a particular behavior. This theory is an extension of the widely applied Theory of Reasoned Action (TRA). The TPB suggests that the proximal determinants of behavior are intentions to engage in that behavior and perceived behavioural control over that behavior. Intentions represent a person's motivation in the sense of his or her conscious plan or decision to exert effort to perform the behavior. Perceived behavioural control is a person's expectancy that performance of the behavior is within his/her control. Control is seen as a continuum with easily-executed behaviours at one end and behavioural goals demanding resources, opportunities and specialized skills at the other. Intentions are determined by three variables. The first is attitudes, which are the overall evaluations of the behavior by the individual. The second is subjective norms, which consist of a person's beliefs about whether significant others think he/she should engage in the behavior. The third is perceived behavioural control (PBC), which is the individual's perception of the extent to which performance of the behavior is within his/her control. In addition, to the extent that PBC reflects actual control, it is predicted to directly influence behavior

Just as intentions are held to have determinants, so the attitude, subjective norm and perceived behavioural control components are also held to have determinants. The attitude component is a function of a person's salient behavioural beliefs, which represent perceived likely consequences of the behavior. Following expectancy-value conceptualizations (Peak, 1955), the model quantifies consequences as being composed of the multiplicative combination of the judged likelihood that performance of the behavior will lead to a particular outcome and the evaluation of that outcome. These expectancy value products are then summed over the salient consequences. Subjective norm is a function of normative beliefs, which represent perceptions of specific salient others' preferences about whether one should or should not engage in a behavior. In the model, this is quantified as the subjective likelihood that specific salient groups or individuals (referents) think the person should or should not perform the behavior, multiplied by the person's motivation to comply with that referent's expectation. Motivation to comply is the extent to which the person wishes to comply with the specific wishes of the referent on this issue. These products are then summed across salient referents. Judgments of perceived behavioural control are influenced by beliefs concerning access to the necessary resources and opportunities to perform the behaviour successfully, weighted by the perceived power of each factor (Ajzen, 1988, 1991). The perceptions of factors likely to facilitate or inhibit the performance of the behaviour are referred to as control beliefs. These factors include both internal control factors (information, personal deficiencies, skills, abilities, and emotions) and external control factors (opportunities, dependence on others, barriers). Ajzen (1991) has suggested that each control factor is weighted by its perceived power to facilitate or inhibit performance of the behaviour. The model qualifies these beliefs by multiplying the frequency or likelihood of occurrence of the factor by the subjective perception of the power of the factor to facilitate or inhibit the performance of the behaviour. So, according to the TPB, individuals are likely to follow a particular health action if they believe that the behaviour will lead to particular outcomes which they value, if they believe that people whose views they value think they should carry out the behaviour, and if they feel that they have the necessary resources and opportunities to perform the behaviour.

Health Locus of Control (HLOC):

The origin of this model can be traced back to Rotter's (1954) social learning theory which states that the likelihood of a behaviour occurring in a given situation is a function of the individual's expectancy that the behaviour will lead to a particular reinforcement and the extent to which the reinforcement is valued. As well as being applied on a specific level, Rotter argued that social learning theory could be applied on a general level such that individuals may have generalized expectancy beliefs which cut across situations. It was at this generalized level that Rotter introduced the distinction between internal and external locus of control orientations, with "internals" believing that events are a consequence of their own actions and thereby under personal control and "externals" believing that events are unrelated to their actions and thereby beyond their personal control.

According to HLOC theory, individuals who have strong internal HLOC beliefs should be more likely to engage in health-promoting behaviours. Conversely, those who believe that their health is due to chance or fate should be less likely to engage in health-promoting behaviours. The prediction of powerful others HLOC is less clear cut. Strong powerful others HLOC beliefs may reflect receptivity to health messages endorsed by health professionals. Alternatively, strong powerful others HLOC beliefs may indicate a strong belief in the ability of health professionals to cure subsequent illnesses and may be unrelated or negatively related to the performance of health-promoting behaviours.

Protection Motivation Theory (PMT):

PMT (Rogers, 1983) was originally developed as a framework for understanding the effectiveness of health-related persuasive communications, although more recently it has also been used to predict health protective behaviour. Roger's (1983) (PMT) outlines the cognitive responses resulting from fear appeals in more detail. It is argued that various environmental (e.g., fear appeals) and intrapersonal (e.g., personality variables) sources of information can initiate two appraisal processes: threat appraisal and coping appraisal. Threat appraisal focuses on the source of the threat and the factors that may increase or decrease the probability of the maladaptive response. Both the perceived severity of the threat and the individual's perceived vulnerability to the threat are seen to inhibit maladaptive responses. However, there may be a number of intrinsic (e.g., pleasure) and extrinsic (e.g., social approval) rewards which may serve to increase the likelihood of maladaptive responses. Coping appraisal focuses on one's ability to cope with the threat and the factors that may increase or decrease the probability of an adaptive response. Both the belief that the recommended action will be effective in reducing the danger (i.e., response efficacy) and the belief that one is capable of performing the recommended action (i.e., self-efficacy) are likely to increase the probability of an adaptive response, although various response costs (e.g., financial cost) associated with performing an adaptive response will serve to inhibit such a response.

Protection motivation results from the two appraisal processes and is a positive function of beliefs about severity, vulnerability, response efficacy, and self-efficacy, and a negative function of beliefs about the rewards associated with the maladaptive response and the response costs of the protective behaviour. Moreover, for protection motivation to be elicited, it is necessary for the rewards associated with the maladaptive response to be outweighed by perceptions of severity and vulnerability, and the response costs of the protective behaviour to be outweighed by perceptions of response efficacy and self-efficacy. Protection motivation, which is usually measured by behavioural intentions, is seen to arouse, direct, and sustain protective behaviour.

Self-Efficacy (SE):

SE is one of the most powerful predictors of health behaviour (Wallston, 1992). It has its origins in Bandura's (1977) social cognitive theory which states that behaviour is a function of both incentives (i.e., reinforcements) and expectancies. Three kinds of expectancies can be identified, these being situation-outcome expectancies which refer to beliefs about how events are connected, outcome expectancies which refer to beliefs about the consequences of performing a behaviour, and self-efficacy expectancies which refer to beliefs about one's ability to perform the behaviour. Thus in order to perform a health behaviour, individuals must value their health (i.e., incentive), believe that their current lifestyle poses a threat to their health (i.e., situation-outcome expectancy), believe that adopting the new behaviour will reduce the threat to their health (i.e., outcome expectancy) and believe that they are capable of performing the behaviour (i.e., self-efficacy expectancy). While all these beliefs are seen to be important in the initiation and maintenance of health behaviour, self-efficacy expectancies are seen to be the most important. Individuals with strong self-efficacy beliefs are believed to develop stronger intentions to act, to expend more effort to achieve their goals, and to persist longer in the face of barriers and impediments (Bandura, 1991).

SE beliefs are therefore believed to play a crucial role in the determination of health behaviour.

According to Bandura (1977, 1982), such beliefs can be conceptualized and measured in terms of three parameters; magnitude, strength, and generality. The first parameter refers to the level of difficulty of the behaviour. Individuals with low-level expectations feel capable of performing only very simple behaviours, whereas individuals with high-level expectations feel capable of performing even the most difficult of behaviours. In this way it is possible to assess individuals' expectations about their level, or magnitude, of performance. The second parameter refers to individuals' confidence that they could perform a specific behaviour, while the third parameter refers to the generality of expectations across situations or domains. The measurement of self-efficacy usually focuses on the strength of the self-efficacy expectation, although it will often incorporate the magnitude of expectation.

SCOPE AND APPLICATION OF PSYCHOLOGICAL PRINCIPLES IN HEALTH, ILLNESS AND HEALTH CARE

Health psychologists can work with people on a one-to-one basis, in groups, as a family, or at a larger population level.

Clinical Health Psychology (ClHP)

ClHP is the application of scientific knowledge, derived from the field of health psychology, to clinical questions that may arise across the spectrum of health care. ClHP is one of many specialty practice areas for clinical psychologists. It is also a major contributor to the prevention-focused field of behavioural health and the treatment-oriented field of behavioural medicine. Clinical practice includes education, the techniques of behaviour change, and psychotherapy. In some countries, a clinical health psychologist, with additional training, can become a medical psychologist and, thereby, obtain prescription privileges.

Public Health Psychology (PHP)

PHP is population oriented. A major aim of PHP is to investigate potential causal links between psychosocial factors and health at the population level. Public health psychologists present research results to educators, policy makers, and health care providers in order to promote better public health. PHP is allied to other public health disciplines including epidemiology, nutrition, genetics and biostatistics. Some PHP interventions are targeted toward at-risk population groups (e.g., undereducated, single pregnant women who smoke) and not the population as a whole (e.g., all pregnant women).

Community Health Psychology (CoHP)

CoHP investigates community factors that contribute to the health and well-being of individuals who live in communities. CoHP also develops community-level interventions that are designed to combat disease and promote physical and mental health. The community often serves as the level of analysis, and is frequently sought as a partner in health-related interventions.

Critical Health Psychology (CrHP)

CrHP is concerned with the distribution of power and the impact of power differentials on health experience and behaviour, health care systems, and health policy. CrHP prioritizes social justice and the universal right to health for people of all races, genders, ages, and socioeconomic positions. A major concern is health inequalities. The critical health psychologist is an agent of change, not simply an analyst or cataloger. A leading organization in this area is the International Society of Critical Health Psychology.

APPLICATIONS

Improving doctor–patient communication

Health psychologists aid the process of communication between physicians and patients during medical consultations. There are many problems in this process, with patients showing a considerable lack of understanding of many medical terms, particularly anatomical terms (e.g., intestines). One area of research on this topic involves "doctor-centered" or "patient-centered" consultations. Doctor-centered consultations are generally directive, with the patient answering questions and playing less of a role in decision-making. Although this style is preferred by elderly people and others, many people dislike the sense of hierarchy or ignorance that it inspires. They prefer patient-centered consultations, which focus on the patient's needs, involve the doctor listening to the patient completely before making a decision, and involving the patient in the process of choosing treatment and finding a diagnosis.

Consultant Health Psychologist:

A consultant health psychologist will take a lead for health psychology within public health, including managing tobacco control and smoking cessation services and providing professional leadership in the management of health trainers.

Principal Health Psychologist: A principal health psychologist could, for example lead the health

Improving adherence to medical advice

Health psychologists engage in research and practice aimed at getting people to follow medical advice and adhere to their treatment regimens. Patients often forget to take their pills or consciously opt not to take their prescribed medications because of side effects. Failing to take prescribed medication is costly and wastes millions of usable medicines that could otherwise help other people. Estimated adherence rates are difficult to measure; there is, however, evidence that adherence could be improved by tailoring treatment programs to individuals' daily lives.

Managing pain

Health psychology attempts to find treatments to reduce or eliminate pain, as well as understand pain anomalies such as episodic analgesia, causalgia, neuralgia, and phantom limb pain. Treatments for pain involve patient-administered analgesia, acupuncture (found to be effective in reducing pain for osteoarthritis ofthe knee), biofeedback and cognitive behavior therapy.

HEALTH PSYCHOLOGIST ROLES

Below are some examples of the types of positions held by health psychologists within applied settings such as the UK's NHS (National Health Service) and private practice.psychology service within one of the UK's leading heart and lung hospitals, providing a clinical service to patients and advising all members of the multidisciplinary team.

Health Psychologist: An example of a health psychologist's role would be to provide health psychology input to a center for weight management. Psychological assessment of treatment, Development and delivery of a tailored weight management program, and advising on approaches to improve adherence to health advice and medical treatment.

Research Health Psychologist: Research health psychologists carry out health psychology research, for example, exploring the psychological impact of receiving a diagnosis of dementia, or evaluating ways of providing psychological support for people with burn injuries. Research can also be in the area of health promotion, for example investigating the determinants of healthy eating or physical activity or understanding why people misuse substances.

Health Psychologist in training / Assistant Health Psychologist: As an assistant / in training, a health psychologist will gain experience assessing patients, delivering psychological interventions to change health behaviours, and conducting research, whilst being supervised by a qualified health psychologist.

CONCLUSION

Health psychology, like other areas of applied psychology, is both a theoretical and applied field. Health psychologists employ diverse research methods. These methods include controlled randomized experiments, quasi-experiments, longitudinal studies, time-series designs, cross-sectional studies, case-control studies, qualitative research as well as action research. Health psychologists study a broad range of variables including cardiovascular disease, (cardiac psychology), smoking habits, the relation of religious beliefs to health, alcohol use, social support, living conditions, emotional state, social class, and more. Some health psychologists treat individuals with sleep problems, headaches, alcohol problems, etc. Other health psychologists work to empower community members by helping community members gain control over their health and improve quality of life of entire communities.

UNIT- II

CENTRAL NERVOUS SYSTEM

DISEASES, ASSESSMENT AND REHABILITATION

Class Presentation

Subject: Behavioural Medicine

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Introduction

Central Nervous System

1. The central nervous system CNS is responsible for integrating sensory information and responding accordingly. It consists of two main components:

   A. The **spinal cord** serves as a conduit for signals between the brain and the rest of the body. It also controls simple musculoskeletal reflexes without input from the brain.
   B. The **brain** is responsible for integrating most sensory information and coordinating body function, both consciously and unconsciously. Complex functions such as thinking and feeling as well as regulation of homeostasis are attributable to different parts of the brain.

2. The brain and spinal cord share some key anatomic features:

   A. Living nervous tissue has the consistency of jelly and requires special protection from physical damage. The entire CNS is encased in bone. The brain is within the cranium, while the spinal cord runs within a canal through the vertebrae.
   B. Within its bony case, the entire CNS is bathed in a cerebrospinal fluid (CSF), a colourless fluid produced by special structures in the brain. CSF provides a special chemical environment for nervous tissue, as well as an additional buffer against physical damage.
   C. The special chemical environment of nervous tissue is maintained by the relatively impermeable membranes of capillaries in the CNS. This feature is known as the blood-brain barrier.
   D. There are two general types of tissue in the CNS:

      - Gray matter consists of nerve cell bodies, dendrites, and axons. Neurons in gray matter organize either in layers, as in the cerebral cortex, or as clusters called nuclei.

- White matter consists mostly of axons, causing it to look white due to the myelin sheathing of the axons.

The highest region of the brain is the cerebrum, which includes both the cerebral cortex that is visible on the outside of the brain as well as other internal structures. The cerebrum is responsible for conscious sensation and voluntary movement, as well as advanced functions such as thinking, learning and emotion.

Any problems in parts or subparts of the central nervous system can lead to significant disturbances in the human behavioural and cognitive functioning among the human beings.

Disturbances of the nervous system, both peripheral and central, are manifested in four basic ways:

1. **Destructive or deficiency phenomena** associated with destruction of nerve tissue (e.g., infarction, tumor, trauma) or processes that depress or block nerve function (e.g., anesthesia).
2. **Irritative phenomena** as represented by seizures and the pins-and-needles or burning paresthesias of peripheral neuropathy, both of which represent excessive neuronal firing secondary to pathologic depolarization.
3. **Release phenomena** as represented by the hyperactive reflexes and spasticity of corticospinal system involvement, the tremor of Parkinson's disease, the excessive emotional responses following bilateral corticobulbar system loss, sedative withdrawal hyperactivity and the appearances of primitive reflex responses that may reappear in dementing illnesses (e.g., with generalized cerebral cortical degeneration).
4. **Compensation phenomena**. There may be appropriate or inappropriate compensations. For example, appropriate compensation would include visual-motor compensation for the nystagmus and vertigo of vestibular disease, circumduction of the paretic leg on walking to avoid tripping, a high-stepping gait to avoid tripping with a foot drop, or a broad-based gait to compensate for ataxia. Inappropriate compensations would include tendon contractures, for example.

Combinations of these disturbances are the rule. The person who has cerebral infarction is hemiparetic, develops spasticity, may develop seizures early or late, and compensates for hemiparesis appropriately with a circumducting gait. If physical rehabilitation is not carried out, s/he can develop a functionally inappropriate flexion contracture of the arm.

Although there are many diseases and disorders related to the problems in central nervous system, we will be discussing about the following here i.e., cerebrovascular (stroke), developmental (cerebral palsy), degenerative (Parkinson's etc), trauma (traumatic brain and spinal cord injury), convulsive (epilepsy), and infectious (AIDS dementia); also will be discussed are the assessment methods for psychological intervention and rehabilitation with such patients.

CEREBROVASCULAR DISEASES

According to the WHO (1970) report, the classification of cerebrovascular diseases (stroke) is partly based on the nature of pathological changes in the brain and partly upon the clinical stage of disease.

I. Pathological changes in the brain
  - *Subarachnoid haemorrhage:* a haemorrhage originating in the subarachnoid space.
  - *Intracerebral haemorrhage:* a haemorrhage originating in the brain parenchyma
  - *Cerebral ischaemic necrosis:* death of brain parenchyma due to lack of blood supply however caused. This includes infarction both haemorrhagic and anaemic (vascular dementia also included).

II. Clinical stage
  - *Transient cerebral ischaemic:* This is focal neurologic deficit commonly lasting for few minutes but never more than 24 hours, leaving no residual deficit, occurring in patients with vascular disease, and frequently showing tendency to reoccur. A premonitory attack of migraine is not included in this category.

Symptoms of stroke

The symptoms of a stroke usually appear suddenly. Initially the person may feel sick, and look pale and very unwell. They may complain of a sudden headache. They may have sudden numbness in their face or limbs, particularly down one side of their body. They may appear confused and have trouble talking or understanding what is being said to them. They may have vision problems, and trouble walking or keeping their balance. Sometimes a seizure (fit) or loss of consciousness occurs.

Depending on what function the damaged part of the brain had, a person may lose one or more of the following functions:

- ability to perform movements — usually affecting one side of the body
- speech
- part of vision
- co-ordination
- balance
- memory and
- Perception.

In chronic cases:

- Hemiplegia
- Hemihypasthesia
- Hemianopia
- Aphasia
- Pseudobulbar paresis
- Dementia
- depression
- Syndrome of aesthetic reaction and emotional lability
- Inattention/personality problems/amnesic syndromes
- Epilepsy

Assessment of the stroke and its consequent dysfunctions

Following scales have been used throughout the researches for assessing the severity of stroke and the various dysfunctions resulting because of it:

- National Institutes of Health Stroke Scale (1989)
- Canadian Stroke Scale (1986)
- Glasgow Coma Scale (1974)
- Rankin Scale (1957), to assess the extent of disability after stroke
- Barthel Index (1965) to assess the daily functioning after stroke.
- Screening Instrument for Neuropsychological Impairments in Stroke (SINS)
- Clock drawing test

It is to be noted that before proceeding towards the neuropsychological testing of stroke patients some of the necessary physical examinations must be concluded, these would give a better picture for diagnosis and prognosis of the disease. Such test includes:

- History of previous episodes
- Physical examination
- Laboratory examination (urine analysis, blood analysis, electrocardiogram)
- Cerebral angiography
- Examination of cerebrospinal fluid
- EEG

PSYCHOLOGICAL INTERVENTIONS AND REHABILITATION

Strokes affect people in different ways depending on the type of stroke and area of the brain affected. Often old skills have been lost, so new ones will need to be learned. It is also important to maintain and improve physical condition whenever possible. Rehabilitation should begin as soon after a stroke as possible and may continue at home.

Following methods have been used over the years for rehabilitation in stroke patients:

A. **Psychoeducation: Psychoeducation** refers to the education offered to people with a mental health condition. Family members are also included. A goal is for the consumer to understand and be better able to deal with the presented illness. Also, the patient's own strengths, resources and coping skills are reinforced, in order to understand that relapse is a part of their recovery, and contribute to their own health and wellness on a long-term basis.

B. **Constraint-Induced-Movement Therapy**: Constraint-induced movement therapy (CIMT) emerged from 19th century primate models of deafferentation and was developed as a clinical intervention by Taub and Uswatte [1], Wolf et al. [2], and others. It has two major components: restraint of the less affected upper extremity (UE) and training of the hemiplegic UE using a shaping paradigm. Shaping consists of practice of skilled tasks calibrated to the patient's motor capacity, advancement of task difficulty as the patient improves, and generous feedback and encouragement.

C. **Gait Retraining**: Hemi-paretic stroke survivors want to walk. Although regaining strength may seem paramount, the latest RCT of the effects of functional strength training versus conventional physiotherapy concluded that leg strengthening has no effect on walking speed, at least in persons with some walking capacity. Even as an adjunct, the added benefit of focused strength training is not clear.Body weight support treadmill training (BWSTT) has received much attention because it facilitates massed practice of gait and allow patients to focus on more normal gait biomechanics.

D. Virtual Reality, Mirror Therapy, and Mental Imagery: Virtual reality (VR) has potential advantages: it can provide massed practice of a skill with many repetitions, task practice can be finely manipulated and made engaging, and sensory input can be controlled. Research in this area is still in the early stages, and studies often are underpowered and lack controls. There have been no direct comparisons of immersive VR with conventional rehabilitation. In mirror therapy (MT), the reflection of the moving unaffected limb is superimposed on the affected limb, creating the illusion of movement in the affected limb. Originally used as a treatment for phantom limb pain, MT now is applied to hemiplegia. As an adjunctive therapy, it is associated with lasting improvement in UE function [23•] and has shown promise for severe distal plegia [24]. However, little is known about the mechanism of MT in stroke. Preliminary results from a study using VR to simulate MT showed increased activation and excitability of the ipsilesional sensorimotor cortex in response to the movement of the unaffected hand [25]. These intriguing findings await replication. Might we dispense with robots, VR, mirrors, and other gadgets entirely? The long-held impression that mental imagery can improve motor performance received scientific support recently, as Page et al. [26] reported that therapist-guided adjunctive mental practice was associated with increased dexterity and changes in patterns of cortical activation. RCTs are ongoing (Table 1).

- **Speech therapy for aphasia: Impairment-based therapies** are aimed at improving language functions and consist of procedures in which the clinician directly stimulates specific listening, speaking, reading and writing skills.
- **Communication-based** (also called consequence-based) **therapies** are intended to enhance communication by any means and encourage support from caregivers. These therapies often consist of more natural interactions involving real life communicative challenges.

Cerebral Palsy

Cerebral palsy is an umbrella term denoting a group of non-progressive,[1][2] non-contagious motor conditions that cause physical disability in human development, chiefly in the various areas of body movement.[3] Scientific-consensus still holds that CP is neither genetic nor a disease, and it is also understood that the vast majority of cases are congenital, coming at or about the time of birth, and/or are diagnosed at a very young age rather than during adolescence or adulthood. It can be defined as a central motor dysfunction affecting muscle tone, posture and movement resulting from a permanent, non-progressive defect or lesion of the immature brain.

TYPES OF CEREBRAL PALSY

Doctors classify cerebral palsy into several types, based on the location and extent of brain damage, the body parts affected, and the kinds of tone and movement difficulties present.

Spastic

Injuries to the cerebral cortex can result in spastic cerebral palsy, which causes abnormally stiff muscles. This condition — the most common type of cerebral palsy — can also cause bone deformities and shortened muscles (contractures). Spastic cerebral palsy is divided into further classifications, depending on which limbs are affected:

■ Diplegia affects the legs (typically both) more than the arms. It's most common in premature babies.

■ Hemiplegia affects one side of the body. It's most common in babies who have strokes or traumatic brain injuries.

■ Quadriplegia affects all four limbs. It's most common in babies who experience an interruption in oxygen supply.

■ Monoplegia affects one limb.

■ Triplegia affects three limbs.

Athetoid

Injuries to the basal ganglia can result in athetoid cerebral palsy, which causes involuntary muscle movements. The movements often interfere with speaking, feeding, grasping, walking and other skills requiring coordination.

Dystonia

Injuries to the basal ganglia also can result in dystonia, which causes fluctuating muscle tone. Although tone is sometimes low, it increases when a person attempts to move or experiences heightened emotions.

Ataxic

Injuries to the cerebellum can result in ataxic cerebral palsy, which causes poor coordination. That, in turn, affects balance, posture and controlled movements. Ataxic cerebral palsy can cause unsteadiness when walking and difficulties with motor tasks.

Mixed

Injuries to multiple brain areas — usually the cerebral cortex and basal ganglia — can result in more than one kind of abnormal muscle tone. For example, someone could have spasticity and dystonia, or dystonia and rigidity. By identifying what type(s) of cerebral palsy a child has, doctors and therapists can recommend treatments. They also can give caregivers a better idea of what the child's future might hold. Some potential problems can be prevented or corrected if addressed early in a child's life.

SYMPTOMS

Primary effects:

- Abnormal muscle tone
- Poor motor control
- Abnormal reflexes
- Balance and movement problems
- Muscle weakness

Secondary effects:

- Inadequate muscle growth
- Malformed bones and joints

Tertiary effects:

These are the coping response that people use-especially during walking-to compensate for cerebral palsy's primary and secondary effects.

COGNITIVE IMPAIREMENTS

- Attention Span problems
- Visual Problems (such as- astigmatism, blurry vision, farsightedness/near sightedness, strabismus, lazy eye etc)
- Hearing loss
- Speech- Language difficulties
- Feeding difficulty, malnutrition, low bone density
- Respiratory and sleep related problems
- Seizures with epilepsy

ASSESSMENT TOOLS FOR DETECTING CEREBRAL PALSY

- Ages and Stages Questionnaire
- BRIGANCE Screens

- Child Development Inventories (such as the Denver II and the Minnesota Child Development Inventory)
- Parents' Evaluations of Development Status
- Modified Checklist for Autism in Toddlers (M-CHAT)

Although these questionnaires are important, they may not consistently identify problems, such as autism, in older children, adolescents, and young adults or in a child any age with a mild condition. However, when used routinely, they can help identify subtle developmental delays that may otherwise be missed.

REHABILITATION FOR CEREBRAL PALSY

A. **Neurosurgical procedures**: The neurosurgical procedures for optimum reduction in spasticity are classified according to the site and the nature of the surgery:

   - *Anatomico-physiological classification*: According to the site involved in maintenance of the muscle tone, the surgical procedures are classified into following types:

     (a) Segmental: The procedure that interrupts teh spinal circuit required in the maintenance of muscle tone in segmental procedure.

     (b) Suprasegmental: The procedure that stimulates or ablates various suprasegmental control sites contributing (nuclei or tracts) to muscle tone maintenance is called as suprasegmental procedure.

B. **Patho-physiological classification:** According to the nature of surgery the procedure are classified into following two types:

   - ***Ablative:*** In this procedure a permanent destructive lesion is created in the neural tissue e.g. fasciculotomy, rhizotomy, thalamotomy, etc. It is also known as neurolytic procedure.
   - *Non*-ablative: In these procedures, with the help of neurostimulation or neurochemicals, a favourable neural response is obtained that can be reversed at any moment of the time e.g. spinal cord stimulation.

All the neurosurgical procedures are selected based on the thorough clinical examinations and need for the habilitation.

C. **Occupational Therapy:** The occupational therapists help improve the fine motor skills- the arm and the hand movements they use to perform such daily tasks as dressing, playing, grooming, eating. The therapist also addresses the visuo-motor and visuo-perceptual skills which are necessary for reading, writing and other work.

D. **Physical Therapy:** Therapist helps in developing gross motor skills like balance, coordination and strength needed for walking and other functions.

These kinds of physical therapy also includes: aqua therapy and hippo therapy:-

i) ***Aqua Therapy***: *Aqua therapy (also known as Aquatic therapy), under the supervision of a trained and certified professional therapist, provides deep, intense exercise within a soothing and comforting environment. This form of therapy promotes physical functioning with the aid of water's restorative and detoxifying properties. Water buoyancy makes aerobic and anaerobic exercises safe and effective by allowing an individual to ambulate freely in a way that doesn't place undue stress on the musculoskeletal system from forces such as gravity and body weight. Aqua therapy takes place in both heated and non-heated environments, although warm water therapy provides a massage effect on muscles, joints and ligaments.*

ii) ***Hippo Therapy:*** *Hippo-therapy is a form of physical, occupational and speech therapy that uses equine (horse) movement to develop and enhance neurological and physical functioning by channelling the movement of the horse. This further develops physical and cognitive abilities. Hippo-therapy is not to be confused with therapeutic horseback riding, in which individuals are taught specific riding skills. Hippo-therapy is built on the concept that the individual and variable gait, tempo, rhythm, repetition and cadence of a horse's movement can influence human neuromuscular development in humans. Horseback riding triggers a series of complex physical and mental reactions; such as making physical adjustments to maintain proper alignment on the horse. Riders must also plan movements to maintain balance on the horse, and be able to interact with the animal.*

E. **Speech and language therapy:** Speech and language pathologists assess the speech and swallowing problems and help people improve their communication skills. If a patient has problems in communicating- augmentation methods for communication could be used such as sigh language, devices such as picture board or voice output devices.

F. **Adaptive equipment's and assistive technology:** A team of therapist, physician, nurses and onsite assistive technical specialist work together to identify, design and help people to get access to equipment that contribute to tone management and foster independent functioning. These equipments include :
   - Adaptive bath chairs and toilet chairs
   - Braces (orthoses), casts and splints to increase stability, improve alignment and reduce muscle contractures, and preserve or even increase –function and range of motion.
   - Specialized beds, powered and manual wheelchairs, walkers for individuals to achieve individual mobility.

G. **Biofeedback in cerebral palsy:** Various biofeedback procedures with modification have been utilized in the treatment of inappropriate movements involved in cerebral palsy. Harris et al. Mounted motion transducers on pendulums suspended within a helmet worn by the patient. The task was to maintain a steady erect head posture. Whenever an inappropriate movement was produced, an auditory signal was presented and a visual feedback from four lights provided information as to the direction of error in space.

Management of spasticity

The treatment of spasticity, like all rehabilitation process, must start with the establishment of specific achievable goals and carefully planned strategy to achieve those goals. There are three potential aims of treatment, to improve function, to reduce the risk of unnecessary complication and to alleviate pain. Following treatment approaches have been used in treatment of spasticity in specific for cerebral palsy:

1. **Positioning and seating:** Correct positioning certainly for the patient is the most important aspect of management. Proper seating is vital. The fundamental principle of seating is that the body should be contained in a balanced, symmetrical and stable

posture which is both comfortable and maximises function. There are many different types of seating systems. All should have the ultimate aim of stabilisation of the pelvis without lateral tilt or rotation, but with a slight anterior tilt so that the spine can adopt normal lumbar lordosis, thoracic kyphosis and cervical lordosis.

2. **Splinting and casting:** The application of splints and casts can prevent the formation of contractures in the spastic limbs and serial casting can improve the range of movement in a joint that is already contracted- a new cast being applied every few days as the range improves.
3. **Oral Medication:** Oral anti-spastic medication has very limited use in the overall management of spasticity. All available agents are limited by their side effects, commonly drowsiness and weakness.

Traumatic Brain and Spinal Cord Injury

Traumatic brain injury (TBI) is leading cause of short term and long term morbidity and mortality. With rapid growth of population without adequate infrastructure and safe transport, the problem is on the increase. Most patients of TBI have a protracted course of rehabilitation. It is the neurobehavioral outcome, rather than the neurological deficits which are really taxing to the patients and the family.

A spinal cord injury is complete if there is no somatic motor or sensory function below the lever of injury. If the arms are spared the patient has paraplegia. If they are involved he has tetraplegia. The level of injury is the lowest intact spinal cord segment. If there is residual function several segments below this, them the injury is incomplete and the patient has either paraparesis or tetraparesis. Use of the terms quadriplegia and quadriparesis should be avoided.

PATHOLOGY OF CLOSED TRAUMATIC BRAIN INJURY

One of the major changes in understanding the closed head injury has been the concept of Diffuse Axonal injury. Widespread damage to the brain due to primary injury per se, not due to secondary insults of herniation or perfusion deficits, is now widely accepted. Though first named so by Adams, it was described by Strich. There is microscopic axonal swelling due to retraction. Grossly there are hemorrhagic and neurotic lesions of corpus callosum, dorsolateral quadrant of rostral pons. Long term cognitive deficits may also result from acute subdural haematomas, contusions and Intracerebral haematomas.

EARLY ASSESSMENTS AND OUTCOMES

A survey of the available literature shows that there is lack of uniformity in defining the terms involved in morbidity of disability. The Glasgow Coma Scale (GCS) is at present the most widely used and accepted scale. It is perhaps inadequate and insensitive for monitoring patients who are likely to deteriorate.

Also the use of X-rays, Computerized tomography CT scans and MRI scans are important in the assessment processes of the TBI and SCI conditions.

CLASSIFICATION OF PATIENT WITH TBI/SCI

- Grade 1- Transient loss of consciousness (<5 min), now alert, oriented without neurological deficit. GCS 14-15
- Grade 2- Previous loss of consciousness but able to follow at least a simple command, other neurological deficit. GCS 9-13
- Grade 3- Previously unresponsive (<5min) and now not following even a simple command. Pupil’s unequal, inappropriate words. GCS <9.
- No evidence of brain function (brain death).

COMPLICATIONS

1. Post traumatic epilepsy
2. Post-concussion syndrome
3. Cognitive impairments
4. Memory impairments such as- retrograde amnesia, post-traumatic amnesia.
5. Behavioural changes including changes in personality changes and changes in interpersonal communication changes, emotional changes.

MANAGEMENT PROCESS FOR TRAUMATIC BRAIN INJURY & SPINAL CORD INJURY

1. **Management of associated injuries**: Associated injuries must be treated well to ensure optimum rehabilitation outcome. Among the more important are the following:
   i) *Brain:* Successful rehabilitation following TBI/SCI is dependent on the total involvement of the disabled person. Impairment of personality, memory, concentration and intellect can profoundly alter outcome. Good executive function is of particular importance in enabling the spinal cord or brain injured person to lead a safe and well-integrated life.
   ii) *Limb*-joints and bones: In cases of spinal cord injuries the patients are more depended on their arms than prior to injury. Joint damage and contractures are frequently very disabling.
   iii) *Peripheral nerve injuries, esp. Brachial plexus:* Paraplegics require both arms for most activities. The affected arm cannot cope so well with the transfers and wheelchair control.

2. **Pain:_** Musculo-skeletal and neurogenic pain are common following spinal cord injury. Cognitive Behavioural Therapy (CBT) for pain helps patients with pain to understand the relationship between one's physiology (e.g., pain and muscle tension), thoughts, emotions, and behaviours. A main goal in treatment is cognitive restructuring to encourage helpful thought patterns, targeting a behavioural activation of healthy activities such as regular exercise and pacing. Lifestyle changes are also trained to improve sleep patterns and to develop better coping skills for pain and other stressors using various techniques (e.g., relaxation, diaphragmatic breathing, and even biofeedback). Hypnosis has also at times being used in managing pain. Apart from the psychotherapeutic techniques, anaesthetics and analgesias are prescribed to the patients but the chances of addiction are high due to which they are moderately administered.

3. **Memory and attention:**
   i) **Direct Attention Training:** The patient needs to improve his ability to focus on the stocking task and resist distractions. The existing evidence supports direct training on the task itself, with a hierarchical progression of increasing attention

demands from simple to complex distracters (M. Sohlberg et al., 2003). There is little evidence that attention training generalizes to novel tasks, so J.B. must be trained on the task he will perform at work.

ii) **External Memory Aids:** There is no direct evidence to support memory "retraining exercises," or the use of internal memory strategies like mnemonics, so these are not recommended. Patients could benefit from the use of an external memory aid such as a map of product locations in the dairy department, perhaps as part of a notebook or portable digital assistant that also allows him to jot down the names of products that need restocking. The existing evidence supports the use of both high- and low-technology aids to support memory in individuals with declarative learning impairments, and also in individuals with problems in executive memory (i.e., the ability to use strategies to search memory) (Sohlberg et al., 2007). These aids must be ecologically relevant and acceptable to the individual.

iii) **Behavior Management:** There are two contexts in which the patients. might need strategies to support positive interactions: when he becomes angry and loses his temper, and in conversation with customers. The existing evidence suggests that both could be addressed with either antecedent behavior management strategies, such as creating verbal scripts and routines (e.g., an "escape" routine when he is losing his temper, greeting routines for customers), or consequence-oriented management such as reinforcement of either positive behaviors or low rates of negative behaviors (Ylvisaker, Turkstra, & Coelho, 2005). Both have supporting evidence, but there is a trend toward increasing use of antecedent-based management in TBI; individuals with TBI often have impaired reasoning, declarative learning, and behavioral control, so punishment might lead to frustration and failure without improving behavior. Antecedent-based management has been critiqued because it entails providing supports that might not be present in all contexts. It is important, therefore, that supports be systematically withdrawn over time to foster independence in diverse contexts (Ylvisaker, Turkstra, & Coelho, 2005).

iv) **Learning New Skills:** The patient will require training to use his new memory aid, behavioral routines, and scripts. The research evidence suggests that individuals like J.B. with declarative learning impairments learn best when training capitalizes on intact procedural memory skills (Ehlhardt et al., 2008). Declarative learning may be achieved in a single trial, includes learning from errors, may be consciously directed and generalized, and is enhanced by strategies such as elaboration and semantic association. By contrast, procedural learning requires multiple repetitions, is probabilistic (i.e., retention is a function of repetition of the correct response), and is dependent on surface features of the learning context rather than semantic associations.

v) **Awareness and Metacognitive Skills:** Individuals with cognitive impairments may realize in retrospect what went wrong, but have trouble figuring out why or what to do so it won't happen again. There is solid evidence that incorporating metacognitive strategies when teaching highly complex activities—such as keeping track of dairy products on the shelves and stocking new items in the appropriate place.

4. **Speech-Swallowing rehabilitation:** The impairment should be assessed and according to the situation it should be treated. For instance person is able to read, speak, write, learn, difficulty in finding words, forming complex sentences, swallowing? Following techniques can be used in such cases:
    a. *Augmentative*-alternative communication: depending upon the severity of impairment either augmentation or alternate methods for communication are used.
    b. *Apraxia/dysarthria:* based on the patients need and severity also the strengths of the patients, the goal of the therapy here is to improve voluntary control through nonspeech and speech exercises (i.e., repeating sounds, sequencing sounds into words).
    c. *Dysphagia:* Based on the results or the dysphagia work-up treatment may include a variety of compensatory techniques. Positioning the patient can compensate for weak structures and increase airway protection .Use of the supraglottic swallow increases airway protection as does a chin tuck position. Adaptive equipment (small-bowled spoon, shortened straw, cups with extended lip, and so forth) is used to control bolus size and allow midline introduction of bolus decreasing labial leakage. Modification of food

consistencies and viscosity of liquids may also be recommended (eg, puree, soft mechanical, thickened liquids).

Epilepsy

Epilepsy is "an occasional, an excessive and a disorderly discharge of nervous tissue" induced by any process involving the cerebral cortex that pathologically increases the likelihood of depolarization and synchronized firing of groups of neurons (John Hughlings Jackson, 1889). There are many potential underlying causes such as metabolic disorders of nerve cells or virtually any disorder that damages cortical tissue including trauma, haemorrhage, ischemia, anoxia, infection, hyperthermia, or the presence of scar tissue relating to prior injury. There are several types of seizures. Broadly, they can be divided into primary generalized seizures and focal onset (localization-related) seizures. In primary generalized seizures, the seizure involves all of the cerebral cortex simultaneously. In focal onset seizures, it involves a localized cluster of neurons having epileptiform activity. While most seizures present with motor correlates, some can present with mainly inhibitory phenomena. The blank, staring episodes of petit mal, the common childhood seizure disorder, are a good example. Seizures are not only recognized by the activity during the main portion of the seizure but also by phenomena that lead up to the clinical seizure (often termed an "aura"), and the condition of the patient after the event (the "post-ictal" state).

Early seizure symptoms (warnings)	**Emotional:**
Sensory/Thought: • Deja vu • Jamais vu • Smell • Sound • Taste • Visual loss or blurring • Racing thoughts • Stomach feelings Strange feelings • Tingling feeling	• Fear/Panic **Physical:** • Chewing movements • Convulsion • Difficulty talking • Drooling • Eyelid fluttering • Eyes rolling up • Falling down • Foot stomping

Seizure symptoms

Sensory/Thought:

- Black out
- Confusion
- Deafness/Sounds
- Electric Shock Feeling
- Loss of consciousness
- Smell
- Spacing out
- Out of body experience
- Visual loss or blurring

After-seizure symptoms (post-ictal)

Thought:

- Memory loss
- Writing difficulty

Emotional:

- Confusion
- Depression and sadness
- Fear
- Frustration
- Shame/Embarrassment

- Hand waving
- Inability to move
- Incontinence
- Lip smacking
- Making sounds
- Shaking
- Staring
- Stiffening
- Swallowing
- Sweating
- Teeth clenching/grinding
- Tongue biting
- Tremors
- Twitching movements
- Breathing difficulty
- Heart racing

Physical:

- Bruising
- Difficulty talking
- Injuries
- Sleeping
- Exhaustion
- Headache
- Nausea
- Pain
- Thirst
- Weakness
- Urge to urinate/defecate

ASSESSMENTS USED FOR EPILEPSY

- Electrocardiograph
- Blood testing
- Positron emission tomography
- Spinal Tap
- Bender visuo motor Gestalt test
- Luria Nebraska Neuropsychological Battery
- WADA test
- Serum prolactin levels

MANAGEMENT AND REHABILITATION IN EPILEPSY

Following treatment and therapeutic methods are used in treatment for epileptic disorders:

A. **Psychopharmacological treatments**: Most commonly used drugs for the treatment of epilepsy are: carbamazepine (common US brand name Tegretol), clorazepate (Tranxene), clonazepam (Klonopin), ethosuximide (Zarontin), felbamate (Felbatol), fosphenytoin (Cerebyx), gabapentin (Neurontin), lacosamide (Vimpat), lamotrigine(Lamictal), levetiracetam (Keppra), oxcarbazepine (Trileptal), phenobarbital (Luminal), phenytoin (Dilantin), pregabalin (Lyrica), primidone (Mysoline), tiagabine (Gabitril), topiramate (Topamax), valproate semisodium (Depakote), valproic acid (Depakene), and zonisamide (Zonegran).
B. **Psychotherapeutic approaches**: These include a wide array of treatment approaches. Depending upon the need of the patient each therapy could be modified and used accordingly.

    1. ***Relaxation therapy*:** Relaxation is a technique to calm your mind by releasing the tension in your body and relaxing your muscles. The idea is that once your muscles are relaxed, your mind relaxes too. You feel calmer and less worked up.People with epilepsy are sometimes advised to use relaxation techniques to

help prevent stress and anxiety, both of which can trigger seizures. Once you learn these techniques, you can relax whenever you feel stressed and prevent a seizure happening. [162] [153]

2. ***Behaviour modification*****:** With this treatment, you work with a therapist to learn ways to change how you behave. The goal is to help you live a full life without ignoring that you have epilepsy or letting it interfere too much with what you do.
3. ***Cognitive behaviour therapy-*** (CBT) to be effective in reducing depression, among people with epilepsy with a depressed affect. Cognitive behaviour therapy is widely used to treat anxiety and depression. It emphasizes the influence of thoughts and their content on emotional state and treatment focuses on changing thought patterns and behaviour. As anxiety and depression are commonly experienced by people with epilepsy,
4. ***Bio-feedback-*** galvanic skin response biofeedback reported significant reduction in seizure frequency. EEG bio-feedback was found to improve the cognitive and motor functions in individuals with greatest seizure reduction.
5. ***Educational interventions-*** Educational interventions were found to be beneficial in improving the knowledge and understanding of epilepsy, coping with epilepsy, compliance to medication and social competencies.
6. ***Family counselling-*** It is considered as adjunctive treatment for epilepsy.

Neurodegenerative/Infectious Disorders of Central Nervous System

These disorders generally are characterized by the loss of neurons and secondary gliosis (scarring) without evidence of major inflammation or necrosis of tissue; a large number of progressive diseases of unknown etiology , and their classification is based on pathologic and clinical findings. Infection of the nervous system can involve the meninges (meningitis) or the brain substance itself (encephalitis), or both (meningoencephalitis). Additionally, infections can be acute or chronic. The organisms that are involved in infection are bacterial, parasitic or viral. Additionally, prions represent an unusual class of infectious agent that can damage the brain. We will discuss each of these scenarios and consider the differential diagnosis. This is true of many of the disorders of the basal ganglia and the rest of the extrapyramidal system. They range from extremely common (Alzheimer disease) to quite rare (the spinocerebellar degenerations). Many are devastating conditions that often affect people in the prime of life. Certainly these are important to discuss in any course on neurology. Some of the commonly studied neurodegenerative and infectious disorders are:

- Alzheimer disease
- Parkinson's disease
- Spinocerebellar ataxia
- Pick's disease
- Muscular dystrophy
- Prion Diseases
- AIDS dementia complex

All the above diseases have deteriorating effects on the structure and functions of the brain, hence leading to severe multifaceted effects on the affected individuals cognitive and behavioural functioning. Almost all the above mentioned diseases share the following symptoms, which are:

Cognitive impairment

- *Executive functioning*
- *Information processing speed*
- *Attention*
- *Memory*
- *Language*
- *Visuospatial/visuoperceptual functioning*
- *General intellectual functioning*

Mood disorders

- Depression and
- Various anxiety disorders

Psychosis

Psychosis is one of the most disabling and distressing symptoms .

Sleep disorders

Specific sleep disorders include insomnia, hypersomnia, parasomnia, and rapid eye movement (REM) sleep behaviour disorder .

Fatigue

Fatigue is one of the most common, distressing and disabling non-motor symptoms.

Neuro-behavioural disorders

Neuro-behavioural problems such as hyper-sexuality, preoccupation with complex motor acts such as disassembling electrical equipment, hypomania and mania, aggression and heightened irritability, an urge to walk considerable distances without purpose, pathological gambling and shopping, and food cravings.

The following are the most common general signs and symptoms of a nervous system disorder. However, each individual may experience symptoms differently. Symptoms may include:

- Persistent or sudden onset of a headache
- A headache that changes or is different
- Loss of feeling or tingling
- Weakness or loss of muscle strength
- Sudden loss of sight or double vision
- Memory loss
- Impaired mental ability
- Lack of coordination
- Muscle rigidity
- Tremors and seizures
- Back pain which radiates to the feet, toes, or other parts of the body
- Muscle wasting and slurred speech

The symptoms of a nervous system disorder may resemble other medical conditions or problems.

ASSESSMENTS USED FOR NEURODEGENRATIVE & INFECTIOUS DISEASES OF THE CENTRAL NERVOUS SYSTEM

Following tools have been used in identifying effects of neurodegenerative disorders:

- PET scan

- SPECT scan
- Mini-mental state examination
- Hachinski Ischemic score
- Clock drawing test
- Modified Mini-mental state examination (3MSE)
- Blessed information, memory, concentration
- Activities of daily living (ADL)
- Blessed Functional Assessment Scale
- Geriatric Depression Scale
- Neuropsychiatric Inventory-Q

These are some of the major assessment tools used in assessing the neuropsychological functioning of the patients affected with degenerative disorders of the brain.

MANAGEMENT PROCESS FOR THE DISEASES

Since the symptoms vary with onset, progress, course and origin; the treatment and management various accordingly. Dealing with severity first and then maintaining further. Here we'll discuss some major therapeutic techniques used by the therapists and physicians and the liaison for the betterment of the individual.

1) **Psychopharmacological approach:** Most of the degenerative and infectious conditions of the nervous system are irreversible and the prognosis is rather poor. Pharmacological methods help in not treating the diseases completely but they halt the fast pacing deterioration in the patient's physical condition. Some of the commonly used medications in such cases are to treat the symptoms such as those of dementia and motor disturbances; these are cholinesterase inhibitors, mimantine, dopamine agonists (Levdopa), MAO-B inhibitors, anticholinergics etc. Apart from these drugs of choice, to reduce the co-morbid conditions of depression, anxiety, hyperactivity, sleep disturbances other complimentary drugs are also used.
2) **Psychotherapeutic approaches:** Other then pharmacological treatment in cases of nervous system diseases it becomes important to make use of the non-pharmacological and psychological treatment methods. It is important to use these alternate medication techniques because the effect of the degenerative conditions

affects the cognitive and behavioural functioning directly. Some of the commonly used therapeutic approaches are discussed below:

A. *Cognitive rehabilitation:* **Cognitive rehabilitation therapy** is a program to help brain-injured or otherwise cognitively impaired individuals to restore normal functioning, or to compensate for cognitive deficits.[1] It entails an individualized program of specific skills training and practice plus metacognitive strategies. Metacognitive strategies include helping the patient increase self-awareness regarding problem-solving skills by learning how to monitor the effectiveness of these skills and self-correct when necessary. There are two approaches usually followed during cognitive rehabilitation therapy these are: the remedial (restorative) approach and the reconstructive approach.

B. *Remedial approach:* This approach involves the following techniques:

   1. *Drill and practice*
   2. *Mnemonic strategy including verbal and visual mnemonics.*
   3. *Link methods*
   4. *Visual story method*

C. *Compensatory approach:* Compensatory strategies include making changes in your environment (home, school, workplace, etc) or adopting different methods of performing activities. As well, compensatory strategies involve making use of devices that help with remembering tasks. For example there are many options for wristwatches and other innovative gadgets that can remind you about a scheduled activity (e.g. a doctor's appointment or visiting a loved one). Other examples of compensatory strategies for memory might be to use an agenda book, a diary or a tape recorder to help to remember things you need to do.

D. *Physical exercise and gait training:* Regular physical exercise with or without physiotherapy can be beneficial to maintain and improve mobility, flexibility, strength, gait speed, and quality of life.[2] In terms of improving flexibility and range of motion for patients experiencing rigidity, generalized relaxation techniques such as gentle rocking have been found to decrease excessive muscle tension. Other effective techniques to promote relaxation include slow rotational movements of the extremities and trunk, rhythmic initiation, diaphragmatic breathing, and meditation techniques.[3] Common changes in gait associated with the disease such as hypokinesia (slowness of movement), shuffling and decreased arm swing are addressed by a variety of strategies to improve functional mobility and safety. Goals with respect to gait during rehabilitation programs include improving gait speed, base of support, stride length, trunk and arm swing movement. Strategies include utilizing assistive equipment (pole walking and treadmill walking), verbal cueing (manual, visual and auditory), exercises (marching and PNF patterns) and varying environments (surfaces, inputs, open vs. Closed.

E. *Occupational Therapy:* Occupational therapists (OT) - Occupational therapy is a health care/rehabilitation profession. OT's work with people of all ages, who are experiencing difficulties in leading independent, productive, and satisfying lives. The goal of an occupational therapist is to assist individuals with performing their activities of daily living. People with Parkinson's Disease may need the help of an occupational therapist for many different reasons. While one person may need help learning various strategies to manage their tremors, another may need help with feeding skills, and yet another may need to learn various coordination exercises to maintain functional use of their hands, and someone else may need a home evaluation so the therapist can make recommendations on equipment needs and home safety, the list is endless. The treatment is individualized to your needs which are based on your goals for therapy as well as the therapists evaluation. If you have lost some independence due to your Parkinson's Disease consult your physician regarding occupational therapy.

F. *Demystifying Infection:* It is important in cases of AIDS dementia complex that the transmission of the infection must be discussed by the occupational therapist in order to clarify any misconceptions. All the routes of infections to spread and preventive methods must be discussed in detail.

G. *Education regarding the stigmatization of illness:* Since HIV/AIDS is a life threatening disease it has been widely stigmatized throughout the corners of the world. It is important for the therapist to first educate the family members and health workers about the problems faced by the patients.

Psychosocial Rehabilitation therapies for the above discussed disorders

It is very important to use the psychosocial rehabilitation techniques with the specified neurological rehabilitation techniques since the affected individual has to be rehabilitating within the society, living and functioning to their fullest potentials. Various therapies are being used in the psychosocial rehabilitation process, some of the important ones are:

- **Behaviour Modification**: **Behavior modification** is the traditional term for the use of empirically demonstrated behavior change techniques to increase or decrease the frequency of behaviors, such as altering an individual's behaviors and reactions to stimuli through positive and negative reinforcement of adaptive behavior and/or the reduction of behavior through its extinction, punishment and/or satiation. It is an application of behavior analysis that does not search for the behavioral antecedent. Behavior modification is now known as Applied behavior analysis (ABA, including Positive behavior support (PBS)) which is more analytical than it used to be.
- **Supportive therapy**: The objective of the therapist is to reinforce the patient's healthy and adaptive patterns of thought behaviors in order to reduce the intrapsychic conflicts that produce symptoms of mental disorders. Unlike in psychoanalysis, in which the analyst works to maintain a neutral demeanor as a "blank canvas" for transference, in supportive therapy the therapist engages in a fully emotional, encouraging, and supportive relationship with the patient as a method of furthering healthy defense mechanisms, especially in the context of interpersonal relationships Trust is very important between patients and the doctors to help patients get better treatment effect.[3] Recent studies suggest that genetics, animal studies and neuroscience may have an impact or play a role in supportive psychotherapy. [4]
- **Family and care givers:** Facilitating communication of the patients and families need and promoting ventilation of the feeling may reduce family’s turmoil. The family should be encouraged to share feelings regarding the illness. The treating team should provide explanation, understanding and support to the family. Education to the family members regarding the nature of illness and prognosis is essential.

- **Group Therapy:** Group therapy approaches used in the rehabilitation settings include education group and family group. In group therapy patients and families have an opportunity to share their experiences, their sense of isolation and stigmatization and educate themselves by learning from the experiences of other members of the group. Members of such groups respond to one another with special sense of understanding and empathy because they share the unique experience of illness and its aftermath.

Therapies for co-morbid disorders such as anxiety and depression:

Most cases of anxiety disorder can be treated successfully by appropriately trained mental health professionals such as licensed psychologists. Research has demonstrated that a form of psychotherapy known as "cognitive-behaviour therapy" (CBT) can be highly effective in treating anxiety and depressive disorders. Psychologists use CBT to help people identify and learn to manage the factors that contribute to their anxiety.

- **Behavioral therapy** involves using techniques to reduce or stop the undesired behaviors associated with these disorders. For example, one approach involves training patients in relaxation and deep breathing techniques to counteract the agitation and rapid, shallow breathing that accompany certain anxiety disorders.
- Through **cognitive therapy**, patients learn to understand how their thoughts contribute to the symptoms of anxiety disorders, and how to change those thought patterns to reduce the likelihood of occurrence and the intensity of reaction. The patient's increased cognitive awareness is often combined with behavioral techniques to help the individual gradually confront and tolerate fearful situations in a controlled, safe environment.
- **Interpersonal therapy** focuses on the behaviors and interactions a depressed patient has with family, friends, co-workers, and other important people encountered on a day-to-day basis. The primary goal of this therapy is to improve communication skills and increase self-esteem during a short period of time. It usually lasts three to four months and works well for depression caused by loss and grief, relationship conflicts, major life events, social isolation, or role transitions (such as becoming a mother or a caregiver).

Neurological deficits lead to handicap. Spontaneous recovery, medical rehabilitation and learning to live with the handicap are key issues in rehabilitation. Access to adequate resources; timely, accurate and relevant information for effective management of neurological disability.

UNIT: III

CARDIOVASCULAR SYSTEM

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2ND YEAR M.PHIL (CLINICAL PSYCHOLOGY)

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CARDIOVASCULAR SYSTEM

The **circulatory system** is an organ system that permits blood and lymph circulation to transport nutrients (such as amino acids and electrolytes), oxygen, carbon dioxide, hormones, blood cells, etc. to and from cells in the body to nourish it and help to fight diseases, stabilize body temperature and pH, and to maintain homeostasis.

HUMAN CARDIOVASCULAR SYSTEM

The essential components of the human cardiovascular system are the heart, blood, and blood vessels. It includes: the pulmonary circulation, a "loop" through the lungs where blood is oxygenated; and the systemic circulation, a "loop" through the rest of the body to provide oxygenated blood. An average adult contains five to six quarts (roughly 4.7 to 5.7 liters) of blood, accounting for approximately 7% of their total body weight. Blood consists of plasma, red blood cells, white blood cells, and platelets. Also, the digestive system works with the circulatory system to provide the nutrients the system needs to keep the heart pumping.

PULMONARY CIRCULATION

The pulmonary circulatory system is the portion of the cardiovascular system in which oxygen-depleted blood is pumped away from the heart, via the pulmonary artery, to the lungs and returned, oxygenated, to the heart via the pulmonary vein. Oxygen deprived blood from the superior and inferior vena cava, enters the right atrium of the heart and flows through the tricuspid valve (right atrioventricular valve) into the right ventricle, from which it is then pumped through the pulmonary semilunar valve into the pulmonary artery to the lungs. Gas exchange occurs in the lungs, whereby CO_2 is released from the blood, and oxygen is absorbed. The pulmonary vein returns the now oxygen-rich blood to the left atrium.

SYSTEMIC CIRCULATION

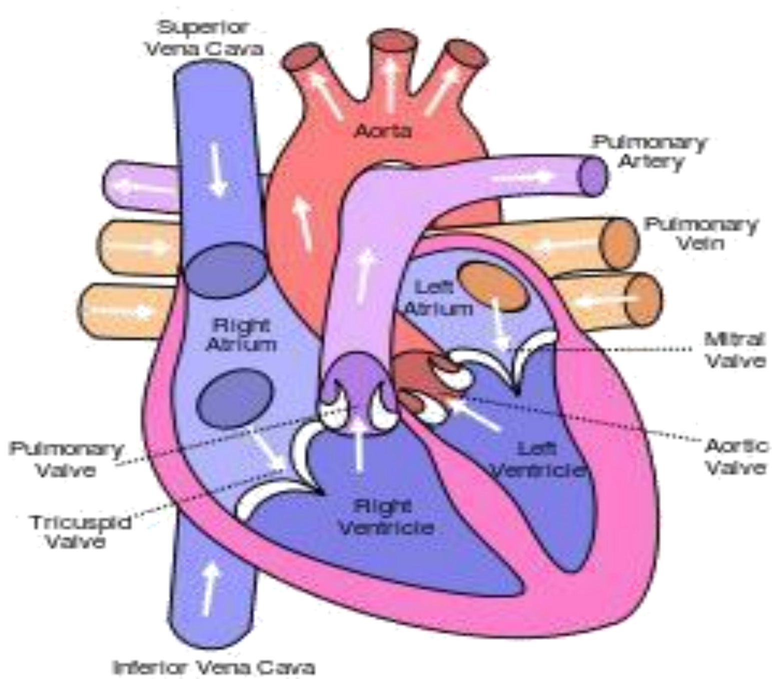
Systemic circulation is the circulation of the blood to all parts of the body except the lungs. Systemic circulation is the portion of the cardiovascular system which transports oxygenated blood away from the heart through the aorta from the left ventricle where the blood has been previously deposited from pulmonary circulation, to the rest of the body, and returns oxygen-depleted blood back to the heart. Systemic circulation is,

CORONARY CIRCULATION

The coronary circulatory system provides a blood supply to the myocardium (the heart muscle). It arises from the aorta by two coronary arteries, the left and the right, and after nourishing the myocardium blood returns through the coronary veins into the coronary sinus and from this one into the right atrium. Back flow of blood through its opening during atrial systole is prevented by the Thebesian valve. The smallest cardiac veins drain directly into the heart chambers.

HEART

The heart pumps oxygenated blood to the body and deoxygenated blood to the lungs. In the human heart there is one atrium and one ventricle for each circulation, and with both a systemic and a pulmonary circulation there are four chambers in total: left atrium, left ventricle, right atrium and right ventricle. The right atrium is the upper chamber of the right side of the heart. The blood that is returned to the right atrium is deoxygenated (poor in oxygen) and passed into the right ventricle to be pumped through the pulmonary artery to the lungs for re-oxygenation and removal of carbon dioxide. The left atrium receives newly oxygenated blood from the lungs as well as the pulmonary vein which is passed into the strong left ventricle to be pumped through the aorta to the different organs of the body.



Heart diseases are complex diseases that may have different manifestations and sometimes unknown etiologies. But there are certain proven risk factors that may lead to heart diseases; like hypertension, diabetes, poor nutrition, lack of exercise and smoking. Stress and depression are the psychological factors that have also been added to the list more recently (Rozanski, Blumenthal & Kaplan, 1999).

According to ICD-10 (1994), Heart diseases or cardiovascular diseases are a class of diseases that affect the heart or the blood vessels. Technically, it refers to any disease that affects the cardiovascular system. *Coronary Heart Diseases* (CHD) or *Coronary artery* or *Ischemic heart disease* is a condition where in the arteries supplying blood to the heart become narrowed due to the deposition of fatty plaque deposits, also known as a process called *arthrosclerosis*. Inadequate blood supply to the heart is also sometimes accompanied by pain or *Angina Pectoris*. *Myocardial Infarction* death of cardiac tissues) or commonly known as Heart Attack is caused when the blood flow to the heart is completely stopped due to entire blockage of arteries. *Arrhythmias* occur when there is an irregular rhythm of the heartbeat. Heart can beat either too fast, too slow or can have an irregular rhythm. Many arrhythmias are harmless but some can be life threatening. If heart rate is abnormal, efficiency of the heart to pump blood is affected which may lead to damage to the brain, heart and other organs.

Surgeries and procedures

- ***Coronary Artery Bypass Grafting (CABG)*** is a surgery that is commonly performed in order to improve the blood flow to the heart which is obstructed due to plaque buildup in the coronary arteries. In this surgery, a healthy artery or vein from the body is grafted to the blocked coronary artery. This grafted artery then bypasses the blocked portion and creates a new pathway for oxygen rich blood to reach the heart muscles.

- ***Percutaneous Coronary Intervention (PCI) or Angioplasty*** is a non-surgical procedure that is used for opening up blocked or narrowed coronary arteries. This procedure can be done in two ways. First, a thin, flexible tube with a balloon fixed at the tip is threaded in to the blocked blood vessel. At this point the balloon is inflated which pushed the plaque to the wall of the arteries and restores the normal blood flow. In the second method, a *Stent*, a small mesh tube, is placed in the coronary artery which supports the inner wall of the artery and helps to keep it open and allow normal blood flow.

- ***Heart Valve Repair or Replacement,*** heart valves are important structural components of the heart that allow blood flow into one heart chamber to another or to the arteries. Each flap has a set of flap called leaflets that open and close rhythmically to regulate blood flow. In some heart conditions, these leaflets do not open as widely as they should due to fusion, thickening or stiffening of the leaflets. In other conditions, theses heart valves might not close tightly resulting in leakage of blood back into the heart chambers instead of moving forward to the arteries. In such conditions, heart valve replacement surgery is performed to restore the functioning of the heart valves.

- ***Arrhythmias Treatment,*** as mentioned earlier, is the irregularity in heartbeat. The first line of treatment for arrhythmias is medicines and if these medications do not work then surgery could be done. Surgery for arrhythmias involves implanting a *Pacemaker* or an *Implantable Cardioventer Defibrillator* (ICD). Pacemakers are devices that placed under the skin of the patients' chest or abdomen and wires from it are connected to the chambers of the heart. Low-energy electrical pulses are emitted that control the heart's rhythm. ICD's are implanted in the same way as pacemakers but in case of this device, it senses irregular arrhythmias and sends an electric shockt o the heart to restore the normal heart rhythm.

- ***Aneurysm Repair:*** Aneurysm occur if the weakened artery wall bulges due to the pressure of blood flow leading to a balloon like protuberance in the artery or the heart muscle. Aneurysms may grow and burst leading to severe consequences. Surgeries are done to repair the weakened artery or heart wall with a patch or graft.

- ***Heart Transplant*** is a surgery in which a diseased heart of a patient is replaced with a healthy heart from a deceased donor. Heart transplants are conducted when the patients are at the end-stage of the disease; where in no treatments seem to give a desired outcome.

RISK FACTORS

1) **Psychosocial risk factors:** Psychological factors including depression, stress, anxiety and hostility influence the development, clinical expression and prognosis of heart disease. Accordingly, it is critical that clinically significant levels of distress are identified, both in hospital and after discharge, to target patients who require specific psychological intervention, in addition to conventional cardiac rehabilitation. Ameliorating distress has been shown to improve adherence to treatment advice, such as modifying psychological defense mechanisms, compliance with medication regimens and attendance at exercise programs.

Depression: Research confirms that while roughly 20 percent of the US population endures an episode of depression in their life, the number soars to 50 percent among folks with heart disease. Long-term investigations report that men and women diagnosed with clinical depression are more than twice as likely to acquire coronary artery disease or experience a heart attack. A number of prospective studies have noted an increased relative risk (1.5 to 2.0) of acute MI and cardiovascular mortality in depressed, but otherwise healthy individuals (Ferketich et al., 2000). The data linking depression with adverse events among patients with established cardiac disease are particularly striking, with a 1998 review noting that 11 of 11 studies reported worsened outcome (Glassman & Shapiro, 1998). The relative risk of future adverse cardiac events and cardiovascular mortality in depressed patients with established cardiac (usually coronary artery) disease is increased approximately three- to four-fold (Lett et al., 2004); (Barth et al., 2004); (van Melle et al., 2004). Between 35–45% of post-MI and post-coronary artery bypass graft surgery (CABG) patients suffer from some level of depression, ranging from a few symptoms to major depressive disorder. Results from two recent intervention trials suggest that much cardiac depression may remit spontaneously within a few months of the cardiac event (Glassman et al., 2002); (Writing Committee for the ENRICHD investigators,2003). Autonomic dysfunction is the physiologic mechanism most often cited as linking depression to increased cardiac mortality (Carney et al., 2001).

Social isolation: Certain chronic aspects of the social environment, including isolation, low social support, and lack of economic and social resources, can increase an individual's risk of developing CAD (Shumaker et al., 1994). Social support refers to the instrumental (i.e., tangible), informational, and emotional support obtained from a person's social ties and community (Cohen & Wills, 1985). In early studies, so-called 'social networks

were measured quantitatively by assessing factors such as the extent of one's participation in group and organizational activities or the number of family members or friends present (Rozanski et al., 1999). Some researchers evaluated the role of living arrangements (alone, married, marital disruption), while others focused on instrumental support such as access to community services and activities. It was shown that a small social network confers a two- to threefold increase in the likelihood of developing CAD over time. It is also imperative to look at the qualitative nature of social support (i.e., amount of perceived emotional support). Low levels of perceived emotional support were shown to confer greater than a threefold increase in the risk of future cardiac events (Blazer, 1982).

Within industrialized societies, cardiac morbidity and mortality are inversely related to socioeconomic status (SES) with disease rates highest among the poorest individuals. Initially, it was assumed that this disparity as due to differences in medical care and standard risk factors such as smoking and high blood pressure, but evidence shows these are only partly to blame (Luepker et al., 1993). This relationship between cardiac outcome and socioeconomic status is observable whether measured by education, income, or occupation. One study (Ruberman, Weinblatt, Goldberg, & Chaudhary, 1984) found that low-SES men were more likely to experience isolation and life stress. These men were also found to have a mortality rate twice as high as their more educated counterparts. It was also found that low SES is associated with increased levels of high-risk behaviors (Winkleby, Fortmann, & Barrett, 1990) and psychosocial risk factors (Barefoot et al., 1991).

Stress: Research has begun to focus on the role that acute stress and. anger may play as triggers for the development of coronary artery disease (CAD; see Krantz, Kop, Santiago, et al., 1996). Previous studies have observed that stressful life events, such as the death of a spouse, often occurred within the 24 hours preceding death among patients who died suddenly from coronary disease. The occurrence of natural disasters and personal traumas has also been correlated with an increase in cardiac events. Several research teams studied the possible physiological mechanisms by which acute stress may trigger coronary events. It was found that acute psychological risk factors may result in impaired dilation of the coronary vessels in coronary patients (Howell et al., 1997), decreases in plasma volume (Patterson, Gottdiener, Hecht, Vargot, & Krantz, 1993), and increased platelet activity and blood clotting tendency (Patterson et al., 1995). These responses may result in an imbalance between cardiac demand and decreased coronary blood supply and may lead to cardiac ischemia (Kop, 1999). Acute psychological f actors may also elicit electrical instability of the myocardium and cause life-threatening arrhythmias (Verrier & Mittleman, 1996). Lown (1987) proposed that ventricular arrhythmias occur in presence of three factors: myocardial electrical instability, an acute triggering event (frequently related to mental stress), and a chronic and intense psychological state.

In addition to the effects of acute or short-term stressors, the possible pathophysiologic effects of chronic stressors were studied in conjunction with CHD risk. Work-related stress is the most widely studied form of chronic stress. Research has sought to elicit which occupations are most stressful and which characteristics of particular occupations lead to an increased likelihood of developing CAD (Karasek & Theorell, 1990).

Several factors were determined to contribute to the amount of stress one experiences on the job. The psychological demands of the job refer to stresses that interfere or tax a worker and make him or her unable to perform at optimal levels. Level of job autonomy or control refers to the ability of the person to influence his or her working conditions, including the nature, speed, and conditions of the work. Job satisfaction includes how many of the worker’s needs are met and the level of gratification attained from the overall work experience (Wells, 1985) Karasek and colleagues (e.g., Karasek & Theorell, 1990) proposed a job demand/control hypothesis in which occupations with high work demands combined with few opportunities to control the work or make decisions (low decision latitudes) are associated with increased coronary disease risk. Other models linking occupational stress to CAD development have been formulated. One such model proposes that work stress is the result of an imbalance between high work demand and low reward (Siegrist, Peter, Junge, Cremer, & Seidel, 1990). This demand-reward imbalance was associated with a 2.15-fold increase in risk for the development of new CAD.

2) Personality

Type A Behavior: This behavior pattern was defined by Rosenman and collegues in 1975 as "an action emotion complex which is exhibited by those individuals who are engaged in relatively chronic struggle to obtain an unlimited number of poorly defined things from their environment in the shortest period of time and if necessary against opposing efforts of other things or persons in this same environment".

The basic characteristics of this pattern are time urgency and impatience, competitive striving for achievement, aggressiveness and easily aroused hostility. Type A behavior pattern is an independent risk factor for the development of coronary disease equivalent in power to smoking or hypertension. A significant interaction was found between type A behavior and age in relation to severity of coronary atherosclerosis after adjusting for standard coronary risk factors. That is more complex than simply being a “workaholic" or having "high job involvement" with which it is often confused. The trait of hostility has received particular attention as a predictor of the development of coronary heart disease. Physiologic autonomic responses to interpersonal stress are altered in hostile individuals, suggesting association between hostility and coronary artery disease. The components of hostility that have been reported to be correlated with chronic heart disease include 3 main categories: cynical thoughts, angry feelings and aggressive behavior.

A hostile person is "one who has little confidence in his fellow men. He/she sees people as dishonest, unsocial, immoral, ugly and mean and believes that they should suffer for their sins". Reductions in tonic vagal cardiac modulation and shifting of autonomic balance in the direction of sympathetic predominance are hypothesized to be the basic of these relations.

Distressed Personality Type: There is abundant evidence that depression, anxiety and pathological anger considered as "negative affectivity" increase the risk for cardiac events in patients with coronary heart disease. Combination of negative affectivity and social inhibition defines a persistent "distressed personality type" and predicts cardiac events independently of established medical risk factors. Psychological distress has been found to elevate resting heart rate and blood pressure, decrease heart rate variability and increase ventricular arrhythmia's and myocardial ischemia.

Generalized anxiety, pervasive fear, specific simple phobias and posttraumatic stress reactions are commonplace in patients who have experienced the diagnosis and severe complications of heart disease. Anxiety is vital, highly differentiated and it originates in one's own body, at the same time representing a response to a biological threat. It is existential and generated by fear for life. Chronic low to moderate intensity anxiety, such as that seen in generalized anxiety disorder may have different effects on both cardiac physiology and cardiac end points than transient, high intensity anxiety such as that seen in phobias and panic disorder. Agoraphobia frequently occurs in patients with cardiac arrhythmias, angina and heart failure developing just as it may in patients with recurrent panic attacks. High levels of phobic anxiety are associated with an increased risk of sudden death, presumably owing to reduced vagal modulation of cardiac rate and increased ventricular tachyarrhythmias.

3) <u>Life-Style and Health Practices:</u> People may believe heart disease correlates solely with physical actions (a lack of exercise, poor diets, smoking, excessive drinking), however, attitudes, emotions, and thoughts are equally significant. Thought processes can accelerate the onset of heart disease and hinder taking concrete strides to promote health. A few may sense a loss of control over their life with taking medication, making time for exercise and giving up favorite foods. Making modifications in everyday life is not easy as it takes training to instill these new practices. To sneak a cigarette or cheat on a diet may satisfy an immediate desire, but will hinder the long-term goal of improving health. Cultivating a healthy lifestyle can diminish the risk of heart disease or manage the condition, even if a higher risk is due to uncontrollable determinants such as sex, family history or age.

Psychological determinants can influence health directly (such as stress causing the release of the hormone cortisol) and indirectly (via behavioral decisions) which can harm or preserve health (exercise, diet and smoking). Health psychologists use a bio-psychosocial guide in recognizing health practices. This method centers on understanding wellness to be the result not only of biological processes (hormonal and endocrine functioning) but also of mental processes such as approaches toward health and how people cope with stress in their lives. Health psychologists then look at factors that relate to socioeconomic status, culture and ethnicity to formulate treatment plans, interventions and prognoses.

HYPERTENSION (HTN)

Hypertension or high blood pressure, sometimes called arterial hypertension, is a chronic medical condition in which the blood in the arteries is elevated. This requires the heart to work harder than normal to circulate blood through the blood vessels.

Blood pressure is summarized by two measurements, systolic and diastolic, which depend on whether the heart muscle is contracting (systole) or relaxed between beats (diastole) and equate to a maximum and minimum pressure, respectively. Normal blood pressure at rest is within the range of 100-140mmHg systolic (top reading) and 60-90mmHg diastolic (bottom reading). High blood pressure is said to be present if it is persistently at or above 140/90 mmHg. Hypertension is classified as either primary (essential) hypertension or secondary hypertension; about 90–95% of cases are categorized as "primary hypertension" which means high blood pressure with no obvious underlying medical cause. The remaining 5–10% of cases (secondary hypertension) is caused by other conditions that affect the kidneys, arteries, heart or endocrine system.

Hypertension is a major risk factor for stroke, myocardial infarction (heart attacks), heart failure, aneurysms of the arteries (e.g. aortic aneurysm), peripheral arterial disease and is a cause of chronic kidney disease. Even moderate elevation of arterial blood pressure is associated with a shortened life expectancy. Dietary and lifestyle changes can improve blood pressure control and decrease the risk of associated health complications, although drug treatment is often necessary in people for whom lifestyle changes are not enough or not effective.

Salt intake is implicated as a factor causing or worsening essential hypertension, because of largely excessive intake of sodium causes the kidneys to increase the blood. Studies indicate that high salt intake may only be related to high blood pressure levels in some cultures and population groups. For e.g., among people who are living in under developed tribal societies, sodium intake is often low, as is the prevalence of hypertension (Daniol and Moellering,1970).

Obesity is another behavioral phenomenon that plays an important role in hypertension. There is an increased prevalence of hypertension in obese persons, although the precise reasons for this remain to be determined (Shapiro,1983). Some have thought that it is merely obese patients consume more sodium, but recent studies have demonstrated that weight loss without salt restrictions can result in significant decrease in blood pressure(Raisen et al, 1978). For this reason, weight loss is an important behavioural method for managing high blood pressure.

HIGH BLOOD PRESSURE AND HYPERTENSIVE HEART DISEASE

Hypertensive heart disease is the No. 1 cause of death associated with high blood pressure. It refers to a group of disorders that includes heart failure, ischemic heart disease, hypertensive heart disease, and left ventricular hypertrophy (excessive thickening of the heart muscle). Heart failure does not mean the heart has stopped working. Rather, it means that the heart's pumping power is weaker than normal or the heart has become less elastic. With heart failure, blood moves through the heart's pumping chambers less effectively, and pressure in the heart increases, robbing your body of oxygen and nutrients. To compensate for reduced pumping power, the heart's chambers respond by stretching to hold more blood. This keeps the blood moving, but over time, the heart muscle walls weaken and are unable to pump as strongly. As a result, the kidneys often respond by causing the body to retain fluid (water) and sodium. The resulting fluid buildup in the arms, legs, ankles, feet, lungs, or other organs, is called congestive heart failure. High blood pressure brings on heart failure by causing left ventricular hypertophy, a thickening of the heart muscle that results in less effective muscle relaxation between heart beats. This makes it difficult for the heart to fill with enough blood to supply the body’s organs, especially during exercise, leading your body to hold onto fluids and your heart rate to increase.

THE MANAGEMENT OF PSYCHOSOCIAL AND BEHAVIOR RISK FACTORS

Modifying Hostility and Type A Behavior: A number of intervention studies have attempted to modify Type A behavior in an attempt to reduce cardiovascular disease risk. Most early studies reported that elements of Type A behavior can be decreased in subjects who are motivated to change (Allan & Scheidt, 1996; Suinn, 1982).

The Recurrent Coronary Prevention Project (RCPP) (Friedman et al., 1986) was the first and most ambitious intervention trial to solely study whether Type A behavior could be modified, and how this modification might impact one's risk of cardiovascular morbidity and mortality. The study looked at a variety of Type A behaviors, including anger, impatience, aggressiveness, and irritability. Over 1,000 patients were assigned to one of three groups: a cardiology counseling treatment group, a combined cardiology counseling and Type A behavior modification group, or a non-treatment control group. The cardiology counseling included training on how to comply with drug, dietary, and exercise regimens as dictated by the participant's physician, counseling on non-Type A psychological problems resulting from the coronary experience, and education about all aspects of cardiovascular disease. Type A counseling included drills to modify various Type A behaviors, discussions on values and beliefs that may cause the behavior pattern, relaxation and stress reduction training to decrease physiological arousal, and changes in work and home demands. After 4.5 years, the final results showed a larger decrease in global Type A behaviors as well as in its components in the Type-A counseling group. Also, rate of recurrent MI was significantly lower in the Type-A counseling group than in either the cardiology counseling or control groups (Friedman et al., 1986).

However, recent evidence points to the fact that much of the reduced cardiac recurrences in the Type A counseling group may be attributed to multiple causes, including increased number of treatment contacts and

increased social support (Mendes de Leon, Powell, & Kaplan, 1991). Hostility is a specific component of Type A behavior that is a significant psychosocial risk factor for cardiovascular disease development. Girdon, Davidson, and Bata (1999) studied the effects of a hostility-reduction intervention on patients with coronary heart disease. Twenty-two highly hostile male coronary patients were randomly assigned to either a hostility intervention group or an information-control group. Those in the intervention group were observed at immediate and two-month follow-ups to be less hostile than controls, as assessed using self-report and structured interviews, and to have significantly lower diastolic blood pressures. Further investigations promise to provide insight into the role of hostility reduction in relation to cardiovascular diseases.

***Interventions to Increase Social Support and Reduce Life Stress:** The Ischemic Heart Disease Life Stress Monitoring Program (Frasure-Smith & Prince, 1987, 1989) was based on prior studies that indicated that periods of increased life stress may precede recurrences of MI (e.g., Rahe & Lind, 1971; Wolff, 1952). Post-MI patients were either assigned to a treatment group (n = 229), which included life stress monitoring and intervention, or a control group (n = 224), which received only routine medical follow-up care. Patients in the treatment group were contacted by phone on a monthly basis and asked to rate 20 symptoms of distress, including insomnia and feelings of depression. If stress levels surpassed a critical level (more than 4 of the 20 symptoms), a project nurse made a home visit to attempt to help the patient assess the cause of the distress and to help the patient cope with the stressors. Over a one-year period, nearly half of the treatment group needed an intervention and received on average five to six hours of counseling, education on heart disease, and emotional and social support. Results showed that during the year of the project there was a 50% reduction in cardiac deaths, a reduction that continued for six months beyond the project's completion. Over seven years following the study, there were fewer MI recurrences among patients in the treatment group (Frasure-Smith & Prince, 1989). The success of the Ischemic Heart Disease Life Stress Monitoring Program could at least partly be attributed to the social and emotional support given to the patients that helped ameliorate depression and feelings of distress, thereby reducing physiological arousal and its negative effects on the cardiovascular system. Specific aspects of the treatment program, including its individualized interventions and treatment based on an individual's stress score, may have also contributed to the program's success.

- ***Cardiac rehabilitation:*** Cardiac rehabilitation aims to reverse limitations experienced by patients who have suffered the adverse pathophysiologic and psychological consequences of cardiac events. Cardiovascular disorders are the leading cause of mortality and morbidity in the industrialized world, accounting for almost 50% of all deaths annually. The survivors constitute an additional reservoir of cardiovascular disease morbidity. In the United States alone, over 14 million persons suffer from some form of coronary artery disease (CAD) or its complications, including congestive heart failure (CHF),angina, and arrhythmias. Of this number, approximately 1 million survivors of acute myocardial infarction (MI), as well as the more than 300,000 patients who undergo coronary bypass surgery annually, are candidates for cardiac rehabilitation.

The major goals of a cardiac rehabilitation program are :

a. Curtail the patho physiologic and psychosocial effects of heart disease.
b. Limit the risk for reinfarction or sudden death.
c. Relieve cardiac symptoms.
d. Retard or reverse atherosclerosis by instituting programs for exercise training, education, counseling, and risk factor alteration.
e. Reintegrate heart disease patients into successful functional status in their families and in society.

* ***Long-Term Lifestyle Changes/ Exercise:*** The Lifestyle Heart Trial, which assessed whether coronary patients could be motivated to and benefit from making and sustaining comprehensive lifestyle changes, is one of the most important intervention studies conducted to date. Ornish and colleagues (1990) randomized 48 patients with moderate to severe coronary heart disease into two groups: an intensive lifestyle change group (n = 28) and a control group (n = 20). The intensive lifestyle change patients were given a lifestyle modification program consisting of several components:

*

1. A 10%-fat vegetarian diet.

2. Stress management training and group support including yoga and meditation in group settings twice a week and individual practice for an hour each day.

3. Smoking cessation.

4. A program to moderate levels of aerobic exercise.

Control group patients were not asked to make lifestyle changes other than those recommended by their cardiologists. The intervention lasted one year and the extent of progression of coronary disease was assessed by comparing coronary angiograms obtained at study onset and at one year. Study results (Ornish et al., 1990) showed that after one year, experimental group participants were able to make and maintain lifestyle changes with bene“cial results, including a 37% reduction in low-density lipoprotein (LDL) cholesterol levels, a 91% reduction in anginal episodes, and a slight reduction in the extent of stenosis (or blockage) in coronary arteries. Controls had very different results, showing only a 6% decrease in LDL cholesterol levels, a 165% increase in reported anginal episodes, and a less significant reduction in the extent of stenosis in coronary arteries. Overall, 82% of participants in the lifestyle intervention group had an average change toward regression of disease. Interestingly, there was a relationship between the extent of adherence to the lifestyle change program and the measured degree of regression of disease, with the most compliant study subjects showing the most improvement in disease status.

Psychotherapy: The primary goal of the psychotherapy is to make patients more psychologically comfortable by minimizing psychological stress and facilitating normal psychological processes. The therapists' aims are to create a

relationship wherein the patient will feel safe to express feelings and will thus have the experience of being accepted. If the patient achieves successful adaptation he develops a realistic insight into the limitations, gives up the powers he had had before the illness and adjusts to the actual situation. Regarding the prognosis, the possibility of evolution of different defense mechanism is of great importance for successful rehabilitation. Some primitive defense mechanisms (regression, denial) can be useful in the beginning of cardiovascular disease and deserve support but in time they must be suppressed because of their counter-effect on rehabilitation. Mature defense mechanisms are developed suppression, humor and sublimation which are more or less prominent depending on individual capacities and traits.

Therapeutic manipulation of defense mechanisms is considered an important factor of adequate adaptation to actual cardiac events. Psychologists should be concerned with determining patients' psychological needs implementing generic stress management and relaxation techniques introducing specific cognitive-behavioral therapies for those exhibiting severe emotional distress and training staff in psychological needs assessment. Psychological intervention programmes that provide short-term support for psychological distress, in conjunction with coronary risk behaviour interventions coupled with exercise, have a positive influence on both physical and psychosocial outcome.

Focal brief psychotherapy may be considered a logical treatment for depression in heart disease patients especially in cases where the depressive syndrome arises in reaction to the development of cardiac disease. In interpersonal psychotherapy, emphasis is placed on working through a focal area of interpersonal difficulty. Difficulties associated with depression may be classified as owing to role transitions, grief or role disputes. It is extremely important to identify those patients who are psychologically predisposed to poor adaptation after a coronary event and also to offer appropriate psychological interventions for both the patients and their partners, as both often suffer considerable distress which can substantially influence the patients convalescence. In cognitive therapy attention is directed to the identification and correction of irrational negative thinking patterns that contribute to feelings of depression.

Precise interventions should begin as soon after the coronary event as possible. The outcome for patients who have suffered a coronary event depends not only on the physical characteristics of the event, but also on the type of person they are and how well they adjust psychologically. Patients' personality is much more stable and much more difficult to change than is mood states such as depression or anxiety. Cognitive therapy approaches and psychosocial interventions play an important role in the treatment of depressed and socially isolated heart disease patients. Separation, difficult relationships, powerful triggers to the emotions of fear, anger, depression and grief may be blocked from direct expression by the patient and have to be dealt with by therapist in a flexible form of supportive counseling tailored to the individual needs.

Some cardiovascular risk factors can be changed through psychological interventions. Hypertension is subjected to some modification, and behavioral interventions have been moderately successful in lowering high blood pressure. Diets that are low in fatty foods helps maintain healthy cholesterol levels are resistant to change through behavioral means. The anger component of hostility can also be modified through practice and training. People can learn to recognize their anger and to express it verbally in a soft, slow voice or to cope with it by a variety of other strategies designed to prevent the anger from turning to range or fury. Because anger has some association with heart disease, people at risk for cardiovascular disease who also have problems with anger control can reduce that risk through these various coping and relaxation strategies.

CONCLUSION

There are many environmental, behavioral, and physiological variables that interact in the development of cardiovascular disorders. Many of the standard CHD risk factors have important behavioral components, and increasing evidence suggests important psychosocial risk factors for CHD, including occupational stress, hostility, and physiologic reactivity to stress. In cardiac patients, the presence of acute stress, low social support, lack of economic resources and psychological depression also appears appear to be important psychosocial risk factors. The identification of psychosocial risk factors for coronary disease have led to several promising behavioral and psychosocial interventions to aid in the treatment and prevention of coronary disease in high-risk individuals.

There also appears to be an important bio-behavioural influences in the development and treatment of essential hypertension. These include excessive salt intake, obesity, and stress. Evidence also indicates that genetic and environmental factors interact in the development of hypertension. However, the modest effects of behavioral stress – reducing techniques such as relaxation training, biofeedback and meditation in lowering blood pressure have proven disappointing.

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UNIT- IV

RESPIRATORY SYSTEM

Class Presentation

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The human respiratory system is a series of organs responsible for taking in oxygen and expelling carbon dioxide. This system includes the lungs, pathways connecting them to the outside environment, and structures in the chest involved with moving air in and out of the lungs. Air enters the body through the nose, is warmed, filtered, and passed through the nasal cavity. Air passes the pharynx (which has the epiglottis that prevents food from entering the trachea). The upper part of the trachea contains the ***larynx***. The vocal cords are two bands of tissue that extend across the opening of the larynx. After passing the larynx, the air moves into the ***bronchi*** that carry air in and out of the lungs. Bronchi are reinforced to prevent their collapse and are lined with ciliated epithelium and mucus-producing cells. Bronchi branch into smaller and smaller tubes known as ***bronchioles***. Bronchioles terminate in grape-like sac clusters known as ***alveoli***. Alveoli are surrounded by a network of thin-walled ***capillaries***. Only about 0.2 µm separate the alveoli from the capillaries due to the extremely thin walls of both structures. The alveoli and the capillaries are responsible for the exchange of oxygen and carbon dioxide. The bottom of the thoracic cavity is formed by the diaphragm. The diaphragm, a dome-shaped muscle at the bottom of the lungs, controls breathing. When a breath it taken, it flattens out and pulls forward, making more space for the lungs. During exhalation, the diaphragm expands and forces air out.

Respiratory movements are controlled by a respiratory center in the medulla of the brain. The functions of this center depend partly on the chemical composition of the blood. For example, if the blood’s carbon dioxide level rises too high, the respiratory center will be stimulated and respiration will be increased. If the carbon dioxide level falls too low, the respiratory center will slow down until the carbon dioxide level is back to normal.

The respiratory system is also responsible for coughing. A large amount of dust and other foreign material is inhaled with every breath. Some of these substances are trapped in the mucus of the nose and the air passages and are then conducted back toward the throat, where they are swallowed. When a large amount of mucus collects in the large airways, it is removed by coughing.

Disorders of the Respiratory System

Diseases and conditions of the respiratory system can be caused by the inhalation of foreign bodies such as cigarette smoke, chemicals, allergens and other irritants. Not all people will develop respiratory ailments as a result of environmental factors, as genetics also play a role in the development of respiratory diseases. Several disorders of the respiratory system – including

1. **Asphyxia, anoxia, and hyperventilation**- have little significance because they are typically short-lived. When they occur on a long-term basis, however, these disorders can have severe effects.

   Asphyxia, a condition of oxygen lack and carbon dioxide excess, may occur when there is a respiratory obstruction, when breathing occurs in a confined space so that expired air is reinhaled, or respiration is insufficient for the body's needs. asphyxia increases respiratory activity.

   Anoxia, a shortage of oxygen alone, is more serious. People suffering from anoxia may rapidly become disoriented, lose a sense of danger and pass into a coma without increasing their breathing.

   Hyperventilation – during periods of intense emotional excitement, people often breathe deeply, reducing the carbon dioxide content of the blood. Because carbon dioxide is a vasodilator (that is, it dilates the blood vessels), a consequence of hyperventilation is constriction of blood vessels and reduced blood flow to the brain. As a result, the individual may experience impaired vision, difficulty in thinking clearly, and dizziness.

2. **Hay Fever:**

   It is a seasonal allergic reaction to foreign bodies – including pollens, dust, and other air- borne allergens- that enter the lungs. These irritants prompt the body to produce substances called histamines, which cause the capillaries of the lungs to become inflamed and to release large amounts of fluid. The result is violent sneezing.

3. **Viral Infections:**

   The respiratory system is vulnerable to a number of infections and chronic disorders. Perhaps the most common cold, a viral infection of the upper and sometimes the lower respiratory tract. The infection that results causes discomfort, congestion, and excessive secretion of mucus. Secondary bacterial infections may complicate the illness. These occur because the primary viral infection causes inflammation of the mucous membranes, reducing their ability to prevent secondary infection.

A more serious viral infection of the respiratory system is influenza, which can occur in epidemic form.

Another infection bronchitis, is an inflammation of the mucosal membrane inside the bronchi of the lung. Large amounts of mucus are produced in bronchitis, leading to persistent coughing.

4. **Bacterial infections:**

   The respiratory system is also vulnerable to bacterial attack by, for example strep throat, whooping cough, and diphtheria. Strep throat an infection of the throat and soft palate, is characterized by edema (swelling) and reddening. Whooping cough invades the upper respiratory tract and moves down to the trachea and bronchi.

   For the most part, strep throat, whooping cough, and diphtheria do not cause permanent damage to the upper respiratory tract. Their main danger is the possibility of secondary infection, which results from lowered resistance. However, these bacterial infections can cause permanent damage to other tissues, including heart tissue.

5. **Pneumonia:**

   There are two main types of pneumonia. Lobar pneumonia is a primary infection of the entire lobe of a lung. Bronchial pneumonia, which is confined to the bronchi, is typically a secondary infection that may occur as a complication of other disorders, such as severe cold or flu.

6. **Tuberculosis and Pleurisy:**

   Tuberculosis is an infectious disease caused by bacteria that invade lung tissue. When the invading bacilli are surrounded by macrophages (white blood cells of a particular type), they form a clump called a tubercle, which is the typical manifestation of this disease.

7. **Lung Cancer:**

   Lung cancer or carcinoma of the lung, is an increasingly common disease. It is caused by smoking and other factors as yet unknown, including possible environmental carcinogens (air pollution) or cancer-causing substances encountered in the workplace (such as asbestos).

The affected cells in the lungs begin to divide in a rapid and unrestricted manner, producing tumor. Malignant cells grow faster than healthy cells; they cord out the healthy cells and rob them of nutrients, causing them to die, and then spread into surrounding tissue.

8. Asthma:

Asthma affects the small, breathing tubes (bronchial tubes) in a person's lungs. Each person with asthma is sensitive to certain "triggers" that can affect the breathing tubes. When someone with asthma is exposed to one or more of their triggers, three things happen:

1. The insides of the breathing tubes swell up (inflammation).
2. The body makes lots of thick, sticky fluid (mucus) inside the breathing tubes.
3. The muscles surrounding the breathing tubes get tight and make the air passages smaller (bronchospasm).

When all of these things happen, it is hard to breathe. This is called an asthma flare-up. Symptoms include coughing, wheezing, chest tightness, and dyspnea.

Asthma is a more severe allergic reaction, which can be caused by a variety of foreign substances, including dust, dog or cat dander, pollens, and fungi. An asthma attack can also be touched off by emotional stress or exercise.

9. Chronic obstructive pulmonary disease:

Chronic obstructive pulmonary disease (COPD) is the intersection of three related conditions — chronic bronchitis, chronic asthma, and emphysema. It is a progressive disease that makes in increasingly difficult for sufferers to breath.

Until recently, there was no commonly accepted definition for COPD (Snider, 1995). The American Thoracic Society (1995), in their guidelines for the treatment of COPD, developed a working definition for COPD that includes three classifications: airflow obstruction, emphysema, and chronic bronchitis. These characteristics are as follows:

(i) Abnormal tests of expiratory flow or airway obstruction are prominent in COPD patients. They are indicative of airflow obstruction, which is progressive in nature. Airflow obstruction has been attributed to the alteration of the small airways and, to some degree, to bronchoconstriction.

(ii) Emphysema is a state of progressive airflow obstruction due to abnormal permanent enlargement of airspaces.

(iii) Chronic bronchitis is the presence of a persistent mucous cough of no known etiology.

Specifically, chronic bronchitis is defined clinically by the persistence of a chronic productive cough for three months or more in each of two successive years in patients in whom other causes of chronic cough have been ruled out. The American Thoracic Society definition excludes patients with either chronic bronchitis or emphysema who do not exhibit persistent airflow obstruction, and includes a subset of asthma patients in whom airflow obstruction is not totally reversible (Snider, 1995). COPD normally follows a progressive course in that patients exhibit a rapid decline in lung function (measured by lung volume), symptoms of cough and dyspnea, more acute chest illness, and later in the disease, hypoxemia (American Thoracic Society, 1995).

Epidemiology of Respiratory Disorders

Prevalence rates for COPD and asthma continue to climb. The American Thoracic Society (1995) estimated that 14 million people, in the USA, suffer from COPD; this represents a 42% increase in prevalence since 1982.

Rates of COPD and asthma vary by gender, age, and race. The American Thoracic Society (1995) reported that COPD prevalence was increasing at a significantly higher rate for women (increasing by 30% in 1985 alone) than for men (a slight increase). The prevalence rate for pediatric asthma is estimated at 7% in the USA, a rate that is increasing more rapidly than for the total population (Centers for Disease Control, 1996a; Evans, 1993).

The Indian Council of Medical Research constituted a National Asthma Task Force with whose sponsorship a multicentric study on respiratory disease epidemiology was undertaken at four centres in India i.e. Chandigarh (the coordinating Centre), Delhi, Bangalore and Kanpur. This study lays the foundation not only for assessment and estimation of the disease burden and management strategies but also for future research in chronic airway obstruction. The mean asthma prevalence in this study is reported in 2.38% of 73605 individuals of over 15 years of age. One or more respiratory symptoms were present in 4.3-10.5% subjects. Female sex, advancing age, usual residence in urban areas, lower socioeconomic status, history suggestive of atopy, history of asthma in a first degree relative and all forms of tobacco smoking were associated with significantly higher odds of having asthma.

Psychological Theories to Asthma

In etymology and clinical manifestation, the act of breathing shares a close relationship with the mind. Thus, inspire can mean to fill with emotion and to draw in air. Psychological distress may become manifest in disrupted breathing, as in the tachypnea seen in anxiety disorders or sighing respirations in the depressed or anxious patient. Disturbances of breathing can likewise perturb any sense of psychic calm, as in the terror of any asthma patient with severe airway obstruction or marked hypoxemia.

The interaction of psychic and somatic influences becomes more complex when one considers the etiology and maintenance of symptoms of pulmonary disorders. For example, self-destructive or depressive feelings may generate or give momentum to a smoking habit, which could eventually result in chronic obstructive pulmonary disease (COPD).

Psychosomatic Theories of Asthma

Historically, an influential literature review on psychogenic factors in asthma (French & Alexander, 194 1) strengthened the view of asthma as psychosomatic in nature. Using data from case studies, French and Alexander concluded that allergic and psychological factors worked in concert to produce asthma symptoms. In addition, they cited evidence of the efficacy of hypnosis in preventing asthma attacks in response to known allergens. According to psychoanalytic theory, individuals with asthma may be experiencing separation anxiety (expressed as a suppressed cry for the mother) that may lead to an attack (Purcell & Weiss, 1970; Turnbull, 1962). According to this view, treatment based on psychoanalytic theory could assist individuals with asthma in working through their unconscious conflicts, resulting in reduced numbers of asthma attacks (French & Alexander, 1941). However, empirical studies have failed to confirm the tenets of the psychosomatic theory of asthma (Creer, 1982).

Over time, this view has been generally superseded by a bio-psychosocial view which emphasizes genetic and biological components that may be responsible for the initial development of asthma (e.g., Matts, 1984; Rees, 1980). However, as discussed above, psychological and social stimuli may also have an effect on the severity of symptoms and the manner in which treatment is conducted. Two psychological models, dyspned suffocation fear theory and cognitive theory , have been extended to the conceptualization of asthma severity and frequency.

Dyspned Suffocation Fear Theory

Ley (1989) proposed the dyspned suffocation fear theory to explain the occurrence of panic attacks. Dyspnea is a commonly reported sensation during a panic attack (de Ruiter, Garssen, Rijken, & Kraaimaat, 1992; Ley, 1989). According to the dyspned suffocation fear theory, the fear that accompanies a panic attack results from severe dyspnea that is induced by hyperventilation. The panic attack itself occurs when an individual feels that he or she has little or no control over the dyspnea (Ley, 1998).

Results indicated that although individuals with asthma or panic disorder report comparable levels of dyspnea, panic symptoms are accounted for by different factors. A significant amount of variance in panic symptoms in individuals with asthma is explained by dyspnea, but not for individuals with panic disorder. In other words, panic symptoms in asthma may be due to dyspnea, but in panic disorder, they are not due to dyspnea levels. This difference in the contribution of dyspnea to the experience of panic symptoms suggests that the dyspnedsuffocation fear theory (rather than explaining panic attacks) may be better utilized to more fully understand dyspnea and its relation to asthma.

Furthermore, **Carr** and Lehrer (1 994) argue that individuals in both the panic disorder and asthma groups reported a level of dyspnea that was significantly greater than controls. Thus, individuals in the panic disorder group did indeed experience a high level of dyspnea. It is possible, however, that this sample was generally composed of individuals experiencing hyper-ventilatory panic attacks. Overall, the results of these studies must be replicated and extended in order to determine the specific role of dyspned suffocation fear in asthma and/or panic disorder.

Cognitive Theory

The cognitive theory of panic proposes that catastrophic misinterpretation of benign bodily sensations results in the experience of a panic attack (Clark, 1986). Extending this theory to asthma, it is possible that individuals with asthma and a tendency to misinterpret symptoms may experience more frequent and more severe asthmatic episodes. Mislabeling non-asthma related bodily sensations as symptoms of asthma may have serious consequences. In studies of asthma inpatients, 19% - 27% consistently mislabeled relatively benign bodily symptoms (e.g., fatigue, worry, irritability, anxiety) as an asthma attack. Those individuals who exhibited this tendency to mislabel symptoms were more likely to be rehospitalized for asthma at a 6-month follow-up. Forty to 84% of the mislabelers were rehospitalized by 6 months, whereas only 29% - 40% of the nonmislabelers were rehospitalized. There was no measured difference between symptom mislabelers and non-mislabelers in lung function (Dirks & Schraa, 1983). In individuals with asthma, the consequences of symptom misinterpretation may be due to the anxiety producing effects of the catastrophic interpretation of benign sensations that may lead to increased bronchoconstriction.

Exacerbants of respiratory disorders

Anxiety and Panic Disorder

Coexisting anxiety or panic disorder probably worsens the course of asthma, and the prevalence of panic disorder and agoraphobia among asthma patients is higher than in the general population. As much as 30 percent of persons with asthma meet criteria for panic disorder or agoraphobia. Panic disorder appears to be under recognized in asthma patients, and its symptoms may be misunderstood by the patient and physician as arising from an exacerbation of asthma. Hyperventilation, a common symptom of anxiety and panic disorder, may trigger an exacerbation of a pulmonary illness. The mechanism may be via brainstem over reactivity to low arterial carbon dioxide (CO_2) levels. Lactate infusion tests, breathing elevated levels of CO_2, and voluntary hyperventilation have been shown experimentally to induce panic attacks in individuals with panic disorder. Voluntary hyperventilation may be the most specific test, and lactate infusion may be the most sensitive.

Repeated experience with severe dyspnea or near-suffocation provides little in the way of cognitive or emotional protection for future episodes. Such events often seem only to traumatize patients and may lead to the development of anxiety disorders or panic disorder. Cohorts of panic disorder patients are more likely than other anxiety disorder patients to have an antecedent history of lung disease.

Patients with asthma and panic disorder seem to have greater anticipatory anxiety about dyspnea and anxiety itself than patients who are not comorbid. The fear of dyspnea may directly trigger asthma attacks. An extremely high level of anxiety predicts increased rates of hospitalization and asthma-associated mortality. Certain personality traits in asthma patients appear to be associated with greater use of corticosteroids and bronchodilators, as well as hospitalizations of greater length than would be predicted from pulmonary function alone. Such traits include intense fear, emotional lability, sensitivity to rejection, and lack of persistence in difficult situations.

Depression and the High-Risk Asthma Patient

Many researchers suggest that depression has a meaningful and negative effect on the course of asthma. Shame and low self-esteem, common symptoms of depression, are seen in many chronically ill patients and are risk factors for a severe course. Patients who feel that they are worth little tend to manage their asthma poorly, and compliance with a medication regimen may suffer. The sleep disturbances of depression could adversely affect self-care, as could the concentration and attention deficits of the mood disorder. Sleep deprivation may reduce a patient's proprioceptive capacity to detect changes in airways resistance and can erode daytime cognitive performance. In a severely depressed patient, unconscious suicidal wishes may find expression in self-neglect. In addition to these behavioral manifestations, the hypothetical parasympathetic dominance of depression could contribute to airways reactivity and constriction.

Other than depression, several other exacerbating psychosocial factors deserve mention. Although there is no asthma-prone personality, personality disorder level psychopathology raises the risk for near-fatal episodes. High levels of aggression, denial of or disregard for asthma symptoms, and psychotic signs are associated with poor outcome or fatal episodes.

Family members of patients with severe asthma tend to have higher-than-predicted prevalence rates of mood disorders, posttraumatic stress disorder (PTSD), substance use, and antisocial personality disorder. How these conditions contribute to the genesis or maintenance of asthma in an individual patient is unknown. The familial and current social environment may interact with a genetic predisposition for asthma to influence the timing and severity of the clinical picture. The interaction between family psychopathology and the clinical course of asthma may be especially insidious in adolescent asthma patients. In turn, the adolescent's developmental need for (and fear of) emotional separation from the family often becomes entangled in battles over medication adherence, as well as other modes of diligent self-care.

Effects of respiratory diseases

Secondary gain

Mosby's Medical Dictionary defines secondary gain as "an indirect benefit, usually obtained through an illness or debility." The concept of secondary gain goes back to the 19th Century and was re-popularized by Dr. Sigmund Freud, who talked about patients "clinging to their disease" as a way to hold on to the benefits their illness provides them.

Some examples of secondary gain

Secondary gain can include anything from getting more attention or sympathy from others to escaping from punishment or responsibilities. Examples of secondary gains for adults include gaining access to pleasurable drugs, getting out of onerous household chores, not having to live up to your career potential or a way to avoid sex and/or having more children. Children also attain secondary gains from illness. Through their illness they get extra attention from mommy or daddy-or to stay home sick from school.

PSYCHOLOGICAL INTERVENTIONS FOR COPD AND ASTHMA

Separate intervention approaches are taken towards COPD and asthma. A number of behavioral techniques are involved in a pulmonary rehabilitation program. The approach with asthma is broader, in part because of the wide array of problems presented by patients with asthma.

Pulmonary Rehabilitation for COPD

Ries (1995) outlined a guide for the pulmonary rehabilitation of patients with moderate to severe disease. Rather than focusing solely on reversing the disease process, the goal of pulmonary rehabilitation is to minimize disability from the disease. The goal could be achieved by enhancing medical therapy, controlling and alleviating symptoms, improving pulmonary functions, and enhancing the quality of life of patients. There are three foci of a pulmonary rehabilitation program:

(i) the patients;
(ii) (ii) a multidisciplinarian approach; and
(iii) (iii) attention to physiology and psychopathology.

The selection of patients was based upon a study of rehabilitation programs conucted by Ries, Kaplan, Limberg, and Prewitt (1995). Six criteria for inclusion were used: (i) patients had symptomatic lung disease; (ii) patients were stable with standard therapy; (iii) patients had functional limitation because of chronic lung disease; (iv) patients were under the care of a primary care provider; (v) patients had no other interfering or unstable medical conditions; and (vi) patients were motivated to be involved in and responsible for their own healthcare.

Components of the comprehensive pulmonary rehabilitation program required a multidisciplinarian team. There were two major classes of components of the program: patient evaluation and program content. Patient evaluation included the following five stages:

Stage 1: Interview. A screening interview was conducted to assess suitability for a pulmonary rehabilitation program. An additional aim of the interview was to establish teamwork between patients and their rehabilitation team.

Stage 2: Medical evaluation. Reviewing the medical history of patients, including laboratory data, helped confirm their disease and its severity.

Stage 3: Diagnostic testing. Pulmonary function evaluation, exercise testing, and the assessment of blood gases at rest and during exercise were performed during this stage.

Stage 4: Psychological assessment, neuropsychological, social, or cognitive problems were detected during this stage. Other information concerning reactions to progressive dyspnea, sexual dysfunction, and social support was obtained.

Stage 5: Goals. After evaluating medical, physiological, and psychological data, realistic goals that were compatible with the patient's disease, needs, and expectations were established.

The second component of a pulmonary rehabilitation program consisted of four areas of program content:

(i) **General education.** Patients were provided education about their disorder. The general philosophy was to teach patients skills to permit them to become allies with their physicians in the management of the patient's respiratory condition.

(ii) **Respiratory and chest physiotherapy instruction.** Each patient's need for respiratory care techniques was assessed and instruction provided on various topics, including: (a) bronchial hygiene; (b) breathing retraining techniques, including pursed lips breathing; (c) proper use of respiratory equipment; and (d) appropriate use of oxygen.

(iii) **Exercise prescription.** Developing a standard program of exercise for each patient was another content area. The aim was to prescribe a specific regimen, consisting of simple, useful, and inexpensive exercises, to improve a patient's exercise tolerance and endurance.

(iv) **Psychosocial support.** The final content area consisted of teaching patients to combat feelings of hopelessness and the inability to cope with their COPD. Significant others of patients were involved in this component of treatment. Significant changes in the condition of COPD patients were reported in several studies, including those conducted by Ries et al. (1995) and Niederman et al. (1991). Ries (1995) suggested that the following findings could be expected as a result of pulmonary rehabilitation of COPD patients: (i) decreases in respiratory and psychological symptoms, and the patient's use of medical resources; (ii) increases in quality of life, physical activity, exercise tolerance, activities of daily living, knowledge about COPD, and independence; and (iii) changes in lung function, possible return to work, and longer survival. As will be noted, these changes represent a gold standard for pulmonary rehabilitation because no single study has produced all of the findings.

Asthma Management

In order to deal effectively with untoward consequences of asthma, two specialists should be involved in the management of this condition: the physician and behavioural therapist.

Three distinct areas are usually considered in management.

- Abnormal pulmonary function must be modified
- Untoward emotional disturbances must be changed
- Maladaptive asthma related to inconsistent family behaviours must be altered.

In tracing psychological approaches to asthma, two stages can be delineated.

Stage 1: Psychotherapy, particularly psychoanalysis

It was pointed out earlier that in the 1940s most psychiatric and psychological treatment was guided by the work of French and Alexander (1941). As the influence of psycho- analysis declined, the relationship of resolving underlying presumed psychological factors to help manage asthma also faded (Renne& Creer, 1985). The focus shifted towards intervention with treatable problems rather than a search for psychological underpinnings to what is a physical problem: asthma.

Stage 2: Behavioral approaches

The rise of behavioral therapy approaches in the 1960s and 1970s shifted the focus of asthma management to resolving asthma-related psychological and behavioral problems. The theme is ongoing as health psychologists investigate, for example, behavioral and psychological aspects of asthma medication (Bender & Milgrom, 1992), or fears generated by the disorder (Creer & Christian, 1976). In the meantime, health psychologists have been successful in developing approaches for treating panic, correct inhaler use, overuse of hospital services, phobic behaviors, and compliance behaviors (Creer, 1979, 1992). Many of the techniques developed by health psychologists, such as using negotiation and contracting, were introduced to effectively manage medication compliance in children with asthma in the 1970s. An example of the use of negotiation and contracting was summarized by Creer (1979):

Over two decades ago, the staff at the National Asthma Center, a residential treatment facility for pediatric asthma, found that more and more children returned to single-parent homes when discharged. As parents often worked, the children were required to treat some attacks by themselves. Hence, it was decided by behavioral and medical scientists to teach children to take their own medications as directed. The procedure involved three steps. First, negotiation took place between psychologists, physicians, and children. Clinical psychologists mediated the discussions and wrote up the negotiated contract. The typical contract, signed by a physician, psychologist, and their patient, specified that when a child demonstrated he or she was taking medications correctly, the youngster would be discharged from the facility. Second, once given medications, psychologists established a procedure whereby children were closely monitored for a period of time to insure that they were taking their medications correctly. Monitoring was gradually faded from the procedure as the children showed they were properly taking their medications. Finally, upon demonstrating that he or she was taking his or her medications as prescribed over a period of time and in different settings, the child, his or her physician, and the youngster's psychologist set the discharge date for the child to return home. Monthly reports from the children and their parents indicated that the procedure was extremely effective in helping the children control their asthma.

These procedures are still state-of-the-art with respect to the management of pediatric asthma. This does not mean that the problem of medication compliance has been solved; however, it remains a major barrier to the proper treatment of both COPD and asthma.
Five approaches can be used under this approach:

1. Relaxation training
2. Biofeedback
3. Operant conditioning
4. Systematic desensitization and
5. Self-management

1. **Relaxation training:**

   When people are anxious, they tend to take rapid, shallow breaths that come directly from the chest. This type of breathing is called thoracic or chest breathing. When you're feeling anxious, you may not even be aware that you're breathing this way.

   Chest breathing disturbs the oxygen and carbon dioxide levels in the body, resulting in increased heart rate, dizziness, muscle tension and other physical sensations. This may signal your body to produce a stress response that contributes to anxiety and panic attacks.

   Relaxation and stress-management techniques can help control asthma symptoms. One relaxation technique that is effective is - "deep breathing relaxation," or "belly breathing."

   Steps instructions :

   1. Put one hand on your belly. Put the other hand on your chest, right in the middle.
   2. When you breathe in, push your belly out. You should feel the hand on your belly move out, but not the hand on your chest.
   3. Close your eyes and push all the air out of your lungs through your mouth while almost closing your lips. Imagine that you are blowing up a balloon.
   4. Next, take a slow, deep breath. Fill your lungs up, and feel your belly rise.
   5. Breathe in and out three times in this deeper way.

2. **Biofeedback:**

   Biofeedback is the name given to a wide variety of procedures wherein some aspects of an individual's physiological functioning is systematically monitored and fed back to that individual, typically in the form of an auditory or visual signal. The individual's task then is to modify that signal in order to change that Bozzay reported an illustrative serious asthmatic child case treated by galvanic skin response. His aim was to teach the child to manage asthmatic episodes alone. This was done by training the child to relax in order to dampen the concomitant emotional overtones resulting from annoying environmental and family stimuli. The findings showed his wheezing subsided and breathing became smoother and easier within 15 minutes of the onset of attacks. Nevertheless, biofeedback has yet to be adopted as a dependable therapeutic approach because of the expense of the equipment and the lack of zeal shown by researchers.

3. **Operant conditioning**

   To deal with maladaptive asthma-related behaviours exhibited by some patients, such as malingering, various secondary gains from the illness and inadequate social behaviours, the implementation of certain measures could result in the effective management of asthmatics, especially children. Positive reinforcement could teach the young child how to use appropriately and effectively the forced expiratory volume (FEV) device, although exaggerated reinforcement over a long period could diminish its worth. This technique does not alleviate asthma severity but helps to eliminate maladaptive secondary gains the patient acquires from the illness.

   Learned maladaptive habit response can be eliminated if it is not reinforced (Moore N 1965). Therefore, parents should not pay too much attention to the frequency of a child's cough which is used to obtain attention and sympathy. Parents might use the time-out procedure to eliminate undesirable learned responses.

4. **Systematic desensitization:**

   Emotional overtone is presumed to be a catalyst in triggering asthmatic attacks. Consequently, instituting systematic desensitization could be of therapeutic value in aborting resultant anxiety which complicates the asthma attack. Some subjects have been trained successfully with systematic desensitization, assertiveness training and progressive relaxation. These subjects have reported a dramatic improvement in controlling the emotional upset caused by exogenous noxious stimuli.

5. **Self-management**

Processes involved in asthma self-management are summarized according to the model of self-management proposed by Creer (1997) and Creer and Holroyd (1997). It consists of the following processes.

(i) **Goal setting**

Setting goals involves patients reaching an accord with their physicians and other medical personnel how best to control a patient's asthma. Typical goals jointly established by physicians and patients include patient's taking prescribed medications as scheduled and avoiding known triggers of their asthma.

(ii) **Information collection**

Self-monitoring the observation and recording of data on oneself is the basis of information collection. In most studies of asthma self-management, a diary is employed. Atypical asthma diary would ask patients to list the symptoms they experienced daily, their highest peak flow readings obtained in the morning and evening, and the medications they took for their asthma. Asthma diaries also serve to prompt patients to monitor their asthma. So as not to overwhelm patients in monitoring their behavior, Creer and Bender (1993) cautioned that target behaviors should be operationally defined, and that patients collect data for only specified periods of time.

(iii) Information processing and evaluation Patients must learn to process and evaluate the information they collect about themselves and their asthma. ***Creer and Holroyd (1997)*** delineated five steps involved in information processing and evaluation: (a) patients must be able to detect changes that occur in the information they observe, record, and process about themselves; (b) a set of objective criteria must be established, such as their personal best peak flow reading, against which patients can evaluate any changes that have occurred in their breathing; (c) patients must evaluate and make judgments about the data they process; (d) patients must evaluate any changes that occur in their breathing in terms of the antecedents that may have produced the change, the behaviors they need to perform to alter any changes, and the consequences of their actions; and (e) asthma patients need to consider the contextual variables in processing and evaluating information about themselves.

(iv)Decision making the correct decision is essential for successful self-management. A study by Creer (1990) analyzed decision making in gold standard physicians and patients. It was determined that both groups used decision making strategies similar to those describe by Arkes (1981) in that they:

(a) avoided pre-conceived notions;

(b) were thoughtful and cautious;

(c) generated a number of management alternatives;

(d) did not misperceive the severity of the situation;

(e) referred to a personal database;

(f) thought in terms of probabilities; and

(g) did not rely on memory.

(v) Action

Action involves the performance of self- management skills to help control asthma. A number of components have been tested in self- management programs for asthma, but self- instruction, including self-statements, is a common thread of these programs.

(vi) Self- reaction

Self-reaction refers to the attention patients direct towards evaluating their performance (Bandura, 1986). On the basis of self-reaction, patients can establish realistic expectations about whether they need more training and expertise. Self-efficacy,the belief that one can adequately perform specific skills in a given situation (Bandura, 1977), influences the performance of self-management skills. The development of self-efficacy scales have permitted this construct to be evaluated in asthma (Tobin, Wigal, Winder, Holroyd, & Creer, 1986). A number of self-management programs have been developed and evaluated for both pediatric and adult asthma. Common outcomes of these programs include a decrease in related hospitalizations, ER visits, school and work absenteeism, asthma medication use, depression, and costs for asthma. At the same time, there have been reported increases in knowledge about asthma, performance of self-management skills, medication compliance, self-efficacy, and quality of life. The positive results have led expert panels on asthma (e.g., National Asthma Education Program, 1991) to recommend such training for all patients with the disorder.

UNIT-V

GASTROINTESTINAL SYSTEM

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GASTROINTESTINAL SYSTEM

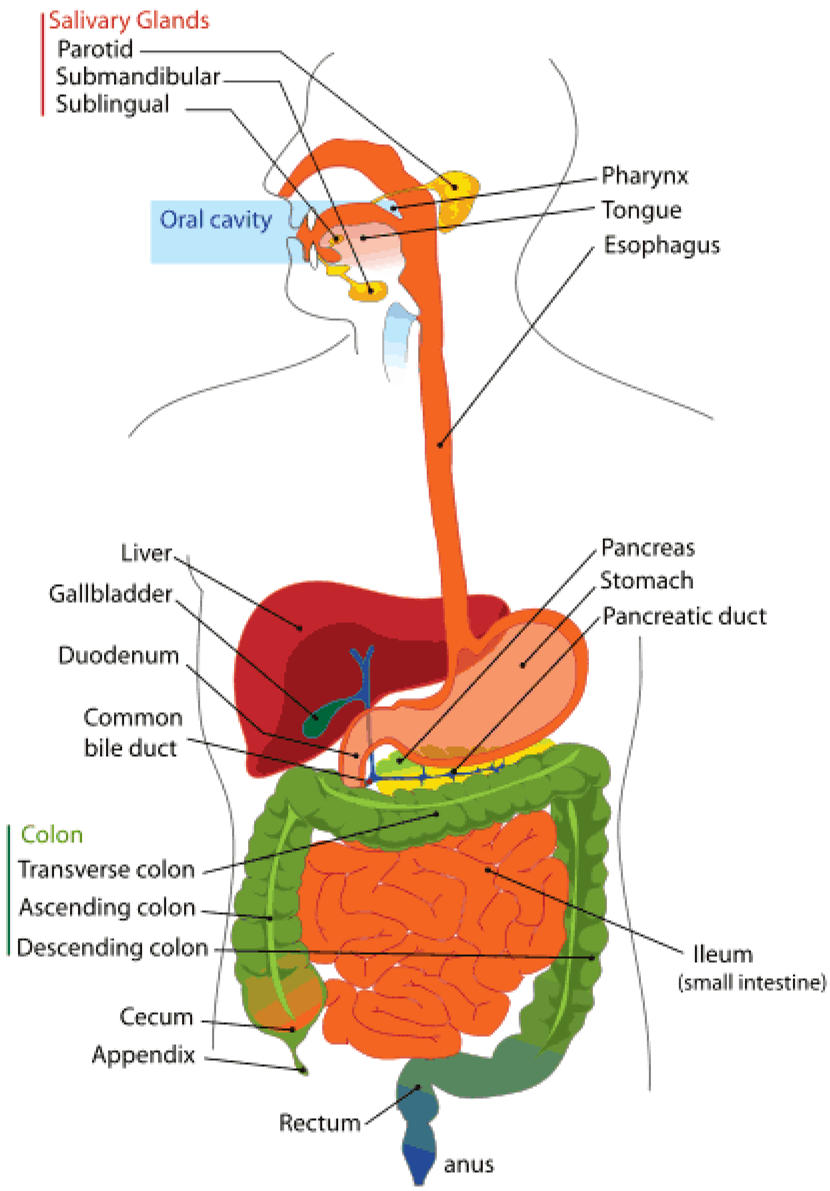
The gastrointestinal (or digestive) system is responsible for taking in food, using nutrients, and eliminating waste products. Nutrients are absorbed from food into the bloodstream and transported to all cells of the body. Nutrients provide energy and contribute to the body's growth and repair.

Gastrointestinal (GI) Tract

The gastrointestinal (GI) tract (also digestive tract, or alimentary canal) is the system of organs in the body that takes in food, digests it for the absorption of nutrients and energy, and expels waste material. The major functions of the GI tract are categorized as four distinct processes:

- **Ingestion** is the consumption of food and other substances through the mouth, as they pass by chewing and swallowing into the GI tract.
- **Digestion** is the process of metabolism by which ingested substances are mechanically and chemically converted for use by the body. Digestion is further categorized into three distinct phases: the cephalic phase in which taste and smell stimulate the nervous system to prepare the body for eating and digestion; the gastric phase in which passage of food into the stomach stimulates the release of gastric juices and pH balancing mechanisms throughout the system; and the intestinal phase in which excitatory and inhibitory reflexes control the passage of partially digested food into and through the intestines.
- **Absorption** is the movement of metabolized nutrients and water from the digestive system into the circulatory and lymphatic capillaries by osmosis, active transport, and diffusion through the cells in the walls and surrounding layers of intestines and their supporting circulatory systems.
- **Excretion** or egestion is the elimination of undigested, mostly solid material from the GI tract by defecation. Fluid products of metabolism throughout the body are also excreted by organ systems not directly part of the GI tract and digestive system, such as the kidneys, skin and lungs.

In addition to processing nutrients as the principal pathways of the digestive system, the GI tract is also a prominent part of the immune system, providing various levels of defense against pathogenic microorganisms and potentially toxic substances throughout the path of digestion. Dysfunction anywhere in the GI tract, whether by disease, trauma, or anatomical anomaly, can result in symptoms or conditions affecting the well-being of the entire individual. Many diseases and disorders of the GI tract can result in feeding difficulties in children and infants.



Digestive System Diagram

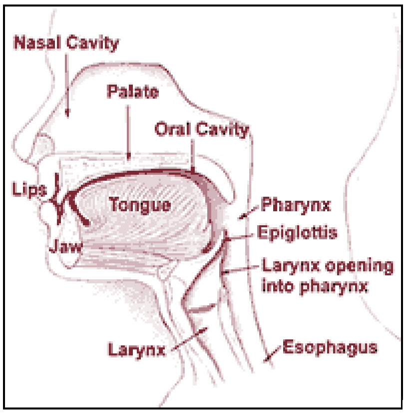
The GI tract is conventionally divided into upper and lower parts, with associated accessory organs.

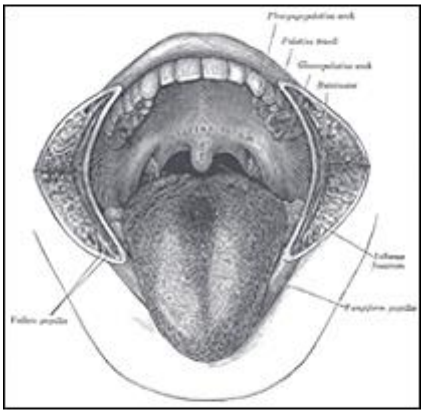
Upper GI Tract

The Upper GI tract consists of the mouth, pharynx, esophagus and stomach. This is where ingestion and the first phase of digestion occur.

The Mouth

- Includes the tongue, teeth, and buccal mucosa or mucous membranes containing the ends of the salivary glands, continuous with the soft palate, floor of the mouth and underside of the tongue.
- Chewing (mastication) is the mechanical process by which food, constantly repositioned by muscular action of the tongue and cheeks, is crushed and ground by the teeth through the muscular action of the lower jaw (mandible) against the fixed resistance of the upper jaw (maxilla).
- Saliva excreted in the oral cavity by three pairs of exocrine glands (parotid, submandibular, and sublingual) is mixed with chewed food to form a bolus, or ball-shaped mass.
- There are two types of saliva: a thin watery secretion that wets the food and a thick mucus secretion that lubricates and causes the food particles to stick together to form the bolus.
- Digestive enzymes in saliva begin the chemical breakdown of food, primarily starches at this point, almost immediately.





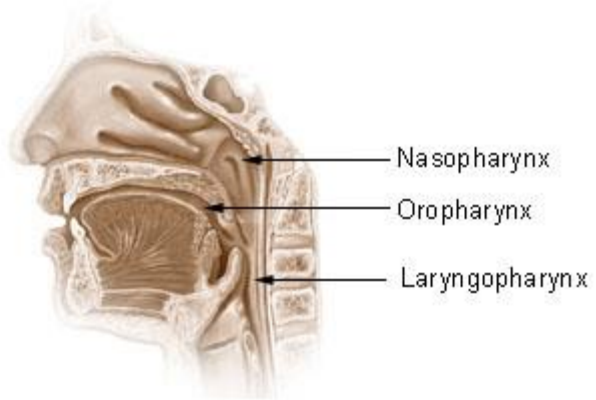
Head and Neck and Isthmus of the Fauces

The Pharynx

- The pharynx is contained in the neck and throat and functions as part of both the digestive system and the respiratory system.
- The human pharynx is divided into three sections:
  - The nasopharynx behind the nasal cavity and above the soft palate;
  - The oropharynx behind the oral cavity and including the base of the tongue, the tonsils, and the uvula

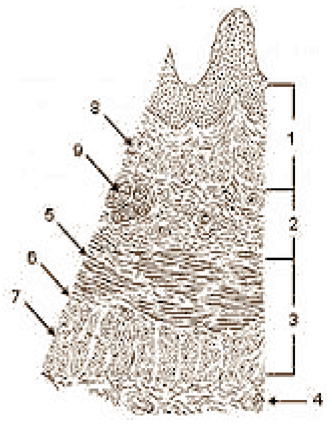
- The hypopharynx or laryngopharynx includes the junction with the esophagus and the larynx, where respiratory and digestive pathways diverge.

- The swallowing reflex is initiated by touch receptors in the pharynx as the bolus of chewed food is pushed to the back of the mouth.
- Swallowing automatically closes down the respiratory or breathing pathway as an anti-choking reflex.
- Failure or confusion of reflexes at this point can result in aspiration of solid or liquid food into the trachea and lungs.



The Esophagus

- The esophagus is the hollow muscular tube through which food passes from the pharynx to the stomach.
- It is lined with mucous membrane continuous with the mucosa of the mouth and into which open the esophageal glands.
- It is surrounded by relatively deep muscles that move the swallowed bolus of masticated food through peristaltic action, piercing the thoracic diaphragm to reach the stomach.



Layers of Esophageal Wall:
1. Mucosa
2. Submucosa
3. Muscularis
4. Adventitia
5. Striated muscle
6. Striated and smooth
7. Smooth muscle
8. Lamina muscularis mucosae
9. Esophageal glands

Esophageal Glands

The Stomach

- The stomach is a hollow muscular organ, located below the diaphragm and above the small intestine that receives and holds masticated food to begin the next phase of digestion.
- Two smooth muscle valves, the esophageal sphincter above and the pyloric sphincter below, keep stomach contents contained.
- The stomach is surrounded by stimulant (parasympathetic) and inhibitor (orthosympathetic) nerve plexuses which regulate both secretory and muscular activity during digestion.
- With a volume of as little as 50 mL when empty, the adult human stomach may comfortably contain about a liter of food after a meal, or uncomfortably as much as 4 liters of liquid.

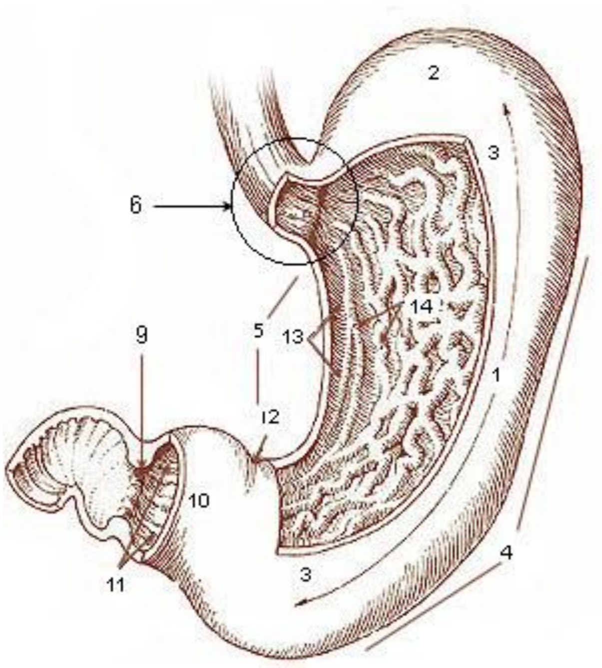


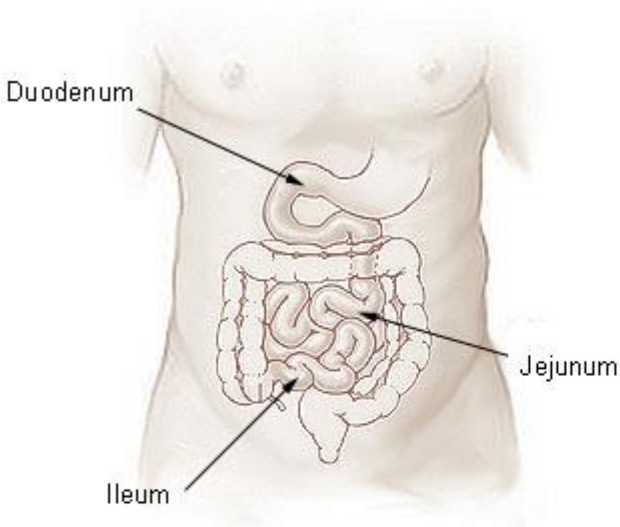
Diagram of the Stomach

Lower GI Tract

The lower GI tract includes the small intestine and large intestine, beginning after the stomach and terminating at the anus. Its function is to complete the digestion and absorption of nutrients and to prepare waste products for elimination from the digestive system.

Small Intestine

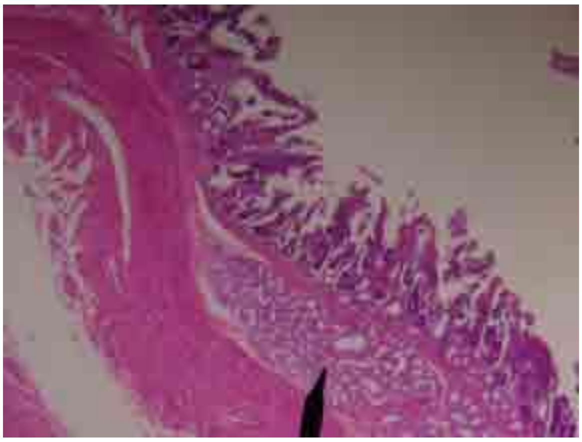
The small intestine is where most digestion takes place. It is structurally divided into three parts: the duodenum, the jejunum, and the ileum. Among humans over five years old, the small intestine tends to vary in length from 4-7 meters (13-23 feet).



Small Intestine

The Duodenum

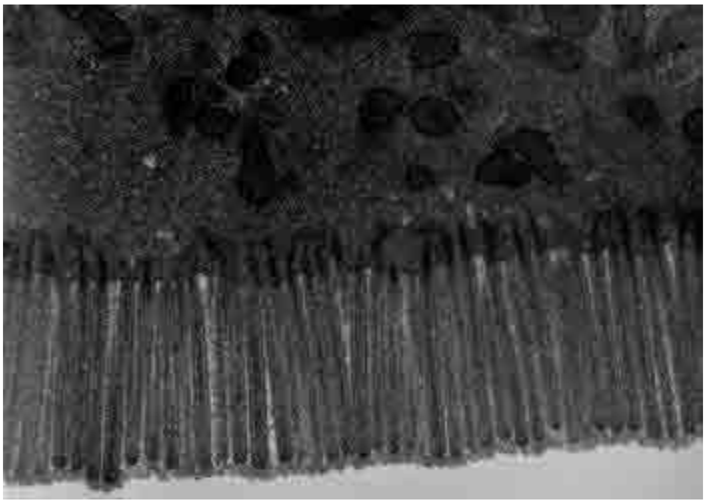
- The duodenum consists of four parts, with the first three forming a "C" shape.
    - The first or superior part of the duodenum begins at the pylors, passing laterally for a short distance before curving into the superior duodenal flexure.
    - The second, or descending, part of the duodenum passes from the superior into the inferior duodenal flexure, and is where the pancreatic and common bile ducts enter the GI tract.
    - The third or inferior horizontal part of the duodenum passes from the inferior flexure, crossing the aorta (major artery) and inferior vena cava (major vein) as well as the spinal column.
    - The forth or ascending part of the duodenum passes over the aorta, and curves past the pancreas to the duodenojejunal flexure.
    - The duodenum is where most of the breakdown of food in the small intestines occurs.
    - It is here that Brunner's glands produce an alkaline secretion to protect the duodenum from acidic chyme entering from the stomach and to activate intestinal enzymes enabling digestion and absorption.



Duodenum-Brunner's Glands

The Jejunum

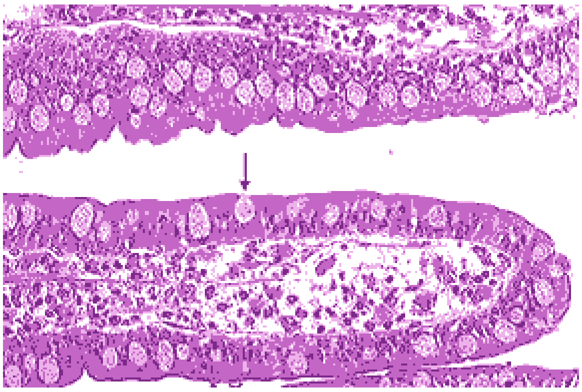
- The jejunum begins at the ligament of Treitz in the duodenojejunal flexure and continues to the ileum.
- The inner surface or mucous membrane of the jejunum is covered by villi (small finger-like structures) much longer than found in the duodenum or ileum, contained in many large circular folds (plicaecirculares) which provide extensive surface area for absorption of nutrients.
    - The villi can increase intestinal absorptive surface area by a factor of 30.
    - The microvilli, extensions of the villi, increase the surface area by an additional factor of 600.
    - Villus capillaries collect amino acids and simple sugars.
    - Villus lacteals or lymphatic capillaries absorb dietary fats.



Microvilli

The Ileum

- The ileum is the final and longest section of the small intestine.
- Both the jejunum and the ileum are suspended by mesentery, a double layer of peritoneum that allows these parts of the intestine to move more freely within the abdomen.
- Like the jejunum, the wall of the ileum has many folds and villi to increase both absorption of enzymes and absorption of nutrients. It also has an increasing number of goblet cells.
- The ileum is responsible for the final stages of protein and carbohydrate digestion, as contents are pushed along by peristaltic waves of smooth muscle contractions.
- There is no absolute demarcation between the jejunum and the ileum, but the ileum tends to have more fat inside the mesentery and has a relatively decreasing diameter.
- Unlike the rest of the small intestine, the ileum has abundant Peyer's patches, lymphoid follicles similar to lymph nodes, which function as an important component of the immune system response to pathogenic organisms in the GI tract.

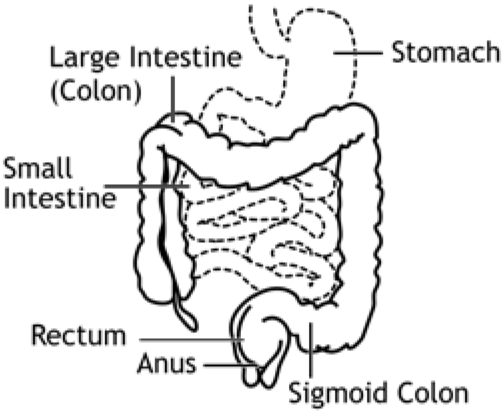


Goblet Cells

Large Intestine

- Also commonly referred to by the name of its longest component, the colon, the large intestine is the last part of the digestive system.
- Its principal function is to absorb remaining water from the waste products of digestion as it compacts the accumulated waste for periodic elimination by defecation.
- While food is not broken down further at this stage, the fluid absorption function of the large intestine does act to gather in vitamins created by beneficial bacteria or flora inhabiting the colon.

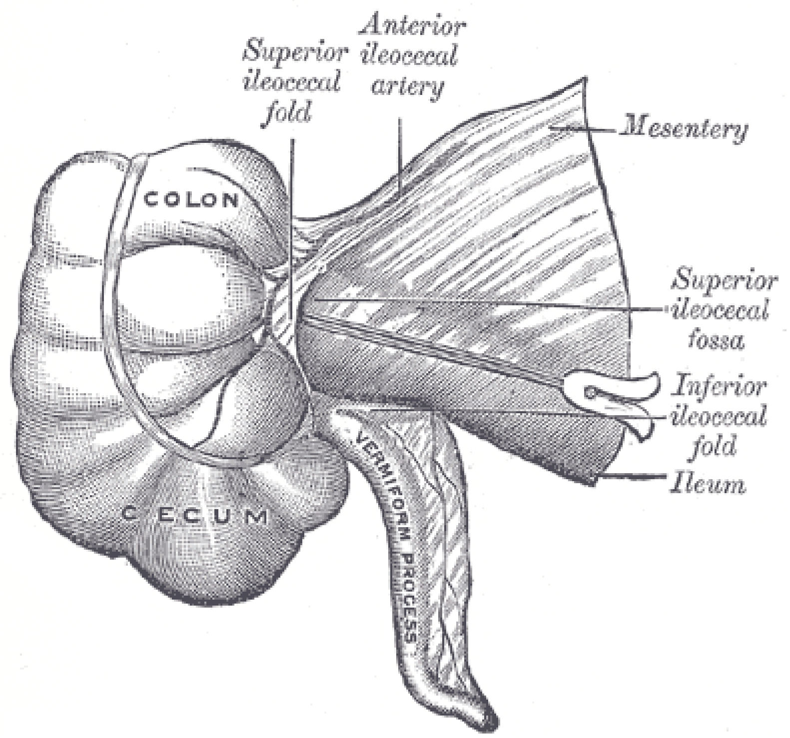
- Instead of the predominance of evaginations of villi found in the small intestine, the large intestine has increased invaginations of glands and an abundance of goblet cells. The large intestine is structurally divided into three parts: cecum, colon, and rectum.



Large Intestine

The Cecum

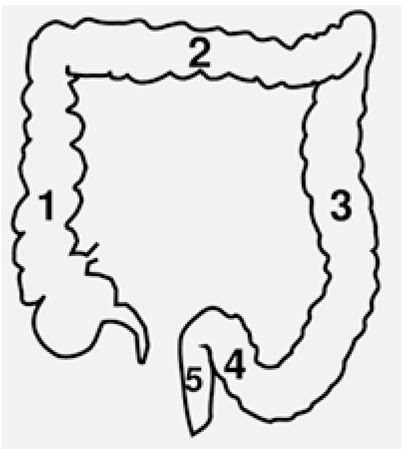
- The cecum is a pouch at the beginning of the large intestine, separated from the ileum of the small intestine by the ileocecal lower right quadrant of the abdomen.
- The cecum is host to a large number of bacteria which aid in the final enzymatic processing of material not completely digested in the small intestine.
- The vermiform appendix is a worm-like cul-de-sac attachment of the cecum, until recently considered entirely vestigial in humans, but now thought to have a role as a haven for the beneficial gut flora, as well as the site of infection-fighting lymphoid cells.



Gray's Cecum

The Colon

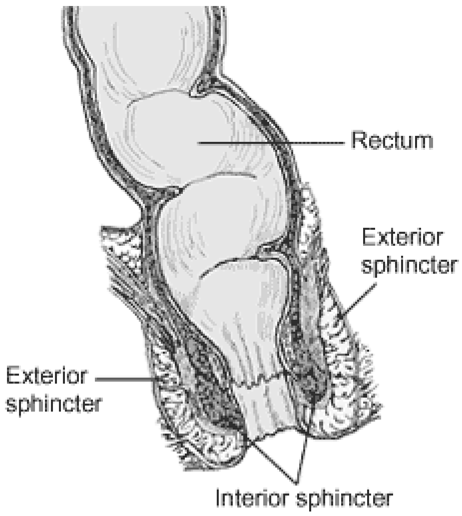
- The colon consists of four parts named for their relative orientation in the abdomen and the rectum:



  - The ascending colon (1)
  - The transverse colon (2)
  - The descending colon (3)
  - The sigmoid colon (4)

  - The rectum (5)

- By the time the chyme has reached the colon, almost all nutrients and most of the water have already been absorbed by the body.
- It is here that the chyme is mixed with mucus and bacteria to become feces. The waste products of bacterial metabolism include some nutrients used by the cells lining the colon for their own nourishment.
- The colon ends at the junction of the sigmoid colon and the rectum.
  - The rectum is the last part of the large intestine, beginning at and continuous with the colon, and terminating at the anus.
  - The rectum provides temporary storage for feces
  - Stretch receptors of the nervous system located in the rectal walls stimulate the desire to defecate. As peristaltic waves propel the feces into the anal canal, external and internal sphincters allow the final exit of waste material from the GI tract.



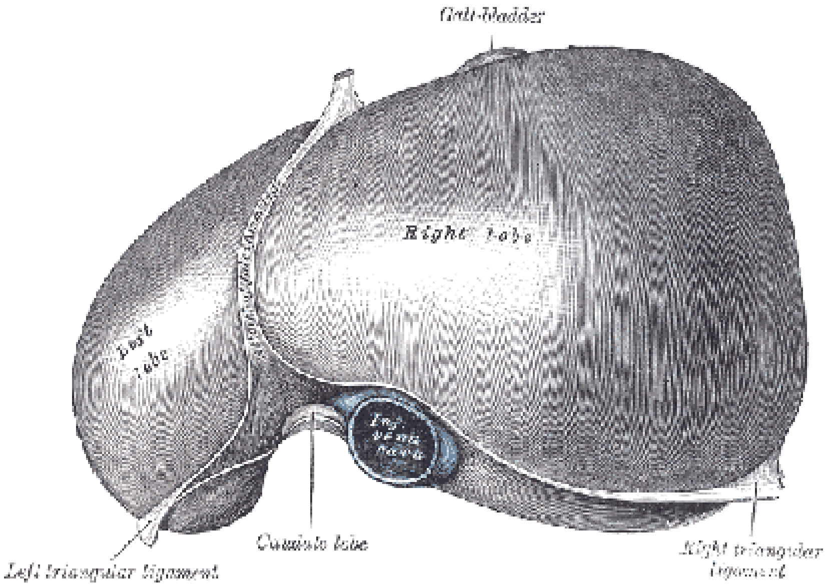
Anorectum

Accessory Organs

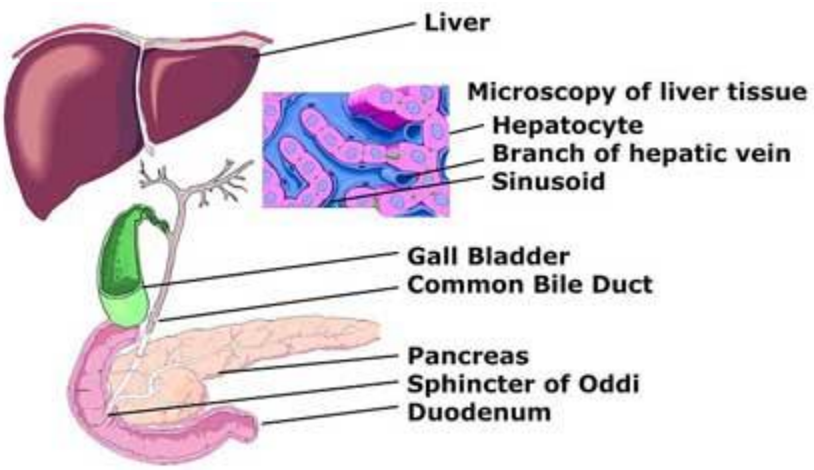
Accessory to the alimentary canal of the GI tract are various secretory, storage, and waste filtering organs and related hormonal glands. Principal among these are the liver, gallbladder, and pancreas.

The Liver

- The liver secretes bile, produced by its hepatocytes, into the duodenum of the small intestine via the biliary system. Bile acts as a kind of detergent, emulsifying fats to promote enzyme action in the intestines.
- Epithelial cells in the liver add a watery solution rich in bicarbonates that act to dilute and neutralize acids at this stage of digestion.
- Cholesterol is also released with the bile and is important for the metabolism of fat soluble vitamins as well as maintenance of normal cell membranes throughout the body.
- Consistent with its major role in metabolism, the liver has a number of functions not strictly related to digestion, such as decomposition of red blood cells, plasma protein synthesis and detoxification.
- The liver is the largest gland in the human body and performs or regulates a wide variety of high-volume reactions involving very specialized tissues.



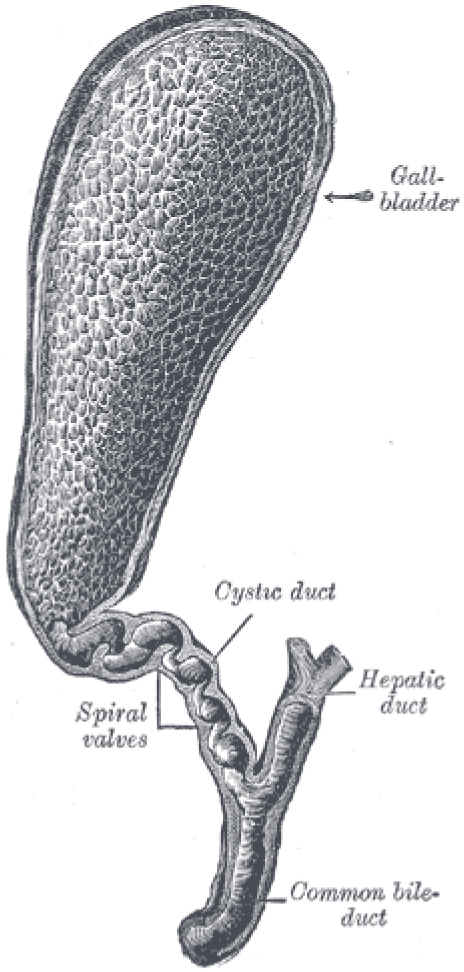
Liver



Biliary System

The Gallbladder

- The gallbladder is connected to the liver and the duodenum by the biliary tract.
- The gallbladder stores the bile (or gall) secreted by the liver until its release is triggered by the digestive process.
- The interior of the gallbladder has a simple columnar epithelial lining characterized by recesses or pouches, which provide the volume for storage.
- The cystic duct connects the gallbladder to common hepatic duct to form the common bile duct.

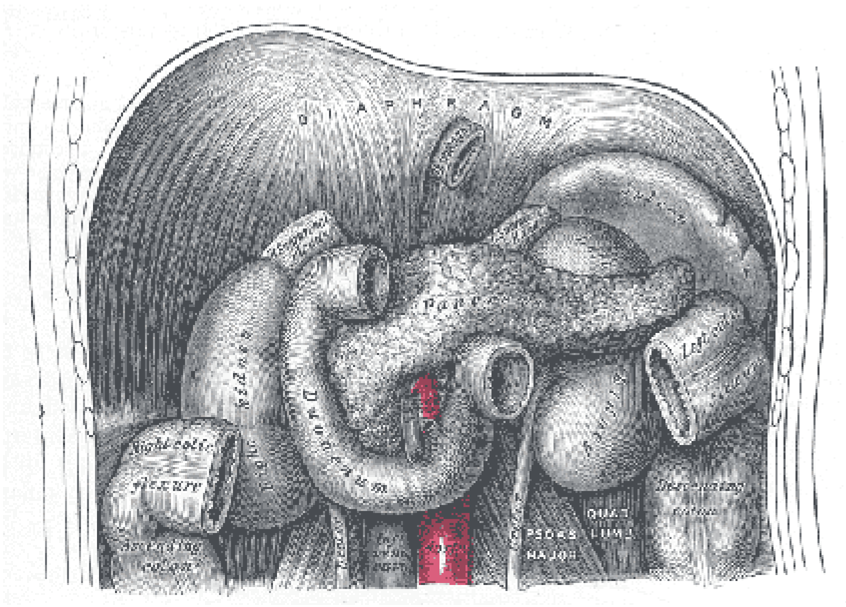


Gallbladder and Simple Columnar Epithelium

The Pancreas

- The pancreas is another relatively large gland that functions as part of both the digestive and endocrine system.
- Its exocrine function is to produce and secrete pancreatic juice rich in digestive enzymes.
- Its endocrine functions include the production of important hormones such as
    - insulin, which helps regulate metabolism at the global and cellular level
    - glucagon, which acts opposite insulin
    - somatostatin, which acts to suppress the release of various other GI hormones and lower the rate of gastric emptying as digestion approaches completion
- The pancreatic duct joins the common bile duct, together entering the major duodenal papilla through the hepatopancreatic ampulla.

- Because the human pancreas is not completely matured until age two, otherwise normal infants will have digestive problems with entire categories of foods until all the appropriate enzymes can be properly secreted by the fully developed pancreas.



Pancreas

GI Histology

In addition to the sequential phases of digestion and the corresponding structure and function of each specific organ in the digestive system, the GI tract is characterized as having four concentric layers of tissue with functional differences in each specialized region.

The Mucosa

- The mucosa (moist linings of mucous membranes) is the first main layer, consists of
    - The primary epithelium, tissues lining the cavities and surfaces of structures throughout the body
    - The lamina propria, containing capillaries, lymph vessels, and glands with ducts opening on to the epithelium
    - The muscularis mucosae, a thin layer of smooth muscle

The Submucosa

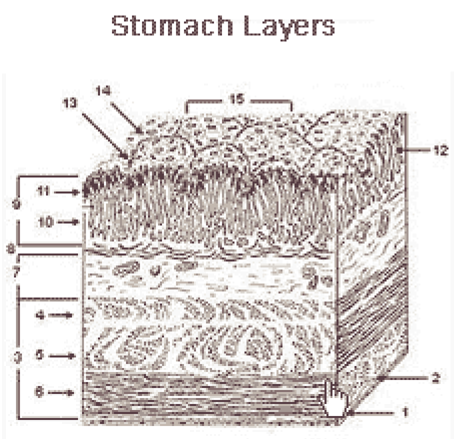
The submucosa consists of fibrous connective tissue, separating the mucosa from the surrounding muscles, and includes fine bundles of nerve plexuses

The Muscularis Externa

- The muscularis externa, external muscle layer, generally has two distinct layers of smooth muscle, the inner (circular) and outer (longitudinal)
- In the stomach, there is a third layer (inner oblique) responsible for the churning or mechanical breakdown of food.
- In the esophagus, part of the external muscle layer is skeletal muscle rather than smooth muscle
- The pyloric and anal sphincters are also formed by the inner layer of the muscularisexterna

The Serosa

The serosa (serous membrane) consists of layers of connective tissue continuous with the peritoneum, which forms the lining of the abdominal cavity and serves as conduit for blood vessels, lymph vessels, and nerves serving the contained organs.



Layers of Stomach Wall:
1. Serosa
2. Tela subserosa
3. Muscularis
4. Oblique fibers of muscle wall
5. Circular muscle layer
6. Longitudinal muscle layer
7. Submucosa
8. Lamina muscularis mucosae
9. Mucosa
10. Lamina propria
11. Epithelium
12. Gastric glands
13. Gastric pits
14. Villous folds
15. Gastric areas (gastric surface)

Section of the human esophagus. Moderately magnified. The section is transverse and from near the middle of the gullet.
a. Fibrous covering.
b. Divided fibers of longitudinal muscular coat.
c. Transverse muscular fibers.
d. Submucous or areolar layer.
e. Muscularis mucosae.
f. Mucous membrane, with vessels and part of a lymphoid nodule.
g. Stratified epithelial lining.
h. Mucous gland.
i. Gland duct.
m'. Striated muscular fibers cut across.

Disorder of gastrointestinal system

It is basically divided into two types:-

1) Functional disorder

2) Structural disorder

"Disease" versus "Illness" and "Structural" versus "Functional"

Although often used interchangeably, the terms disease and illness actually connote two fundamentally distinct concepts. Disease is defined as pathology that can be objectively seen on blood tests, x-rays, endoscopies, or other diagnostic tests. For example, on endoscopy esophagiis is seen as inflammation of the esophagus; on blood test diabetes is signified by a high blood-sugar. Illness, on the other hand is what one feels. The patient experiences symptoms and feels ill. Illness is subjective and cannot be easily measured on routine tests.

When illness results from a visible disease (pathology or physiologic abnormality) it is said to be "structural." For example, consider chest pain that is caused by a myocardial infarction or shortness of breath caused by pneumonia. However, often patients experience illness without underlying disease. For instance, on MRI patients with chronic headaches may have a completely normal brain structure yet may still suffer from persistent symptoms.

The same phenomenon often occurs in the gastrointestinal tract. While gastrointestinal symptoms such as nausea, abdominal pain, or constipation may be sometimes caused by a demonstrable anatomical cause (e.g., peptic ulcer, bowel obstruction, or diverticulitis),quite frequently these symptoms have no underlying anatomical or biochemical bases.

5Instead, these symptoms are thought to relate to disorders of gastrointestinal function, such as abnormal intestinal motility (dysmotility or abnormal contractility of the gut), abnormal intestinal perception (visceral hypersensitivity or increased sensitivity to gastrointestinal sensations), and/or abnormal brain-gut communication. Hence the term, "functional" GI disorders.

However, despite these findings:

1) These functional abnormalities can often only be detected through expensive and invasive tests that are not available in clinical practice.

2) The severity of the functional abnormalities varies from patient to patient and does not correlate with the severity of symptoms.

3) Whether these abnormalities are the cause or the result of the functional GI disorder is sometimes unclear.

These disorders are as follows:-

- **Irritable Bowel Disease :-**

  IBS is a common, sometime disabling (Corney & Stanton, 1990), and poorly under-stood functional condition of the lower GI tract. It is most often characterized by abdominal pain and change in bowel habit (diarrhea and/or constipation) occurring in the absence of abnormalities on the appropriate physical and laboratory investigations. However, individuals may often suffer other GI symptoms such as bloating, flatulence, nausea, belching, abdominal distention, mucous stools, abdominal rumbling and gurgling, and loss of appetite and/or weight. Non-gastrointestinal symptoms, including headache, palpitation, dyspnea, fatigue, weakness, muscular pains and disorders of anxiety and mood are also often reported.

  In 1988, the Thirteenth International Congress of Gastroenterology met in Rome, Italy, to attempt a more positive diagnostic classification of IBS. Their decision was based in part on the work of Manning and his colleagues (Manning, Thompson, Heaton, & Morris,1978), who had compared the prevalence of 15 symptoms in patients with IBS to that of individuals suffering from organic GI disease, and found that six primary symptoms are as follows:-

  (i) visible abdominal distention

  (ii) pain relieved by defecation

  (iii) more frequent stools with pain onset

  (iv) looser stools with pain

  (v) passage of mucus, and

  (vi) a subjective sensation of incomplete evacuation differentiated IBS from bowel disease.

The Rome Criteria for IBS are as follows:-

At least three months continuous or recurrent symptoms of:

1. Abdominal pain or discomfort which is

   (a) relieved by defecation,

   (b) and/or associated with a change in frequency of stool,

   (c) and/or associated with a change in consistency of stool; and
2. Two or more of the following, on at least a quarter of occasions or days:

   (a) altered stool frequency (either more than three bowel movements per day or fewer than three bowel movements per week

(b) altered stool form (lumpy/hard or loose/watery stool),

(c) altered stool passage (straining, urgency, or feeling of incomplete evacuation),

(d) passage of mucus,

(e) bloating or feeling of abdominal distention.

The Rome Criteria, as they have come to be known, are currently the most commonly accepted and utilized for IBS.

- **Functional bowel syndrome (FBS) :-**

  Since the classification of IBS in 1988, another team has constructed a similar system for all the functional gastrointestinal disorders, and has placed IBS in the category of the functional bowel disorders. Thompson et al. (1992) define these disorders, propose diagnostic criteria, and discuss their differentiation from IBS.

- **Functional abdominal bloating (FAB)** is a disorder characterized by a feeling of fullness, bloating and/or abdominal distention. Audible bowel sounds and expulsion of gas may also be present. Symptoms must have been present for at least three months, and must not be related to dietary factors. Individuals with FAB do not meet diagnostic criteria for IBS or other functional bowel disorders.

- **Functional constipation (FC)** is defined as persistent symptoms of difficult, infrequent, or seemingly incomplete defecation, which has no other apparent etiology. Symptoms must be present again, for at least three months, individuals must report straining upon defecation at least 25% of the time, lumpy or hard stools, a sensation of incomplete evacuation, and two or fewer bowel movements in a week. Abdominal pain is not present, and individuals do not meet criteria for IBS.

- **Functional diarrhea (FD)** is defined by as the frequent and/or urgent passage of unformed stool must include two or more of the following symptoms present for at least two months: unformed stool passage 75% of the time; three or more bowel movements per day, at least half of the time; and increased stool. Patients do not report abdominal pain or hard or lumpy stools. They experience urgency and may occasionally soil themselves.

- **Functional abdominal pain (FAP),** is described as six months of ongoing pain that is experienced in the abdomen, but bears no relationship to the function of the GI tract, or other physiological systems (e.g., reproductive system, etc.). As with all functional GI disorders, organic disease is conspicuously absent.

- **Constipation** is the difficult passage of stools (bowel movements) or the infrequent (less than three times a week) or incomplete passage of stools. Constipation is usually caused by inadequate "roughage" or fiber in the diet, or a disruption of the regular routine or diet. Constipation causes a person to strain during a bowel movement. It might include small, hard stools, and sometimes causes anal problems such as fissures and hemorrhoids. Constipation is rarely the sign of a more serious medical condition.

- **Inflammatory Bowel Disease (IBD) :-**

  Inflammatory bowel disease (IBD) is a descriptive label which subsumes two painful land sometimes life-threatening conditions, Crohn's disease and ulcerative colitis. Crohn's disease is best characterized by inflammation that can penetrate the layers of the entire gastrointestinal tract, often resulting in bowel obstruction and/or infection of extra intestinal systems. Ulcerative colitis manifests similarly, but involves only the mucosa of the large bowel. IBD are chronic, episodic course, with symptoms including severe abdominal pain, loss of appetite, vomiting, and diarrhea. Complications of exacerbations can include electrolyte imbalance and anemia due to vomiting, anorexia, bloody diarrhea, and mineral malabsorption.

- **Peptic Ulcer Disease (PUD) :-**

Peptic ulcers are craters or open sores in the lining of the upper gastrointestinal tract. They include **duodenal ulcers** (those that are located in the top of the small intestine or duodenum) and **gastric ulcers** (those found in the stomach). Peptic ulcers are common and usually occur singly. But it is possible to have two or more, or even both duodenal and gastric ulcers at the same time. Duodenal ulcers are more common than gastric ulcers. Peptic ulcers are caused by acid and pepsin (an enzyme) produced in the stomach.

- **Duodenal Ulcer symptoms:**

✓ Pain that awakens patients from sleep
✓ Burning or gnawing sensation in the upper abdomen
✓ Pain in the back, lower abdomen or chest area may occasionally occur

- ✓ Pain that occurs when the stomach is empty (about two hours after a mean or during the night)
- ✓ Relief frequently occurs after eating

❖ **Gastric Ulcer symptoms:**

- ✓ Gastric ulcer pain may be less severe than duodenal ulcer pain and is noticeably higher in the abdomen
- ✓ Eating may increase pain rather than relieve pain
- ✓ Pain is described as aching, nagging, cramping or dull
- ✓ Other symptoms may include nausea, vomiting and weight loss
- ✓ Some ulcers may produce no symptoms at all.

However, occasional painless bleeding, anemia (low blood count), or the passage of black, tarry stool may be the first sign of peptic ulcer disease.

➢ **Esophageal Disorders :-**

Esophageal disorders represent chronic symptoms typifying esophageal disease that have no readily identified structural or metabolic basis. Although mechanisms responsible for the disorders remain poorly understood, a combination of physiologic and psychosocial factors likely contributes toward provoking and escalating symptoms to a clinically significant level. Several diagnostic requirements are uniform across the disorders:

- ✓ Exclusion of structural or metabolic disorders potentially responsible for symptoms is essential.
- ✓ An arbitrary requirement of at least 3 months of symptoms with onset at least 6 months before diagnosis is applied to each diagnosis to establish some degree of chronicity.
- ✓ Gastro esophageal reflux disease(GERD) must be excluded as an explanation for symptoms.
- ✓ A motor disorder of the types with known histopathologic bases (eg, achalasia, scleroderma esophagus) must not be the primary symptom source.

<u>Epidemiology of Gastrointestinal Disorders :-</u>

Prevalence of IBS, identified by the Rome Criteria, was approximately 17%. Contrary to most published reports, this investigation did not uncover a gender difference in bowel symptom reporting. The prevalence of functional bowel disorders other than IBS was as follows: functional abdominal pain 26%, chronic constipation 17.4%, and chronic diarrhea 17.9%.

Drossman et al. (1993) examined the prevalence, and found that 69% of the entire sample (5430responders) reported having experienced at least one functional bowel syndrome in the previous three months, and based on the data received, estimated that9%of the nation may suffer from IBS, 3.0% from functional constipation, 1.6% from functional diarrhea, 32.1% from functional abdominal bloating, and 2% from chronic functional abdominal pain.

Findings on 15%–30% of coronary angiograms performed in patients with chest pain are normal. Although once considered a diagnosis of elderly women, chest pain without specific explanation was reported twice as commonly by subjects 15–34 years of age than by subjects older than 45 years of age in a pain is not well established.

Between 7% and 8% of respondents from a householders survey reported dysphagia that was unexplained by questionnaire ascertained disorders. Less than 1% report frequent dysphagia. Functional dysphagia is the least prevalent of these functional esophageal disorders.

➢ **Structural disorders**

Structural disorders are those in which the bowel looks abnormal and doesn’t work properly. Sometimes, the structural abnormality needs to be removed surgically. The most common structural disorders are those affecting the anus, as well as diverticular disease and cancer.

❖ **Anal disorders**

✓ **Hemorrhoids** are swollen blood vessels that line the anal opening caused by chronic excess pressure from straining during a bowel movement, persistent diarrhea, or pregnancy. There are two types of hemorrhoids: internal and external.

- **Internal hemorrhoids** are normal structures cushioning the lower rectum and protecting it from damage by stool. When they fall down into the anus as a result of straining, they become irritated and start to bleed. Ultimately, internal hemorrhoids can fall down enough to prolapse (sink or protrude) out of the anus.

- **External hemorrhoidal** are veins that lie just under the skin on the outside of the anus. , after straining, the external hemorrhoidal veins burst and a blood clot forms under the skin. This very painful condition is called a pile.

- ✓ **Anal fissures** are splits or cracks in the lining of the anal opening. The most common cause of an anal fissure is the passage of very hard or watery stools. The crack in the anal lining exposes the underlying muscles that control the passage of stool through the anus and out of the body. An anal fissure is one of the most painful problems because the exposed muscles become irritated from exposure to stool or air, and leads to intense burning pain, bleeding, or spasm after bowel movements.

- ✓ **Perianal abscesses** can occur when the tiny anal glands that open on the inside of the anus become blocked, and the bacteria always present in these glands cause an infection. When pus develops, an abscess forms.

- ✓ **Aanal fistula** often follows drainage of an abscess and is an abnormal tube-like passageway from the anal canal to a hole in the skin near the opening of the anus. Body wastes traveling through the anal canal are diverted through this tiny channel and out through the skin, causing itching and irritation. Fistulas also cause drainage, pain, and bleeding. They rarely heal by themselves and usually need surgery to drain the abscess and "close off" the fistula.

<u>**Etiologies of Gastrointestinal Disorders**</u>

- ❖ **Physiological mechanisms for IBS and the functional bowel disorders**
- ✓ **Abnormal gut motility :-**According to Whitehead (1992b), increased motility in the intestinal tract is characteristic of most IBS patients. However, this motility is neither consistent, nor is it necessarily abnormal. Griffin (1985) summarized a number of different bowel motility patterns that have been observed in IBS. For example, cramping abdominal pain is often associated with large and small intestinal changes in the duration and frequency of increased intraluminal pressure and contractions; parasympathomimetic agents tend to provoke increased colonic muscle activity; increased resting colonic muscle activity is observed in diarrhea-predominant IBS, and the opposite pattern is often observed when constipation predominates.
- ✓ **Anxiety and the brain-gut link :-** Fossey and Lydiard (1990) have hypothesized that neuroanatomical connections between the hypothalamus, the origin of information transmitted to the gut via parasympathetic and sympathetic pathways and the GI tract may be indirectly responsible for IBS symptoms, through primary pathology at the CNS level [which] may secondarily impact on the gastrointestinal system via neurohumoral processes,

resulting in gastrointestinal distress. Unfortunately, however, it is unclear to what the authors refer by primary pathology at the CNS level. Because high activity of the locus coeruleus, a pontine noradrenergic nucleus, has been correlated with vigilance and attention to novel or fear provoking stimuli, and less activation with behaviors such as sleep, grooming, and feeding this supports the notion that IBS is related to the physiological aspects of clinical anxiety.

❖ **Psychological/sociocultural explanations for IBS and the functional bowel disorders:-**

✓ **Abnormal pain perception:-**Early studies of lower intestinal function found that colonic distention produced pain found that balloon insertion and inflation into the colons of both IBS patients and healthy but constipated control subjects produced significantly different pain perception in the IBS patients than in the controls.

✓ **Physical and sexual abuse history:-**Another potentially etiologically important issue in IBS and other functional gastrointestinal disorders is sexual and physical abuse history. A representative study was conducted who assessed both IBS and IBD patients with a structured psychiatric interview and a sexual trauma interview. The results indicated that the prevalence of severe lifetime sexual abuse was significantly higher in IBS patients (32% vs. 0%), as was any sexual abuse (54% vs. 5%).

✓ **Maladaptive psychological and/or behavioral characteristics:-** Psychological and/or behavioral characteristics as explanatory for IBS and other psycho physiological disorders encompasses a very large area, including personality, specifically the traits of neuroticism (Latimer, 1983) and negative affectivity depressive and anxious symptomology and abnormal illness behavior, concept first defined by Whitehead, Winget, Fedoravicius, Wooley, and Blackwell (1982) as frequent visits to physicians, multiple somatic complaints, and disability disproportionate to physical findings.IBS patients were significantly higher than those of patients with organic GI disease and healthy controls, and that IBS patients. Such patients have bodily preoccupation, hypochondriacal beliefs, and disease phobia. The authors concluded that IBS patients exhibit abnormal attitudes that are not related solely to depression or to the experience of physical symptoms, but reflect distress about illness disproportionate to reality.

✓ **Psychological disturbance or disorder:-** Patients had suffered significantly more lifetime diagnoses of major depression, generalized anxiety disorder, panic disorder, somatization disorder, and phobia, than had the IBD patients. The majority of these individuals had experienced the psychiatric disorders prior to the onset of GI symptoms.

- ✓ **Health beliefs and illness behaviours**

  Patients with more severe IBS commonly believe that their bowel symptoms indicate serious gut disease/cancer, and attend selectively to abdominal sensations, dismissing other information (e.g., from a doctor) that contradicts such beliefs.
- ✓ Patients with IBS report many non-gastrointestinal disorders. They make two to three times as many visits to physicians for non-gastrointestinal complaints and report missing an average of 13.4 days from work or usual activities due to illness compared with 4.9 days for the whole sample. These patterns of somatization and of health care seeking behavior may be learned during childhood.

- ❖ **Abnormal Motility:-**In healthy subjects, strong emotion or environmental stress can lead to increased motility in the esophagus, stomach, small intestine, and colon. The GIDs, however, are characterized by having an even greater motility response to stressors (psychological or physiological) when compared to normal subjects.These motor responses are partially correlated with bowel symptoms, particularly vomiting, diarrhea and constipation, but are not sufficient to explain reports of chronic or recurrent abdominal pain.
- ❖ **Early Family Environment:-**The aggregation of GIDs in families is not only genetic. What children learn from parents may contribute to the risk of developing an GID. In fact, children of adult patients with IBS make more health care visits (and incur more health care costs) than the children of non-IBS parents.
- ❖ **Inflammation:-**IBS have increased activated mucosal inflammatory cells. This information appears to relate to other clinical observations that about one third of patients with IBS or dyspepsia describe that their symptoms began after an acute enteric infection, and also, up to 25% of patients presenting with an acute enteric infection will go on to develop IBS-like or dyspeptic symptoms the mucosa of these individuals typically have increased inflammatory cells and inflammatory cytokine expression. It is likely that mucosal inflammation may, at least in part, be a determinant of visceral hypersensitivity and sensitization.
- ❖ **Genetic factors:** Research by Gregory and Rosen (1965) has demonstrated that the brothers of ulcer patients are about twice as likely to have ulcers as comparable members of the general population. Have also been reported for close relatives and these frequencies are specific to given reactions-that is the relatives of bronchial asthma cases show an increased frequency of bronchial asthma but not of other psychosomatic disorders.

More recent studies of learning in the autonomic system, however indicate that increased incidence of specific psychosomatic conditions in given families could result from common experience and learning.

- **Somatic weakness:** The organ affected may be one that is especially vulnerable. Factors as diverse as heredity, illness, or prior trauma may produce somatic weakness in a particular organ system, making it more vulnerable to stress than others. The person who has inherited or developed a "weak" stomach presumably will be prone to gastrointestinal upsets during anger or anxiety.
- **Personality characteristics and inadequate coping patterns:** The work of Flanders Dunbar (1943, 1954) and a number of other early investigators raised the hope of identifying specific personality factors associated with particular psychosomatic disorders, for example, rigidity, high sensitivity to threat, and proneness to chronic underlying hostility among those who suffer from hypertension. If it were possible to delineate ulcer types, hypertensive characters, accidental prone personalities and so on, such findings would of course be of great value in understanding, assessing, and treating psychosomatic disorders- and perhaps even in preventing them.

  More recent research evidence, however, suggests that such an approach is oversimplified. For example, although Kidson (1973) found hypertensive patients to be significantly more insecure, anxious, sensitive, and angry than a nonhypertensive control group, a sizeable number of control-group members also showed these characteristics. Similarty, Jenkins (1974) identified what he called "Type A" people- those who strive dillegently to achieve a time-conscious, tense, unable to relax, and active.
- **Kinds of stress:** Approaching the problem from the standpoint of stress rather than personality factors, Alexander (1950) hypothesized that each type of psychosomatic disorder could be associated with a particular kind of stress. He concluded that peptic ulcer, for example, are typically associated with frusturation of the needs for love and protection. Presumably, the frusturation of these needs would give rise to such emotions as anxiety and anger, and these emotions, in turn would trigger excessive secretions of stomach acid- leading eventually to peptic ulcers. Subsequent research, however, has failed to demonstrate a consistent relationship between particular disorders and particular types of stress. Rather it would appear that a wide range of stress situations can lead to a given type of disorder-and, conversely, that a wide range of disorders can result from a given type of stress.

- **Interpersonal relationships:** Destructive effects that stressful interpersonal patterns- including marital unhappiness, divorce, and bereavement- may have on personality adjustment. Such patterns may also influence physiological functioning.
- **Learning in the automatic nervous system:** While Pavlov and other investigators demonstrated that autonomic responses can be conditioned as in the case of salivation- it has long been assumed that we could not learn to control such responses voluntarily. Not only can autonomic reactivity be conditioned involuntarily via the classical Pavlovian model, but operant learning in the autonomic nervous system can also take place.Thus, the hypothesis developed that psychosomatic disorders may arise through accidental conditioning and reinforcement of such patterns. " A child who is repeatedly allowed to stay home from school when he has an upset stomach may be learning the visceral responses of chronic indigestion".

<u>**Psychological Intervention for Gastrointestinal Disorders**</u>

- **Psycho education:-**Indicate that the GIDs are very real and the intestine is overly responsive to a variety of stimuli such as food, hormonal changes, medication, and stress. Pain resulting from spasm or stretching of the gut, from a sensitive gut, or both, can be experienced anywhere in the abdomen and can be associated with changes in GI function leading to symptoms (eg, patient greater responsibility and control regarding the treatment, and improve pain tolerance.

- **Brief Psychodynamic Psychotherapy:-**Svedlund, Sjodin, Ottosson, and Dotevall (1983) reported one of the first controlled trials of psychotherapy for IBS. In 101 patients recruited from a medical clinic, they compared standard medical care (bulking agents, anti-spasmodic and tranquilizing medications) to standard medical care plus brief (10 sessions) psychodynamically oriented psychotherapy,focused on the identification of stress and emotions contributing to bowel symptoms and then the development of effective coping strategies. The results indicated that the psychotherapy condition was more effective than that of standard medical care on reducing reports of abdominal pain, bowel dysfunction,and somatic complaints. Interestingly, symptoms of anxiety and depression improved similarly in both conditions.

- **Hypnotherapy:-** Hypnotherapy is a state of unusual concentration on the suggestions of the therapist and a willingness to follow their instructions Whorwell *et al* have reported well constructed controlled trials of hypnotherapy in IBS. The technique is focused around a specific "gut directed" hypnosis protocol where the patient is taught to assert control over gut function and imagery whilst in an hypnotic state. Patients are given a simple account of intestinal smooth muscle physiology and hypnotised in a standard manner. The patient is then requested to place their hand on the abdomen and to sense both a positive feeling of abdominal warmth and increased control over gut function. During hypnosis, visualisation is also employed, using the analogy of a gently flowing river and a gently flowing bowel to reinforce a positive bowel image. In 1984, Whorwell, Prior, and Faragher conducted a study in which they compared home-based audiotaped hypnotherapy, defined as suggestions of general relaxation and mental imagery focused on warmth and relaxation of the gut and abdominal area, specifically, to a medication placebo and physician support.

- **Cognitive Behavioral Therapy:**

  This is a structured form of psychotherapy that is usually conducted individually but can be administered in group format. The treatment usually consists of a course of 6–12 sessions that focus on the present situations in which symptoms occur rather than the patient's history. CBT is based on the theory that maladaptive thoughts are the causes of psychological symptoms such as anxiety and depression, which in turn cause or exacerbate physical symptoms. An example would be a patient who believes that eating in a public place will always cause them to have diarrhea and other embarrassing symptoms (a catastrophizing maladaptive thought), which might lead the patient to both avoid social interactions (self defeating behavior) and to become anxious when dining in a restaurant. The anxiety and autonomic arousal caused by this maladaptive thought may actually trigger diarrhea. The therapist aims to help the patient recognize maladaptive thoughts and self-defeating behavior patterns that are adversely affecting life functioning, symptom experience, and mental well-being. Therapy tasks commonly include increasing awareness of the association between stressors, thoughts, and symptoms; examining and correcting irrational beliefs; countering automatic negative thoughts; observing and problem-solving factors that exacerbate symptoms; and identifying and adopting alternative, more effective coping strategies to handle challenging life situations and deal with gastrointestinal symptoms. In between therapy visits, patients are typically asked to complete homework assignments related to the treatment tasks. It should be noted that the relative emphasis on individual treatment components varies a lot. Some interventions that fall under the general umbrella of CBT are mostly or exclusively either cognitive or behavioral in nature (ie, they either focus on changing thought patterns or on learning and practicing healthy behavior patterns).

- **Stress Management:-**Decreasing your stress levels and learning effective stress management techniques may be beneficial for your disease outlook and may decrease the severity of your gastrointestinal symptoms. Here are some helpful suggestions to manage stress.

  - **Eat a well-balanced diet: -** Good nutrition is the key to both good physical and mental health. Inadequate nutrition increases stress on your body and decreases its ability to heal. Choose foods wisely and, in addition to reducing stress, your body will love you for it!
  - **Become a better breather** by learning to breathe more slowly and deeply from your abdomen. Stress can cause shallow breathing, which means that your body won't get enough oxygen to fully relax unless you consciously make an effort to breathe deeply.

- **Watch your 'self-talk'** because much of our anxiety is self-induced, meaning that we often get ourselves wound up worrying about worst-case scenarios or blowing small incidents out of proportion.
- **Monitor your negative thoughts** to see how often you fret about things such as making mistakes or losing your job. Try to substitute a negative thought with a positive, but realistic one.
- **Get physical** because exercise is a well-known tension reducer. Take care to increase exercise slowly and assess your body's tolerance to this as you do. Use caution though, as high-impact exercises might exacerbate gastrointestinal symptoms in persons with gastroesophageal reflux disease (GERD), hiatus hernia, Crohn's disease, and ulcerative colitis.
- **Become a better time manager** so time doesn't manage you. Many of us underestimate the amount of time it will take to do something, which means we're often running late. Try keeping a time management log for a week to get a better idea of how much time various tasks actually take.
- **Learning to say no** is a very good idea. Thinking you can 'do it all' creates unnecessary pressure. Learn how to set boundaries for yourself. Politely, yet firmly, turn down additional responsibilities or projects for which you don't have the extra time or energy.
- **Take time out for yourself.** Our minds and bodies require a certain amount of variety or else our overcharged nervous systems will keep speeding right into the next day. Try to take at least one day off each week to do something you really enjoy, whether it's reading, listening to music, or just hanging out with friends.
- **Have a good belly laugh.** Laughter is a natural stress reliever that helps to lower blood pressure, slow your heart and breathing rate, and relax your muscles.

➢ **Anger Management: -** It is the process of managing one's anger that is the primary goal of counseling people to effectively deal with anger. The goal is not to eliminate anger. Anger is a natural and healthy emotion. After a client acknowledges he or she is angry, a counselor can help the client learn how to reduce the emotional and physiological arousal that anger causes and learn to control its effects on people and the environment. To be more effective, practitioners should attempt to understand the extent and expression of the anger, the specific problems resulting from the anger, the function the anger serves, the underlying source of the anger, and the domain the problems occur in (e.g. emotional, physiological, or cognitive) before choosing interventions for the client. Specific strategies and skills as well as some additional considerations in helping clients manage anger.

- **Relaxation therapy:-** Given that IBS is a disorder of brain-gut and mind-body interactions, many individuals find symptom relief and an improved sense of well-being when they incorporate simple relaxation techniques into their daily lives. Although stress is inevitable, if not managed well, it can become detrimental to one's physical and emotional health. Thus, a regular practice of deep relaxation is associated with several health benefits including:
- a reduction of generalized anxiety,
- increased energy levels and productivity,
- improved concentration and memory,
- improved sleep,
- decreased fatigue,
- increased sense of self-confidence, and
- reduced muscle tension.

- **Biofeedback Therapy:-**

Biofeedback is a behavioural technique that uses visual or auditory cues to teach patients to alter physiological responses. With biofeedback, physiological events which are not normally appreciated by the patient are sensed by a technological interface and amplified to give the subject visual or auditory feedback. The biofeedback loop is based on the polygraph ("lie detector") which monitors tiny changes in electodermal conductivity occurring in response to stress and relaxation. Changes in cutaneous electrical activity are electronically transformed into a computerised animation of the gut shown on the computer screen. This animation can be controlled by the patient who learns to manipulate the computerized representation of bowel movement using a combination of mental and physical relaxation. In a study of computer aided gut directed biofeeback, 40 IBS patients who were refractory to conventional treatment underwent 4 half hour biofeedback sessions. Eighty percent of the patients learned to achieve progressively deeper levels of relaxation, and in 50%, the technique was reported helpful in controlling bowel symptoms.

- **Diet:-**

  Eating causes rhythmic contractions of the colon. Normally, this may cause a person to have a bowel movement 30 to 60 minutes after a meal. In a person with IBS, the urge to defecate may come sooner and may be accompanied by pain, cramps, and diarrhea.

  Many people say their IBS symptoms are triggered by eating certain foods. As a result, treatment includes figuring out which foods are the culprits and avoiding them when you eat. Changes in diet reduce IBS symptoms in 50 to 70 percent of people.

Foods that commonly trigger IBS symptoms include:

- **Dairy products**
- **Caffeine**
- **Fatty foods**
- **Vegetables, like beans or broccoli, that cause gas**
- **Foods containing the sweeteners sorbitol and fructose**
- **Wheat cereals**
- **Alcohol**

Before changing your diet, take note over the course of several days which foods seem to cause problems. You may want to consult a dietitian to help you adhere to healthful eating strategies, such as:

- **Drinking six to eight glasses of water a day**, especially if you have diarrhea. Drinking carbonated beverages can increase discomfort from gas.

- **Eating more fiber**. Dietary fiber often helps reduce IBS symptoms in both patients who have constipation as well as those who have diarrhea. Whole-grain breads and cereals, fruits, and vegetables are good fiber sources. Starting a high-fiber diet may cause gas and bloating for a few weeks. Fiber supplements such as bran, psyllium derivatives, or polycarbophil (20 to 30 grams/day) may help relieve constipation and may also reduce diarrhea.
- **Eating smaller meals more often or eating smaller portions**. Large meals can cause cramping and diarrhea.
- **Consuming probiotics**, such as yogurt or acidophilus supplements. Some patients find they help reduce symptoms. Research suggests that adding "good" bacteria may help return the

balance of the microflora in the bowel to normal or prevent disease-causing bacteria from attaching to the bowel wall.

- **Life-Style Changes: -** It is important to try to stop smoking, since smoking has been linked to ulcer formation, reduced healing, and ulcer recurrences. Also try to minimize stress in your life. Stress may worsen ulcer symptoms. Patients with a history of peptic ulcer disease may have frequent recurrences during their lifetime minimize the risks and prevent painful symptoms.

DISCUSSION:

Due largely to research conducted in a variety of disciplines, knowledge about such gastrointestinal disorders as IBS, PUD, IBD, and FBD, has increased tremendously in the past 100 years. The contribution of the medical, nutritional, and behavioural sciences has greatly improved the assessment, diagnosis, classification, and management of these. However, much remains to be done. Classification of etiological and maintaining variables, as well as the establishment of effective primary and secondary intervention strategies for all of these disorders represent important goals for clinical psychological science in the twenty-first century.

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UNIT- IV

GENITOURINARY/RENAL/REPROUDCTIVE SYSTEM

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Genitourinary System:

The genitourinary system or euro genital system is the system that includes all organs involved in reproduction and in the formation and voidance of urine. Or, in anatomy, it is the organ system of the reproductive organs and the urinary system. These are grouped together because of their proximity to each other, their common embryological origin and the use of common pathways, like the male urethra. Also, because of their proximity, the systems are sometimes imaged together.

Organs involved in Genitourinary System:

All the organs involved in reproduction and the formation and release of urine includes:

For urine

- Kidneys
- Urethra
- Bladder
- Ureter

Reproduction (in women's)

- Ovaries
- Uterus
- Fallopian tubes
- Vagina
- Clitoris

Reproduction (in men's)

- Testes
- Seminal Vesicles
- Prostate
- Seminal Duct
- Penis

Working and Functions of Genitourinary system:

Following are the functions of this system:-

Urinary System:

It consists of two kidneys, two ureters, the urinary bladder, and the urethra. Tubules in the kidneys are intertwined with vascular networks of the circulatory system to enable the production of urine. After the urine is formed, it is moved through the ureters to the urinary bladder for storage. Micturition, or voiding of urine from the urinary bladder, occurs through the urethra.

- The urinary system, along with the respiratory, digestive, and integumentary systems, excretes substances from the body. For this reason, these systems are occasionally referred to as excretory system.
- The urinary system maintains the composition and properties of the body fluid that establishes the internal environment of the body cells. The end product of the urinary system is urine, which is voided from the body during micturition (urination).
- After Metabolic processes, produce cellular wastes must be eliminated if homeostasis is to be maintained. Just as the essential nutrients are transported to the cells by the blood, the cellular wastes are removed through the circulatory system to the appropriate excretory system.
- The urinary system is the principal system responsible for water and electrolyte balance. Electrolytes are compounds that separate into ions when dissolved in water. Electrolyte balance is achieved when the number of electrolytes entering the body equals the number leaving. Hydrogen ions, for example, are maintained in precise concentration so that an acid-base, or pH, balance exists in the body.
- A second major function of the urinary system is the excretion of toxic nitrogenous compounds—specifically, urea and creatinine
- Other functions of the urinary system include the elimination of toxic wastes that may result from bacterial action and the removal of various drugs that have been taken into the body.
- All of these functions are accomplished through the formation of urine by the kidneys. Blood to be processed by a kidney enters through the large renal artery. After the filtration process, it exits through the renal vein. The importance of filtration of the blood is demonstrated by the fact that during normal resting conditions the kidneys receive approximately 20% to 25% of the entire cardiac output. Every minute, the kidneys process approximately 1,200 ml of blood.

Reproductive system:

- The reproductive system is a collection of organs that work together for the purpose of producing a new life. Scientists argue that the reproductive system is among the most important systems in the entire body. Without the ability to reproduce, a species dies.
- The major organs of the reproductive system include the external genitalia and internal organs, including gonads that produce gamete, which is a cell that fuses with another cell during conception in organisms that reproduce sexually. Substances such as fluids, hormones, and pheromones are also important to the effective functioning of the reproductive system.
- The male reproductive system consists of two major parts: the testes, where sperm are produced, and the penis. In humans, both of these organs are outside the abdominal cavity. Having the testes outside the abdomen facilitates temperature regulation of the sperm, which require specific temperatures to survive. If the testicles remain too close to the body, the higher temperature will likely harm the spermatozoa, making conception more difficult or impossible. The testes are carried in an external pouch known as the scrotum, where they normally remain slightly cooler than body temperature to facilitate sperm production.
- The two major parts of the female reproductive system are the vagina and uterus — which act as the receptacle for semen — and the ovaries, which produce the female's ova. The vagina is attached to the uterus through the cervix, while the Fallopian tubes connect the uterus to the ovaries. In response to hormonal changes, one ovum, or egg — or more in the case of multiple births — is released and sent down the Fallopian tube during ovulation. If not fertilized, this egg is eliminated as a result of menstruation.
- The fertilization of the ovum with the sperm occurs only at the ampullary-isthimic junction, which is why intercourse does not always result in pregnancy.
- At the time of conception, the ovum meets with spermatozoon, where a sperm may penetrate and merge with the egg, resulting in fertilization. While the fertilization usually occurs in the oviducts, it can also happen in the uterus itself. The egg then becomes implanted in the lining of the uterus, where it begins the processes of embryogenesis and morphogenesis. When the fetus is mature enough to survive outside of the womb, the cervix dilates and contractions of the uterus propel it through the birth canal.

Renal or urinary system, its problems, causes and treatment, failure and dialysis :

Renal or Urinary System:

The urinary system, also known as the renal system, which is a group of organs that work together to produce, store, and release urine. Urine is the liquid waste material excreted from the body. The organs that work together in this system include the two kidneys each consists of millions of functional units called nephrons, bladder, ureters and urethra. It is also known as the urinary or the excretory system .The kidneys are a vital part of the renal system. They are located in the back portion of the abdominal cavity, with one on either side. Perhaps the most well-known function of the kidneys is to transport urine into the tubes known as ureters before it exits the body. These organs also have several other important functions, however, such as helping to regulate blood pressure by secreting the enzyme renin. They also work to regulate the pH balance in the human body as well as the balance of electrolytes such as sodium and potassium and controls red blood cell production by secreting the hormone erythropoietin.

Problems caused by Renal system: Micturition /Voiding problems:

Micturition, also known as urination and voiding, is a process that allows the body to excrete excess water and get rid of some substances that are potentially harmful to the body. Excess water and other water-soluble substances exit the body in the form of urine. The act of micturition is supposedly a voluntary process but for some people, especially the very young and very old as well as those who are suffering from conditions affecting the brain and the spinal, it becomes uncontrollable. This is because micturition is under the control of the nervous system, which serves as pathway through which the impulse to urinate travels. Once an individual is ready to urinate, urine will flow out of the body through an opening in the urethra.

The micturition reflex, or the urge to urinate, often involves sending of signals between the urinary bladder, the spinal cord, and the brain. When the walls of the urinary bladder are stretched to their limit due to the presence of urine, nerves located in the bladder walls will send a signal that it is ready to release the accumulated fluid. This signal will travel to the spinal cord, and up to the brain. When the brain receives this signal, it will send down a command for the muscular walls of the bladder to contract and expel the urine out of the body. In circumstances where a person is stuck in traffic, the brain will normally relay a message to the bladder to delay the passing out of urine.

Symptoms of micturition problems include:

- **Urinary incontinence** : where he is unable to hold back urination.
- **Urinary frequency:** where he experiences the urge for urination more often than normal.
- **Urinary retention**: where an individual is unable to empty the bladder completely. Chronic urinary retention is frequently asymptomatic - a patient is able to urinate, but may experience lower urinary tract symptoms (LUTS), related to storage and voiding difficulties. This is in contrast to acute urinary retention, a medical emergency, which is painful and the patient is unable to urinate despite a full bladder. Chronic urinary retention, whilst not immediately life-threatening, can lead to **hydronephrosis** and **renal impairment** and puts the patient at risk of acute-on-chronic retention, so requires diagnosis and treatment.

Other symptoms are:

- **Anuria :** where there is no urine output at all.
- **oliguria :** where there is very low urine output.

These are often due to the presence of underlying medical conditions such as **prostate problems**, pregnancy, and urinary tract infections. Patients with traumatic spinal cord injury, especially those resulting in the paralysis of muscles from the neck down, also often experience problems in micturition.

Treatment:

Treatment options are based on the underlying cause of the voiding dysfunction, severity of symptoms, and findings from the physical, laboratory, and medical test results. Treatment may consist of one or more of the following approaches. Your doctors will discuss which specific method(s) will be tried with your child.

- **Managing constipation.** Proper management of constipation through the use of enemas, laxatives, and dietary fiber intake can reduce urinary wetting and urinary tract infections. Parents are encouraged to keep an elimination diary on the child. Over time, the stool softeners can be removed and the child remains on a high fiber intake.
- **Eliminating bladder irritants.** Your doctor may recommend increasing your child's water intake to dilute the urine and eliminating caffeine, carbonated beverages, citrus juices, and chocolate - products thought to irritate the bladder and may make voiding uncomfortable for your child.

- **Treating urinary tract infections (UTI).** A short course of antibiotics can be used in children with recurrent urinary tract infections.
- **Incorporating behavioral interventions.** Behavioral interventions are tools and techniques children and their parents can use to gain control over voiding dysfunction. The goals of behavioral interventions are to help your child remain continent and empty the bladder effectively.
- **Surgery.** Sometimes, though rarely, surgery is needed to correct an underlying anatomical problem that is the cause of the voiding dysfunction.

ENURESIS:

Enuresis is a condition that has been described since 1500B.C the term derived from Ancient Greek word refers to a repeated inability to control urination, more commonly called bed-wetting and involves the voluntary or involuntary release of urine into bedding, clothing or other inappropriate places. Use of the term is usually limited to describing individuals old enough to be expected to exercise such control .Enuresis commonly affects young children. Many cases of enuresis clear up by themselves as the child matures, although some children need behavioural or physiological treatment in order to remain dry. In adults, loss of bladder control is often referred to as urinary incontinence rather than enuresis; it is frequently found in patients with late stage Alzheimer's diseases or other form of dementia

Classification:

1. Primary enuresis refers to children who have never been successfully trained to control urination. This represents a fixation.
2. Secondary enuresis refers to children who have been successfully trained but revert back to wetting in a response to some sort of stressful situation. This represents a regression.

Types of enuresis include:

1. Nocturnal enuresis (bedwetting during night)
2. Diurnal enuresis (day time bedwetting)
3. Mixed enuresis - Includes a combination of nocturnal and diurnal type. Therefore, urine is passed during both waking and sleeping hours.

Current DSM-IV-TR Criteria:

a. Repeated voiding of urine into bed or clothes (whether involuntary or intentional)

b. Behavior must be clinically significant as manifested by either a frequency of twice a week for at least 3 consecutive months or the presence of clinically significant distress or impairment in social, academic (occupational), or other important areas of functioning.
c. Chronological age is at least 5 years of age (or equivalent developmental level).
d. The behavior is not due exclusively to the direct physiological effect of a substance (such as a diuretic) or a general medical condition (such as diabetes, spina bifida, a seizure disorder, etc.).

All these criteria must be met in order to diagnose an individual.

Causes:

The causes of enuresis are not so clear. DSM-IV-TR does not distinguish between children who wet bed involuntarily and those who voluntarily release urine. For the majority of bed wetters, there is no single clear physical or psychological explanation for enuresis.

After age 5, wetting at night—often called **bedwetting** or sleep wetting—is more common than daytime wetting in boys. Experts do not know what causes nighttime incontinence. Young people who experience nighttime wetting tend to be physically and emotionally normal, Day time incontinence that is not associated with urinary infection or anatomic abnormalities is less common than nighttime incontinence and tends to disappear much earlier than the nighttime versions. This form of incontinence occurs more often in girls than in boys. Along with this research findings suggest that voluntary and involuntary enuresis have different causes.

Causes and Symptoms:

Symptoms: The symptoms of enuresis are straightforward- a person urinates in inappropriate places or in inappropriate times.

Following are the diurnal, nocturnal voluntary and involuntary causes of enuresis:

(i) **Slower physical development:** Between the ages of 5 and 10, incontinence may be the result of a small bladder capacity, long sleeping periods, and underdevelopment of the body's alarms in the brain that signal a full or emptying bladder. This form of incontinence will fade away as the bladder grows and the natural alarms become operational.

(ii) **Excessive output of urine during sleep:** Normally, the body produces a hormone that can slow the making of urine. This hormone is called **antidiuretic hormone**, or ADH. The body normally produces more ADH during sleep so that the need to urinate is lower. If the body does not produce enough ADH at night, the making of urine may not be slowed down, leading to bladder overfilling. If a child does not sense the bladder filling and awaken to urinate, then wetting will occur.

(iii) **Anxiety:** Experts suggest that anxiety-causing events occurring in the lives of children ages 2 to 4 might lead to incontinence before the child achieves total bladder control. Anxiety experienced after age 4 might lead to wetting after the child has been dry for a period of 6 months or more. Such events include angry parents, unfamiliar social situations, and overwhelming family events such as the birth of a brother or sister. Incontinence itself is an anxiety-causing event. Strong bladder contractions leading to leakage in the daytime can cause embarrassment and anxiety that lead to wetting at night.

(iv) **Genetics:** Certain inherited genes appear to contribute to incontinence. In 1995, Danish researchers announced they had found a site on human chromosome 13 that is responsible, at least in part, for nighttime wetting. If both parents were enuretic, 77% of their children are too; if only one parent was enuretic, then 44% of their offspring are also. Experts believe that other, undetermined genes also may be involved in incontinence.

(v) **Obstructive sleep apnea:** Nighttime incontinence may be one sign of another condition called obstructive sleep apnea, in which the child's breathing is interrupted during sleep, often because of inflamed or enlarged tonsils or adenoids. Other symptoms of this condition include snoring, mouth breathing, frequent ear and sinus infections, sore throat, choking, and daytime drowsiness. In some cases, successful treatment of this breathing disorder may also resolve the associated nighttime incontinence.

(vi) **Structural problems:** Finally, a small number of cases of incontinence are caused by physical problems in the urinary system in children. A condition known as urinary reflux or vesicoureteral reflux, in which urine backs up into one or both ureters, can cause urinary tract infections and incontinence. Rarely, a blocked bladder or urethra may cause the bladder to overfill and leak. Nerve damage associated with the birth defect spina bifida can cause incontinence. An **ectopic ureter**, a misplacement of the ureter outside the bladder, can also commonly cause incontinence. In these cases, the incontinence can appear as a constant dribbling of urine.

(vii) **An overactive bladder:** Muscles surrounding the urethra (the tube that takes urine away from the bladder) have the job of keeping the passage closed, preventing urine from passing out of the body. If the bladder contracts strongly and without warning, the muscles surrounding the urethra may not be able to keep urine from passing. This often happens as a consequence of urinary tract infection and is more common in girls.

(viii) **Infrequent voiding:** Infrequent voiding refers to a child's voluntarily holding of urine for prolonged intervals. For example, a child may not want to use toilets at school or may not want to interrupt enjoyable activities, so he or she ignores the body's signal of a full bladder. In these cases, the bladder can overfill and leak urine. Additionally, these children often develop urinary tract infections (UTIs), leading to an irritable or overactive bladder.

(ix) **Other causes:** Some of the same factors that contribute to nighttime incontinence may act together with infrequent voiding to produce daytime incontinence. These factors include a small bladder capacity, constipation and food containing caffeine, chocolate or artificial coloring. Sometimes overly strenuous toilet training may make the child unable to relax the sphincter and the pelvic floor to completely empty the bladder. Retaining urine (incomplete emptying) sets the stage for urinary tract infections.

(x) **Rare Causes:** The proposed condition **PANDAS** (pediatric autoimmune neuropsychiatric disorders associated with streptococcal infections) has been used to describe a set of children who have a rapid onset of **OCD** and/or tic disorders following a streptococcal infection, with a link to other symptoms such as enuresis. A broader classification of this hypothesis, PANS, has been proposed which states that some patients suffer these symptoms in response to mycoplasma or lyme disease or even viruses rather than streptococcal. PANS is an acronym for **Pediatric acute-onset neuropsychiatric syndrome.** This hypothesis describes children who have abrupt, dramatic onset of **obsessive-compulsive disorder** (OCD) or anorexia nervosa coincident with the presence of two or more neuropsychiatric symptoms. It is believed that these children experience a rise in dopamine levels as a result of cross-reactive anti-neuronal antibodies. The rise in dopamine can cause such side effects as enuresis, bed-wetting, and urinary urgency.

(xi) Involuntary enuresis is much more common than voluntary enuresis. Involuntary enuresis may be categories as either primary or secondary. Primary enuresis occurs when young children lack bladder control from infancy. Most of these children have urine control problems only during sleep; they do not consciously, intentionally, or maliciously wet the bed. Researcher suggest that children who are night time only bed wetters may have a nervous system that is slow to process the feelings of a full bladder. Consequently, these children do not wake up in time to relieve themselves. In other cases, the childs enuresis may be related to sleep disorders.

(xii) Children with diurnal enuresis wet only during the day. They appear to be two types of day time wetter's. One group seems to have difficulty controlling the urge to urinate. The other group consciously delays urinating until they lose control. Some children have both durinal and nocturnal enuresis.

(xiii) Secondary enuresis occurs when the child has stayed dry day and night for at least six months, then returns to wetting. Secondary enuresis usually occurs at night. Many studies have been done to determine if there is a psychological component to enuresis. Researchers have found that secondary enuresis is more likely to occur after a child has experienced a stressful life event such as the birth of a sibling, divorce or death of parent, or moving to a new house.

(xiv) Enuresis or urinary incontinence in elderly adults may be caused by loss of independent control of body functions resulting from dementia, bladder infections, uncontrolled diabetes, side effects of medication, and weakened bladder muscles. Urinary incontinence in adults is managed by treatment of the underlining medical condition, if one is present; or by the use of adult briefs with disposable liners.

Treatments

Treatment for enuresis is not always necessary. About 15% of children who have enuresis outgrow it each year after age six. When treatment is desire, a physician will rule out obvious physical causes of enuresis through a physical examination and medical history. Several different treatment options are then available.

Behavior Modification: Behavior modification is often the treatment of choice for enuresis. It is inexpensive and has a success of about 75%. The child's bedding includes a special pad with sensor that rings a bell when the pad becomes wet. The bell wakes the child, who then gets up and goes to the bathroom to finish emptying his bladder. Over time, the child becomes conditioned to waking up when the bladder feelsfull. Once this response is learned**,** some children continue to wake themselves help from without the alarm, while others are able to sleep all night remain dry. A less expensive behavioral technique involves setting alarm clock to wake the child every night after few hours of sleep, until the child learns to wake up spontaneously. In trials, this method was as effective as the pad-and-alarm system. A newer technique involves an ultrasound monitor worn on the child's pajamas. The monitor can sense bladder size, and sets off an alarm once the bladder reaches a predetermined level of fullness. This technique avoids having to change wet bed pads.

Other behavior modifications that can be used alone or with the pad-and-alarm system include:

(i) Restricting liquids starting several hours before bedtime.

(ii) Waking the child up in the night to use the bathroom.

(iii) Teaching Urinary retention techniques

(iv) Giving the child positive reinforcement for dry nights and being sympathetic and understanding about wet nights.

Treatment with medication: There are two main drugs for treating enuresis.

Imipramine, a tricyclic antidepressant, has been used since the early 1960. It is not clear why this antidepressant is effective in treating enuresis when other antidepressants are not.

Desmopressin acetate (DDAVP) Has been widely used to treat enuresis since the 1990s. It is available as a nasal spray or tablet. Both impiramine and DDAVP are very effective in preventing bed-wetting, but have high relapse rates if medication is stopped.

a. **Alternative therapies:** Some success in treating bed-wetting has been reported using hypnosis works, the results are seen within four to six sessions. **Acupuncture** and massage have also been used to treat enuresis, within inconclusive results

b. **Psychotherapy:** Primary enuresis does not require psychotherapy. Secondary enuresis, however, is often successfully treated with therapy. The goal of the treatment is to resolve the underlining stressful event that has caused a relapse into bed wetting. Unlike children with involuntary enuresis, children who intentionally urinate in inappropriate places often have other serious psychiatric disorders. Enuresis is usually a symptom of another disorder. Therapy to treat the underlining disorder is essential to resolving enuresis.

c. **Growth and development:** Many children overcome incontinence naturally (without treatment) as they grow older. The number of cases of incontinence goes down by 15 percent for each year after the age of 5.

d. **Moisture alarms:** At night, **moisture alarms,** also known as bedwetting alarms, can awaken a person when he or she begins to urinate. These devices include a water-sensitive sensor that is clipped on the pajamas, a wire connecting to a battery-driven control, and an alarm that sounds when moisture is first detected. For the alarm to be effective, the child must awaken or be awakened as soon as the alarm goes off. This may require having another person sleep in the same room to awaken the bedwetter. Bed-wetting alarms have been around since 1938, when O. H. Mowrer and

e. W. M. Mowrer first invented the "bell and pad". This behavioral training is one of the safest and more effective treatments. By twelve weeks, the child will most likely have mastered his nighttime bladder control.

f. **Medications:** Nighttime incontinence may be treated by increasing ADH levels. The hormone can be boosted by a synthetic version known as **desmopressin**, or DDAVP, which recently became available in pill form. Patients can also spray a mist containing desmopressin into their nostrils. Desmopressin is approved for use by children. There is difficulty in keeping the bed dry after medication is stopped, with as high as an 80% relapse rate.

Another medication, called **imipramine**, is also used to treat **sleep wetting**. It acts on both the brain and the urinary bladder. Unfortunately, total dryness with either of the medications available is achieved in only about 20 percent of patients.

If a young person experiences incontinence resulting from an overactive bladder, a doctor might prescribe a medicine that helps to calm the bladder muscle, such as **oxybutynin**. This medicine controls muscle spasms and belongs to a class of medications called **anticholinergics**.

(i) **Dry-bed training:** During a visit to a doctor's office, parents can be instructed in bladder retention control training by having their child drink more and more fluids during the day and delay urination for longer periods of time, attempting to strengthen bladder control. They are also encouraged to enforced hourly wakings for trips to the bathroom during the night and to develop a cleanup routine for the child (possibly including cleaning more than just the sheets), in addition to positively reinforcing dry nights (nights without urination incidents).

(ii) **Bladder training and related strategies :** Techniques that may help **daytime incontinence** include:

(iii) Urinating on a schedule, such as every 2 hours (this is called timed voiding)

(iv) Avoiding caffeine or other foods or drinks that may contribute to a child's incontinence.

Prognosis

Enuresis is a disorder that most children outgrow. For those who do receive treatment, the overall success rate of behavioral therapy is 75%. The short term success rate with drug treatment is even higher than with behavioral therapy. Drugs do not, however, eliminate the enuresis. Many children who take drugs to control their bed-wetting relapse when the drug are stopped.

Prevention

Although enuresis cannot be prevented, one side effect of the disorder is the shame and social embarrassment it causes. Children who wet may avoid sleepovers, cram, and other activities where their bed- wetting will become obvious. Loss of these opportunities can cause a loss of self-esteem, social isolation and adjustment problems.

End Stage Renal Disease (ESRD)

End-stage renal disease is referred when the kidneys permanently fail to work. And, renal failure refers to temporary or permanent damage to the kidneys that result in loss of normal kidney function. There are two different types of renal failure--acute and chronic. Acute renal failure has an abrupt onset and is potentially reversible. Chronic renal failure progresses slowly over at least three months and can lead to permanent renal failure. The causes, symptoms, treatments, and outcomes of acute and chronic are different.

Acute renal failure	Chronic renal failure
Myocardial infarction. A heart attack may occasionally lead to temporary kidney failure.	**Diabetic nephropathy.** Diabetes can cause permanent changes, leading to kidney damage.
Rhabdomyolysis. Kidney damage that can occur from muscle breakdown. This condition can occur from severe dehydration, infection, or other causes.	**Hypertension.** Chronic high blood pressure (hypertension) can lead to permanent kidney damage.

Decreased blood flow to the kidneys for a period of time. This may occur from blood loss or shock.	**Lupus (SLE).** A chronic inflammatory/autoimmune disease that can injure the skin, joints, kidneys, and nervous system.
An obstruction or blockage along the urinary tract.	A prolonged urinary tract obstruction or blockage.
Hemolytic uremic syndrome. Usually caused by an E. coli infection, kidney failure develops as a result of obstruction to the small functional structures and vessels inside the kidney.	**Alport syndrome.** An inherited disorder that causes deafness, progressive kidney damage, and eye defects.
Ingestion of certain medications that may cause toxicity to the kidneys.	**Nephrotic syndrome.** A condition that has several different causes. Nephrotic syndrome is characterized by protein in the urine, low protein in the blood, high cholesterol levels, and tissue swelling.
Glomerulonephritis. A type of kidney disease that involves glomeruli. During glomerulonephritis, the glomeruli become inflamed and impair the kidney's ability to filter urine. Glomerulonephritis may lead to chronic renal failure in some individuals.	**Polycystic kidney disease.** A genetic disorder characterized by the growth of numerous cysts filled with fluid in the kidneys.
Any condition that may impair the flow of oxygen and blood to the kidneys such as cardiac arrest.	**Cystinosis.** An inherited disorder in which the amino acid cystine (a common protein-building compound) accumulates within specific cellular bodies of the kidney, known as lysosomes.
	Interstitial nephritis or pyelonephritis. An inflammation to the small internal structures in the kidney.

Symptoms of renal failure:

The symptoms for acute and chronic renal failure may be different. The following are the most common symptoms of acute and chronic renal failure. However, each individual may experience symptoms differently. Symptoms may include:

Acute	Chronic
• Hemorrhage • Fever • Weakness • Fatigue • Rash • Diarrhea or bloody diarrhea • Poor appetite • Severe vomiting • Abdominal pain • Back pain • Muscle cramps • No urine output or high urine output • History of recent infection (a risk factor for acute renal failure) • Pale skin • Nosebleeds • History of taking certain medications (a risk factor for acute renal failure) • History of trauma (a risk factor for acute renal failure) • Swelling of the tissues	• Poor appetite • Vomiting • Bone pain • Headache • Insomnia • Itching • Dry skin • Malaise • Fatigue with light activity • Muscle cramps • High urine output or no urine output • Recurrent urinary tract infections • Urinary incontinence • Pale skin • Bad breath • Hearing deficit • Detectable abdominal mass • Tissue swelling • Irritability • Poor muscle tone • Change in mental alertness • Metallic taste in mouth

• Inflammation of the eye • Detectable abdominal mass Exposure to heavy metals or toxic solvents (a risk factor for acute renal failure)	

Renal failure diagnosis:

In addition to a physical examination and complete medical history, diagnostic procedures for renal failure may include the following:

a. **Blood tests.** Blood tests will determine blood cell counts, electrolyte levels, and kidney function
b. Urine tests
c. **Renal ultrasound (also called sonography).** A noninvasive test in which a transducer is passed over the kidney producing sound waves which bounce off the kidney, transmitting a picture of the
d. organ on a video screen. The test is use to determine the size and shape of the kidney, and to detect a mass, kidney stone, cyst, or other obstruction or abnormalities.
e. **Kidney biopsy.** This procedure involves the removal of tissue samples (with a needle or during surgery) from the body for examination under a microscope; to determine if cancer or other abnormal cells are present.
f. **Computed tomography scan (also called a CT or CAT scan).** A diagnostic imaging procedure that uses a combination of X-rays and computer technology to produce horizontal, or axial, images (often called slices) of the body. A CT scan shows detailed images of any part of the body, including the bones, muscles, fat, and organs. CT scans are more detailed than general X-rays. Contrast CT usually cannot be done when there is kidney failure.

Treatment for acute and chronic renal failure:

Specific treatment for renal failure will be determined by your doctor based on:

(i) Your age, overall health, and medical history

(ii) Extent of the disease

(iii) Type of disease (acute or chronic)

(iv) Underlying cause of the disease

(v) Your tolerance for specific medications, procedures, or therapies

(vi) Expectations for the course of the disease

(vii) Your opinion or preference

Treatment may include:

1. Hospitalization
2. Administration of intravenous (IV) fluids in large volumes (to replace depleted blood volume)
3. Diuretic therapy or medications (to increase urine output)
4. Close monitoring of important electrolytes such as potassium, sodium, and calcium
5. Medications (to control blood pressure)
6. Specific diet requirements
7. Dialysis
8. Kidney transplantation

In some cases, patients may develop severe electrolyte disturbances and toxic levels of certain waste products normally eliminated by the kidneys. Patients may also develop fluid overload. Dialysis may be indicated in these cases.

Treatment of chronic renal failure depends on the degree of kidney function that remains. Treatment may include:

9. Medications (to help with growth, prevent bone density loss, and/or to treat anemia)
10. Diuretic therapy or medications (to increase urine output)
11. Specific diet restrictions or modifications
12. Kidney transplantation
13. Dialysis

Dialysis:

Dialysis is a procedure that is performed routinely on persons who suffer from acute or chronic renal failure, or who have ESRD. The process involves removing waste substances and fluid from the blood that are normally eliminated by the kidneys. Dialysis may also be used for individuals who have been exposed to or ingested toxic substances to prevent renal failure from occurring. There are two types of dialysis that may be performed, including the following:

14. Peritoneal dialysis. Peritoneal dialysis is performed by surgically placing a special, soft, hollow tube into the lower abdomen near the navel. After the tube is placed, a special solution called dialysate is instilled into the peritoneal cavity. The peritoneal cavity is the space in the abdomen that houses the organs and is lined by two special membrane layers called the peritoneum. The dialysate is left in the abdomen for a designated period of time which will be determined by your doctor. The dialysate fluid absorbs the waste products and toxins through the peritoneum. The fluid is then drained from the abdomen, measured, and discarded. There are three different types of peritoneal dialysis: continuous ambulatory peritoneal dialysis (CAPD), continuous cyclic peritoneal dialysis (CCPD), and intermittent peritoneal dialysis (IPD).

CAPD does not require a machine. Exchanges often referred to as "passes," can be done three to five times a day during waking hours. CCPD requires the use of a special dialysis machine that can be used in the home. This type of dialysis is done automatically, even while you are asleep. IPD uses the same type of machine as CCPD, but treatments take longer. IPD can be done at home, but usually is done in the hospital.
Possible complications of peritoneal dialysis include an infection of the peritoneum, or peritonitis, where the catheter enters the body. Peritonitis causes fever and stomach pain. Your diet for peritoneal dialysis will be planned with a dietitian, who can help you choose meals according to your doctor's orders. Generally:

- You may have special protein, salt, and fluid needs.
- You may have special potassium restrictions.
- You may need to reduce your calorie intake, since the sugar in the dialysate may cause weight gain.

15. Hemodialysis. Hemodialysis can be performed at home or in a dialysis center or hospital by trained healthcare professionals. A special type of access, called an arteriovenous (AV) fistula, is placed surgically, usually in your arm. This involves joining an artery and a vein together. An external, central, intravenous (IV) catheter may also be inserted, but is less common for long-term dialysis. After access has been established, you will be connected to a large hemodialysis machine that drains the blood, bathes it in a special dialysate solution which removes waste substances and fluid, then returns it to your bloodstream.

Hemodialysis is usually performed several times a week and lasts for four to five hours. Because of the length of time hemodialysis takes, it may be helpful to bring reading material, in order to pass the time during this procedure. During treatment you can read, write, sleep, talk, or watch TV. At home, hemodialysis is done with the help of a partner, often a family member or friend.

Possible complications of hemodialysis include muscle cramps and hypotension (sudden drop in blood pressure). Hypotension may cause you to feel dizzy or weak, or sick to your stomach. Side effects are avoided by following the proper diet and taking medications, as prescribed by your doctor. A dietitian will work with you to plan your meals, according to your doctor's orders. Generally:

a) You may eat foods high in protein such as meat and chicken (animal proteins).
b) You may have potassium restrictions.
c) You may need to limit the amount you drink.
d) You may need to avoid salt.
e) You may need to limit foods containing mineral phosphorus (such as milk, cheese, nuts, dried beans, and soft drinks).

1. When a person’s kidneys fail due to damage or disease, dialysis treatment is used as a replacement for kidney function. Dialysis is a procedure in which a person’s body is cleaned of impurities and toxins, a task which his or her kidneys would perform if they were still functional. People who undergo regular dialysis treatment are usually in end-stage renal failure and have no more than 10% to 15% kidney function remaining.
2. Dialysis treatment is crucial for people suffering from kidney failure. Without functional kidneys, they can no longer remove salts, waste, and water, and their bodies cannot maintain safe levels of sodium, potassium, and other minerals. Dialysis also helps to control blood pressure, which can rise or fall dangerously due to an imbalance of salts and minerals.
3. For some people, dialysis treatment is only a short-term measure, required as a result of acute kidney failure, or damage or disease that causes temporarily impaired kidney function. Others need regular dialysis as a result of chronic kidney failure. In these cases, the kidneys are permanently damaged, and only a successful kidney transplant will end the need for chronic dialysis.
4. There are two types of dialysis treatment. These are called **hemodialysis** and peritoneal dialysis. Each procedure works slightly differently, but operates on the same principles of replacing kidney function by removing waste products from the blood.

5. The process of hemodialysis uses an artificial kidney, called a hemodialyzer, to remove the waste products and fluids that build up in the blood. To allow blood to flow through the artificial kidney, a dialysis patient must undergo a minor surgical procedure that creates an access point in an arm or leg. For the average person, dialysis treatment occurs three times a week for around four hours per session. The actual frequency and time depends on the amount of kidney function an individual person has remaining, how quickly waste products build up in the blood, and other factors.
6. The second type of dialysis is called peritoneal dialysis. Rather than removing the blood from the body in order to remove waste, the blood is cleaned while still inside the body. A doctor first creates an access point with a minor surgical procedure that places a catheter in the abdomen. At each peritoneal dialysis session, the catheter point is slowly filled with a solution called the dialysate, which then fills the interior of the abdominal cavity. Waste products in the blood filter through arteries and veins into the dialysate via osmosis.
7. While dialysis can replace the work that the kidneys do, dialysis is not itself a cure for kidney failure; rather it is a treatment that manages the condition. Many people who require chronic dialysis can live fairly normal lives, apart from the need to undergo the treatment several times per week. As the dialysis procedure is improved, it is likely that people requiring this treatment will be able to live just as long as people with functional kidneys.

Psychological Interventions:

End-stage renal disease is progressive, and the disturbances it brings are progressive. Any intervention has to be tailored to the progress of the disease itself, with the individual's level of physical, psychological, and social functioning as the central focus. The concurrent physiologic, psychological, and social stresses demand cognitive effort from the patient in coping. Researchers suggest that patient adherence to a medical regime is significantly related to high social desirability and a shorter length of time on dialysis. Patients with ESRD are empowered for self-care in matters of drug administration and PD management, but adherence to treatment requirements must be a voluntary act of submission, with consent for the adjustment and adaptation to the illness and treatment. Psychosocial intervention is best started as early (at diagnosis) and demands continuous effort.

8. **Nursing Implications:** End-stage renal disease has a characteristically downward trajectory. Patients have to come to terms with their current physical condition. Psychosocial nursing interventions should attempt to facilitate adjustment to changes in the course of the illness and to normalize social interaction and lifestyle by preventing medical crises, controlling symptoms, and incorporating the PD treatment regimes into daily living. Knowledge can significantly minimize a patient's anxiety. It is crucial that nurses have the skills to provide clear information, to help patients identify their goals in the course of treatment, and to assist with problem-solving for optimal physical functioning. Assessment: Assessment determines the patient's needs, identifies problems and potential problems, and collects information for a treatment plan so that appropriate support can be rendered. The assessment therefore focuses on the effect of the illness on the patient. Useful information includes the patient's lifestyle, patterns of daily living, personality, strengths and interests, normal coping patterns, understanding of the current illness, perception of treatment regimes, recent life stresses or changes, and major issues raised by the disease. By listening to the patient and the family in the course of discussion, nurses can identify the observable psychosocial interferences consequent to the disease and the needs for assistance. At the same time, information on the expected course and likely outcome of the disease can be provided.

9. **Encouragement:** The role of the health care professional is to encourage and, where possible, to enable patients to accept responsibility for their health and well-being and to fulfill their obligations within the family and society. As well as providing knowledge and clarifying misconceptions, nurses can encourage patients to accept the personal limitations consequent to the illness and its treatment. When a patient is encouraged to perform self-care, better self-esteem and power to maintain health are established. When open discussion and awareness of the mutual situation is encouraged between patients and their partners, positive and understanding attitudes are reinforced. The perception of emotional support has a documented association with better physical and mental health in dialysis patients.

10. **Life Enhancement:** Dependent patients may adapt to treatment regimes more easily, but excessive dependence can create extreme demands on caregivers and can impede rehabilitation. Some patients

may achieve a secondary gain from the illness, and some may enjoy the role of "patient." Nurses can facilitate a patient's adaptations to treatment requirements by maximizing the patient's strengths and supporting the patient in the treatment environment at home, while making judicious use of available resources. Individualizing treatment and minimizing its complexity may encourage adherent behavior. Frequent assessment, education, motivation, reinforcement, encouragement, and teaching concerning self-management and self-monitoring will, at the very least, maximize the patient's comprehension of the illness and personal motivation for adherence— an especially important requisite for living with a chronic condition. Participation by patients in patient support networks, rehabilitation activities, physical exercise schedules, and educational programs may help individuals to establish new supportive relationships, to achieve social recognition and appreciation, and to overcome social isolation, which has been discovered to be associated with treatment noncompliance.

11. **Compliance**—adherence to treatment requirements in a therapeutic regimen—is a critical concept for chronic renal failure patients and for their caregivers alike. Compliance changes over time. Significant differences related to social support have been found, especially when the demographic variable of education is controlled.

12. **SUPPORT TO FAMILY:** Family members play an important role in the wellbeing of the PD patient. They should not be neglected in the process of patient care. A change in the pattern of family life (integrating the lives of family members more flexibly with the patient's life) may be necessary to meet the patient's PD needs. The patient and the family should be encouraged to share their feelings in a trusting relationship and to make flexible adjustments to cope with the course of the patient's illness. Previous studies have revealed that sadness, guilt, and loss were pervasive and prevalent in partners—a unique perspective on the negative impact that dialysis can have on couples. The mobilization of community services is useful to reduce stressors.

13. **Nursing Intervention:** Local experience in the Renal Unit, Tuen Mun Hospital, Hong Kong SAR, illustrates practicable interventions. In our unit, for all adult ESRD patients who will start on PD, psychosocial intervention starts with a predialysis briefing and continues with subsequent encounters at the predialysis assessment, break-in education, teaching and learning sessions, telephone visits, home visits, and counseling sessions. Through the collaborative efforts of health care professionals

and the patient support group, rehabilitation activities are coordinated with the goals of enhancing peer support and optimizing mental strengths, physical fitness, and social strengths. Activities include community networks, voluntary visits to new patients, a tuck shop, production of a CD for peer support, health talks, maintenance of a Web site **(www.hk-doctor.com/kpa/),** Tai Chi, gateball, overnight camps, outings, dinners, mah-jongg competitions, karaoke competitions, and participation in the rehabilitation programs organized by the Hong Kong Society of Nephrology. During the period February 1993 – August 2002, we followed 694 patients. Most maintained a positive attitude toward life and got as much out of each day as they possibly could by integrating PD with life. Four patients and 3 relatives were found to be experiencing depressive symptoms that necessitated psychiatric treatment. One patient and 3 relatives committed suicide.

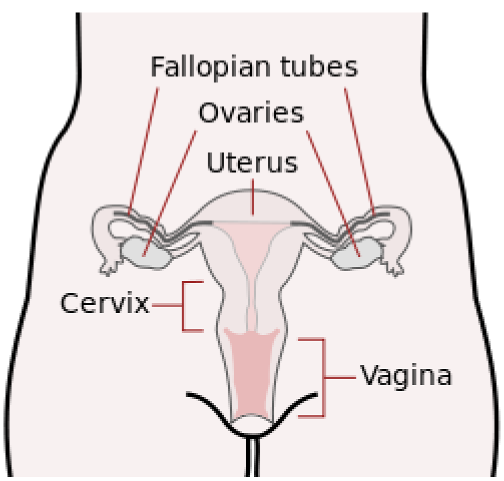
14. **SUPPORT TO STAFF:** Dialysis professionals can have a major impact— positive or negative—on how patients deal with the life changes caused by PD. Prospects for renal rehabilitation increase if the dialysis staff convey a positive attitude and show belief in the patient's potential to achieve rehabilitation goals—particularly goals that the patient is able to control, such as adherence to treatment requirements, exercise, and self-care. However, not uncommonly, staffs are working in emotionally-laden situations, such as those involving difficult behavior on the part of a patient or caregiver, deteriorating conditions in patients with whom a good rapport has been established, or failure of treatment. The experience of anxiety related to job demands and of spiritual distress related to an inability to find meaning or purpose in professional and personal life is frequently mentioned by staff. Renal team members can be given opportunities (in meetings or conferences, for example) to evaluate the causes of stress, to develop ideas, to share ideas with peers, and to create opportunities to honor or encourage members of the team. Some ways to divert attention from stress are to seek humor in experiences, to learn from patients to accept limitations while remaining professional in demeanor and presentation, and to take appropriate time away from work to rest and toplay.

Reproductive system:

The **human female reproductive system** (or **female genital system**) contains two main parts: the **uterus**, which hosts the developing fetus, produces vaginal and uterine secretions, and passes the male's **sperm** through to the fallopian tubes; and the **ovaries**, which produce the female's egg cells. These parts are internal; the vagina meets the external organs at the **vulva**, which includes the **labia, clitoris** and **urethra**. The **vagina** is attached to the uterus through the **cervix**, while the **uterus** is attached to the **ovaries** via the **Fallopian tubes**. At certain intervals, the **ovaries** release an **ovum**, which passes through the **Fallopian tube** into the uterus.

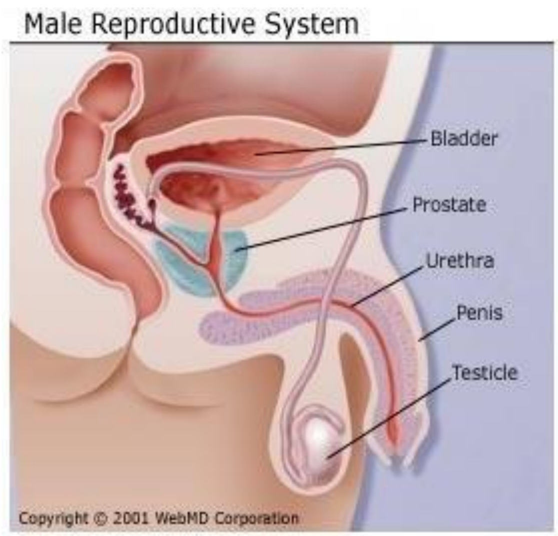
If, in this transit, it meets with **sperm**, the sperm penetrate and merge with the egg, **fertilizing** it. The egg releases certain molecules that that are essential to guiding the sperm and these allow the surface of the egg to attach to the sperm's surface then the egg can absorb the sperm and **fertilization** begins. The **fertilization** usually occurs in the **oviducts**, but can happen in the **uterus** itself. The **zygote** then implants itself in the wall of the uterus, where it begins the processes of **embryogenesis** and **morphogenesis**. When developed enough to survive outside the womb, the **cervix** dilates and contractions of the uterus propel the **fetus** through the birth canal, which is the **vagina**.

The **ova** are larger than **sperm** and have formed by the time a female is born. Approximately every month, a process of **oogenesis** matures one **ovum** to be sent down the **Fallopian tube** attached to its **ovary** in anticipation of **fertilization**. If not fertilized, this egg is flushed out of the system through **menstruation**.



The **human male reproductive system** consists of a number of **sex organs** that form a part of the **human reproductive** process. In this type of reproductive system, these sex organs are located outside the body, around the **pelvic** region.

The main male sex organs are the **penis** and the **testes** which produce semen and **sperm**, which, as part of **sexual intercourse, fertilize** an **ovum** in the female's body; the fertilized ovum (**zygote**) develops into a **fetus**, which is later born as a **child**.



Sexual dysfunction or **sexual malfunction** is difficulty experienced by an individual or a couple during any stage of a normal **sexual activity**, including physical pleasure, **desire**, preference, **arousal** or **orgasm**. According to the **DSM 5**, sexual dysfunction requires a person to feel extreme distress and interpersonal strain for a minimum of 6 months (excluding substance or medication-induced sexual dysfunction). Sexual dysfunctions can have a profound impact on an individual's perceived quality of sexual life.

A thorough sexual history and assessment of general health and other sexual problems (if any) are very important. Assessing (performance) **anxiety, guilt, stress** and **worry** are integral to the optimal management of sexual dysfunction. Many of the sexual dysfunctions that are defined are based on the **human sexual response cycle**, proposed by **William H. Masters and Virginia E. Johnson,** and then modified by **Helen Singer Kaplan Sexual dysfunction** can take many forms — it's not limited to erectile dysfunction or lack of interest in sex, often referred to as a low libido. Sexual dysfunction can involve pain during intercourse, an inability to maintain an erection, or difficulty experiencing an orgasm.

Though there are many causes of diminished libido and sexual dysfunction in men and women, there are also many ways to increase libido and rekindle the joy of sex once you identify the problem.

Sexual Dysfunction in Women:

Sexual dysfunction in women is grouped into different disorders: sexual pain, problems with desire, arousal problems, and orgasm difficulty. Changes in hormone levels, medical conditions, and other factors can contribute to low libido and other forms of sexual dysfunction in women.

Specifically, sexual dysfunction in women may be due to:

15. **Vaginal dryness.** This can lead to low libido and problems with arousal and desire, as sex can be painful when the vagina isn't properly lubricated. Vaginal dryness can result from hormonal changes that occur during and after menopause or while breastfeeding, for example. Psychological issues, like anxiety about sex, can also cause vaginal dryness. And anticipation of painful intercourse due to vaginal dryness may, in turn, decrease a woman's desire for sex.
16. **Low libido.** Lack of sexual desire can also be caused by lower levels of the hormone estrogen. Fatigue, depression, and anxiety can also lead to low libido, as can certain medications, including some antidepressants.
17. **Difficulty achieving orgasm. Orgasm** disorders, such as delayed orgasms or inability to have one at all, can affect both men and women. Again, some antidepressant medications can also cause these problems.
18. **Pain during sex.** Pain is sometimes from a known cause, such as vaginal dryness or endometriosis. But sometimes the cause of painful sex is elusive. Known as vulvodynia or vulvar vestibulitis, experts don't know what's behind this mysterious type of chronic, painful intercourse. A burning sensation may accompany pain during sex.

Sexual Dysfunction in Men

The types of sexual dysfunction men may experience include:

19. **Erectile dysfunction (ED). ED** can be caused by medical conditions, such as diabetes or high blood pressure, or by anxiety about having sex. Depression, fatigue, and stress can also contribute to erectile dysfunction.
20. **Ejaculation problems.** These include premature ejaculation (ejaculation that occurs too early during intercourse) and the inability to ejaculate at all. Causes include medications, like some antidepressants, anxiety about sex, a history of sexual trauma (such as a partner being unfaithful), and strict religious beliefs.
21. **Low libido.** Psychological issues like stress and depression, as well as anxiety about having sex also can lead to a decreased or no sexual desire. Decreased hormone levels (particularly if testosterone is low), physical illnesses, and medication side effects may also diminish libido in men.

Causes Sexual Problems:

There are many factors which may result in a person experiencing a sexual dysfunction. These may result from emotional/psychological or physical causes.

- **Physical causes:** Many physical and/or medical conditions can cause sexual problems. These conditions include **diabetes, heart disease**, neurological diseases, hormonal imbalances, **menopause**, chronic diseases such as kidney disease or liver failure, and alcoholism and **drug abuse**. In addition, the side effects of certain medications, including some **antidepressant** drugs, can affect sexual desire and function.
- **Psychological causes:** These include work-related stress and **anxiety**, concern about sexual performance, marital or relationship problems, **depression**, feelings of guilt, and the effects of a past sexual trauma.
    a. Female sex is very psychological. There may be so many psychological causes. Female sexual desire naturally fluctuates over the years. Highs and lows commonly coincide with the beginning or end of a relationship or with major life changes such as pregnancy, menopause or illness. If you are concerned by a low or decreased sex drive there are life style changes and sex techniques that may put you in the mood more often.
    b. According to some studies more than 40% of females complain of low sexual desire at some point. Still researchers acknowledge that it is difficult to measure what is normal and what is

not.

c. Similarly if your sex drives is weaker that it once was your relationship maybe stronger than ever. There is no magic number to define low sex drive. It varies from woman to woman. Causes for low sex drive are based on a complex interaction of many components affecting intimacy including physical well being, emotional well being, experiences, beliefs, life styles and current relationships. Alcohol and drugs, surgery, fatigue, menopause, pregnancy and breast feeding and then **physiological causes** such as anxiety, depression, physical appearance, low self esteem etc. are also causes of low sex drive.

d. There can be many reasons of this condition in men also, such as hormonal imbalances, stress, drugs, sleep deprivation etc.

e. **Hormone Imbalances (Testosterone):** Testosterone levels are the most common root of libido issues in men. As the principle male sex hormone, testosterone maintains and drives your libido. When testosterone levels get too low, a man can lose much of his sex drive. Luckily, through testosterone hormone therapy, men can establish a more normal level, helping to reinstate their sex drive.

f. **Sleep:** The amount of sleep a man gets may also affect his libido. Sleep is necessary to help the body recharge and stay energized. A regular loss of sleep can result in a man being too tired to have sex. Getting at least eight hours of uninterrupted sleep a night can quickly boost a lowered libido.

g. **Stress:** Whether it be job, money, or relationship related, stress can have a big impact on your libido. When your body becomes stressed it releases two main hormones called adrenaline and cortisol. These hormones are fine in normal doses, but if produced by the body too frequently it can lead to chronic stress. Eventually these hormones begin to interfere with testosterone, causing a lowered libido.

h. **Alcohol:** You may be tempted to believe that alcohol increases your libido. However, chronic alcohol use can actually lower your libido. When you drink alcohol it depresses many important parts of your body, including your hypothalamus and pituitary gland. These parts of your body regulate the function of your gonads. Once they become depressed by alcohol they can cause a noticeable drop in libido. Alcohol may also affect the blood vessels handicapping the body’s ability to achieve a firm erection.

i. **Disease:** Certain disease may also cause a noticeable drop in libido. These diseases include diabetes, cardiovascular disease, Parkinson’s disease and anemia. Diseases that specifically attack the cardiovascular system are especially prominent causes of libido problems in men as they can affect the ability to achieve an erection.

Medication: Certain medications may also cause a drop in libido for men. Antidepressants and muscle relaxants are common causes of lowered libido in men. Illegal drugs can also lower a man's libido, especially marijuana, cocaine and heroin. Please contact a good urologist for complete check up and appropriate management and/or referral to endocrinologist or psychologist.

Treatment:

The ideal approach to treating sexual problems involves a team effort between the patient, doctors, and trained therapists. Most types of sexual problems can be corrected by treating the underlying physical or psychological problems. Other treatment strategies focus on the following:

- **Providing education.** Education about human anatomy, sexual function, and the normal changes associated with aging, as well as sexual behaviors and responses may help a woman overcome her anxieties about sexual function and performance.
- **Enhancing stimulation.** This may include the use of erotic materials (videos or books), **masturbation**, and changes to sexual routines.
- **Providing distraction techniques.** Erotic or non-erotic fantasies, exercises with intercourse, music, videos, or television can be used to increase relaxation and eliminate anxiety.
- **Encouraging non-coital behaviors.** Non-coital behaviors (physically stimulating activity that does not include intercourse), such as sensual massage, can be used to promote comfort and increase communication between partners.
- **Minimizing pain.** Using sexual positions that allow the woman to control the depth of penetration may help relieve some pain. The use of vaginal lubricants can help reduce pain caused by friction, and a warm bath before intercourse can help increase relaxation.
- Treatment depends on the cause of the sexual dysfunction. Medical causes that are reversible or treatable are usually managed medically or surgically. Physical therapy and mechanical aides may prove helpful for some people experiencing sexual dysfunction due to physical illnesses or disabilities.
- Sildenafil (Viagra) may be helpful for men who have difficulty attaining an erection. The medication increases blood flow to the penis. It must be taken 1 to 4 hours before intercourse. Men who take nitrates for coronary heart disease should not take sildenafil.
- Mechanical aids and penile implants are an option for men who cannot attain an erection and find sildenafil is not helpful.

- Women with vaginal dryness may be helped with lubricating gels, hormone creams, and -- in cases of premenopausal or menopausal women -- with hormone replacement therapy. In some cases, women with androgen deficiency can be helped by taking testosterone. **Kegel exercises** may also increase blood flow to the vulvar/vaginal tissues, as well as strengthen the muscles involved in orgasm.
- Vulvodynia can be treated with numbing cream, biofeedback, or low doses of certain antidepressants that also treat nerve pain. Surgery has not been successful.
- Behavioral treatments involve many different techniques to treat problems associated with orgasm and sexual arousal disorders. Self-stimulation and the Masters and Johnson treatment strategies are among the many behavioral therapies used.
- Simple, open, accurate, and supportive education about sex and sexual behaviors or responses may be all that is required in many cases. Some couples may benefit from joint counseling to address interpersonal issues and communication styles. Psychotherapy may be required to address anxieties, fears, inhibitions, or poor body image.

Sex–therapy:

- ➢ Stop-start [Do-Redo method]: Men should stimulate the penis near the point of ejaculation and then stop for 30-60 seconds and again stimulate it. Repeat the process for five to six times.
- ➢ **Squeeze: -** Stimulation to penis should be done near the point of ejaculation and then firmly squeeze around the penis just below the head. It has effect of preventing ejaculation.

Other than that these methods also can apply:

- ➢ Share your worries by communicating with your partner
- ➢ Spend more time with other form of sexual intimacy like kissing, cuddling, licking and massaging.
- ➢ Seek a doctor's help.

Infertility:

Infertility is the inability of a person, animal or plant to reproduce by natural means. It is usually not the natural state of a healthy adult organism, except notably among certain **eusocial** species (mostly **haplodiploid** insects).

In humans, infertility may describe a woman who is unable to conceive as well as being unable to carry a pregnancy to **full term.** There are many biological and other causes of infertility, including some that **medical intervention** can treat. Infertility rates have increased by 4% since the 1980s, mostly from problems with fecundity due to an increase in age. About 40% of the issues involved with infertility are due to the man, another 40% due to the woman, and 20% result from complications with both partners.

Women who are **fertile** experience a natural period of fertility before and during **ovulation**, and they are naturally infertile during the rest of the **menstrual cycle. Fertility awareness** methods are used to discern when these changes occur by tracking changes in cervical mucus or **basal body temperature**.

The World Health Organization defines infertility as : Infertility is "a disease of the reproductive system defined by the failure to achieve a clinical pregnancy after 12 months or more of regular unprotected sexual intercourse (and there is no other reason, such as breastfeeding or postpartum **amenorrhoea**). Primary infertility is infertility in a couple who have never had a child. Secondary infertility is failure to conceive following a previous pregnancy. Infertility may be caused by infection in the man or woman, but often there is no obvious underlying cause.

Primary and secondary infertility:

Primary infertility: It is defined as when a couple faces challenges with their first pregnancy. In other words, they have trouble conceiving or maintaining a pregnancy. The term is also used when a woman has never had a successful birth of a child or when a man has never successfully impregnated a woman.

Secondary infertility: It is the inability to get pregnant despite frequent, unprotected sex — for at least a year in women under age 35 or six months in women age 35 and older — by a couple who have previously had a pregnancy. Secondary infertility shares many of the same causes of primary infertility.

The causes of infertility can be both physical and emotional, and can be experienced by both men and women.

Causes:

Female Infertility:

1) Defects of the uterus and cervix (fibroids, polyps, birth defects)
2) Hormone imbalance or deficiencies, often related to age
3) **Ovarian cysts** and **polycystic ovary syndrome** (PCOS)
4) Pelvic infection or **pelvic inflammatory disease** (PID) caused by infections like **tuberculosis**
5) Scarring from sexually transmitted disease or endometriosis
6) Tumor

7) Long-term (chronic) disease, such as diabetes age-related factors
8) uterine problems
9) previous **tubal ligation**
10) Endometriosis
11) advanced maternal age
12) Autoimmune disorders
13) Clotting disorders
14) Obesity
15) Excessive exercising, eating disorders or poor nutrition
16) Exposure to certain medications or toxins
17) Heavy use of alcohol

Additionally, there can be egg-related problems, such as egg production in the ovaries, movement of the eggs from the ovary to the uterus, attachment of the eggs to the uterine lining, and survival of the egg or embryo once it has attached to the lining.

Male Infertility:

a. Genetic abnormalities
b. Hormone deficiency or taking too much of a hormone Impotence
c. Infections of the testes or the epididymis, the tube that stores and carries sperm
d. Older age
e. Previous chemotherapy
f. Previous scarring due to infection (including sexually transmitted diseases), trauma or surgery
g. Environmental pollutants
h. Exposure to high heat for prolonged periods
i. Radiation exposure
j. Retrograde ejaculation (dry orgasm)
k. Smoking
l. Heavy use of alcohol, marijuana, or cocaine
m. Use of certain prescription drugs

Additionally, there can be sperm-related issues including a decrease in the number of sperm produced by the man, a blockage of the sperms' release, or sperm that do not function properly

Psychosociologcial impact:

a. In many cultures, inability to conceive bears a stigma. In closed social groups, a degree of rejection (or a sense of being rejected by the couple) may cause considerable anxiety and disappointment. Some respond by actively avoiding the issue altogether; middle-class men are the most likely to respond in this way.

b. The consequences of infertility are manifold and can include societal repercussions and personal suffering. Advances in assisted reproductive technologies, such as IVF, can offer hope to many couples where treatment is available, although barriers exist in terms of medical coverage and affordability. The medicalization of infertility has unwittingly led to a disregard for the emotional responses that couples experience, which include distress, loss of control, stigmatization, and a disruption in the developmental trajectory of adulthood.

c. Infertility may have profound psychological effects. Partners may become more anxious to conceive, increasing **sexual dysfunction.** Marital discord often develops in infertile couples, especially when they are under pressure to make medical decisions. Women trying to conceive often have clinical depression rates similar to women who have heart disease or cancer. Even couples undertaking IVF face considerable stress.

d. The emotional losses created by infertility include the denial of motherhood as a rite of passage; the loss of one's anticipated and imagined life; feeling a loss of control over one's life; doubting one's womanhood; changed and sometimes lost friendships; and, for many, the loss of one's religious environment as a support system.

e. Emotional stress and marital difficulties are greater in couples where the infertility lies with the man.

Treatment:

Treatment depends on the cause of infertility, but may include counselling, fertility treatments, which include in vitro fertilization. According to **ESHRE** recommendations, couples with an estimated **live birth rate** of 40% or higher per year are encouraged to continue aiming for a spontaneous pregnancy. Treatment methods for infertility may be grouped as medical or complementary and alternative treatments. Some methods may be used in concert with other methods. Drugs used for women include clomiphene citrate, human menopausal gonadotropin, follicle-stimulating hormone, human chorionic gonadotropin, gonadotropin-releasing hormone analogs, aromatase inhibitor, metformin. counselling, fertility treatments, which include in vitro fertilization medical or complementary and alternative treatments.

UNIT-VII

DERMATOLOGY

POST GRADUATE INSTITUE OF BEHAVIORAL AND MEDICIAL SCIENCES, RAIPUR, C.G.

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DERMATOLOGY

INTRODUCTION

As an easily noticed and touched organ, the skin has a special place in psychiatry. With its responsiveness to emotional stimuli and ability to express emotions such as anger, fear, shame, and frustration, and by providing self-image and self-esteem, the skin plays an important role in the socialization process, which continues from childhood to adulthood (Domonkos, 1971; Koblenzer, 1983). Psychodermatology describes an interaction between dermatology and psychiatry and psychology. A relationship between psychological factors and skin diseases has long been hypothesized. Psycho dermatology addresses the interaction between mind and skin Psychiatry is more focused on the 'internal' non-visible disease, and dermatology is focused on the 'external' visible disease..

ROLE OF STRESS AND ANXIETY IN PSYCHODERMATOLOGICAL CONDITION

Stress and other psychological factors trigger the formation and exacerbation of many dermatological diseases (Van Moffaert, 1992; Koo and Pham, 1992). Every person has a shock organ that is sensitive to stress, which is defined by environmental and genetic factors, and this shock organ is the skin in people who display dermatological symptoms under stress. Panconesi suggested naming dermatological diseases that are activated and whose symptoms are exacerbated by emotional stressors a "dermatological stress disorders" (2000).

There is much research on the role of stress in dermatological diseases, and they are categorized as:

I. Environmental factors that cause stress;
II. Subjective experiences towards specific situations
III. Biological responses to stress (Cohen, 1995)

Studies define stress along three general categories:

1) Major stressful life events (e.g., change of employment, major personal illness and financial problems)

2) Psychological or personality difficulties, and

3) Lack of social support.

Research has shown that the subjective experience of stress is more important than the actual stress and the conditions that cause stress. Stress is the first reason that comes to mind, excluding in the formation of gastrointestinal ulcers and various studies has shown that gastrointestinal ulcers and hemorrhages occur more frequently in situations that cause social stress, such as earthquakes and economic crises. This can be explained by the decrease in regional blood flow and increase in gastric acid secretion following stress (Levenstein, 1999). In the same way, in a controlled study with 1500 people following the great earthquake in Japan Kodama found that atopic eczema had increased when compared with normal controls (1999).

Biological responses to stress vary from person to person; vasoactive peptides, lymphokines, and chemical mediators are secreted after stress and inflammation develops as a result of their influence on the immune system.

Stress sets off several physiological reactions in the body that can affect the skin. It causes the release of hormones like cortisol that thicken hair follicle cells and increase oil production—the perfect recipe for acne. Stress can also trigger neuropeptide, chemicals unleashed from nerve endings in the skin that leave it red or itchy, and encourage T cells (the skin's infection fighters) to overreact, making the skin turn over too quickly and flake or scale. Then there are the blood vessels: Under stress they become more reactive, either clamping shut (so skin looks pale or sallow) or opening too widely (causing the skin to flush).

Stress, emotional trauma, bereavement, divorce, redundancy, depression – all these have a psychological effect on the brain and the nervous system which, in turn, affects the skin. Actually having a chronic skin condition like eczema or psoriasis can, in itself, create psychological fluctuations via anxiety and depression which can make the skin condition worse. An estimated 20 per cent of psoriasis sufferers also have depression. During periods of anxiety or stress, the adrenal gland produces more of a hormone called cortisol, which affects the body’s immune system. This in turn can cause the skin’s own defenses to either weaken, as in the case of eczema, or go into overdrive, as in the case of psoriasis. There is growing scientific evidence to prove that psychology and skin conditions are directly linked-

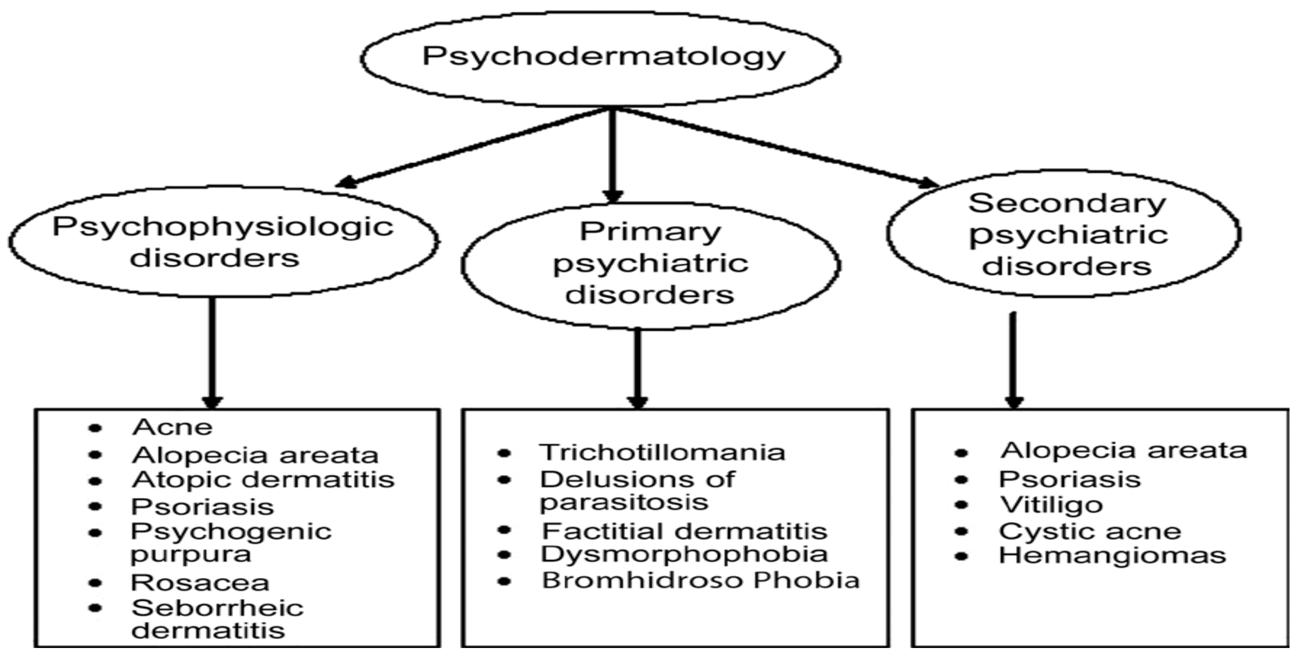
A study published in the British Journal of Dermatology found that almost 40 cent of psoriasis sufferers recalled stressful events in the month before their condition got noticeably worse, though in some cases it could take just two days to bring on an attack.

- A study in the "The International Society of Dermatology" found that people with psoriasis reported a lower quality of life and higher than normal stress levels.

- A study in the American Journal of Pathology. Researchers found that the immune cells in skin can over-react when levels of stress rise, resulting in inflammatory skin diseases.

Psycho dermatology is divided into three categories according to the relationship between skin diseases and mental disorders.

VARIOUS PSYCHODERMATOLOGICAL CONDITIONS AND ROLE OF STRESS AND ANXIETY IN PSYCHO DERMATOLOGICAL CONDITION



PSYCHOPHYSIOLOGIC (PSYCHOSOMATIC) DISORDERS

Caused by skin diseases triggering different emotional states (stress), but not directly combined with mental disorders (psoriasis, eczema).

Here psychiatric factors are instrumental in the etiology and course of skin conditions. The skin disease is not caused by stress but appears to be precipitated or exacerbated by stress.

1. **Atopic dermatitis**

   The onset or exacerbation of atopic dermatitis often follows stressful life events. Symptom severity has been attributed to interpersonal and family stress, and problems in psychosocial adjustment and low self-esteem have been frequently noted. Adults with atopic dermatitis are more anxious and depressed compared with clinical and healthy control groups. Children with atopic dermatitis have higher levels of emotional distress and more behavioral problems than healthy children or children with minor skin problems.

   Psychosocial morbidity in atopic dermatitis

   Psychological stress may be an acquired factor affecting the expression of atopic dermatitis. Atopic individuals with emotional problems may develop a vicious cycle between anxiety/depression and dermatologic symptoms. In one direction, anxiety and depression are frequent consequences of the skin disorder. The misery of living with atopic dermatitis may have a profoundly negative effect on health-related quality of life (HRQOL) of children and their families. Teasing and bullying by children and embarrassment by adults and children can cause social isolation and school avoidance. The social stigma of a visible skin disease, frequent visits to doctors and the need to constantly apply messy topical remedies all add to the burden of disease. Lifestyle restrictions in more severe cases can be significant, including limitations on clothing, staying with friends, owning pets, swimming or playing sports. The impairment of quality of life caused by childhood atopic dermatitis has been shown to be greater than or equal to that of asthma or diabetes.

2. **Urticaria**

   Severe emotional stress may exacerbate preexisting urticaria. Increased emotional tension, fatigue, and stressful life situations may be primary factors in more than 20% of cases and are contributory in 68% of patients. Difficulties with expression of anger and a need for approvals from others are also common. Patients with this disorder may have symptoms of depression and anxiety, and the severity of pruritus appears to increase as the severity of depression increases. Cold urticaria may be associated with hypomania during winter and recurrent idiopathic urticaria with panic disorder.

While the etiology is unknown in 79% of urticaria patients, it is also known that psychological factors have a direct effect on the development of the disease in 11%-21% of the patients and plays a facilitating role in 24%-68% (Champion, 1969; Michealsson, 1969). Whatever the reason is, severe emotional stress exacerbates urticaria and traumatic events such as earthquakes may cause the disease (Arnold, 1990, Stewart, 1989). While 51% of urticaria cases begin with stressful life events, this percentage is 77% in cholinergic urticaria and 82% in dermographism (Czubalski, 1977). Depression is also frequent in these patients and it was reported that as the severity of depression increases, scratching of urticaria plaques increases (Gupta, 1994). When personality type and disease are matched, it was reported that these patients can not express their anger and hostility sufficiently and that they seek approval from others (Juhlin, 1981). While Unal et al. (1991) reported that, anxiety and depression are three times more frequent in chronic urticaria patients than the general population.

Acne Excoriee

The habitual act of picking at skin lesions, apparently driven by compulsion and psychological factors independent of acne severity, has been reported in the perpetuation of self-excoriation. Most patients with this disease are females with late onset acne. Psychiatric co morbidity of acne excoriee includes body image disorder, depression, anxiety, obsessive-compulsive disorder (OCD), delusional disorders, personality disorders, and social phobias. Immature coping mechanisms and low self-esteem have also been associated.

Herpes Simplex Virus, Herpes Zoster, and Human Papillomavirus Infections

There is increasing evidence that stress has a role in recurrent herpetic infection. In one study, experimentally induced emotional stress led to herpes simplex virus reactivation. Other studies have demonstrated an inverse correlation between stress level and present CD4 helper/ inducer T lymphocytes, thus contributing to herpes virus activation and recurrences. It has also been suggested that stress-induced release of immunomodulating signal molecules (e.g., catecholamine, cytokines, and glucocorticoids) compromises the host's cellular immune response leading to reactivation of herpes simplex virus. Relaxation treatment and biofeedback may reduce the frequency of recurrences. Herpes zoster has been associated with chronic child abuse, and severe psychological stress of any sort may depress cell-mediated immune response.

PRIMARY PSYCHIATRIC DISORDERS RESPONSIBLE FOR SELF-INDUCED SKIN DISORDERS.

Primary psychiatric disorders are encountered less often than psycho physiologic disorders. These disorders have received little emphasis in the psychiatry or dermatology literature, even though they may be associated with suicide and unnecessary surgical procedures. Most of these disorders occur in the context of somatoform disorder, anxiety disorder, factitious disorder, impulse-control disorder or eating disorder.

- **Trichotillomania** **(**The pulling out of one's own hair)

  In dermatological terms, trichotillomania is the pulling out of one's own hair; in terms of a psychiatric definition, impulsiveness accompanies the hair pulling behavior (Stein, 1995). It was reported that in trichotillomania patients, a severe distress forms immediately before the hair pulling behavior and distress decreases, while pleasure and satisfaction is felt afterwards. Generally, hair on the forehand and temporal regions, eyebrows, eyelashes, beard, and pubic hairs are pulled, and these may then be sucked on and swallowed (Gupta, 1987). It was categorized among not otherwise defined impulse control disorders in DSM IV; the etiology of trichotillomania is accepted to be variable and it is known that the rate of comorbidity with other diseases is high, the most frequent being obsessive-compulsive disorder, and sometimes it may not meet DSM - IV criteria (Mc Elroy, 1994). Among other causes of Trichotillomania are simple habit, reaction to stress, mental retardation, depression, anxiety, and delusions in rare cases. Childhood trauma and emotional neglect may play a role in the development of this disorder. In delusional cases, patients believe that there is something in the hair root, and when pulled out, it will disappear and normal hair will grow. This rare condition is called trichophobia. The identification of the underlying etiology and designing the treatment accordingly is essential. Trichotillomania is one of the rare dermatological diseases that show diagnostic symptoms in the histopathological examination of the skin. This change is called trichomalacia and is only observed in Trichotillomania patients (Lachapelle, 1977). Forty-three percent of the cases deny that they pull out their hair and with his to pathological examination the correct diagnosis can be made in these cases (Cristenson, 1991).


- **Obsessive-compulsive disorder**

  Patients usually present to dermatologists because of skin lesions resulting from scratching, picking, and other self injurious behaviors. They typically have an increased level of psychiatric symptomatology compared with age and sex matched controls taken from the general population of dermatology patients, and many patients experience negative stigmatization in their daily life. Common behaviors include compulsive pulling of scalp, eyebrow, or eyelash hair; biting of the nails and lips, tongue and cheeks; and excessive hand washing. It has been found that OCD in child and adolescent dermatology patients most commonly presents as trichotillomania, onychotillomania and acne excoriée.

- **Dermatitis artifacta (Factitial dermatitis)**

  Artifact dermatitis are the lesions entirely by the actions of the fully aware patient on the skin, hair, nails or mucosa caused by the patient using nails, the butt of a cigarette, sharp objects, or chemicals (Gupta, 1987a; Fabish, 1980), no rational motive for this behavior. Although patients deny it. Lesions separate from the surrounding normal tissue with exact geometrical borders and may replicate many skin disorders. Severe personality disorders, obsessive-compulsive disorder, depression, psychosis, mental retardation, and Munchausen’s syndrome are among the psychiatric disorders that accompany artifact dermatitis (Gupta, 1993a; Stein, 1992). As these patients cannot control their self-images and moods, they have difficulty in maintaining interpersonal relationships. They engage in self-mutilating behavior with feelings of emptiness and anger. These lesions are a cry for help in response to the stress associated with undeveloped coping mechanisms. It was reported that the rate of manifestation of the lesions after situations that cause severe stress, such as illness, accident, and bereavement, was 19%-33%, and that the lesions regress as the stressful situation disappears (Sneddon, 1975). In addition to mutilating their skin, suicidal tendency, substance use, or compulsive eating are frequent in borderline personality disorder patients.

Diagnosis: The peculiar looks of the lesions and the fact that patients cannot explain how they were formed make the diagnosis easy (Hollender, 1973). Direct confrontation of the patient is not suggested (Spraker, 1983). Early diagnosis can prevent the illness from becoming chronic.

This is an artifactual skin disease caused The condition is more common in women than in men (3 : 1 to 20 : 1). The lesions are usually bilaterally symmetrical, within easy reach of the dominant hand, and may have bizarre shapes with sharp geometrical or angular borders, or they may be in the form of burn scars, purpura, blisters and ulcers. Erythema and edema may be present. Patients may induce lesions by rubbing, scratching, picking, cutting, punching, sucking or biting or by applying dyes, heat or caustics. Some patients inject substances, including feces and blood. Reported associated conditions include OCD, borderline personality disorder, depression, psychosis and mental retardation. And most of the patients with factitious dermatitis have some sort of personality disorder and they often use some means to damage his or her own skin, such as burning cigarettes, chemicals or sharp instruments.

- **Psychogenic excoriation**

  Psychogenic excoriation occurs in 2% of dermatology patients mostly in women. It is an uncommon psycho dermatological condition, which responds well to serotonin reuptake inhibitors and behavioral therapy. It is characterized by excessive scratching or picking of the skin. The lesions are usually found on face, upper limbs and upper back. It is a chronic disorder with a high rate of psychiatric co morbidity. Major depressive syndrome was the most common psychiatric disorder found in the PE group.

- **Neurotic Excoriations**

  As per DSM-IV-TR classification an impulse-control disorder not otherwise specified, neurotic excoriations are self-inflicted lesions that typically present as weeping, crusted, or lichenified lesions with postinflammatory hypopigmentation or hyperpigmentation. The preferred term for this disorder is *pathologic skin picking*. Usual sites are the extensor aspects of extremities, scrotum, and perianal regions. Repetitive scratching, initiated by an itch or an urge to excoriate a benign skin lesion, produces lesions. The behaviors of these patients sometimes resemble those with OCD.

- **Delusions of parasitosis** *(* Ekbom’s syndrome)

  The most common form of monosymptomatic hypochondriacal psychosis encountered among patients with skin problems is called delusions of parasitosis. Delusional parasitosis is a syndrome in which the patient has the false belief that he is infested by parasites or organism; and they often elaborate on how these organisms reproduce, move and spread under their skin or even exit the skin. It may occur as the sole psychologic disturbance, or it may be associated with an underlying psychiatric disorder or physical illness. The psychiatric differential diagnosis include schizophrenia, psychotic depression, psychosis in patients with florid mania or drug-induced psychosis, and formication without delusion, in which the patient experiences crawling, biting and stinging sensations without believing that they are caused by organisms. Patients with delusion of parasitosis often present with the matchbox sign, in which small bits of excoriated skin, debris or unrelated insects or insect parts are brought in matchboxes or other containers as a proof of infestation.

- **Dysmorphophobia**

  This condition is also called body dysmorphic disorder or dermatological non-disease. Patients with this condition are rich in symptoms but poor in signs of organic disease. Self-reported ‘complaints’ or ‘concerns’ usually occur in three main areas: Face, scalp and genitals. Facial symptoms include excessive redness, blushing, scarring, large pores, facial hair and protruding or sunken parts of face. Other symptoms are hair loss, red scrotum, urethral discharge and herpes and AIDS phobia. Strategies to relieve the anxiety due to the perceived defects may include camouflaging the lesions, mirror checking, comparison of ‘defects’ with the same body parts on others, questioning/ reassurance seeking, mirror avoidance and grooming to cover up ‘defects’. Women are more likely than men to be preoccupied with the appearance of their hips or their weight, to pick their skin, to camouflage defects with makeup and to have comorbid bulimia nervosa. Men are more likely than women to be preoccupied with body build, genitals and hair thinning and to be unmarried and to abuse alcohol. Patients with body image disorders, especially those involving the face, may be suicidal. Associated comorbidity in dysmorphophobia may include depression, impairment in social and occupational functioning, social phobias, OCD, skin picking, marital difficulties and substance abuse.

Psychogenic pruritus

In this disorder, there are cycles of stress leading to pruritus as well as of the pruritus contributing to stress. Psychologic stress and comorbid psychiatric conditions may lower the itch threshold or aggravate itch sensitivity. Stress liberates histamine, vasoactive neuropeptide and mediators of inflammation, while stress-related hemodynamic changes (e.g., variation in skin temperature, blood flow and sweat response) may all contribute to the itch-scratch-itch cycle. Psychogenic pruritus has been noted in patients with depression, anxiety, aggression, obsessional behavior and alcoholism. The degree of depression may correlate with pruritus severity.

SECONDARY PSYCHIATRIC DISORDERS CAUSED BY DISFIGURING SKIN

Ichthyosis, acne conglobata, vitiligo, which can lead to states of fear, depression or suicidal thoughts.

This category includes patients who have emotional problems as a result of having skin disease. The skin disease in these patients may be more severe than the psychiatric symptoms, and, even if not life-threatening, it may be considered 'life-ruining'. Symptoms of depression and anxiety, work-related problems and impaired social interactions are frequently observed.

- **Psoriasis**

  Stress has long been reported to trigger psoriasis. Psoriasis is associated with a variety of psychological difficulties, including poor self-esteem, sexual dysfunction, anxiety, depression and suicidal ideation. Psoriasis is associated with substantial impairment of health-related quality of life (HRQOL), negatively impacting psychological, vocational, social and physical functioning. The most common psychiatric symptoms attributed to psoriasis include disturbances in body image and impairment in social and occupational functioning. Quality of life may be severely affected by the chronicity and visibility of psoriasis as well as by the need for lifelong treatment.

  Five dimensions of the stigma associated with psoriasis have been identified:

  (1) Anticipation of rejection

  (2) Feelings of being flawed

(3) Sensitivity to the attitudes of society

(4) Guilt and shame

(5) Secretiveness.

Depressive symptoms and suicidal ideation was frequently associated in psoriasis. In general, psychological factors, including perceived health, perceptions of stigmatization and depression are stronger determinants of disability in patients with psoriasis than are disease severity, location and duration. In a recent prospective study of patients with psoriasis, the frequency of psychiatric disturbance decreased with improvement in the clinical severity and symptoms of psoriasis. The emotional effects and functional impact of the disease are not necessarily proportionate to the clinical severity of psoriasis.

- **Alopecia areata**

The role of psychological factors in the pathogenesis of alopecia areata (AA) has long been the subject of debate. The influence of psychologic factors in the development, evolution and therapeutic management of alopecia areata is well established. Acute emotional stress may precipitate alopecia areata, perhaps by activation of over expressed type 2b corticotrophin-releasing hormone receptors around the hair follicles, and lead to intense local inflammation. Release of substance P from peripheral nerves in response to stress has also been reported, and prominent substance P expression is observed in nerves surrounding hair follicles in alopecia areata patients. Substance P degrading enzyme neutral endopeptidase has also been strongly expressed in affected hair follicles in the acute-progressive as well as the chronic-stable phase of the disorder. Comorbid psychiatric disorders are also common and include major depression, generalized anxiety disorder, phobic states and paranoid disorder.

- **Vitiligo**

Vitiligo is a specific type of leukoderma characterized by depigmentation of the epidermis. In some studies, patients with vitiligo have been found to have significantly more stressful life events compared with controls, suggesting that psychologic distress may contribute to onset. Links between catecholamine-based stress, genetic susceptibility and a characteristic personality structure have been postulated. Psychiatric morbidity is typically reported in approximately one-third of patients, but, in one study, 56% of the sample had adjustment disorder and 29% had depressive disorders. Patients with vitiligo are frightened and embarrassed about their appearance, and they experience discrimination and often believe that they do not receive adequate support from providers. Younger patients and individuals in lower socioeconomic groups show poor adjustment, low self-esteem and problems with social adaptation. Most patients with vitiligo report a negative impact on sexual relationships and cite embarrassment as the cause.

MISCELLANEOUS

This group includes disorders or symptoms that are not otherwise classified.

Cutaneous Sensory Syndromes

Patients with these syndromes experience abnormal skin sensations (e.g., itching, burning, stinging, and biting or crawling) that cannot be attributed to any known medical condition. Examples include glossodynia, vulvodynia, and chronic itching in the scalp. These patients often have concomitant anxiety disorder or depression.

- **Psychogenic Purpura Syndrome**

  This condition, also known as auto erythrocyte sensitization syndrome (Gardner-Diamond syndrome), is seen primarily in emotionally unstable adult females. These patients present with bizarre, painful, recurrent bruises on extremities, frequently after trauma, surgery, or severe emotional stress. The exact mechanism is unclear; however, hypersensitivity to extravasated red cells, autoimmune mechanisms, and increased cutaneous fibrinolytic activity has been implicated in the pathogenesis. These patients may have overt depression, sexual problems, and feelings of hostility, obsessive-compulsive behavior, borderline personality disorder, and factitious dermatitis. Without a correct diagnosis, these individuals can receive treatments that are neither necessary nor effective. The diagnosis can be made in a patient who has atypical history and clinical picture of the syndrome and in whom a skin test with use of the patient's blood reveals a positive reaction.

- **Pseudopsychodermatologic Disease**

  These patients may have bizarre skin symptoms without obvious physical findings or a subclinical skin disease that has been eradicated or modified by scratching. Psychodermatologic disease can mimic other skin disorders, and medical disorders can mimic psycho dermatologic conditions. Localized bullous pemphigoid lesions, for example, have been mistaken for dermatitis artefacta; multiple sclerosis, folliculitis, hypothyroidism, and vitamin B12 deficiency have been initially diagnosed as delusions of parasitosis.

Suicide in Dermatology Patients

Suicide has been reported in patients with longstanding debilitating skin diseases and must be considered when evaluating these patients. Even clinically mild to moderate severity skin disease may be associated with significant depression and suicidal ideation. Cotterill and Cunliffe reported suicides in 16 patients (7 men and 9 women) with body image disorder or severe acne.

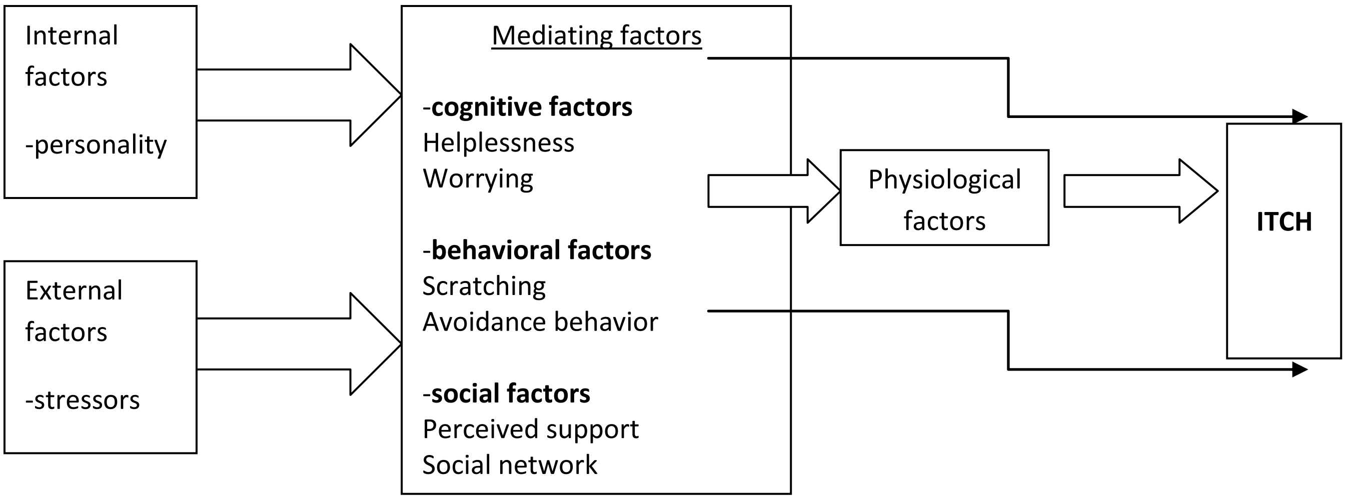
Itch is a major feature of many skin diseases, which adversely affects patient's quality of life. Besides disease severity, psycho physiological factors have been proposed to influence the itch sensation. Ex: atopic dermatitis, contact dermatitis, urticaria and psoriasis

Apart from the severity other factors have also been proposed to influence itch intensity such as sweating, skin dryness, or physical effort, psychological factors have regularly been described to influence itch intensity.

One of the most striking examples of the influence of psychological factors on itch came from the observation that itch (and the scratching response) could be aggravated by showing individuals itch-related pictures of fleas, mites, scratch marks, allergic reactions, etc. high among patients suffering from itch and that negative emotions can increase the level of itch.

Biopsychosocial factors can best be shown in a diathesis-stress model, which is based on the hypothesis that internal vulnerability factors (diathesis), such as personality, interact with external environmental factors (such as major life events and other stressors) to trigger a disease or itch.

BIOPSYCHOLOGICAL MODEL OF ITCH



TREATMENT

PSYCHOLOGICAL INTERVENTIONS OF PSYCHO DERMATOLOGICAL CONDITION

Nonpharmacologic management of psycho dermatologic conditions includes both structured and unstructured interventions that may ameliorate skin disorders, reduce psychological distress, and improve the functional status of the affected individual. Many approaches are available. Cognitive behavioral psychotherapy, behavior modification, insight-oriented psychotherapy, family therapy, and supportive psychotherapy are common interventions used for dermatologic disorders; it can be accomplished in a one-to-one relationship setting or can be carried out in a group setting. Psychotherapeutic interventions have demonstrated improvements in eczema, psoriasis, neurotic excoriation, BDD, and trichotillomania.

Cognitive-behavioral Therapy

Cognitive–behavior therapy (CBT) is a treatment approach that aims to change maladaptive ways of thinking, feeling and behaving through the use of cognitive and behavioral interventions. This model takes the view that it is not situations in and of themselves that are stressful, but rather the perception that one takes of them that makes them so. According to the cognitive model, the beliefs that patients hold about their condition often influence how they cope with and adapt to it. A common feature in the beliefs of people with emotional difficulties is that they have negative and irrational content. These perceptions are often the result of distortions in processing, such as 'cognitive errors' (Beck, 1976, 1993).

CBT focuses on examining and trying to challenge dysfunctional beliefs and appraisals which may be implicated in a person's low mood or avoidance of certain situations or behaviors'. Consequently, targeting cognitions and maladaptive behavior are the key areas of CBT interventions for facilitating change. According to this approach, beliefs are considered as hypotheses to be tested rather than assertions to be uncritically accepted. Therapist and client take the role of 'investigators 'and develop ways to test beliefs, such as 'Others do not like me because of my eczema' or 'I won't be happy anymore because of my vitiligo'. Success at challenging these beliefs involves providing evidence that they are erroneous, and underscored by anxiety and depression (Beck, 1993).

CBT has been successfully applied to various skin conditions. For example, Horne et al. (1989) used cognitive–behavioral therapy along with standard medical treatment in treating three patients suffering with atopic eczema. All three showed a post-treatment reduction in symptom severity, an increase in their ability to control the disorder and a decrease in their reliance on medication. Four controlled studies have also used a cognitive–behavioral approach with psoriasis patients (Price et al., 1991; Zacharie et al., 1996; Fortune et al., 2002; Fortune et al., 2004). Findings have shown adjunctive cognitive–behavioral interventions result in a reduction of psychological distress and in the clinical severity of the condition. Additionally, Papadopoulos et al. (1999) compared two matched groups of vitiligo patients, one of which received CBT while the other received standard medical treatment alone. Results suggested that patients could benefit from CBT in terms of coping and living with vitiligo. There was also preliminary evidence to suggest that gains made through CBT influences the progression of the condition. Finally, Ehlers et al. (1995) employed CBT with patients with atopic dermatitis and found significant reductions in anxiety, frequency of scratching and itching as well as cortisone use.

Behavior Therapy

Behavior therapy incorporates applications derived from learning theory (classical and operant conditioning) and employs them to the treatment of persistent, maladaptive, learned habits. It is based on the assumption that since autonomic responses can be learned, they can also be unlearned via different methods. Among behavior therapy techniques are systematic desensitisation, assertiveness and social skills training, behavior analysis, relaxation training (e.g. autogenic and progressive muscle relaxation, biofeedback) habit-reversal training and imagery. The aim of these techniques is to progressively diminish maladaptive behavioural responses by repeatedly inhibiting the anxiety by means of competing responses (Wolpe, 1980). A behavior analysis is conducted where the clinician collects information about the relationship between stimuli and behavioral responses in order to understand the role of anxiety.

Diverse behavioral therapeutic strategies have been applied, either separately or in combination with other psychological techniques to dermatological conditions.

Systematic desensitisation is an appropriate technique for the treatment of dermatoses which feature anticipatory anxiety (Van Moffaert, 1992). The fear and apprehension that patients with skin disease may feel about them may be challenged by this technique. Through graded exposure, the patient enters situations that they may fear and avoid. The *habit- reversal technique* is a common strategy used to inhibit scratching and it has been reported to have some success with skin disorders, such as eczema and psoriasis (Ginsburg, 1995). It involves self monitoring for early signs and situational cues of scratching and practicing alternative responses, such as clenching the fists (Ehlers et al., 1995). *Relaxation* has beneficial effects on skin disorders because it reduces stress levels. It is a useful way to help patients prepare for anxiety-provoking situations or to cope with stressful social predicaments. Relaxation can be used on its own as a means to reduce anxiety or tension or can be paired with imagery. There are various different techniques, such as progressive muscle relaxation or autogenic relaxation training.
Imagery with skin disease patients is employed in order to help them cope with anxiety relating to their condition. Imagery is a useful technique for helping the patient to visualise the feared situation while in a relaxed state (Papadopoulos & Bor, 1999). *Assertiveness and social skills training* is appropriate for patients with cutaneous conditions that attract attention from others, such as staring or personal questions. Interventions focus on improving social skills and ways of

expressing emotions, thus helping patients deal more effectively with the reactions of others and learn a more positive mode of social functioning (Robinson et al., 1996).

Many techniques used for stress management program: self observation, relaxation training, time management and problem-solving etc.

Self-Observation:

A daily diary format is used, with patients being asked to keep a record of how they responded to challenging or stressful events that occurred each day. A particular stress (e.g. argument with spouse) may precipitate a sign or symptom (e.g. pain in the neck).

Behavioral modifications include identifying triggers that lead to maladaptive behavior ex: hair pulling behavior,scratching along with replacing the behavior with a competing activity such as clenching fists or pulling on a Koosh Ball

Magnet stones (magnetite's) to reduce tension

Self-monitoring: This involves systematically observing and discriminating when the behavior occurs, recording the responses, and evaluating one's own behavior.

Habit reversal: This is a set of procedures taught to a child that includes the following components: increasing awareness of the habit; teaching a competing response to practice when the child feels the urge to engage in the habit, in situations where the habit historically occurs or for 1 minute after the occurrence of the habit; practicing stress and anxiety reduction procedures on a daily basis; and support and encouragement from parents.

Competing reaction training: This is a component of habit reversal that is occasionally used alone; a child is taught a socially appropriate alternative behavior or response and encouraged to practice it on a daily basis when they feel the urge to engage in the habit, in situations where the habit historically occurs or for 1 minute after the occurrence of the habit

Group therapy

Understanding and addressing social factors can be extremely helpful for patients and their families coping with skin disease. For example, atopic dermatitis can result in diminished quality of life. Providing education regarding the etiology and pathogenesis of atopic dermatitis as well as realistic expectations can improve adherence to treatment regimens while simultaneously addressing the anxiety, frustration, shame, and depression that is frequently associated with the disease.16 Weber et al demonstrated improvement of pruritus (decreased frequency of pruritus) and improved quality of life (both leisure and personal relationships). Group therapy is a mode of intervention that helps individuals with a common problem enhance their social functioning though group exercises. Group members are given the opportunity to share their experiences, feelings and difficulties in a safe atmosphere under the auspices of a group facilitator. Using a combination of instruction, modeling, role-play, feedback and open discussion, members of the group are encouraged to discover more about the interaction process. In most cases 6–12 clients meet with their therapist at least once a week for about 2 hours. Usually groups are organized around one type of problem (such as coping) or type of client (such as psoriasis patients).

Through group interaction, ineffective and immature ways of coping are discouraged, positive attitudes are fostered and feelings, such as loneliness and isolation, that many patients experience, diminish. Moreover, group members can bolster one another' self-confidence and self-acceptance, as they come to trust and value one another, and develop group cohesiveness. Group therapy allows participants to try out new skills in a supportive environment and members learn from one another. Thus this offers features not found in individual treatment.

Various approaches, such as social skills training to group therapy have been tried with patients with skin disorders (Robinson et al., 1996). Patients with chronic skin conditions, such as psoriasis or eczema are known to benefit from group therapy and such therapy has increased their confidence in coping with them disease (Ehlers et al., 1995; Seng & Nee, 1997; Fortune et al., 2002).

Muscle Relaxation

Numerous guided imagery and progressive relaxation programs are available. Their goal is to reducing muscle and psychic tension and producing pleasant images and sensations. Physiologically, the techniques are effective in reducing sympathetic reactivity and enhancing parasympathetic activity. Relaxation therapy has been used to treat acne, acne excoriée, psoriasis, eczema, hyperhidrosis, and neurotic excoriations. Edmund Jacobson in 1938 developed a method called progressive muscle relaxation Patients were taught to relax muscle groups, such as those involved in "tension headaches" When they encountered and were aware of situations that caused tension in their muscles, the patients were trained to relax. A recent study has confirmed a long-held belief that acne worsens under stress. Therefore, regular practice of a stress management technique may decrease the occurrence and severity of acne.

Hypnosis

Early researchers in Psychodermatology experimented with the use of hypnosis (Van Moffaert, 1992).Hypnosis brings about changes in physiological parameters, such as skin conductance, skin temperature and vasomotor reactions all of which can be decisive in the etiology of skin diseases (Van Moffaert, 1992). Neurodermatitis, chronic urticaria and viral warts are skin diseases with which hypnosis has been successfully used (Barber, 1978). Hypnotic techniques and the trance state have been used since ancient times to assist in healing.. Hypnosis has proved beneficial for some patients with eczema, alopecia areata, hyperhidrosis, verruca, urticaria, rosacea, lichen planus, pain syndromes, and vitiligo. Benefits in reducing scratching in eczema can be long-lasting, and more highly hypnotizable subjects may achieve greater benefits. Hypnosis has been used to facilitate resolution of psychogenic excoriations in acne excoriée. Recent studies in patients with alopecia areata (including several with ophiasis distribution) demonstrated that hypnosis promotes excellent regrowth in approximately 50% of treated patients and improvements in depression and anxiety in almost all patients. Efficacy has also been demonstrated in treating warts. Hypnosis is best performed by qualified clinicians with appropriate training.

Biofeedback

Biofeedback is a pleasant and noninvasive intervention allowing the patient to gain control over measurable physiological reactions that are often considered automatic and beyond voluntary control. Blood pressure, heart rate, skin temperature, galvanic skin response (sweating), and muscle are frequent targets of treatment. Use of electromyography for measurement of muscle tension and hand temperature for measurement of blood flow is common. The patient is given feedback in real time, and simple techniques are taught to enable the patient to consciously modify and control these physiological activities.

The physiological modifications learned during biofeedback training can lead to positive changes in skin and emotional functioning. Virtually anyone can learn and benefit to some degree from biofeedback training. The goal of training is to master self-regulatory techniques that can be used quickly and inconspicuously throughout the day to manage stress and modulate physiological reactivity. Biofeedback training typically uses auditory or visual feedback cues for the patients. Structured breathing techniques (square-box breathing or deep and rhythmic breathing), progressive muscle relaxation, and guided imagery techniques are often used to induce a more relaxed state. This more relaxed and healthful state is accompanied by more relaxed musculature and increased blood flow to the extremities. The patient is made aware of the more relaxed state by slowing or quieting of auditory tones or decreased frequency of tones. Sometimes visual cues are used with soothing color change or decreased height of histograms. Patients can actually observe the physiological benefits of simple stress management techniques. This physiological validation coupled with greater awareness of proprioceptive cues can be an excellent motivator for use of the techniques outside the training sessions. Biofeedback is used widely throughout medicine with success in the treatment of skin disease, hypertension etc. Benefits have been seen in patients with rosacea, acne, eczema, urticaria, and psoriasis. Biofeedback of skin temperature by temperature –sensitive strip or by thermocouple can be used for relaxation, dyshidrosis and Reynaud's syndrome.108-110 HRV biofeedback can also help reduce the stress response that tends to exacerbate many inflammatory skin disorders

Biofeedback training is usually best performed by trained psychologis or other medical professionals.

Meditation

Meditation describes a state of concentrated attention on some object of thought or awareness. It usually involves turning the attention inward to a single point of reference. Concentrative type medication focuses the attention on the breath, an image, or a sound (mantra) to still the mind and allow a greater awareness and clarity to emerge. There is a centering or focusing on a single thing. The simplest form of concentrative meditation involves sitting quietly and focusing one's attention on the breath. Both yoga and meditation practitioners believe that there is a direct correlation between one's breath and one's state of mind. If a person is anxious, frightened, agitated, or distracted, the breath will tend to be shallow, rapid, and uneven. On the other hand, when the mind is calm, focused, and composed, breaths will tend to be slow, deep, and regular.

Counseling

- Information giving (psycho education)
- Supportive counseling to family

How can therapy help in the treatment and management of dermatology patients?

Therapy can help dermatology patients to:

- Come to terms with their conditions.
- Explore treatment options and facilitate decision-making.
- Examine difficulties they are experiencing with their condition and gain insight into what factors maintain those difficulties.
- Explore and challenge dysfunctional appraisals, beliefs and assumptions.
- Identify useful coping strategies.
- Facilitate interaction skills.
- Examine issues that may be indirectly linked to the skin condition.
- Challenge and cope with anticipatory anxiety and depression.

Group therapy, especially social and assertiveness skills training can help dermatology Patients:

- Encounter difficulties in social situations.
- Discuss their problems with others who can empathies.
- Develop a better understanding through the others' experiences of their condition.
- Allow members to acquire and develop a variety of skills and put them in practice with other members.
- Serve as a means of emotional and social support for skin patients.

CONCLUSION

Psycho dermatologic disorders are conditions involving interaction between the mind and the ski It is suggested to use a biopsychological model, which takes into account the psychological (e.g. psychiatry comorbidity such as major depression and the impact of skin disorder on the psychological aspects of quality of life) and social (e.g. impact upon social and occupational functioning) factors. The treatment of psychodermatological disorders should be carried out through the multidisciplinary approach, including family physician, dermatologist, psychiatrist and psychologist. It is very important to educate dermatologists in the diagnostic procedures and therapy of psychiatric disorders, which sometimes coexist with the skin disease. The cooperation of the dermatologist, psychologist and a psychiatrist in order to increase the life quality of the patients is of utmost importance. The management of psycho dermatologic disorders requires evaluation of the skin manifestation and the social, familial and occupational issues underlying the problem.

"Sickness is not just an isolated event, nor an unfortunate brush with nature. It is a form of communication – the language of the organs – through which nature, society, and culture speak simultaneously." (Synnott, 1993)

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UNIT-VIII

ONCOLOGY

Class Presentation

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Syllabus

Psychosocial issues associated with cancer- quality of life, denial, grief reaction to bodily changes, fear of treatment, side effects, abandonment, recurrence, resilience, assessment tools and goals of interventions for individual and family, and therapy techniques.

Introduction

"Oncology" is a branch of medicine that deals with cancer. The word "onco" means bulk, mass, or tumour while "-logy" means study. Psychosocial oncology is a specialty in cancer care concerned with the understanding and treatment of the social, psychological, emotional, spiritual, and functional aspects of cancer, At all stages of the disease from trajectory to prevention through to bereavement. Psychosocial oncology involves a holistic approach to cancer care that addresses a range of human needs that can improve or optimize the best possible quality of life for individuals and their networks affected by cancer.

What is cancer?

Cancer is a generic term for a large group of diseases that can affect any part of the body. Cancer is the uncontrolled growth and spread of cells. One defining feature of cancer is the rapid creation of abnormal cells that grow beyond their usual boundaries, and which can then invade adjoining parts of the body and spread to other organs. This process is referred to as metastasis. Metastases are the major cause of death from cancer. Cancer can start almost anywhere in the human body, which is made up of trillions of cells. Normally, human cells grow and divide to form new cells as the body needs them. When cells grow old or become damaged, they die, and new cells take their place. When cancer develops, however, this orderly process breaks down. As cells become more and more abnormal, old or damaged cells survive when they should die, and new cells form when they are not needed. These extra cells can divide without stopping and may form growths called tumors. Other terms used for cancer are malignant tumours and neoplasms.

Among various diseases, cancer has become a big threat to human beings globally. As per Indian population census data, the rate of mortality due to cancer in India was high and alarming with about 806000 existing cases by the end of the last century. Cancer is the second most common disease in India responsible for maximum mortality with about 0.3 million deaths per year. In spite of good advancements for diagnosis and treatment, cancer is still a big threat to our society (Kotnis et al, 2005). This is the second most common disease after cardiovascular disorders for maximum deaths in the world (Jemal et al, 2007). It accounts for about 23 and 7% deaths in USA and India, respectively. The world's population is expected to be 7.5 billion by 2020 and approximations predict that about 15.0 million new cancer cases will be diagnosed; with deaths of about 12.0 million cancer patients (Brayand et al, 2006). The prevalence of cancer in India is estimated to be around 2.5 million, with about 8, 00,000 new cases and 5, 50,000 deaths per annum (Nandakumar, 1990-96).

CLASSIFICATION BY STAGE

Staging is the process of finding out how much cancer is in a person's body and where it's located. Cancer staging is used to determine how extensive the cancer is, which, in turn, is a measure of prognosis (ACS, 2004). The determination of disease stage is generally based on the American Joint Committee on Cancer's TNM system, whereby *T* stands for tumour size and spread to local tissue, *N* represents lymph node involvement, and *M* describes the presence and extent of distal metastasis (Greene et al., 2002).The American Joint Committee on Cancer (AJCC) and the International Union for Cancer Control (UICC) maintain the *TNM classification system* as a tool for doctors to stage different types of cancer based on certain standards. It's updated every 6 to 8 years to include advances in our understanding of cancer.

In the TNM system, each cancer is assigned a letter or number to describe the tumor, node, and metastases.

- **T** stands for the original (primary) **tumor**. It gives information about aspects of the original (primary) tumor, such as its size, how deeply it has grown into the organ it started in, and whether it has grown into nearby tissues.
- **N** stands for **nodes**. It tells whether the cancer has spread to the nearby lymph nodes

- **M** stands for **metastasis**. It tells whether the cancer has spread to distant parts of the body

Not all cancers are staged using the TNM system. Some cancers grow and spread in a different way. For example, many cancers in or around the brain are not staged using the TNM system, since these cancers tend to spread to other parts of the brain and not to lymph nodes or other parts of the body. Staging systems other than the TNM system are often used for Hodgkin disease and other lymphomas, too, as well as for some childhood cancers.

RISK FACTORS FOR CANCER

1. **Behavioural Risk Factors**

Tobacco Use:

There is strongest evidence for the association between behaviour and cancer development is the consistently strong association between cigarette smoking and lung cancer. Tobacco use is also associated with the development of cancers of the head and neck, pancreas, kidney, bladder, and cervix (WHO, 1995). The consumption of tobacco is the leading cause of cancers in India the regular use of tobacco via smoking, chewing, snuffing etc. in some areas of the country, which is responsible for 65 to 85% cancer incidences in men and women, respectively. The various cancers produced by the use of tobacco are of oral cavity, pharynx, esophagus, larynx, lungs and urinary bladder. It has been observed that women in Bangalore are known to have the highest rates of cancers of esophagus in the world (around eight per 100,000). Contrarily, men in Bhopal have the highest rate of tongue cancer in the world (nine per 100,000) (Bobba et al. 2003). Smoking is the most notorious factor for the causation of lung cancer (Hammond et al, 1958). Approximately, 87 and 85% males and females have been found to have lung cancer due to tobacco smoking in the form of bidi (a thin South Asian cigarette type structure filled with tobacco flake and wrapped in a tendu leaf, tied with a string at one end) (Behera et al, 2004) and cigarette in India (Notani et al, 1974).

Diet and Exercise

Dietary patterns may modify risk of cancer by both contributing to the development of cancer and protecting against cancer (Winters, 1998). High intake of dietary fat may be associated with the occurrence of colorectal, breast, prostate, and possibly gynecologic and pancreatic cancer. Very high intake of nitrates and salt (especially from salting, smoking, and pickling for food preservation) has been associated with incidence of stomach cancer. High intake of fruits and vegetables, conversely, may be associated with decreased occurrence of colorectal, breast, stomach, esophageal, pharyngeal, and oral cancer (WHO, 1995).

Substance Abuse

Alcohol consumption has been considered as one of the major causes of colorectal cancer as per a recent monograph of WHO (Baan et al, 2007). Associations between alcohol use and development of cancer of the head, neck, and liver have been clearly demonstrated (Lundberg & Passik, 1998). Annually, about 9.4% new colorectal cancer cases are attributed to the consumption of alcohol, globally (Parkin et al, 2002). An increased risk of 10% was observed with consumption of more than two drinks per day, which suggests a causative role of alcohol consumption in colorectal cancer (Toriola et al, 2008).

Sexual behaviours:

The primary risk factor for development of cervical cancer is infection with the Human Papilloma Virus (HPV), which may be transmitted through sexual contact (Auchincloss &McCartney, 1998). Sexual behaviours associated with increased risk of HPV infection and cervical cancer include the following: young age at first intercourse, multiple sexual partners, and having a sexual partner who has had multiple partners (Auchincloss&McCartney,1998).

2. **Risk factors beyond personal control**

Radiation

In the developed and developing countries, the radiations are also notorious carcinogens. About 10% cancer occurrence is due to radiation effect, both ionizing and non-ionizing (Belpomme et al. 2007). The major sources of radiations are radioactive compounds, ultraviolet (UV) and pulsed electromagnetic fields. The main series of cancers induced by exposure to the adequate doses of the carcinogenic radiations include thyroid, skin, leukemia, lymphoma, lung and breast carcinomas. The most common source of ionizing radiation exposure is Radon, which is a radioactive element. High risk of breast cancer among girls at puberty is due to chest irradiation of X-rays (used for diagnostic and therapeutic purposes). The major risk factor for various types of skin cancers viz. basal cell carcinoma, squamous cell carcinoma and melanoma is the exposure to ultraviolet light, which is a non- ionizing radiation (Anand et al, 2000).

Miscellaneous pollutants

It is estimated that about 90% cancer is owing to the environmental contaminants (Anand et al, 2000). Various types of cancers are believed to be due to ill effects of the polluted environment. The risk of lung cancers is increased by a number of outdoor pollutants such as poly aromatic hydrocarbons. Long term exposure to PAHs (polyaromatic hydrocarbons) in air was found to increase the risk of deaths associated with lung cancer. Indoor environmental pollutants such as volatile organic compounds and pesticides increase the risk of leukemia and lymphoma, brain tumors, Wilm's tumors, Ewing's sarcoma and germ cell tumors. An increased risk of cancer has been observed in people using chlorinated water for drinking purposes for a long time. N-Nitroso compounds (mutagenic in nature) are formed from nitrates present in drinking water and increase the risk of lymphoma, leukemia, and colorectal cancer and bladder cancers (Belpomme et al, 2007).

3. **Inherent Risk Factors**

Inherent risks for cancer include such factors as **family history**, **ethnic background** and **age**. Only about 1% to 2% of cancers are inherited. Cancer is almost never inherited, but family history and a genetic predisposition play a major role in its development. A women who has a mother or sister with breast cancer has a two to three fold chance of developing that disease. About one third of all women with breast cancer have a family history of the disease. Ethnic background also is a factor ; compared with European Americans, African Americans have a significantly higher rate of mortality risk for cancer. These differences are not due to biology but to differences in socioeconomic status, knowledge about cancer and attitudes towards the disease. The strongest risk factor for cancer is advancing age. The older one becomes, greater that person`s risk for cancer.

CANCER TREATMENT

Cancer treatment most commonly involves one or more of the following: surgery, chemotherapy, radiation therapy, and hormonal therapy. Surgery and radiation therapy are typically localized treatments that aim to remove or destroy an identifiable tumor(s). Traditional chemotherapies (e.g., alkylating agents) operate systemically in the body by interfering with DNA activity among rapidly dividing cells, thereby inhibiting cell growth and promoting cell death (ACS, 2006h). Hormonal therapy in cancer refers to treatments that block the effects or levels of certain hormones (e.g., estrogen or androgen) that can contribute to the growth and spread of hormone-sensitive cancers (ACS, 2006f, 2006d). Common examples include Tamoxifen, an anti-estrogen agent for breast cancer, and Lupron, an anti-androgen drug, for prostate cancer.

Other cancer treatments include biological therapy (also known as immunotherapy or biological response modifiers), which functions to enhance the ability of the immune system to fight with cancer and infections (ACS, 2006i). More recently, a number of targeted therapies have received FDA approval for use in antineoplastic (i.e., anti- cancer) treatment. For example, monoclonal antibodies are laboratory-derived antibodies that are used to carry cancer-toxic substances to targeted cancer cells to which they bind (ACS, 2006g).

Bone marrow transplant (BMT) is a procedure whereby stem cells, found in the bone and from which blood cells derive, are purposely destroyed by chemotherapy and/or radiation and then replaced by a donor's or one's own purged stem cells (ACS, 2004a). This procedure is used for patients with hematologic (i.e., blood-born) malignancies such as leukemia or lymphoma, and for those who are no longer able to make the blood cells needed to carry oxygen, fight infection, and prevent bleeding due to previous chemotherapy- and/or radiation-induced stem cell damage.

As cancer is not a single disease but rather one with many different presentations, complications, and courses, treatment is often similarly complex. It often includes multiple modes of treatment (e.g., surgery, chemotherapy, and radiation therapy), may involve multiple lines of treatment (i.e., lack of response to one type of chemotherapy leads to treatment with another), and is often supplemented with toxicity control treatments to manage side effects (e.g., pain medications, anti-emetics). Further, the field of therapeutic oncology is replete with experimental treatments to identify more effective means to manage and potentially cure cancer.

Among patients with advanced disease, where treatment options are often limited, many opt to undergo experimental treatment by participating in a Phase I or II Clinical Trial (ACS, 2006c). These trials seek to identify the safety and effectiveness of a drug that has only been tested in lab and animal studies (Phase I) or minimally in humans (Phase II). Examples of clinical trials currently underway include anti-angiogenesis agents aimed at blocking the formation of new blood vessels around tumors (ACS, 2006a), and gene therapy approaches whereby genetic material (e.g., "suicide genes") is introduced into cancer cells to induce cell death, replace altered/missing genes with healthy ones, or make cancer cells more sensitive to chemotherapy or radiation therapy (ACS, 2005).

Side Effects of Treatment

Given the aggressive, multimodal, and systemic nature of many cancer protocols, patients can experience a range of short-term, long-term, and late effects of treatment (Souhami & Tobias, 2005).Chemotherapy is a cancer treatment that uses drugs to destroy cancer cells. It is also called "chemo". It is a non targeted systemic treatment, chemotherapy attack healthy as well as cancer cells often cause the most pervasive side effects through damage to rapidly dividing cells. These include cells in the digestive track, hair follicles, and skin. As a result, nausea, vomiting, diarrhea, appetite changes, hair loss, and dry skin are common short-term effects. A serious complication of chemotherapy is suppression of the immune system, which occurs as a result of toxicity to the rapidly dividing white and red blood cells. Such immune suppression can lead to infection, anemia, inability to receive further toxic treatment, and in rare cases, death. Cardiotoxicity and peripheral neuropathies (nerve damage) can also infrequently occur.

Fatigue (Cella, Lai, Chang, Peterman, & Slavin, 2002) and pain (Goudas, Bloch, Gialeli-Goudas, Lau, & Carr, 2005) are two commonly reported effects of cancer treatment. Weight changes, insomnia, and psychological distress are also relatively wide spread. Both men and women can experience sexual impairment as a result of treatments that directly or indirectly affect the sexual organs and/or hormones (Schover, 1998). For example, a majority of women who experience premature menopause from surgical removal of their ovaries or radiation/chemotherapy-induced ovarian failure, report sexual changes including vaginal dryness, loss of sexual desire, and pain during intercourse (i.e., dyspareunia; Muscari Lin, Aikin, & Good, 1999). Recently, a phenomenon described as *chemobrain* has been identified, suggesting that chemotherapy may produce cognitive impairment, including short-term memory loss and concentration difficulties (Jansen, Miaskowski, Dodd, Dowling, & Kramer, 2005).

The targeted therapies, including surgery and radiation therapy, are more localized in effect. For example, wound infections or skin burns may present as a typical outcome. It should be noted that, in rare cases, radiation as well as chemotherapy can contribute to secondary cancers, often years after treatment. To significantly reduce this risk, there are limitations on the lifetime dose of treatment a patient can receive.

Bone marro transplant patients who receive bone marrow from a donor can experience graft versus- host disease, a condition in which the donor’s tissue attacks that of the patient’s, resulting in mild to severe inflammation of the internal organs or skin.

While depression and menopausal symptoms are potential effects of hormonal treatments, flulike symptoms are more typical among those receiving biological therapy or monoclonal antibodies.

Though this list appears daunting, progress in supportive care has helped to mitigate many of these side effects and substantially improve and preserve quality of life (QOL). In particular, pharmacologic and behavioral interventions have resulted in greater management of nausea, pain, infection, anemia, insomnia, anxiety, and depression. As a reflection of these improvements, surveys of patients in 1983 and 2000 regarding the severity of chemotherapy side effects indicate that physical side effects— at the forefront of patients' concerns years ago—have been surpassed by psychosocial concerns related to family, work, and social life (Carelle et al., 2002). On the other hand, certain issues, particularly hair loss and fatigue, have remained a consistent concern over time.

Morbidity (Psychiatric) associated with treatment modalities

There are mainly three forms of treatment available (surgery, chemotherapy and radiotherapy) associated with psychiatric morbidity. Psychiatric morbidity associated with cancer therapies ranges from 18 to 40%.

- **Surgery**

Surgery often generates fear of the procedure and over the lost of body parts. Mastectomy is the surgical treatment which has been studied extensively. Anxiety depression and sexual problems were found in a 'substantial minority' of patients who had undergone this treatment (Morris and Greer 1997). Maguire et al (1980) done study on women with breast cancer followed up one year after surgery. The anxiety symptoms noted were persistent tension, inability to relax, palpitations and panic attacks. Roughly one third of the patients had sexual problems. They had either stopped intercourse or ceased to enjoy it. Husbands mastectomy patients also reported sexuality and intimacy as the severely areas following the surgery (Wellisch et.,al 1978).

Other problems in this area were disturbances in the body image and a feeling of personal inadequacy (Morris, 1979).

Mastectomy, permanent colostomy, maxillofacial surgery and hysterectomy have reported to produce immense psychological impact on patients, like depressive illness, psychosexual problems and social problems (isolation, loneliness, and decreased social visits), drinking, and occasionally suicide.

- **Radiotherapy**

Radiation therapy (RT) is associated with highly unpleasant side effects. The side effects include nausea, vomiting and increasing fatigue. A prospective study of patients receiving radiotherapy had shown significant psychiatric problems in the first three months (Schmale, Morrow, Davis et al. 1982). In a study on levels of anxiety and depression in patients receiving radiation treatment in a cancer hospital in india, found anxiety and depressive disorders were detected frequently, both prior to treatment and later during follow up. Frequency of anxiety increased significantly after initiating radiotherapy, but later reduced during follow- up assessment after a few months (Chaturvedi, Chandra, Channabasavanna et al. 1996).

- **Chemotherapy**

Chemotherapy can produce nausea and vomiting as the immediate effects. Though various chemotherapeutic agents vary in this emetogenic potency, almost all have side- effects. After an initial episode of nausea and vomiting, 15 to 65% of patients develop anticipatory nausea and vomiting (Lederberg, Holland,1989). Sometimes it become so severe that they cannot continue treatment. Patient may also develop a conditioned response when exposed to sights and smells reminiscent of chemotherapy experience. Lung cancer patients receiving palliative chemotherapy were found to have fewer depression and communication problems than those receiving no treatment at all (Hughes 1985). Maguire et al (1980) noted that chemotherapy caused fatigue, nausea and irritability, along with adverse effect on family and sexual life.

Psychological reactions and adjustment to diagnosis and treatment

Diagnosis of cancer evokes far greater emotional reaction than diagnosis of any other disease, regardless of mortality rate or treatment modality. Shock and disbelief are the commonest intial responses, followed by anger, depression and feeling of loss and grief. The normal reaction can vary from person to person. The intensity and duration of emotional distress and the degree to which interferes with patients life seems to determine whether the emotional responses in normal or abnormal.

When doctor is enclosing information on cancer patient because of existential threat has to use series of adaptive defences to withhold psychological stability. The very first encounter with a diagnosis of malignant disease arouses more intense emotional reactions than with any other disease. This leads to creating defence mechanisms with which doctors should be familiar and should acknowledge in therapeutic process (Tope et al. 1993). Usual accompanying psychological symptoms are **fear of body image changes, disabilities, addictions and death.** Patients' first reaction is fear of death or fear of separation from others and himself, and psychiatric disorders, communication with family etc. That can lead to developing panic attacks or other disorders. Person confronted with death goes through many different phases and states such as **phase of denial, phase of anger, phase of bargaining, phase of depression and finally acceptance**. The usual defence mechanisms among oncology patients **are regression, denial, projection and suppression**. Success of defences does not only depend on ego–strengths forming during development of patients' personality but also on actual object relationships like family relationships and relations with physician (Gregurek 2006). Good communication skills are extremely important for suitable care of oncology patients (Hagerty et al. 2005). Different ways of communicating patients' diagnosis can produce different emotional reactions.

Psychological consequences of cancer diagnostics and treatment can be very significant. **On the physical level**, cancer can cause **great changes in body image** and in the way patients perceive their body. Oncology patients have various psychological problems such as emotional liability, changes in future perspectives, feelings of solitude, abandonment, marginalisation, stigmatisation, interpersonal problems, an all these problems can occur during different disease stages and during treatment with variety of psychological consequences (Braš 2008). The role of liaison psychiatrist on oncology departments consists of two components: helping patient from the diagnosis till the end of treatment and collaborating with medical team (Bloch & Kissane 2000). Assignments of medical stuff is to identify negative emotions and overcoming it and openly showing it among colegues and consequently reducing feelings of guilt and discussion on uniting all the actions that insure patients' better psychological and somatic state (Fawzy et al. 2003).

PSYCHOSOCIAL FACTORS AND CANCER

Psychosocial factors can affect cancer in a number of ways. Psychosocial variables also indirectly affect the initiation of cancer through consumption of fatty diet or exposure to stress. Psychosocial factors are also involved in the progression of cancer after it is initiated. For example, stress exposure and certain ways of coping may affect progression of cancer. There is considerable evidence suggesting that cancer patients suffer from substantial and long-term psychological distress associated with different forms of cancer and its medical treatment [Spijker and Trijsburg (1997).The psychosocial management of adjustment problems experienced by people with cancer seems to be an obvious requirement for a more effective treatment of the disease. Nevertheless, there is an ongoing debate about whether and to what extent psycho oncological care can be effective in patients suffering from cancer.

QUALITY OF LIFE:

The term quality of life (QOL) is used to evaluate the general well-being of individuals and societies. According to the World Health Organization (WHO), quality of life is defined as individual perception of life, values, objectives, standards, and interests in the framework of culture. Quality of life (QOL) typically involves the assessment of several dimensions: physical well-being, emotional well-being, social well-being, and functional well-being (Andersen, 1993). Cancer can produce many different symptoms, some subtle and some not at all subtle. Some symptoms of cancer affecting quality of life in patients would be cancer type and stage, as some types of cancer do not present any symptoms until they are in advanced stages, time since diagnosis, patient acceptance, and intensity of the disease and the level of psychological distress experienced by caregivers **(**Tavoli, 2008). A number of illness-related factors exist that can affect quality of life. The amount of distress experienced by an individual with cancer has been related to quality of life. Depression and anxiety symptoms are common in these patients and impair the patient's quality of life, comfort level and treatment compliance, which ultimately can affect the patient's survival.

Some psychological interventions have been designed to improve QOL in cancer populations. Carlson (2003) used a mindfulness-based intervention to improve QOL in breast and prostate cancer patients. Following the intervention, there was a significant increase in global QOL compared with pre-intervention measurements. This study also found a reduction in depression, anxiety, emotional• irritability, and cognitive disorganization. Speca, Carlson, Goodey, and Angen (2000) examined the effects of a mindfulness based intervention on stress and QOL in a heterogeneous cancer sample. They found that the intervention reduced mood disturbances and stress among both male and female cancer patients.

Denial

Denial is a defence mechanism by which people avoid the implications of an illness. It is a common reaction to chronic illness that has been observed among heart patients (Krantz & Deckel, 1983), stroke patients (Diller, 1976), and cancer patients (Meyerowitz, 1983).

DENIAL IN CANCER PATIENTS:

Denying that it is really a cancer, and hope, despite all odds, that it will all be made well, or it will be discovered that it was a misdiagnosis - this sort of delusion has been observed to last for several years in some patients. Some others just accept "the death sentence" and die. Not all who die have given up. Not every dying person comes to terms with what is happening in a way that is clear to the outside observer.

HOW TO MANAGE DENIAL:

1. It is important for all clinicians to understand their contribution to denial to ensure that the patient receives factual, clear information regarding the disease and its implication. We should make sure that the patient's denial is not due to lack of information, lack of understanding or lack of agreement with medical recommendations (Cousins, 1982; Shelp & Perl, 1985).

2. We have to assess denial carefully, including how and when it is used by the patient, the benefits and risks, the patient's usual coping style, the function it serves, and its significance for the patient and for others. Denial may be expressed in various ways, such as downward comparison, minimization of illness, or lack of emotional response. If these responses are present, it signals that the patient is frightened. Instead of confronting denial or challenging assumptions, it is best to provide support for patients to discuss issues concerned as they are able and at their own pace.

3. Distinguish between a fact being denied (e.g. diagnosis of cancer) and implications of the fact denied (e.g. cancer will not return). The former may interfere with necessary treatment; the latter may maintain morale in a difficult situation.

4. It should be addressed directly when denial compromises patients' safety, such as not reporting symptoms, or not complying to treatment. When denial inhibits actions of importance, for example, making decisions on treatment, planning realistically, and communicating honestly with loved ones, it is also important to intervene.

5. We have to adopt a non-confrontational approach, with respect for the essentially protective nature of it. We have to explore and validate before attempting to change anything. Instead of judging the denial as "good" or "bad", it is more helpful to ask "Is this reaction helping the patient cope with his challenges?" We must understand that it serves a healthy purpose in many instances, and we need to demonstrate that we are prepared to allow patients to have their own responses to their own life situations.

6. If denial is causing significant problems, direct confrontation may only increase the use of denial. For example, we should not tell a patient directly in this way: "It's the cancer… it's killing you… you should make use of your limited time to complete your family business and let your loved one say good-bye to you." In general, respect and support for the patient, including his defences, yields far more therapeutic benefits than trying to change or override such defences. Even when the defences are extremely maladaptive, encouraging their positive aspects and building on them as strengths is more likely to be beneficial than direct confrontation and attack; such as encouraging a patient to use pain medications to increase independence and control over symptoms. In this way, the need for control is acknowledged, but the denial of the meaning of pain is not confronted directly. We can gently explore his or her understanding of the cause of his or her pain and weakness: "What is your understanding of the cause of this pain? What do you think is making you so weak? How can I best help you?"

A good policy for breaking bad news is to move slowly and let the patient determine the pace. At times, a more urgent discussion is required because of rapid clinical deterioration. When a more direct approach is indicated by clinical urgency, we must retain as much respect as possible for the patient's personal choices about coping, while expressing our concerns in as balanced and clear a way as possible: " I know you try to maintain as optimistic a view as possible about the cancer. I want to support your hopes about this. At the same time, I want to be sure that we have covered the decisions that need to be made if that is not how things turn out. We need to discuss…"

7. Last but not least, emphasise to patients that they will not be abandoned. They will be supported and cared for, whether the medical news is good or bad.

GRIEF REACTION TO BODILY CHANGES:

Body image reflects a direct personal perception and self-appraisal of one's physical appearance, whereby negative thoughts and feelings related to one's body indicate a disturbance of body image and lead to dissatisfaction with one's self (stokes R, Frederick-Recascino C, 2003). Body image is one of the most profound psychological consequences from cancer treatments affecting patients with a variety of disease sites. The scars and physical disfigurement serve as reminders of the painful experience of cancer and its treatment. The stress and depression that may be a result of body image concerns can further impact other areas of the Patient`s and family`s life, such as sexual intimacy, psychological disorders, and self-esteem. In women who have had breast surgery, concerns range from distress over scars to feelings of decreased sexual attractiveness and restrictions of use of certain items of clothing. A high personal investment in one's body image can act as a source of self-worth (Sarwer D, Cash T, 2008). Because women generally have a focus on body image-related evaluation and investment (McKinley N, 1998), a diagnosis of breast cancer is likely to further exacerbate this propensity (Lazarus, 1991). Indeed, the loss of a breast is inherently linked to a woman's identity, sexuality and sense of self (Manderson L, Stirling L, 2007) with approximately one-third of breast cancer survivors expressing distress that is directly related to disturbed body image after successful cancer treatment (Scott J, Halford K, Ward B.2004). Furthermore, long-term patterns of weight gain after cancer treatment are common (Makari-Judson G, Judson C, Mertens W.,2007, Helms R, O'Hea E, Corso M..2008), creating additional bodily challenges.

Difficulties in this area may manifest themselves as:

1. Significant emotional difficulty in accepting physical scarring, cosmetic prostheses or cosmetic results of reconstructive surgery aimed at restoring physical appearance;
2. Loss of intimacy and sexuality;
3. Loss of confidence and self-esteem, withdrawal and/or social isolation;
4. Preoccupation with perceived physical defects, feelings of self-loathing, shame and/or inadequacy, anger, loss of confidence, anxiety and depression;
5. Neglect of personal care; loss of appetite with associated weight loss *or* overeating and weight gain.

Interventions generally involve an assessment of mood, coping and adjustment behaviors, the grieving process and the personal meaning of the loss or change in physical appearance to the patient (and possibly the spouse and family). This may include an exploration of their personality, self-perceptions and self-worth, significant life roles and goals, and previous significant losses or changes. Interventions are aimed at facilitating the adjustment and grieving process. Interdisciplinary working with disciplines such as CNSs, Physiotherapists, Occupational Therapists, Speech and Language Therapists, Sexual Therapists or Palliative Care teams may be required.

Cancer Recurrence

Cancer recurrence is defined as the return of cancer after treatment and after a period of time during which the cancer cannot be detected. (The length of time is not clearly defined.) The same cancer may come back in the same place it first started or somewhere else in the body. For example, prostate cancer may return in the area of the prostate gland (even if the gland was removed), or it may come back in the bones. In either case it’s a prostate cancer recurrence.

There are different types of cancer recurrence:

• Local recurrence means that the cancer has come back in the same place it first started.

• Regional recurrence means that the cancer has come back in the lymph nodes near the place it started.

• Distant recurrence means the cancer has come back in another part of the body, some distance from where it started (often the lungs, liver, bone marrow, or brain).

Patient reaction to recurrence

Over 1.2 million individuals are diagnosed with cancer recurrence each year and over half will progress rapidly and die of their disease (Jemal et al., 2005). Cancer recurrence and advanced cancer produce multiple psychological responses including depressive symptoms and difficulties with disability (Mahon & Casperson, 1997). At least at the time of recurrence diagnosis, stress is the predominant emotional response, and equivalent to that reported at the time of initial diagnosis (Andersen et al., 2005).

➢ **Resilience and Cancer**

Resilience is an "inner strength" that helps you bounces back after stressful situations. When you are resilient, you may recover more quickly from setbacks or difficult changes, including illness. Developing resilience begins with simple actions or thoughts that you practice, such as planning for what you'll do next and learning to accept change. Being resilient doesn't mean that you find it easy to deal with difficult or stressful situations or that you won't feel angry, sad, or worried during tough times. But it does mean that you won't feel so overwhelmed. You'll be less likely to give up and more likely to cope with stressful situations in healthy ways. Despite all the psychosocial problems associated with cancer, many people clearly have cancer experiences that they weather quite well from a psychological standpoint, adjusting successfully to major changes in their lives. The positive adaptation to the cancer experience can be enhanced by feelings of control or self-efficacy in response to the cancer experience. People who are able to experience a sense of personal control over their cancer, its treatments or their daily activities cope more successfully with cancer. They do have control over emotional reactions and physical symptoms and it lead them to have good wellbeing.

CLINICAL ASSESSMENT OF CANCER PATIENTS

Although the most direct and thorough means of identifying distress levels and psychosocial

needs of cancer patients is individual clinical assessment, it is impractical to conduct thorough interview-based assessments of all patients at high-volume oncology clinics, particularly given that only one-third of the patients would be expected to require psychosocial services (Zabora, 1998). **Zabora (1998) identified five self-report instruments** commonly used to screen for distress among cancer patient populations: the Brief Symptom Inventory (BSI; Derogatis, 1993), the Profile of Mood States (POMS; McNair, Lorr & Droppleman, 1971), the General Hospital Questionnaire (GHQ; Goldberg, 1978), the Hospital Anxiety and Depression Scale (HADS; Zigmond&Snaith,

1983), and the Medical Outcomes Study Short Form (SF-36;Ware et al., 1993). With the exception of the SF-36 (which, as a general quality of life measure, incorporates assessment of both psychological and physical symptoms) these instruments focus on psychological symptoms such as global distress, anxiety, and depression (Gotay & Stern, 1995).

Screening instrument assessing both psychological and physiological symptoms of cancer patients is the ***Psycho-Oncology Screening Tool***, or POST, which was developed by the Psycho-Oncology Program of the University of Florida Department of Clinical and Health Psychology. ***The instrument identifies the distress levels, fatigue and pain levels, and perceived needs of patients, incorporating elements of both distress and needs surveys.*** The one-page instrument (which can be completed in five minutes) is composed of four sections: distress and discomfort symptom level visual analogue scales, a depressive symptoms checklist, a social concerns checklist, and interest in psychosocial services questions. Among the initial 569 POST respondents, over one-third indicated interest in psychosocial services. Compared to the uninterested respondents, the interested respondents displayed significantly higher levels of anxiety, confusion, depression, anger, total distress, depressive symptoms, and social concerns (Durning et al., 2002). These results suggest that oncology patients with high levels of distress may tend to be amenable to receiving clinical services.

Once patients choose to pursue psychological services (either because they have been referred for services by health care providers, because their responses to a screening instrument prompted a referral, or because they independently sought services) more individualized and thorough assessments are warranted. As with any psychological assessment, the clinical interview may provide the most relevant and comprehensive clinical information. Interview topics relevant to cancer patient populations is as follows.

Areas to assess when interviewing cancer patients

1. Medical status (stage, treatment, prognosis)

2. History and course of current and past medical conditions

3. Current medications

4 .Cancer risk factors (behavioral, genetic, environmental)

5. Psychosocial history:

Education/employment

Current family situation

Family of origin

6. Alcohol and substance abuse use (past and present)

7. Knowledge and beliefs surrounding cancer diagnosis

8. Current stressors:

Identify cancer and non-cancer related stressors

Assess current coping strategies

Social support (family, friends, membership in religious and other organizations)

9. Psychological functioning/psychiatric history:

Assess current mood (objective and subjective ratings)

Past psychological history (therapy or counselling, psychiatric diagnosis and/or medications)

Family history of psychiatric illness

Site-specific concerns (e.g., body image and arm edema for breast cancer patients) may be addressed through instruments designed specifically for those purposes. For example, the *Functional Assessment of Cancer Therapy system* (FACT; Cella et al., 1993) includes modules assessing concerns associated with particular cancer sites; modules are also available for assessing side-effects of treatment (such as fatigue and nausea) and additional issues such as spirituality.

Common questionnaires used in oncology populations

1. Overall distress ---------

- ✓ Profile of Mood States (POMS) McNair et al. (1971)
- ✓ Brief Symptom Inventory (BSI) Derogatis (1993)
- ✓ Symptom Checklist 90-R (SCL-90) Derogatis (1983b)
- ✓ General Hospital Questionnaire Goldberg (1978) (GHQ)
- ✓ Hospital Anxiety and Depression Scale (HADS) Zigmond & Snaith (1983)

2. Adjustment to illness

- ✓ Psychosocial Adjustment to Illness Scale (PAIS)Derogatis (1983)

3. Anxiety

- ✓ State-Trait Anxiety Inventory (STAI) Spielberger et al. (1983)
- ✓ Impact of Events Scale (IES) Horowitz, Wilner & Alvarez (1979)

4. Depression

- ✓ Beck Depression Inventory (BDI) Beck et al. (1961)
- ✓ Center for Epidemiological Studies Depression (CESD) Radloff (1977)
- ✓ Quality of life Medical Outcomes Studies SF-36 Ware et al. (1993)
- ✓ Functional Assessment of Cancer Therapy (FACT) Cella et al. (1993)
- ✓ Fatigue Multidimensional Fatigue Symptom Inventory (MFSI) Stein et al. (1998)
- ✓ Functional Assessment of Cancer Therapy Fatigue (FACT-F) Cella (1997)

5. Pain

- ✓ McGill Pain Questionnaire Graham et al. (1980)
- ✓ Brief Pain Inventory (BPI) Daut, Cleeland & Flanery (1983)
- ✓ Memorial Pain Assessment Card Fishman et al. (1987) (MPAC)

PSYCHOSOCIAL INTERVENTIONS FOR CANCER PATIENTS

In 1998, approximately 60 000 new cancer patients were diagnosed in the Netherlands (Visser *et al*, 2002). In that same year, 37 000 patients died of this disease (Visser *et al*, 2002). About half of all patients cannot be cured and receive treatment with a palliative intent.

Clearly, the emotional impact of a cancer diagnosis is devastating and characterised by shock, disbelief, anger, anxiety, depression and difficulty in performing activities of daily living. A similar response occurs at each transitional point of the disease that is, beginning treatment, recurrence, treatment failure and disease progression (Pasacreta and Pickett, 1998). Although it is obvious that many patients with cancer experience emotional distress, van't Spijker *et al* (1997) found that percentages for depression varied from 0 to 46%, for anxiety from 0.9 to 49% and for general psychological distress from 5 to 50%. These data do not refer to patients in a specific stage of cancer, which may account for the wide variation in prevalence rates. Less variation in prevalence rates of emotional distress is found in patients with advanced disease. In this population, emotional distress and depression, in particular, appear to be a common problem (Zabora *et al*, 1997; Massie and Popkin, 1998). Hotopf *et al* (2002) estimated that the prevalence of depression ranged from 15% for major depression to at least 30% for all depressive disorders (including minor depression).

Moreover, several studies (Slevin *et al*, 1996; Sanson-Fisher *et al*, 2000; Soothill *et al*, 2001) report that patients in an advanced stage of the disease have high levels of psychosocial needs that are not properly met. Professional caregivers appear to be selective in their receptiveness of patients' needs, focus on physical problems and to a much lesser extent on emotional problems and psychosocial needs.

In fact, Redd (1995) suggests that an important factor responsible in part for the birth of psychosocial oncology as a field was the publishing of certain studies that underscored the successful use of behavioral procedures to control the anticipatory side effects of cancer chemotherapy, such as nausea and vomiting (e.g., Morrow & Morrell, 1982).

A wide array of therapeutic interventions has been applied to address the emotional and physical needs of patients with cancer. These can be delivered in individual or group modalities. In practice, there is some degree of overlap or combinations within intervention categories. The major intervention approaches are herein described.

Goal of intervention for individual:

Decreasing Distress and Improving Positive Psychosocial Adjustment

The diagnosis of cancer creates a great deal of disruption in the lives of cancer patients and their families. Additionally, problems that may have been present but manageable prior to the diagnosis of cancer may suddenly appear overwhelming. Therefore, stress-management interventions focusing on identifying and decreasing sources of stress and developing more effective coping techniques can be extremely beneficial to almost all newly diagnosed cancer patients. A number of studies have found that relaxation and stress-management interventions can impact distress and positive gains in cancer patients (Antoni et al., 2001; Fawzy et al., 1990a; Larsson & Starrin, 1992). It is important that psychologists or other medical staff working with cancer patients pay close attention to levels of distress and work closely with psychiatrists (preferably who specialize in oncology) who can prescribe and monitor psychotropic medications. Early identification and management of depression among cancer patients is important so that patients may receive optimal treatment for both psychological and physical symptoms (Berney et al., 2000; Lloyd-Williams, 2000).

Controlling Treatment-related Responses and Behaviors

Interventions that instruct patients in self-help techniques to help to control treatment side effects can have an enormous impact on quality of life as well as adherence to treatment (Redd, Montgomery & DuHamel, 2001). Psychological interventions targeting specific treatment-related responses or behaviors have been proven effective in a number of studies (e.g., Andersen&Tewfik,

1985; Burish, Snyder&Jenkins, 1991; Burish et al., 1987; Decker, Cline&Gallagher, 1992;Vasterling et al., 1993). Because chemotherapy may induce various gastrointestinal symptoms, patients may develop anticipatory nausea; similarly, because radiotherapy procedures may induce fear reactions (including claustrophobia related to the treatment machines), patients may develop anticipatory anxiety. If such symptoms increase in severity, a patient may become non-compliant with treatment. Therefore, the interventions focused on direct treatment-related effects may be effective in improving not only the presenting symptoms but also compliance with future treatment.

Controlling Physical Effects of Disease and Treatment

Very few studies have specifically examined the effectiveness of non-pharmaceutical interventions for the treatment of cancer-related fatigue (CRF). Suggested psychological interventions for CRF include education about fatigue and sleep hygiene (Fortin&Kirouac, 1976), stress-reduction/relaxation and psychotherapy (Cimprich, 1993), and exercise. However, there are no empirically tested guidelines that suggest the appropriate type or amount of exercise for cancer patients with CRF (Dimeo, Rumberger & Keul, 1998; Schwartz, 1998; Schwartz et al., 2001; Segal et al., 2001). Pharmacological treatment of cancer pain may involve the use of non-opioid analgesics for mild to moderate pain and opioid analgesics for moderate to severe pain, with adjuvant analgesics (e.g., antidepressants, neuroleptics, psychostimulants, anticonvulsants, corticosteroids, and oral anesthetics) accompanying the opioid or non-opioid analgesics when indicated (Breitbart & Payne, 1998). Non-pharmacological interventions recommended for treating cancer-related pain and associated psychological symptoms include individual and group interventions composed of any of the following techniques: psychotherapy, cognitive-behavior therapy, relaxation exercises, imagery/distraction exercises, hypnosis, and/or biofeedback (Breitbart & Payne, 1998).

Changing Health-related Behaviors

As described above, primary prevention programs are aimed at modifying health-related behaviours (such as smoking, diet, or exercise) that are associated with increased vulnerability to the development of cancer. Similar lifestyle modification programs can be tailored to cancer patients. Programs may assist patients in developing new health behaviors that will decrease the probability of recurrence or the development of new cancers as well as limiting the development of other comorbid conditions, such as coronary heart disease, diabetes, or pulmonary disease (Demark-Wahnefried et al., 2000; Pinto, Eakin & Maruyama, 2000). Although there are fewstudies that include health behavior interventions for cancer patients, the results of dietary studies (Chlebowski et al., 1993; Kristal et al., 1997; Nordevang et al., 1992; Pierce et al., 1997) and exercise (Dimeo, Rumberger & Keul, 1998; Stoll, 1996) are promising. Unfortunately, research examining smoking cessation programs for cancer patients has shown relatively low long-term quit rates (Andersen, 2002).

Enhancing Health Outcome

Because psychological distress is associated with neuroendocrine and immune functioning, interventions have been designed to improve functioning of these systems. Cancer patients are vulnerable to profound changes in distress, fatigue, pain and general quality of life that may modulate neuroendocrine functions associated with disease progression (Andersen, Kiecolt-Glaser & Glaser, 1994; Andersen et al., 1998; Herbert & Cohen, 1993a, 1993b; Ironson, Antoni & Lutgendorf, 1995). The chronic challenge of dealing with cancer may play a role in neuroendocrine dysregulation that can lead to changes in immunity that may, in turn, contribute to disease progression (Turner-Cobb et al., 2000). Psychosocial interventions may impact stress-related variables and/or facilitate positive adaptations that may alter neuroendocrine and immune functions. It has been hypothesized that improvements in neuroendocrine and immune system functioning resulting from psychosocial interventions may lead to improved survival for the patients participating in the interventions. Although there are a few studies that have noted changes in immune or neuroendocrine measures related to psychological interventions (Cruess et al., 2000; Fawzy et al., 1990b; Gruber et al., 1993), evidence of a direct impact of psychosocial interventions on health outcomes is less compelling. To date, the handful of studies reporting on the impact of individual and group interventions on survival has lead to both positive (Fawzy et al., 1993; Kuchler et al., 1999; Ratcliffe, Dawson&Walker, 1995; Richardson et al., 1990; Spiegel et al., 1989) and negative findings (Cunningham et al., 1998; Edelman et al., 1999; Goodwin et al., 2001). Nevertheless, almost all of the studies examining the impact of a group intervention on patient survival have identified significant psychological and quality of life benefits (Spiegel, 2001).

Educational Interventions

The goal of educational interventions is to reduce cancer patients` distress and improve their sense of control that may be undermined by lack of knowledge and feelings of uncertainty.

For example, Messerli, Garamendi, and Romano (1980) argued that a patient`s fear, anxiety, and distress would decrease as a function of increased medical knowledge and information accessibility.

With these types of interventions, patient education has involved a variety of venues, including written materials, films, audiotapes, videotapes, and lectures. The protocols studied included topics covering technical aspects of the disease and its treatment, potential side effects, navigating the medical system, and the physician-patient relationship. An early study investigating the benefits of an educational approach was conducted by Jacobs, Ross, Walker, and Stockdale (1983).

Patients with Hodgkin`s disease participating in the education sample were mailed a 27-page booklet that included disease-related information. Three months later, compared to a no-education control, these individuals were found to show a decrease in depressive and anxiety symptoms, as well as an increase in their knowledge about Hodgkin`s disease.

More recently, Hack et al. (1999) conducted a multicenter study whereby patients were provided the choice to receive an audiotape of the initial consultation session with their oncologist. Such an approach was hypothesized to impact positively on the physician-patient relationship, as well as to provide the cancer patient with the opportunity to review the information discussed during the consultation. Although a trend was observed regarding a decrease in anxiety for patients who chose to receive the audiotape, this change was not statistically significant. However, at a six-week follow-up assessment, patients receiving the tape recalled significantly more information and were found to report a higher degree of satisfaction with the physician-patient relationship.

Management of psychosocial problem in cancer

The diagnosis of cancer causes stress on any individual, who relates both to symptoms of the disease and to the psychological meaning attached to cancer. The patient’s ability to manage these stresses depend on prior level of emotional adjustment, the threat the cancer poses to attainment of age appropriate goals (e.g. cancer, starting a family), the presence of emotially supportive person (confiding ties) in the environment and variables determined by the desease itself. When emotional distress associated with having cancer exceeds what is ‘expected’ or ‘normal’, a psychiatric disorder may have developed and should be assessed.

❖ **Psychopharmacological Management**

- Tricyclic antidepressants (TCA)
- Lithium carbonate
- Monoamine oxidase inhibitors
- Psychostimulants

- Benzodiazepines
- Electroconvulsive therapy

❖ **Psychosocial methods of treatment**

While the importance of psychosocial factors in cancer is increasingly recognized, psychological interventions are still often seen as peripheral to the major task of the oncology team. Psychological intervention, with the emphasis on fostering a positive attitude, helping the patients to comply and cope with the treatment, and reducing emotional distress, can it fact complement and traditional medical treatment. Moreover, the methods and approach the counsellor/psychotherapist choose, must be very specific & based on the need of the patient, not a general psychotherapeutic approach. The focus should be on quick impact of the approach.

The types of psychological interventions used in cancer patients are many and diverse, like behaviour therapy, cognitive therapy and supportive therapy. The aims of psychological interventions are:

- To reduce anxiety, depression and other emotional distresses.
- To improve the mental adjustment to cancer by imparting a positive attitude.
- To promote a sense of personal control over cancer and its treatment.
- To improve patients communications with the spouse or other family members.
- To encourage the open expression of emotions.

During psychological intervention for cancer related problems, is should be acknowledge that a cancer patient, after all, is a psychologically normal individual under severe stress. Hence, the therapy should be brief and directed towards current problems. There should be also be an active attempt to identify and make use of the personal strengths.

Behaviour therapy

Behavioural techniques are used mainly in early stages of the cancer. Emotional distresses, like anxiety and depression and other specific problems, like chemotherapy related anticipatory nausea and cancer pain are effectively treated using these these techniques. The main behavioural techniques used are relaxation exercise, systematic desensitization, positive imagery, hypnosis and biofeedback (Chaturvedi, Kumar, Ramachandra 1996).

- ***Relaxation exercises***

This is useful for anxiety and related problems like pain, nausea and sleep disturbances. It is in fact a simple, effective and rapid method of controlling anxiety. Relaxation training is usually done as deep muscle relaxation (DMR) or cue controlled relaxation (CCR). The premise of DMR is that muscle tension is in some way related to anxiety and that reduction in muscular. CCR can be practiced after mastering DMR. This uses calming instructions, with or without an imagery, when the patient is faced with an anxiety provoking situation.

Breathing exercise and meditation are other relaxation techniques used. In breathing exercise, the patient focuses his attention on breathing. This is used for generalization of relaxation through self control to settings triggering the aversive reactions.

- ***Systematic desensitization***

This technique of graded exposure is found to be effective for chemotherapy- related conditioned responses like nausea, vomiting and anxiety. This controls the nausea and vomiting by reducing the autonomic arousal and producing a physical state imcompatibale with the characteristic of nausea and vomiting.

- ***Positive imagery***

In this technique an emotional state counteractive to anxiety is induced by relaxation training or hypnosis, and then anxiety evoking stimulus are introduced. Reciprocal inhibition of anxiety occurs due to creation of a stimulus control.

Cancer patients generally having a feeling of loss of control over the treatment, which can cause emotional distress and reduce drug compliance. Imagery technique may be useful here. In this technique, the patient in a relaxed state conjures up the image of cancer cells being destroyed by treatment and body defences. Treatment is visualized as a friendly and harmful image, which skills week and ineffectual cancer cells. For each patient Imagery tasks have to the modified according to the problems. This shift the patient’s role from a passive one of being the recipient of the treatment o an active role of taking part in his own treatment.

- ***Behavioural task and activity scheduling***

The loss of control resulting from cancer and its treatment, can generalize to other areas of patients life and result in hopelessness, helplessness and social withdrawal. Appropriately reinforced activity schedule can foster a fighting spirit and cultivate positive attitude. In the initial stages when the patient has severe emotional distress related to a particular time or situation, distraction may be a useful technique. Patient is asked to indulge in pleasurable activities for distraction. The activity scheduling is formed after considering the individual characteristics. The reinforcement should also be appropriate for the individual.

- ***Biofeedback***

Biofeedback makes use of sensitive electronic equipment to monitor physiologic activity such as muscle activity, skin resistance or peripheral temperature in the form of visual auditory signals. These signals are then used for management of anxiety and other related problems.

- ***Distraction***

Distraction is a useful mood limiting technique that most patients can use. It is helpful to talk with patients about their experiences. Methods that can be used here include asking patients to notice any variability in mood and whether this is linked to distracting activ- ities (i.e. behaviour). The patient can be introduced to the idea of 'thought stopping', that is techniques and actions that limit difficult and uncomfortable thoughts. While this focuses on cognitive processes the techniques are often behavioural as they involve the patient in changing something about what they do and/or their routines in order to have the effect of stopping the uncomfortable thoughts.

- ***Modelling***

This method involves the use of *in vivo* or videotape demonstrations of successful coping during invasive diagnostic/treatment procedures to teach behavioural coping skills. It is most commonly used with children. One example of this method is the use of a film in which a child scheduled for repeated bone marrow aspirations describes thoughts and feelings that he/she often experiences and then demonstrates behavioural coping skills to manage his/her fear and distress.

Cognitive Therapy

According to the cognitive model of adjustment to cancer, it is the perceptions, interpretations and evaluations that the individual makes about the cancer, which determines his emotional and behavioural responses. As in depression, existence of the cognitive traid in adjustment to cancer has also been postulated (Moorey, Greer 1984).

The components of the cognitive traid are:

- View of the diagnosis,
- Perceived control and
- View of the prognosis

The traid influences the personal reactions to the stress. After an initial period of turmoil, most patients develop a relatively stable adjustment style. The cognitive schema of an individual selects information from environment which is congruent with his coping strategy and filters out information which are at variance with it. Thus a patients with fighting spirit will attend to more optimistic facts while a hopeless patients will ignore this and search for negative aspects.

- ✓ ***Thought record diary:***

A significant part of CBT is the self-monitoring of thoughts using a diary of feelings and behaviour and carrying out behavioural experiments to see what actually works between therapy sessions. It is quite usual for the therapist to provide some patient education which includes talking through how thoughts affect mood. Thoughts are recorded and emotions described in a diary. The diary can be used to link coping techniques to changes in thoughts and mood and can be reviewed with patients in the sessions.

- ✓ ***Cognitive restructuring.***

Cognitive restructuring interventions are used to alter beliefs and attitudes that may contribute to the patient's distress. This technique is similar to that used by many modern psychotherapists who often encourage the patient to reframe stressful life events as less threatening and under his/her control. In this way, the patient is encouraged to “restructure” his/her thoughts and beliefs. This method involves the patient and clinician reviewing thoughts, feelings, and beliefs about medical treatment/procedures in order to identify those that elicit fear and distress. The patient is then encouraged to consider other ways of viewing the fearful event(s) that might help reduce feelings of distress and anxiety. An interesting example of the use of cognitive restructuring is reported by Chen et al.,who studied children undergoing lumbar puncture in the treatment of leukemia. To reduce the children's distress, Chen et al. had them recall and then more realistically appraise their responses to their most recent lumbar puncture. The aim was to enhance the children's confidence and beliefs regarding their ability to cope effectively with the pain and distress of lumbar punctures. The intervention was successful, resulting in reductions in both pain and distress.

Thinking errors applied to cancer patients

1) **Black or white**: Viewing situations, people or self as entirely bad or entirely good – nothing in between. Examples ‘My family never understands my needs. They never get it!’
2) **Exaggerating**: Making self-critical or other-critical statements that include terms like never, nothing, everything or always. Example: ‘I always get the worst possible side-effects from chemo.’
3) **Filtering**: Ignoring the positive things that occur to and around self but focusing on and accentuating the negative. Eg. ‘I’m sure that Tamoxifen will give me clots travelling to my lungs.’
4) **Discounting:** Rejecting positive experiences as not being important or meaningful. Eg. ‘My oncologist was reassuring, but he’s just trying to lift my morale.’
5) **Catastrophising:** Blowing expected consequences out of proportion in a negative direction. Eg. ‘Although they say my prognosis is good, I’m sure that I’ll die from this cancer.’
6) **Judging:** Being critical of self or others with a heavy emphasis on the use of should have, ought to, must, have to and should not have. Eg: ‘My doctor should not keep me waiting for my appointment. He ought to know I keep to precise time.’
7) **Mind reading:** Making negative assumptions regarding other people’s thoughts and motives. for eg: ‘That worried look on that nurse’s face must mean I’m in trouble with this cancer.’
8) **Forecasting:** Predicting events will turn out badly. ‘This cancer is destined to return. I’m doomed.’
9) **Feelings are facts**: Because you feel a certain way, reality is seen as fitting that feeling. ‘I am very sad about getting these chemo side-effects and so this illness is certain to turn out badly for me.’
10) **Labelling:** Calling self or others a bad name when displeased with a behaviour. ‘I’m such a fool when I forget to ask my doctor the questions I have written out in my pocket.’
11) **Self-blaming:** Holding self responsible for an outcome that was not completely under control. ‘Getting cancer is my fault as I’ve let too much stress affect my life.’

‘Negative Thoughts Trigger Negative Feelings’ worksheet in Adult Psychotherapy Homework Planner. Jongsma, A.E (Eds.), 2004

A 2006 study of nearly two hundred women with early-stage breast cancer showed that a ten-week cognitive behavioral therapy group course helped the women reduce social disruption and improve their outlook, sense of well-being, and ability to relax, even up to one year after the therapy. A similar effect was shown in a 2006 study of men who had been treated for early-stage prostate cancer with surgery or radiation. The group of men who had the ten-week cognitive behavioral therapy stress management course had better quality of life afterward than those who did not.

Supportive Therapy

Supportive therapy can be done in individual as well as group setting.

Individual supportive therapy:

Here on to one setting helps to develop a relationship of trust, so that patient can talks freely. Efforts should be made to reduce denial but maintain hope. Meaningful activities should be encouraged for long as possible. Listening understanding and sometimes, just sitting quietly with the patient were also described as elements of therapy.

Group supportive therapy

Support group are frequently employed in psychosocial intervention to provide emotional support. Groups of cancer patients, recovered cancer patients those who have undergone surgery, chemotherapy or radiotherapy, spouses, or family members can be very effective in alleviating distress among participants. Obviously, groups are not the answer for every cance patient; at the same time, there is some evidence to suggest that support groups associated with better psychological adjustment to cancet in some.

The two main functions of supportive therapy: are emotional support and provision of information regarding the disease and its treatment. The stress of having cancer and the mode of its treatment create the need for emotional support. The patient also need clear, unambiguous information about the disease and its treatment, which should be impaired sensitively. Further, the empirical literature suggests that group therapy protocols that focus primarily on providing peer support and emphasize the shared expression of emotions are less effective than either educational protocols (e.g., Helgelson, Cohen, Schulz, & Yasko, 1999) or programs teaching coping skills (Edelman, Craig, & Kidman, 2000).

Crisis Intervention

This is another model of psychotherapy with seems to be useful in some cancer patients. A crisis may be precipitated by increased anxiety, emotional distress and a breakdowm of problem solving skills. There should also be evidence that the patient is motivated to change with regard to coping behaviours.

Interventions with Families of Cancer Patients

Comprehensive reviews of research on the psychosocial functioning of the spouses, family members, and/or caregivers of adults with cancer (e.g., Lewis, 1986; Northouse, 1984; Sales 1991) suggest that these significant others display levels of distress similar to those of the patients. While many display only slightly elevated distress, some display more significant psychological problems. Furthermore, the significant others' adjustment may be affected by the course of the disease, interactions with and needs of the patients, and levels of external support. Therefore, patients who are identified as in need of psychosocial services may also have spouses or family members who could benefit from services. Descriptive studies of interventions with significant others of cancer patients suggest that the interventions may produce positive results (Berger, 1984; Carter & Carter, 1994; Cohen, 1982; Cohen & Wellisch, 1978; Reele, 1994; Sabo, Brown & Smith, 1986; Walsh-Burke, 1992). Although controlled studies (e.g., Christensen, 1983; Goldberg & Wool, 1985; Heinrich & Schag, 1985; Toseland, Blanchard & McCallion, 1995) have yielded few significant results, the interventions appear to have been associated with improved functioning for some participants.

- ✓ **Family Therapy**

When a family member is diagnosed with cancer, all members of the family are affected in some way. Cohen and Wellisch (1978) proposed that the cancer diagnosis can be viewed as an "accent" on the family's typical mode of functioning, which may result in either increased or decreased engagement among family members. When presenting for family therapy, the family may identify the cancer diagnosis as the primary problem, overlooking how all family members' reactions to the diagnosis may affect the functioning of the family (Cohen, 1982). The family therapist must assess the family's developmental level, unique style, and patterns of interaction in order to best assist the family in adjusting to changes precipitated by the cancer diagnosis. The therapist should also consider how the roles of each family member have changed since the diagnosis.

- ✓ **Couples Therapy**

While focusing on the needs of loved ones diagnosed with cancer, spouses may overlook their own needs. In couple's therapy, the needs of both the patient with cancer and his or her spouse should be directly addressed; this may require effort to shift focus from the patient to the spouse or to the relationship of the couple. In fact, the focus of the intervention may be learning to balance the needs of the patient and spouse so that both can provide support for each other. At the same time, consideration should be given to issues specific to the patient's diagnosis and treatment. For example, couples therapy following mastectomy for breast cancer may focus on issues of body image and sexuality that affect both the patient and her partner (Christensen, 1983).

Conclusion

Oncology offers a unique opportunity for psychologists to contribute to the care and well-being of a varied population that includes those at risk for the development of cancer as well as those with a diagnosis of cancer. Due to the importance of lifestyle factors in the development of cancer and psychological factors in early detection behavior, clinical health psychologists can assist in the design and implementation of behavior change strategies to improve both primary and secondary prevention. The fact that a high number of cancer patients report elevated distress and decreased quality of life indicates that there is ample opportunity for clinical health psychologists to provide an array of psychosocial interventions. Additionally, clinical health psychologists can serve a vital role in consultation and in education of medical staff involved in the care of cancer patients. Lastly, there are numerous opportunities for clinical health psychologists to become involved in research within the area of psycho-oncology.

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UNIT-IX

HIV/AIDS

BY RIJU RAJ

M.PHIL 2ND YEAR

POST GRADUATE INSTITUE OF BEHAVIORAL AND MEDICIAL SCIENCES, RAIPUR, C.G.

Contents:

1. Brief Introduction

HIV causes AIDS. HIV stands for **Human Immunodeficiency Virus**. It breaks down the immune system which protects the body against disease. HIV causes people to become sick with infections that normally wouldn't affect them.

AIDS stands for **Acquired Immune Deficiency Syndrome**. It is the most advanced stage of HIV disease. Some people develop HIV symptom shortly after being infected. But usually takes more than 10 years. There are several stages of HIV disease. The first HIV symptoms may include swollen glands in the throat, armpit or groin. Other early HIV symptoms include slight fever, headaches, fatigue and muscle aches. These symptoms may last for only a few weeks.

Symptoms of HIV/AIDS:

AIDS symptoms appear in the most advanced stage of HIV disease. In addition to a badly damaged immune system, a person with AIDS may also have thrush —

1. a thick, whitish coating of the tongue or mouth that is caused by a yeast infection and sometimes accompanied by a sore throat severe or recurring vaginal yeast infections,
2. chronic pelvic inflammatory disease severe and frequent infections periods of extreme and unexplained tiredness that may be combined with headaches, lightheadedness,

3. Dizziness, quick loss of more than 10 pounds of weight that is not due to increased physical exercise or dieting but bruising more easily than normal long periods of frequent diarrhea,
4. frequent fevers and/or night sweats swelling or hardening of glands located in the throat, armpit, or groin periods of persistent, deep, dry coughing, increasing shortness of breath
5. appearance of discolored or purplish growths on the skin or inside the mouth unexplained bleeding from growths on the skin, from the mouth, nose, anus, or vagina, or from any opening in the body frequent or unusual skin rashes,
6. Severe numbness or pain in the hands or feet, the loss of muscle control and reflex, paralysis, or loss of muscular strength confusion, personality change, or decreased mental abilities.

Causes of HIV/AIDS:

The most common ways HIV is spread are by:

- Having sex with an HIV infected person
- sharing needles or syringes with someone who has HIV/AIDS
- being deeply punctured with a needle or surgical instrument contaminated with HIV
- getting HIV-infected blood, semen, or vaginal secretions into open wounds or sores
- Babies born to women with HIV/AIDS can get HIV from their mothers during birth or from breastfeeding.
- Infected blood transfusion

However HIV is not transmitted by simple casual contact such as kissing, sharing drinking glasses, or hugging.

2. Model of HIV disease service program in India:

Shortly after reporting the first AIDS case in 1986, the Government of India established a National AIDS Control Program (NACP) which has become the Department of AIDS under Ministry of Health and Family Welfare.

In 1991, the scope of NACP was expanded to focus on blood safety, prevention among high-risk populations, raising awareness in the general population, and improving surveillance. A semi-autonomous body, the National AIDS Control Organization (NACO), was established under the Ministry of Health and Family Welfare to implement this program. This “first phase” of the National AIDS Control Program lasted from 1992 -1999. It focused on initiating a national commitment, increasing awareness and addressing blood safety. It achieved some of its objectives, notably increased awareness. Professional blood donations were banned by law. Screening of donated blood became almost universal by the end of this phase. By 1999, the program had also established a decentralized mechanism to facilitate effective state-level responses. States such as Tamil Nadu, Andhra Pradesh, and Manipur demonstrated a strong response and high level of political commitment, many other states, such as Bihar and Uttar Pradesh, have yet to reach these levels.

The second phase of the NACP began in 1999 and ended in March 2006. Under this phase, India continued to expand the program at the state level. Greater emphasis was placed on targeted interventions for the most at risk populations, preventive interventions among the general population, and involvement of NGOs and other sectors and line departments, such as education, transport and police. In order to induce a sense of urgency, the classification of states has focused on the vulnerability of states, with states being classified as high and moderate prevalence (on the basis of HIV prevalence among high risk and general population groups) and high and moderate vulnerability (on the basis of demographic characteristics of the population).

While the government’s response has been scaled up markedly over the last decade, major challenges remain in raising the overall effectiveness of state-level programs, expanding the participation of other sectors, and increasing safe behavior and reducing stigma associated with HIV-positive people among the population.

The Third Phase of NACP program has dramatically scaling up targeted interventions in order to achieve a very high coverage of the most at risk groups. Under this phase, surveillance and strategic information management also receive a big boost. A partnership with civil society organizations was at paramount in the implementation of the program with special focus on involvement of community in the program planning and implementation.

On completion of NACP III, government of India has realized their strengthens and with the help of development partners and donor agencies, NACO has conducted consultations with all the stakeholders including the representatives from civil societies, community representatives, non-health departments and experts from public health and designed the program activities for NACP IV. In future the focus of this phase will be primarily on scaling up prevention through NGOs and sustaining the efforts and results gained in last 3 phases and integration with the health systems response to the epidemic e.g. through provision of ART, STI services, and treatment of opportunistic infections through the National Rural Health Mission.

3. Pre and post test counseling:

Both **pre- and post-test counseling** are essential because it is important to have a clear understanding of what the test is and what its implications may be, in order to be able to make informed choices.

Pre test counseling:

In the pre test, the client:

- Are informed about the test.
- Made aware of the reason why HIV testing is being recommended
- Informed about the current contact of the plan and how the client will obtain the result
- Supported till the result is obtained in reducing his/her stress where he/she is has experienced an event that is associated with a higher likelihood that subsequent infection may have occurred.
- Answer to the client's queries and needs

Posttest counseling:

In case of HIV test **negative** result:

- The client must interpreted the test result that shows no infection
- The client must be recommended for further test so that to be in safe side
- The client must be reeducated once again regarding the ways he/she can be further infected.

In case of HIV test **positive** result:

- The result must be shared to the client in a straight forward and simple manner.
- The client must be provided time to consider the result

- Ensuring that the client understands the result
- Providing support and empathy for emotions that the client might express in response to the test result.
- Discussing immediate concerns that the client may have such as partner, families, impact of results, and disclosure of results.
- Describing follow-up services available including treatment, care, counseling and other community-based services.
- Discussing ways to prevent forward transmission of HIV.
- Client's and their family member's doubt must be clarified.

4. Psychosocial issues of HIV/AIDS:

Following are the ***psychosocial issues*** related to HIV and HIV affected people:

- **Stigmatization**
- **Discrimination**
- **Social isolation**

Psychological issues of HIV/AIDS:

- **Sense of grief and loss**
- **Alteration of personality**
- **Anxiety**
- **Depression**
- **Substance abuse**
- **Suicide**

Psychosocial intervention:

Psycho social intervention mainly focuses on Practical Support and Assistance that ensure:

- Increasing social network
- Spending time with friends & family
- No discrimination in work settings and other community setting

5. Psychological assessment and interventions:

- ***Personal interviews*** are carried out to assess or to reveal patient's current feeling regarding the disease, his outlook and his approach to deal with this and his opinion regarding his family and other social associates.
- ***Family counseling:*** In this each member of the family is included in the sessions both individually and in group to discuss their attitude towards HIV/AID and there feeling towards the current situation of the infected member. Further the level of psychological stress the family members are going through is also assessed.
- ***Professional counseling*** that includes ***Individual Therapy*** and ***Support Groups*** and ***CBT***
- ***Anger*** and ***anxiety management***, ***problem solving***, ***Solution Focused Therapy***.
- ***Education*** that includes learning to manage the disease and still enjoy life

6. Highly active anti- retroviral treatment (HAART):

Antiretroviral agents have greatly improved the prognosis of patients infected with HIV. There has also been a dramatic decrease in the complications of HIV infection. The development of drug resistance is reduced by using a combination of drugs. The standard treatment for HIV infection is called ***HAART- Highly Active Antiretroviral Therapy*** which usually includes two nucleoside reverse transcriptase inhibitors (NRTIs) with either a non-nucleoside reverse transcriptase inhibitor (NNRTI) or one or two protease inhibitors (PIs).

Following are the combinations of drugs that are used:

Nucleoside Reverse Transcriptase Inhibitors

- These inhibit the RNA-dependent DNA polymerase (reverse transcriptase) which HIV uses to convert viral RNA into DNA before its incorporation into the cell genome.
- NRTIs should be used with caution in patients with chronic hepatitis B or hepatitis C (there is greater risk of hepatic side-effects), in hepatic impairment, renal impairment and in pregnancy.
- Side-effects include gastro-intestinal disturbances, headaches and blood disorders (including anaemia, neutropenia, and thrombocytopenia).

Protease Inhibitors:

- These inhibit HIV enzyme required to produce mature infectious viral particles by cleaving structural proteins and enzymes from their precursors. They are potent inhibitors of HIV replication and work synergistically with nucleoside drugs.

- They reduce HIV viral load and increase CD4 counts more effectively than nucleoside analogues, especially when used in triple therapy.
- PIs are associated with gastro-intestinal disturbances, headaches, hyperglycaemia (caution in diabetes), increased risk of bleeding (especially in haemophilia), hepatic impairment, lipodystrophy and metabolic effects.

Fusion Inhibitors

- Enfuvirtide, which inhibits HIV from fusing to the host cell, is licensed for managing infection that has failed to respond to a regimen of other antiretroviral drugs. It is used with other antiretroviral drugs and is administered by subcutaneous injection twice daily
- *Other Antiretrovirals* like Maraviroc is the first CCR5 receptor antagonist licensed for the treatment of HIV infection. Raltegravir is an integrase inhibitor and is indicated in combination with other antiretroviral drugs for HIV infection resistant to first-line HAART. Eviplera® is a new one-tablet formulation, used in antiretroviral-naïve patients with HIV-1.

7**. Neuropsychological problems of HIV/AIDS:**

Types of **Central Nervous System Disorder**:

a. Dementia

b. ***Cerebral Toxoplasmosis***, also known simply as toxoplasmosis, is the most common central nervous system infection in HIV patients. It is caused by protozoa called Toxoplasma gondii, which lives in the soil and in animal feces. In HIV patients and other people with suppressed immune systems, however, the bacteria can cause **brain abscess** (tissue damage and the accumulation of pus)—the symptoms of which vary depending on the location of the infection in the brain. Usual symptoms of toxoplasmosis include speech difficulties, seizures, confusion, and lethargy, which develop over the course of days to weeks.

c. ***Cryptococcal Meningitis*** is a type of infection that is caused by a fungus. The course of the illness is usually slow and may develop over days or months.

d. ***Progressive Multifocal Leukencephaly*** is an infection caused by a rare virus. A patient with PML may suffer from dementia (a broad range of cognitive problems, including memory loss, poor judgement, etc.), facial weakness, visual problems, and a loss of coordination. The symptoms vary from person to person and generally reflect which area of the brain is affected.

e. ***Central Nervous System Lymphoma*** is the second most common nervous system abnormality in HIV patients. Primary lymphoma generally only develops in the central nervous system when the immune system is suppressed. Primary lymphoma—as opposed to metastatic lymphoma—is cancer that originates in the lymphatic system and has not spread from some other part of the body.

Types of Peripheral Nervous System Disorder:

a. Neuropathy

Neuropathy, also known as peripheral neuropathy, is disease in the peripheral nerves—the nerves that lead to and from the spinal cord and connect with all the various parts of the body. It is very common in HIV patients, usually in the later stages of HIV disease. It can manifest itself in several different ways.

Distal symmetric neuropathy is the most common form of HIV-related neuropathy. It affects the feet first and then the hands and it affects both sides of the body equally. Patients often feel a strange tingling and painful burning sensation that can spread up the legs and arms. Some patients feel numbness or weakening in the arms and legs.

Acute Inflammatory Demyelinating Neuropathy involves the nerve root (where the root connects with the spinal cord) and the myelin sheath that surrounds and protects the nerves. The onset of this kind of neuropathy is usually very rapid, sometimes developing within hours to days.

b. ***Myopathies*** are neurological disorders that involve the skeletal muscles—muscles that are connected to bones, like the biceps in the upper arm and quadriceps in the thigh. There are many different types of myopathies (including, for example, the muscular dystrophies), but the most common type in people with HIV is **polymyositis (PM)**. Its symptoms are in the form of muscular aches, cramping, and tenderness, and extreme muscle weakness. Weakness primarily affects the neck, arms, and upper portion of the legs—making it difficult to stand up from a sitting position. Many patients also experience fever, malaise (general bodily discomfort), and loss of appetite.

Treatment:

Central Nervous System Disorders:

Some neurological problems can be treated with medication(s). Anti-dementia drugs can be prescribed to relieve confusion and slow the progression of mental decline. Neurological infections can be treated with antibiotics. There is no known treatment for PML till today. If there is an AIDS-related tumor in the brain or spinal cord, radiation therapy or steroids may be helpful, although the prognosis is poor.

Peripheral nervous system:

Treatment of the various peripheral nervous system disorders usually focuses on relieving the pain and other symptoms. Drug therapy is often used to treat neuropathic pain. Typical medications include tricyclic antidepressants such as amitriptyline (Elavil), anticonvulsants such as gabapentin (Neurontin), and analgesics such as tramadol (Ultram).

Acute inflammatory demyelinating syndrome often requires immunotherapy or plasmaphoresis as part of its treatment. Immunotherapy involves injecting a specific protein into the blood to stop the abnormal immune response that is causing the neuropathy. Plasmaphoresis involves removing some of the blood, separating the cells from the plasma, and then re-injecting the cells back into the body.

8. Issues related to prevention/ spreading of awareness:

Although global attention to HIV and AIDS remains strong, particularly regarding treatment initiatives, until recently HIV-prevention has garnered scant attention. Treatment alone will not reverse the epidemic, and current prevention efforts have not been successful in halting HIV transmission.

Issues:

- lack of capacity, such as health human resources and infrastructure;
- disjointed programs, such as prevention programs not integrated into institutions or other health-related services;

- reliance on ineffective interventions, such as abstinence-based programs inadequate implementation of interventions and approaches proven to be effective, such as harmreduction;
- lack of coordination among stakeholders;
- and the challenge of stigma; the ethical challenges involved in research;
- the difficulty of sustaining political support for prevention programs, compared with interventions with shorter time frames and faster results, such as treatment.

Improving HIV prevention issues:

- It is critical that countries understand how the epidemic is affecting them specifically to ensure prevention interventions are appropriate and cost-effective. This includes **gathering information** about HIV infection rates among different population groups within a given country. Ongoing country-level surveillance of the epidemic is essential for countries to plan and adjust their prevention strategies accordingly.
- As part of "knowing the epidemic" there is a **need to increase HIV testing**. It is estimated that just 11% of the world's population is aware of their HIV status. Research has shown that people who are aware that they are HIV positive decrease their risk behaviors. It is essential that there be widespread and easy access for HIV-testing as part of comprehensive HIV prevention programming. It must also be emphasized that support for testing should not negate the need for privacy, confidentiality, and consent—safeguards must be in place to ensure that these rights are respected.
- Developing **effective preventive models** that includes features like adequate human resource and institutional Capacity; a focus on interventions that are locally relevant, evidence-based, and targeted to the appropriate population; a comprehensive approach, including mass media campaigns to increase awareness and programs that build self-esteem and life skills such as safer sex negotiation; the involvement of multiple sectors, including communities which are affected by HIV/AIDS; initiatives to address stigma.
- **Comprehensive preventive** measures should be included to encompass structural interventions, successful prevention efforts that require a diversity of biomedical and behavioral methods that will provide individuals with a range of options, and have the potential to further decrease the risk of HIV infection if used in combination. Such

measures should focus on interventions to prevent sexual transmission, interventions to prevent blood-borne transmission, interventions to prevent mother-to-child transmission.

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UNIT-X

PAIN: PHYSIOLOGICAL AND PSYCHOLOGICAL PROCESS

POST GRADUATE INSTITUE OF BEHAVIORAL AND MEDICIAL SCIENCES, RAIPUR, C.G.

CONTENT

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WHAT IS PAIN?

Pain is felt by all humans except those who are comatose, unconscious, anesthetized, or congenitally analgesic, but may be suppressed or inhibited under certain circumstances. *Webster's 1966 International Dictionary* provides an adequate psychophysiological description of pain, without psychoanalytical considerations, “Pain (Middle English, peine; Latin, prena, penalty, punishment), 1.originally penalty, 2. the sensations one feels when hurt, mentally or physically, especially distress, suffering, great anxiety, anguish, grief, etc.: opposed to pleasures 3. A sensation of hurting, or strong discomfort, in some parts of the body, caused by an injury, disease, or functional disorder, and transmitted through the nervous system." Definition 2 covers the feelings of pain modified, amplified, or distorted by central psychic and emotional influences, and 3 refers to the sensation of pain evoked in response to peripheral stimulation of the nerve endings concerned.

The International Association for the Study of Pain’s (IASP’s) definition of pain is **‘an unpleasant sensory and emotional experience associated with potential or actual tissue damage’. (**Derived from the Latin word for punishment)

Pain involves the total experience of some noxious stimulus which is influenced by current context of the pain, previous experience, learning history and cognitive process.(*Feuerstein et al.,1987).* These mechanisms are an integral part of the body’s defences, providing early warning of impending damage and triggering physiological (via nociception) and behavioural (via subjective pain) responses to avoid or minimize that damage.

PREVALENCE

The National Institute of Health identified chronic pain as the costliest medical problem in America, impacting nearly 100 million individuals (Byrne & Hochwarter, 2006). More recently, according to the Centre for Disease Control and Prevention’s annual report, one in four adults say they suffered from a day-long episode of pain in the past month, and one in ten adults reported pain lasting 1 year or more (CDC, 2006). Over 20% of all medical visits and 10% of all drug sales are pain related (Max, 2003). In occupational contexts, chronic pain is not only a significant source of absenteeism but also a major factor in reducing productivity while at work. Indeed, approximately half of all employees experience pain while on the job, with individuals whose work involves repetitive movement or heavy lifting being impacted in greater numbers (Byrne & Hochwarter). It has been estimated that pain results in $79 billion annually in lost worker productivity. A recent study estimated that the total health care expenditures for back pain alone reached over $90.7 billion in 1998 (Xuemei, Pietrobon, Sun, Liu, & Hey, 2004).

Overall, the total direct and indirect costs of chronic pain in the United States have been estimated to be between $150 and $260 billion annually (Byrne & Hochwarter). These statistics led the 108th U.S. Congress to formally declare the 10-year period beginning January 1, 2001, the "Decade of Pain Control and Research" (CDC). In India low back pain is growing with the prevalence rate of 40 to 69 % (Kumar, 1999). Nearly 25% of the adult subjects suffer from chronic musculoskeletal (MS) pain or discomfort. In 2006 the prevalence rate of pain disorder was found at 22% while another study in 2012-end put it at 14%.

HISTORY OF PAIN

Since Aristotle pain has been classified not as a perception but as a mood state, and so excluded from the five senses. For most of history, pain has been regarded largely in mechanical terms. For example, Sextus reports the Epicurean claim that **'it is impossible for what is productive …of pain not to be painful'**, which implies linear causality.

Descartes, in particular, argued that pain is evidenced by the withdrawal of the relevant body part from the noxious stimulus, as a result of nerve action. These models regard the pain mechanism and the subsequent behavioural response as distinct from the individual's subjective experience of pain. This mechanistic behaviourism continues to have a pervasive influence on scientific thinking, and the distinction between the mechanistic response to stimulation and the subjective experience distinction still lingers in the current distinction between nociception and pain.

TYPES OF PAIN

Acute or Chronic

Pain can be classified as either **acute or chronic.** While both may differ in terms of duration, a more helpful distinction is to regard ***acute pain*** as that which serves to protect after injury and promote healing, and ***chronic pain*** as a disease of pain which does not serve this function. Generally, healthcare professionals regard acute pain as an appropriate symptom of various disease states and procedures, which can be treated by removing the cause of the pain, and managed in the interim with appropriate treatments such as analgesic medications.

In some cases, however, 'acute pain' fails to resolve after the expected period, so that the pain itself becomes a disease state (that is, chronic).

Organic or Psychogenic Pain

Psychogenic pain, also called **psychalgia**, is that is caused, increased, or prolonged by mental, emotional, or behavioural factors. Headache, back pain, or stomach pain are some of the most common types of psychogenic pain. It may occur, rarely, in persons with a mental disorder but more commonly it accompanies or is induced by social rejection, broken heart, grief, love sickness, or other such emotional events.

Medicine refers also to psychogenic pain or psychalgia as a form of chronic pain under the name of *persistent somatoform pain disorder* or *functional pain syndrome*. Causes may be linked to stress, unexpressed emotional conflicts, psychosocial problems, or various mental disorders. Some specialists believe that psychogenic chronic pain exists as a protective distraction to keep dangerous repressed emotions such as anger or rage unconscious.

Organic pain refers to any pain resulting from a disorder, abnormality or chemical imbalance in an organ system, namely the human body. Consequently, organic pain is an extremely broad term, covering pain causes that range in diversity from heartburn to multiple sclerosis. Unlike non-organic pain, organic pain has a traceable and identifiable cause in a specific organic system.

Examples of Organic Pain

Examples of gastrointestinal and urinary system organic pain include.

(i) Constipation
(ii) Crohn's disease (an inflammatory disorder related to the digestive tract)
(iii) Gastrointestinal infections
(iv) Heartburn
(v) Sexually transmitted diseases
(vi) Ulcerative colitis (a condition that causes ulcers to arise in the colon and rectum)
(vii) Ulcers
(viii) Urinary tract infection.

Remember, though, that organic pain is not confined to the gastrointestinal and urinary tract systems. Nearly any system or bodily organ can be affected by organic pain, including (but not limited to) the:

(i) Circulatory system
(ii) Musculoskeletal system
(iii) Neurological systems
(iv) Skin

Nociceptive or Neuropathic pain

Pain can be further divided into two broad categories:

- *Nociceptive* pain
- *Neuropathic* pain.

There are two types of ***Nociceptive*** pain: ***somatic and visceral pain.***

Somatic pain is caused by the activation of pain receptors on the surface of the body, such as the skin (cutaneous tissues) or tissues that are deeper, such as muscle (musculoskeletal tissues).

When pain occurs in the musculoskeletal tissues, it is called *deep somatic pain.*

Deep somatic pain is usually described as “dull” or “aching” but localized. This type of pain is often expressed by people who overdo it and strain muscles when performing physical activity or exercising. Surface somatic pain is usually sharper and may have a burning or pricking quality. Common causes of surface somatic pain include postsurgical pain or pain related to a cut or burn.

Viscera refer to the internal areas of the body that are enclosed within a cavity. ***Visceral pain*** *is* caused by activation of pain receptors resulting from infiltration, compression, extension, or stretching of the chest, abdominal, or pelvic viscera. Visceral pain is not well localized and is usually described as “pressure-like, deep squeezing.” Examples of visceral pain include pain related to cancer, bone fracture, or bone cancer.

Neuropathic pain is a neurological disorder resulting from damage to nerves that carry information about pain. Neuropathic pain is reported to feel different from somatic or visceral pain and is often described using words such as “shooting,” “electric,” “stabbing,” or “burning.” It may be felt travelling along a nerve path from the spine into the arms and hands or into the buttocks or legs. Examples of neuropathic pain conditions include *phantom limb* *pain, post-herpetic neuralgias, and other painful neuropathies* (e.g., diabetes or alcohol related).

Specific Chronic Pain Conditions

Pain can occur in many parts of the body, each with its own prevalence, patterns, and presenting characteristics. However, there are a number of pain conditions that are more common than others.

1- **Chronic low back pain (CLBP)** is the most common chronic pain condition, impacting 15 to 45% of adults annually and at least 70% of adults over a lifetime *(Andersson, 1997).* **Back pain** is the most common cause of job-related disability and a leading contributor to missed work. Most low back pain follows injury or trauma to the back, but pain may also be caused by degenerative conditions such as arthritis or disc disease (protruding, herniated, or ruptured disc), sciatica, osteoporo- sis, or other bone diseases. Chronic low back pain is often associated with affective distress and disability; however, research indicates that cognitive-behavior therapy (CBT) can be an effective treatment for chronic low back pain (*Hoffman, Papas, Chatkoff, & Kerns, 2007).*

2-Headaches represent another large category of painful conditions. The majority of headaches are so-called ***primary headaches.*** These include tension **headache and migraine headache. Tension headaches** are the most common and affect 38 to 78% of people *(Rasmussen, Jensen, Schroll, & Olsen, 1991).* The pain is typically located in the forehead, neck, and shoulder areas, and many people describe the feeling as having a tight band around their head. Migraine headaches affect 18% of women and 6% of men *(Lipon, Stewart, Diamond, Diamond, & Reed, 2001).*

3-Fibromyalgia syndrome (FMS) consists of a set of unexplained physical symptoms with general pain and hypersensitivity to palpation at specific body locations called *tender points.* In addition, patients with FMS often report a range of functional limitations and psychological dysfunction, including persistent fatigue, sleep disturbance, stiffness, headaches, irritable bowel disorders, depression, anxiety, cognitive impairment, and general malaise sometimes referred to as *fibro fog* (*Baumstark & Buckelew, 1992).* Fibromyalgia syndrome occurs predominately in adults and has a female to male ratio of 7 to 1 in those seeking treatment.

MODELS OF PAIN

1-The Gate Control Theory :

The Gate Control Theory, which was initially described in 1965 by Ronald Melzack and Patrick Wall (*Melzack & Wall, 1965)*, suggested that the experience of pain was not simply the result of the interpretation of nerve impulses sent directly from sensory neurons to the brain. Rather, the theory suggested that the impulse pathway was more complex and allowed the opportunity for the impulses to be modulated by other incoming stimuli before reaching the brain.

According to the theory, modulation of the signal occurs at a site in the *dorsal horn* of the spinal cord, where a type of "gate mechanism" exists. The gate opens and closes depending on feedback from other nerve fibers in the body, including descending neural impulses from the brain such as those related to an individual's thoughts or mood (e.g., anxiety or depression). When the gate is open, more sensory information regarding pain is allowed to be transmitted to the brain, but when the gate is closed, less information is transmitted to the brain. Thus, the theory had a significant impact on the study of pain because it recognized that psychological factors can have important roles in the experience of pain.

2-Biopsychosocial Model:

In contrast to the traditional biomedical model, which assumed that all illness was a function of biological malfunctions, the biopsychosocial model emphasizes the dynamic and reciprocal relationships between the social, biological, and psychological domains of physical health problems *(Engel, 1977).* Consistent with more general systems theory *(von Bertanlanffy, 1968),* the model notes that a change in one domain (e.g., the biological domain in the case of a chronic painful condition) necessarily results in changes in the other domains (e.g., psychological and social domains).

Biopsychosocial models suggest that pain is not just a biological process involving the transmission of sensory information about tissue damage to the brain, but is the product of the interactions among biological, psychological, and social factors. All of these factors have an impact on a person's experience of pain, including the intensity, duration, and consequences of pain. For example, when pain persists over time, a person may develop negative beliefs about his or her pain (e.g., "this is never going to get better," "I can't deal with this pain") or negative thoughts about himself or herself (e.g., "I'm worthless because I can't work," "I'm never going to get better"). As pain continues, a person may avoid participating in certain activities for fear of further injury or exacerbating the pain (e.g., work, social activities, or hobbies). As the person withdraws and becomes less active, his or her muscles may weaken, he or she may gain or lose weight, and his or her overall physical conditioning may decline.

The biopsychosocial model has been commended for emphasizing that the social domain should be attended to in terms of its impact on the experience of chronic pain (Kerns & Jacob, 1995). For example, changes such as shifting family roles, loss of income, and increased family and marital distress can have negative effects on pain and disability.

Limitation: However, the model's specific influence on the chronic pain field has been limited. The model has failed to contribute to specific theoretical refinements about mechanisms of transaction, particularly the potential influence of the social (and family) domain on the development and perpetuation of the chronic pain condition or its associated problems *(Kerns, Otis, & Wise, 2002).*

3-Cognitive-Behavioral Transactional Model :

Building upon cognitive-social learning theory and behaviorism, a cognitive-behavioral transactional model of the role of families in the course of chronic illness has been described by *Kerns, Otis, and Wise* (2002).The cognitive-behavioral transactional model emphasizes the importance of social support and the family in the development and maintenance of chronic pain. The model suggests that interactions related to pain all take place within a social and family learning environment that selectively reinforces coping attempts and outcomes in terms of optimal pain management, continued constructive activity, and emotional well-being, or, conversely, reinforces increased pain, disability, and affective distress.

Consistent with a cognitive-behavioral perspective on chronic pain (*Turk, Meichenbaum,& Genest, 1983)*, the model hypothesizes that the family plays an active role in seeking out and evaluating information about the painful condition itself and the specific challenges it poses, as well as in making judgments about the family's and its members' capacities and vulnerabilities in meeting the challenges. It is on the basis of these appraisals that the family and its members make active decisions about alternative responses, act upon their decisions, and evaluate the adequacy of the response.

Central to the transactional model is the additional notion that the family's response and its perceived effects, in turn, shape future appraisals of stress and challenge in a dynamic and reciprocal fashion. Perceptions of failed efforts to manage the painful condition will likely enhance the intensity of the perceived threat of the condition, perhaps contributing to a heightened level of perceived pain, increased disability, and affective distress. Conversely, perceptions of success in coping will likely moderate the experience of pain, increase confidence in the family's ability to respond effectively in the future, and reinforce the repeated use of similar strategies.

4-Cognitive-Behavioral Fear-Avoidance Model :

Vlaeyen and Linton (2000) have proposed a cognitive-behavioural, fear-avoidance model of chronic pain to explain the role of fear and avoidance behaviours in the development and maintenance of chronic pain and related functional limitations. According to this model, there are two opposing responses an individual may have when experiencing pain. One response is that an individual may consider pain to be nonthreatening and consequently engage in adaptive behaviours that promote the restoration of function.

In contrast, pain may be interpreted threatening, a process called *catastrophizing.* Vlaeyen and Linton proposed that catastrophizing contributes to a fear of pain and may lead to avoidance of activities that may elicit pain, guarding behaviours (i.e., behaviours performed with the goal of protecting a site of pain such as bracing while walking), and hypervigilance to bodily sensations. Consistent with principles of operant reinforcement, as activities associated with pain are avoided and feelings of fear subside, avoidance behaviours are negatively reinforced. As an individual becomes more depressed and inactive, the cycle of pain is fuelled even further, and fear and avoidance is further increased. Thus, avoidance has the potential to increase disability and negative mood and ultimately contribute to the experience of pain. Previous research supports a relationship between fear avoidance and chronic pain *(Asmundson & Taylor, 1996; Crombez, Vlaeyen,Heuts, & Lysens, 1999)*.

PHYSIOLOGICAL PROCESS IN PAIN

Scientists have distinguished among three kind of pain perception.

1-The first is mechanical *Nociception* (pain perception) that results from the mechanical damage to the tissue of the body.

2-The second is *Thermal damage* or the experience of pain due to temperature exposure.

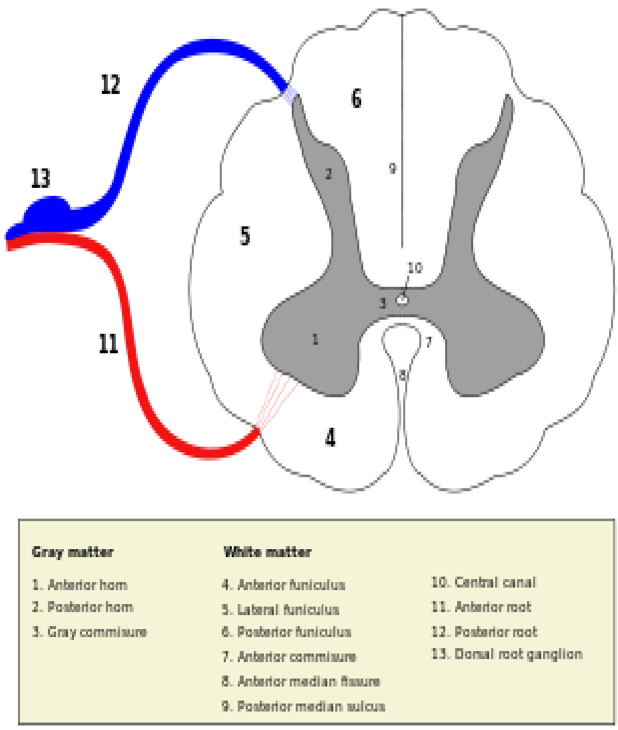
3-The third is referred to as *polymodal nociception,* a general category refer to pain that triggers chemical reactions from tissue damage.

Nociceptor in the peripheral nerves first sense injury and in response release chemical messenger, which are conducted to the spinal cord where they are past directly to the reticular formation and thalamus and in to the cerebral cortex. These region of the brain ,in turn, identify the site of injury and send message down to the spinal column which lead to muscle contraction which can help block the pain and changes in other bodily functions such breathing.

Two major types of peripheral nerve fibers are involved in nociceptor.

- A-delta fibers are small myelinated fibers that transmit sharp pain.
- C- fibers are unmyelinated nerve fibers, involved in polymodal pain , that transmit dull or aching pain (Myelination increases the speed of transmission so sudden and intense pain is more rapidly conducted to the cerebral cortex than is the slower dull aching pain of the C- fibers.)

Peripheral nerve fibers enter the spinal column at the dorsal horn. The **posterior horn** (**posterior cornu**, **dorsal horn**, **spinal dorsal horn**) of the spinal cordis the dorsal (more towards the back) grey matter of the spinal cord. It receives several types of sensory information from the body, including light touch, proprioception and vibration this information is sent from receptors of the skin, bones, and joints through sensory neurons whose cell bodies lie in the dorsal root ganglion.



Sensory aspects of pain are heavily determined by activity in the A delta fibers, which project onto areas in the thalamus and the sensory areas of the cerebral cortex.

The motivational; and affective element of pain is influenced by C fibers. The experience of pain then is determined by the balance of activity in these nerves which reflects the pattern and intensity of stimulation.

ROLE OF NEUROTRANSMITTER IN PAIN:

OPIOIDS:

Opioid receptors modulate nociceptive input at many sites in the CNS. Microinjection of opioid agonists into these regions results in complex alterations in pain behaviour, reflecting both inhibition and disinhibition (the inhibition of an inhibitory interneuron) of pathways involved in nociception.

SEROTONIN AND NOREPINEHRINE

A dysfunctional serotonin and norepinephrine system that promotes depressive symptoms is likely to also have dysfunctional descending serotonin and norepinephrine pathways. This would have the effect of intensify pain signals and explain why depressed patients may also feel chronic pain including headaches, lower back and neck pain, joint pain, as well as fatigue and tiredness. And, just like chronic depression, chronic pain can itself alter the functioning of the nervous system and perpetuate itself.

In the case of depression concentrations and output of the two neurotransmitters become erratic, leading to a dysregulated signaling system. The pathways of serotonin and norepinephrine in the brain begin in the brain stem and project to various brain regions, including the frontal cortex, hypothalamus, and limbic regions. Symptoms of depression are associated with the frontal cortex and limbic regions. Serotonin and norepinephrine pathways also extend downward from the brain stem into the spinal cord. These descending pathways have many functions, including suppressing the body's input from minor pain, like aching muscles and joints. It's important that these minor pain signals be cut off—otherwise we would be so distracted that we would be unable to function normally.

OTHER NEUROTRANSMITTERS

Other transmitters that appear to be important in altering pain behaviour are *acetylcholine, GABA, thyrotropin-releasing hormone, and somatostatin.*

PSYCHOSOCIAL FACTORS AND PAIN

Pain can affect more than just a person's back or knee; it can impact every aspect of one's life including the lives of significant others. In turn, the responses of significant others and culture in which a person with pain lives can have an impact on how they experience pain. Developmental factors such as cognitive ability, prior experiences with pain, and beliefs and expectations about pain can also play a role.

Therefore, it is important to consider the potential impact of these types of factors when conducting a pain assessment, conceptualizing a case, or when creating a pain treatment plan. The following section provides a brief review of these factors and their relationship to pain.

Behavioural Influences on Pain

Reinforcement: Pain behaviours can be positively reinforced. They may also be maintained by escape from noxious stimulation through the use of drugs, rest, or avoidance of undesirable activities such as work. In addition, well behaviours (e.g., working) may not be positively reinforced by significant others, allowing pain behaviours to be more rewarding. Health care professionals may reinforce pain and pain behaviour by their responses. Physicians who prescribe medication when they observe pain behaviour may, ironically, contribute to the occurrence of future pain behaviours. Patients learn that complaints elicit responses from physicians, and if these responses provide some relief of pain, then they may repeat these behaviours in order to obtain the desired outcome. With the anticipated outcome of pain relief, patients are likely to pay attention to and consequently report higher levels of pain. The alternative for the physician is to prescribe routine time-contingent medication that is not dependent on reported level of pain.

Classical (respondent) conditioning also plays a central role in pain. For example, cancer patients who experience nausea and vomiting following chemotherapy feel nausea when confronted with neutral cues previously paired with chemotherapy, such as doctors, nurses, the hospital, and even patients' clothes (*Carey & Burish, 1988*). It is common to observe cancer patients long in remission who report nausea as soon as they see their doctors' faces, even years after completion of treatment.

Over time, fear of pain may become associated with an expanding number of situations and behaviors, and many neutral or pleasurable activities may be avoided because they elicit or exacerbate pain.

Another way in which principles of learning influence pain is through **social learning.**

There is evidence that some pain behaviours may be acquired by **observational learning.** For example, children learn how to interpret and respond to symptoms and physiological processes from their parents and social environment. But children can learn both appropriate and inappropriate responses to injury and disease and, as a result, be more or less likely to ignore or over-respond to symptoms.

<u>Cognitive Influences on Pain</u>

Studies have consistently demonstrated that patient's attitudes, beliefs, and expectancies about their plight, themselves, their coping resources, and the health care system influence their experience of pain, activity, disability, and response to treatment *(Flor & Turk, 1988).*

Beliefs about the meaning of pain and ability to function despite discomfort can make marked differences to patients' lives. For example, a cognitive representation that one has a very serious, debilitating condition, that disability is a necessary aspect of pain, that activity is dangerous, and that pain is an acceptable excuse for neglecting responsibilities will likely result in maladaptive responses.

Through a process of stimulus **generalization**, patients may avoid more and more activities, and soon become more physically deconditioned and disabled.

Spiegel and Bloom (1983) reported that ratings of pain severity by cancer patients could be predicted not only by their use of analgesics and affective state but also by their interpretations of pain. Patients who attributed their pain to a worsening of their disease experienced more pain despite levels of disease progression comparable to patients with more benign interpretations.

Expectation: Expectation can significantly influence pain perception. Practitioners can communicate positive and negative expectation to their patients and actually affect (positively or negatively) their patient's individual perception of pain *(Melzack and Wall, 1982).* Practitioners expectation for the good (or the bad) sequel of a treatment have been found to be transmitted quite effectively to the patients.

Once beliefs and expectancies about a disease are formed they become stable and are very difficult to modify. People tend to avoid experiences that could invalidate their beliefs and guide their behavior in accordance with these beliefs, even in situations where the belief is no longer valid.

In chronically ill populations, people's beliefs about different aspects of pain and psychological factors appear to predict health care use better than the number or severity of physical symptoms. For example, pain-related beliefs have been found to be associated with psychological functioning, physical functioning, coping efforts, pain behaviours, and response to treatment.

Catastrophic thinking—experiencing extremely negative thoughts about one's plight and interpreting even minor problems as major catastrophes—appears to be a particularly potent way of influencing pain and disability. In a prospective study,

Percieved control: A sense of control, particularly a belief in one's ability to control pain, affects, its perception. The ability to control aspects of a painful episode, such as its timing, is an effective coping act. One experiment demonstrated that giving volunteer response warning prior to receiving cold-pressure pain (0, 30, or 180 second) actually decreased the level of pain they reported. Having a higher level of self efficacy a buffer in raising pain threshold and tolerance for pain. Arthritis patients were assessed for their level of arteries self efficacy and subjected to thermal pain .those with the highest ratings on efficacy had the highest pain threshold and tolerances and rated the pain episode as the least unpleasant

Self-efficacy—a personal conviction that one can successfully execute a course of action to produce a desired outcome in a given situation—is a major mediator of therapeutic change

It is important to remember that coping behaviours are influenced by people's beliefs that the demands of a situation do not exceed their coping resources. For example, Patients' coping behaviour will be highly related to their self-efficacy expectations, which in turn appeared to be determined by their expectancies regarding levels of pain that would be experienced.

Affective Influences on Pain

The interactive roles of sensory processes and affective states are supported by an overwhelming amount of evidence. The affective components of pain include many different emotions, but they are primarily negative. **Depression,** anxiety and anger have received the greatest amount of attention in chronic pain studies.

Research suggests that 40–50% of chronic pain patients suffer from depression (*Romano & Turner, 1985).* It is not surprising that chronic pain patients are depressed.

Anxiety: Anxiety tends to be a magnifier of pain. Anxiety interferes with the relaxation needed to cope with pain and serves to worsen anticipation about pain (known as Anticipatory anxiety) Being anxious has been found to increase the fear of pain and the likelihood that one will attempt to avoid it. That is important in medical setting because anxious patients may avoid necessary medical procedure or be combative toward their health professionals.

Frustrations related to persistence of symptoms, unknown etiology, and repeated treatment failures along with anger toward employers, insurers, the health care system, family, and themselves, all contribute to the general dysphoric mood of patients. Internalization of angry feelings is strongly related to measures of pain intensity, perceived interference, and frequency of pain behaviours (*Kerns, Rosenberg &Jacob, 1994).*

The precise mechanisms by which anger and frustration exacerbate pain are not known. One reasonable possibility is that anger exacerbates pain by increasing autonomic arousal. Anger may also block motivation for and acceptance of treatments oriented toward rehabilitation and disability management rather than cure. Yet rehabilitation and disability management are often the only treatments available for these patients. It is important to be aware of the role of negative mood in chronic pain patients because it is likely to affect treatment motivation and adherence to treatment recommendations. For example, patients who are depressed and who feel helpless may have little initiative to adhere; patients who are anxious may fear engaging in what they perceive as physically demanding activities; and patients who are angry at the health care system are not likely to be motivated to respond to recommendations from yet another health care professional.

Influence of age on pain

Pain and Older Adults

Research indicates that older adults with chronic pain often report similar levels of pain intensity when compared to younger adults with chronic pain *(Harkins & Price, 1992).* Although reductions in visual acuity, auditory sensitivity, and increases in reaction time are highly prevalent with old age, there does not appear to be a significant loss in sensitivity to painful stimuli. Although older adults experience similar sensory acuity for pain as younger adults, older adults tend to report less pain-related negative affect and suffering.

One explanation for this observation may be that older adults' reaction to pain has been influenced by their socialization history (*Whitbourne & Cassidy, 1996*). For older adults, the presence of pain may be viewed as an expected part of growing older. In addition, the fact that a person is older may mean that they had previous exposures and more experience with painful conditions and is less affected by their presence. Given these issues, the older adult may be less likely to present with significant pain complaints during a clinical assessment or treatment, as they may assume that pain is a natural part of growing older.

Physical and financial limitations often prevent older individuals from engaging in outside activities that would provide opportunities to develop supportive emotional relationships with others. Social support networks have been found to help alleviate the effects of stress, promote effective health behaviours, and influence health outcomes.

Pain in Children and Adolescents

Brief episodes of acute pain related to routine injuries and illnesses in childhood are common, with 15% of healthy school-aged children reporting brief episodes of pain *(Chambliss, Heggen, Copelan, & Pettignano, 2002).* Children's typical responses to acute pain are usually short lived and normal activity is often quickly resumed, as is typically observed with adults. However, chronic pain in children, often associated with an underlying disease, a traumatic injury, or an ongoing trauma causing sustained injury can result in unnecessary suffering of the child and family, disruption of the family routine, and restriction of the child's daily activities, thereby increasing the risk of long-term disability (*Caffo & Belaise, 2003)*. In fact, chronic pain in childhood can often result in somatic and psychiatric dysfunction, with studies showing that children experiencing chronic pain are more likely than other children to complain of anxiety, to demonstrate hypochondriacal beliefs, to engage less frequently in social activities, and to experience higher levels of generalized anxiety (*Campo, DiLorengo, & Chiappetta, 2001)*. It has been suggested that children's pain is more plastic than that of adults, such that psychosocial factors may exert an even more powerful influence on children's pain perception than on adults' pain perception (*Mc-Grath & Hillier, 2002)*.

The presentation of chronic pain in children may also differ from that of adults, and there are numerous factors that may influence the child's experience of pain, including child factors (e.g., cognitive level, or temperament), cognitive factors (e.g., expectations about treatment efficacy), behavioural factors (e.g., child's distress responses, avoidance of activities), and emotional factors (e.g., anticipatory anxiety, depression; *McGrath & Hillier, 2002).* While some of these factors are stable for a child (e.g., temperament), other factors change progressively, (e.g., age, cognitive level, physical state, and family learning). Child factors and situational factors (e.g., level of control over situation) may interrelate to shape how children generally interpret the various sensations caused by tissue damage. For example, as children grow, they learn ways to express pain and ways to cope with pain, and their experience is certainly shaped by their family, culture, and interactions with caregivers and peers. Children's ongoing physical growth may also play a role in their ability to recover more quickly than adults from injury. Pain behaviour in children has also been found to vary as a function of the child's developmental level. Older children will be able to describe the location, intensity, duration, and sensation of pain, whereas younger children may not be able to distinguish pain from other negative affective states (*Tarnowski & Brown, 1999)*. Pain behavior in children has also been found to differ depending on the presence or absence of a caregiver during a painful medical procedure, Parents' attitudes and expectations, their anxiety levels, and whether they are overly protective and reinforcing of dependence are variables that may affect children's ability to successfully cope.

Due to the number of parental variables that may influence child coping, there is a need to assess characteristics of the parent, child, and parent-child interactions when assessing pain in children.

Family Factors and Pain

Interests have developed among many chronic pain researchers in exploring the ways in which family interactions can impact the experience and course of chronic pain conditions.

Research supports the hypothesis that positive attention from a spouse (e.g., making the patient comfortable in a chair, or taking away duties) contingent on a patient's expressions of pain is associated with higher reported levels of pain and pain behaviours *(Block, Kremer, & Gaylor, 1980; Kerns, Haythornthwaite, Southwick, & Giller, 1990)*, higher frequency of observed pain behaviors *(Paulsen & Altmaier, 1995; Romano et al., 1992)*, and reports of greater disability and interference (*Flor, Turk, & Rudy, 1989; Turk, Kerns, & Rosenberg, 1992)*. In addition, there is evidence that a high frequency of negative responding to pain from a spouse (e.g., yelling, complaining, or name calling) is reliably associated with depressive symptom severity and other demonstrations of affective distress (*Kerns et al., 1990; Kerns, Southwick, et al., 1991).* There is also evidence that level of global marital satisfaction and gender *(Flor et al., 1989; Turk et al., 1992)* and depressive symptom severity and level of pain (*Romano et al., 1995)* may serve to moderate these relationships. The health care provider, in collaboration with the family, can develop a pain management plan that targets identified problems and hypothesized factors contributing to the patient's experience of pain. Optimally, the health care provider's treatment plan and its implementation should remain consistent with the family's treatment goals. The structure and functioning of a family system is likely to vary across individuals and over the course of time. Just as siblings, parents, and teachers can reinforce pain behaviours and disability in children, so too can spouses, adult children, and caregivers influence the experience of pain in older adults *(Kerns, Otis, & Stein, 2002)*.

Health care providers are encouraged to be flexible in their definition of "family," as they may find it necessary at times to include individuals outside of the nuclear family who possess high reinforcement potential (e.g., friends, neighbours, and health care providers).

Culture and Diversity Issues

Of particular importance in understanding the influence of the family on the experience of chronic pain is the cultural background and context of the family. Culture, defined as the behavioural and attitudinal norms of a group of people and the systems of meaning in which they take place, shapes a person's (or family's) beliefs and behaviours related to illness, health care practices, help-seeking behaviours, and their receptiveness to medical interventions. Culture also shapes efforts to make sense of symptoms and suffering (*Kirmayer, Young, & Robbins, 1994*). Cultural factors related to the experience of pain can influence pain expression, the language used to describe pain, coping responses, beliefs about pain and suffering, and perceptions of the health care system *(Lasch, 2000)*. Culture can also influence the types of treatments that are considered acceptable. Within each cultural group, variations in symptom attribution may affect the clinical presentation, course, and outcome of many disorders, including pain disorders *(Kirmayer et al.)*. Certainly, pain is affected by our own past experiences and the social world in which we live (*Morris, 1999)*. Culture is defined as a shared system of values, beliefs, and learned patterns of behaviours and is not simply defined by ethnicity. Culture is also shaped by such factors as proximity, education, gender, age, and sexual preference *(Low, 1984)*.

Gender difference

Men and women, boys and girls tend to differ somewhat in their experience of and reactions to the pain. Many studies have examined sex difference in threshold and tolerance for the pain. *Berkely* (1997) points out women are generally found to have slightly lower pain threshold, lower pain tolerance, and greater ability to make fine discrimination among painful stimuli. Man and woman do seems to have different attitude toward the pain due to their early experiences(*Fearon et al.1996)* .Women tend to report more pain than man ,different measure of coping with pain, and different response to treatment. The partial explanation of this variation could be hormonal. Although both men and women have estrogens, progesterone and testosterone but the level of each hormone differ in them and it tends to fluctuate monthly in case of women.

Personality difference

The personality factor most related to pain perception is neuroticism, which involves propensity to experience anxiety, emotional liability, insecurity and reactivity. Being high in neuroticism is associated not only with perception of pain but also with more pain behaviours *(Lauver and Johanson ,1997).* Another trait associated with pain perception is private body consciousness which refers to tendency to pay close attention to physical sensation.

One important personality variable relating individual reaction to painful stimuli is the one that classifies people as Augmenters or Reducers .Those who are most tolerant of pain tend to be perceptual; reducers they see stimuli around them as part of the larger "field" in which the stimuli are embedded. Reducers tend to have a body image with the definite boundaries they are extroverted, have low level of anxiety and minimize stimulation *(Sternbach, 1968*).

Augmenters characteristically perceive stimulation as greater than average. Also termed "sensitizers" they respond to external stimuli directly by trying to do something to deal with it .Reducers, on the other hand, play down external stimulation and deny it *(Goldstein, 1973)* not all research verifies perceptual; reactance in pain perception however.

MANAGEMENT OF THE PAIN

Pharmacological control of Pain: The traditional and most common method of controlling pain is through the administration of drugs. In particular, Morphin (named after Morpheus, the Greek god of sleep) has been the most popular painkiller for decades.(Melzack & Wall,1982)Some drugs ,such as local anesthetics ,can influence transmission of pain impulse from the peripheral receptors to the spinal cord. The injection of drugs such as spinal blocking agents that block the transmission of pain impulse up the spinal cord is another method. Paracetamol (acetaminophen), or a non steroidal anti-inflammatory drug (NSAID) such as ibuprofen are preferable in mild pain. Fentanyl is convenient for chronic pain management. Oxycodone is used across the Americas and Europe for relief of serious chronic pain. Pentazocine, dextromoramide and dipipanone are not recommended in new patients except for acute pain where other analgesics are not tolerated or are inappropriate, for pharmacological and misuse-related reasons. Amitriptyline is prescribed for chronic muscular pain in the arms, legs, neck and lower back. While opiates are often used in the management of chronic pain, high doses are associated with an increased risk of opioid overdose

Methadone can be used for either treatment of opioid addiction/detoxification when taken once daily or as a pain medication and as such is usually administered on an every 12-hour or 8-hour dosing interval.

Some antidepressant and antiepileptic drugs are used in chronic pain management and act primarily within the pain pathways of the central nervous system, though peripheral mechanisms have been attributed as well.

Surgical control of Pain: It involves cutting or creating lesion in the so called pain fibers at various points in the body so that pain sensation can no longer be conducted. Some surgical techniques attempt to disrupt the conduct of pain from the periphery to the spinal cord whereas others are designed to interrupt the flow of pain sensation from the spinal cord upward to the brain. But sometimes the effects are short lived and can ultimately worsen the problem it damages the nervous system.

Sensory control of Pain: One of the oldest known techniques of pain control is counter irritation, a sensory method. It involves inhibiting pain in one part of the body by stimulating or mildly irritating another area.

Behavior Therapy

Behavior therapy talks about conditioning between pain and pain related behaiour and how and when it is reinforced. The goal of behavior therapy is to mitigate excessive problematic pain-associated behaviors (e.g., excessive medication usage, limping) and increase those adaptive behaviors occurring infrequently or not at all (e.g., walking, exercise, self-care, work). Behavior therapy is found to be less well controlled in outpatient settings. The assistance of others is needed to ensure that environmental contingencies are systematically applied at home or in other relevant settings.

Steps involved in behavior therapy

Step 1: Define problem behaviors (operants) that warrant attention (e.g., medication use, excessive reclining, and avoidance of activities).

Step 2: Determine the relationship between operants and environmental consequences (i.e., identify reinforcers and the temporal contingencies that exist to maintain these).

Step 3: Assess whether the link between operants and reinforcers is modifiable (i.e., identify whether reinforcers can be modified so that they become contingent on desired [adaptive] behaviors, identify how frequently reinforcers should be applied after desired behaviors occur, identify what quotas might need to be established, identify those persons who should be involved in the contingency management process [e.g., spouse, significant other]).

Step 4: Establish how the systematic disruption of problem behaviors and consequences can be conducted (i.e., how to extinguish undesired behaviors [which reinforcers should be with held and when]).

Step 5: Establish transferability to home and work (i.e., consider to what extent the contingencies can be translated into the home, work, or any setting in which it becomes necessary to maintain these desired behaviors; perform follow-up assessments; determine if contingencies need to be modified in other settings; determine if the newly learned behaviors have been extinguished and whether these can be reinstated).

Limitaions:

- ✓ Behavioral modifications might not be sustained. Setbacks may occur once the desired outcome is achieved or the incentive is gone. Alternative reinforces (e.g., praise, attention) may need to be set in place to maintain the behaviors. The behavioral changes can dissipate in settings (e.g., home, work) in which reinforcement patterns are less systematic or consistent.
- ✓ In addition, despite the changes acquired in behavior therapy but what is modified is the overt behavior, not the perception of pain.
- ✓ The behavioral approach is simplistic—specifically, the idea that the person engages in behavior because he or she comes to expect a particular outcome. Thus, behavior therapy fails to factor in those qualities of being human that influence and dictate behavior. For example, expectation, anticipation, thinking, planning, and remembering can also influence behavior and mediate pain-related behavior and perception (Seligman 1990).

Cognitive-Behavioral Therapy

CBT is focusing upon the correction of distorted thinking processes and the development of strategies (coping) with which to deal effectively with pain, its effects, and psychosocial stressors. CBT is effective in a number of different chronic pain problems, including low back pain, headache, fibromyalgia, osteoarthritis, rheumatoid arthritis, and temporomandibular joint disorders (Turner and Chapman, 1982).

CBT focuses on internal appraisals of pain and disability by examining and addressing the cognitions, emotions, and behaviours associated with pain and pain-related activities and modification of maladaptive cognitions and beliefs (schemata) and the development of effective coping strategies. CBT therapy is structured, with clear agendas set by the patient and therapist focusing on prominent areas of concern for the patient, conducted in about 8–12 fifty-minute sessions. Here the therapist is directive, guiding the use of homework treatments, outlining exercises, and assessing the efficacy of the modalities employed, yet this role remains flexible, with the patient's input guiding any shifts undertaken in the therapy. CBT assumes a collaborative effort between the patient and therapist (Fishman and Loscalzo, 1987).

CBT aims at giving directions and instructions to the patient to reappraise thoughts and events occurring in their life experiences. Faulty appraisals and misattributions are reframed and replaced with those that are reality-based (i.e., less irrational).

Homework assignments help the patient and therapist identify those situations, moods, feelings, and thought processes associated with pain.

Cognitive restructuring & coping skill training:

Cognitive restructuring, an interactive process involving the Socratic method, is used to teach patients to identify and modify maladaptive, negatively distorted thoughts that may lead to negative feelings, such as depression, anxiety, and anger. The patient is encouraged to examine irrational, self-defeating thoughts and discriminate between these and more rational alternatives. Coping skills training is aimed at helping patients develop a repertoire of skills for managing pain and stress and providing patients with a general set of problem-solving or coping skills that can be used in a wide range of situations that induce pain. The therapist helps the patient develop a broader range of effective coping strategies by examining existing coping strategies, determining their effectiveness, and facilitating the development of a broader range of strategies. CBT has the advantage of broad applicability in a number of situations. It is a relatively low-cost intervention and appears to be cost-effective.

Distraction:

Individual who are involved in intense activities such as military or sports can be oblivious to painful injuries. These are extreme examples of a commonly employed pain technique- distraction. By focusing attention on an irrelevant and attention getting stimulus or by distracting oneself with high level of activity one can turn attention away from pain. They are two quite different mental strategies to control discomfort- first, by focusing on another activity or to focus directly on the event but to reinterpret the experience.

Limitation It requires sustained active patient participation. Also, therapists need to have specialized training in CBT in order to use it effectively.

Mindfulness:

Mindfulness, breathing, imagery and visualization are effective for stress, anxiety, depression and pain. The practice of mindfulness develops ability to control attention and regulate emotions. Being aware of the present moment is a powerful method of directing our focus on what truly exists around us as opposed to thinking about the past or future.

Using mindfulness to change the outcome means that you become aware of the process and that it is a thought that is driving the physical responses and behavior. As human beings we have the unique ability to think about our own thinking and to be able to imagine and play out scenarios in our head. So it help us to overcome the painful feeling.

Various techniques of Mindfulness:

- Imagery
- Visualization

Supportive therapy:

In supportive therapy therapist undertake a warm, reflective, and empathic approach to reduce patient distress and reassure the patient that he or she is understood and that the magnitude of his or her plight is appreciated. The therapist emphasize that modification and improvement in functioning are essentially the patient’s responsibility. This emphasis can enhance the patient’s sense of personal control and self-efficacy, which is critical for overcoming the tendency of the patient to succumb to powerlessness and helplessness. Therapist can give advice for strategies to reduce discomfort, improve sleep, address medication use, and develop pain-modulating interventions (e.g., relaxation training). These advices fosters the notion of the patient's responsibility and autonomy by allowing the patient to select those aspects of treatment that are most appealing, thus possibly facilitating patient compliance (Miller and Sanchez 1994).

Ongoing follow-up with supportive therapy is required. Follow-up reinforces the notion that the patient and physician or clinician are working in concert to effectively mitigate pain and optimize adaptive functioning while restoring pleasure and balance and bolstering relationships in the patient's life. With this reinforcement, the patient can avoid the potential pitfall of viewing himself or herself as passive and helpless in the treatment process.

<u>Group therapy:</u>

Groups lead to the mutual sharing of experiences, provide an opportunity to learn from the experiences of others, foster education about treatment strategies, and foster coping strategies. Thus Group therapy can serve to diffuse the sense of isolation experienced by the pain patient. The shared experiences of persons in the group can offer the patient a sense of being fully understood, often without the requirement of having to explain his or her experiences or emotions to others who do not share the same level of physical pain.

Groups can be of either ***inclusive or exclusive.*** Some groups are exclusive—that is, organized around particular disorders (e.g., groups for persons with fibromyalgia, arthritis, or cancer pain). Others can be inclusive—broader and open to persons with recurrent or chronic pain of various sorts. The purposes of group therapy include reducing feelings of isolation (i.e., universalization) and providing a forum in which mutual support, information exchange, advice, modeling of effective coping, and abreaction of emotional experiences are possible. The structure of the group sets the stage for and dictates the focus and perspective of the participants so the rules of the group process need to be delineated early. If patients are allowed to ventilate distress and can respond to the support and advice offered by peers, attendance in therapy sessions can be a source of inspiration and empowerment. A special benefit arising from group process is *abreaction*, whereby repressed, emotionally laden experiences are brought to conscious awareness (i.e., re experienced) and in the process, insight is gained.

Other techniques or intervention

Acupuncture:

Acupuncture has been in existence in China for more than 2000 years. In Acupuncture treatment long thin needles are inserted into specially designated areas of the body that theoretically influences the areas in which a patient is experiencing the disorder. Although the main goal of acupuncture is to treat the illness, it is also used in pain management because it appears to have an analgesic effect. Researchers believe that Acupuncture works partly as a sensory method of controlling pain as well as it is associated with other psychologically based techniques for pain control.

Biofeedback:

Biofeedback refers to a procedure in which physical parameters (e.g., muscle tension) are continuously monitored and fed back to the patient, who then attempts to alter the physical parameter. For example, an individual attempting to regulate and modify the degree of muscle tension in forehead muscles would have electrodes placed on the forehead. The electrical signals from these electrodes would be relayed to a monitor and presented in any of a number of formats (e.g., visual, auditory). The patient, attending to the signal, would then use the information presented to develop strategies to reduce muscle tension.

Biofeedback from electromyography assists the patient in learning to reduce muscle tension; the levels of measured muscle tension are signaled back to the patient for modification. This technique is useful in tension headache, temporomandibular joint disorders, fibromyalgia, and other myofacial pain disorders. Thermal biofeedback monitors skin temperature to give the patient an indicator of the degree of peripheral vasodilation.

Relaxation and Imagery Training:

Relaxation and imagery (R&I) has been employed in both acute and chronic pain and has been successfully implemented in the treatment of tension headache, migraine headache, temporomandibular joint pain, chronic back pain, and myofascial pain syndrome. Progressive muscle relaxation (PMR) is the most common approach used. imagery involves talking the patient through vivid images that are particularly comforting and relaxing. Just as imagining the taste and aroma of a craved meal or an erotic thought can induce a dramatic constellation of physiologic responses, so too, it is thought, can guided imagery modify physiologic reactions to pain. The patient is asked to imagine a place that he or she finds most relaxing in order to set the stage for being in an emotional state that would be inconsistent with anxiety or tension. Deep breathing exercises involve the direction of deep breathing in a manner that potentiates a deepened relaxation.

Hypnosis:

It is a self-induced state brought on by the patient with the assistance of the hypnotist. The mechanism of pain relief brought on by hypnosis is unclear. Questions arise as to whether it is the pain (unpleasant sensations) that is removed or, rather, the reactions to normally painful stimuli. Current conceptualizations view hypnosis as a form of focused attention that is useful in managing acute and chronic pain states, including headache, fibromyalgia, back pain, trigeminal neuralgia, arthritis, phantom limb pain, and cancer pain. Analgesia produced in response to hypnosis was thought to be brought on by modification of attention control systems (i.e., anterior frontal cortex) within the brain.

Analgesia produced through hypnosis is an active process involving the patient's full cooperation. Imaging technology has supported that the process of hypnosis produces signal changes in areas of the brain concerned with sensation and perception (i.e., the sensory cortex and thalamus) and areas of the brain where sensory information is integrated. Hypnotic responsiveness varies considerably from person to person but appears to be a stable trait. Because some patients are less susceptible to hypnosis than others, hypnosis might not be a suitable intervention for every patient with complex pain. Hypnosis requires active psychological engagement with the patient in order to be effective.

Vocational Rehabilitation:

Patients with pain can experience significant losses, including the loss of work. Beyond the obvious resultant loss of income, and perhaps medical coverage, the loss of work can imply several other losses for the patient depending on the meaning of work for that person. For many persons, work is a source of self-identity, power, influence, and a social network. Loss of work, therefore, can mean the feared loss of control and a feared loss of usefulness.

Vocational rehabilitation may serve an instructional role, helping the patient develop skills that will be suitable for other kinds of work. Vocational rehabilitation also may be helpful in fostering the patient's independence, autonomy, and self-efficacy and may assist the patient particularly when there is financial need, ineligibility for disability or compensation, and the need for insurance and medical coverage. Vocational rehabilitation can help the patient go beyond the boundaries of his or her illness.

Many state agencies provide vocational and educational services for persons with disabilities that can help patients adapt to a work setting. Within such programs, the patient's interests, skills, aptitudes, limitations imposed by physical and psychiatric conditions, and employment needs are assessed. On the basis of these assessments, work training and supervision are provided, with eventual job placement being the ultimate goal.

DISCUSSION:

As the importance, intensity, complexity and cost of pain have been seriously recognized pain is now taken more seriously in the medical as well as psychological perspective. Now it is recognized as an important psychological issue in its own right rather than the inconvenient symptom it was once regarded to be. Till now, even with growing technology and development in the health care system, the management of pain is difficult. We are still going through the causes, models and perspective on pain and pain related behaviour. As a mental health professional it's our responsibility to frame a person, came with pain or pain related disorder, in his psychological world. Psychological management of pain along with pain killers found to be very effective in these cases as pain leads to severe emotional problems as person feels dependent, helpless and frustrated. A substantial literature exists documenting the relationship between chronic pain and comorbid conditions such as depression anxiety and addictive disorders (Brown, Patterson, Rounds, & Papasouliotis, 1996). Depression, in particular, has been noted to be a common factor in the experience of pain, with prevalence rates estimated to be 30 to 54% in chronic pain samples (Banks & Kerns). Studies that have evaluated behavioural intervention in comparison with non treatment, have found that the behavioural interventions reduce report of pain disability and psychological distress. These interventions improve psychological and social functioning as well.

The most effective pain treatment programme involves a combination of behavioural, cognitive, physical and analgesic approaches to pain. ***Pain clinics*** can be quite effective in achieving the goal of pain reduction. These clinics aims at managing pain with multidimensional programme which involves initial evaluation, individualized treatment, and many other components like involvement of family and relapse prevention are also involved.

CONCLUSION:

Pain insists upon being attended to, it has been identified as a cause of global disability in human beings which can be a result of physiological as well as psychological factors. In history pain has been identified as a productive and protective phenomenon. There are different factors playing important role in the perception of pain. Person's age, gender, family and other factors are crucial in the development of pain behaviour. Not only these factors but our bodily functions, at the level of nervous system and neurotransmitters, are important to understand the causes of pain. Individual's cognition, belief, kinds of reinforcement, family dynamics, all collaborate together to decide the occurrence, persistence and management of pain and pain behaviour. As pain behaviour and its perception is closely related to our psychological world and in turn influencing it too so the role of a clinical psychologist becomes very crucial in pain management. Pain management programme appears to play an important role in helping chronic pain which includes multidisciplinary interventions. Most important thing is to fight with pain, as Lance Armstrong, the famous cyclist after winning the battle with Cancer truly said "***Pain is temporary, quitting lasts forever.***"

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UNIT-XI

TERMINALLY ILL

PRESENTEE:

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DEFINITION : TERMINALLY ILL

A status assigned to a person who has been diagnosed with an illness and is expected to die within a certain time frame, usually six months. Terminal illness is a disease that cannot be cured or adequately treated and that is reasonably expected to result in the death of the patient within a short period of time. This term is more commonly used for progressive diseases such as cancer or advanced heart disease than for trauma. In popular use, it indicates a disease that eventually ends the life of the sufferer.

A patient who has such an illness may be referred to as a terminal patient, terminally ill or simply terminal. Often, a patient is considered terminally ill when their estimated life expectancy is six months or less, under the assumption that the disease will run its normal course. The six- month standard is arbitrary, and best available estimates of longevity may be incorrect. Consequently, though a given patient may properly be considered terminal, this is not a guarantee that the patient will die within six months. Similarly, a patient with a slowly progressing disease, such as AIDS, may not be considered terminally ill because the best estimates of longevity were greater than six months. However this does not guarantee that the patient will not die unexpectedly early. In general, physicians slightly overestimate the survival time of terminally ill cancer patients, so that, for example, a person who is expected to live for about six weeks would likely die around four weeks.

MEDICAL DEFINITION OF DEATH AND DYING:

The medical definition of death is a major medical and legal issue, and an important issue in declaring a person legally dead. The specific criteria used to pronounce legal death are variable and often depend on certain circumstances in order to pronounce a person legally dead. Controversy is often encountered due to the conflicts between moral and ethical values. A state defined in the US by the Uniform Determination of Death Act, 1981, as that which occurs in an individual who has sustained either

(1) Irreversible cessation of circulatory or respiratory functions, or

(2) Irreversible cessation of all functions of the entire brain, including the brain stem.

(Segen's Medical Dictionary. © 2012 Farlex,)

SPIRITUAL DEFINITION OF DEATH:

Spiritual Death is the separation of the soul (spirit) from God.

Spiritual death is "a state of being in which the human soul is separated from God and has not been enlivened by his Spirit."

RELIGIOUS DEFINITION OF DEATH:

There are many different religions and belief systems across the world For Hindus, death represents a spiritual opportunity to attain oneness with God. Death is a component of the natural life cycle - life, death and rebirth.

In the history of Christianity, death has been defined generally as the separation of the immortal soul from the mortal body.

The whole life of a Muslim constitutes of a trial and test by means of which his final destiny is determined. For him, death is the return of the soul to its Creator, God, and the inevitability of death and the Hereafter is never far from his consciousness. This serves to keep all of his life and deeds in perspective as he tries to live in preparedness for what is to come. For Muslims, the concept of death and the afterlife in Islam is derived from the holy *Qur'an*, the final revealed message from God.

_“ hen the human’s brain has stopped functioning completely, drugs and a respirator are keeping his heart beating and his or her lungs inflating.”Usually, the same way it has traditionally been defined in all cultures: by a lack of vital signs. Most world religions lack a clear doctrinal statement that certifies when, exactly, the moment of death can be said to have occurred. For most of human history, there was no need for one since prior to the invention of life-support equipment, the absence of circulation or respiration was the only way to diagnose death.

This remains the standard of death in most religions. By the early 1980s, however, the medical and legal community also began to adopt a second definition of death—the irreversible cessation of all brain functions and some religious groups have updated their beliefs.

PSYCHOLOGY OF DYING AND BEREAVED FAMILY:

A dying patient typically passes through five successive stages. Denial and Isolation, Anger, Bereavement, Depression and Acceptance. Bereavement is highly stressful and is associated with impaired immune functioning and a raised mortality, particularly from cardio vascular disease. Grief the psychological response to Bereavement is a process with its own successive stages of denial, pining, depression and finally acceptance. Grief occurs after all forms of loss, not only those involving death and takes similar if less extreme forms in all cases.

Kubler-Ross published and legitimized the study of death while promulgating more humane and sensitive treatment of the terminally ill. Her extensive interviews with 400 dying patients reveal that patients know without being told that they are dying; that they need to talk about it; and that they need to maintain hope, even if there is no hope of cure. Kubler-Ross's work reminds clinicians that fatal illness is not a sudden event, that dying patients retain their humanity, and that practitioners can provide support to patients by assuring them that they will maintain their relationship with them during the dying process. Kubler-Ross postulates five stages that many dying patients pass through from the time they first become aware of their fatal prognosis to their actual death.

1. ***Denial***. "No, not me!" is the dying patient's common initial response. If it doesnot interfere with treatment, denial can mitigate the initial overwhelming anxiety.

2. ***Anger***."Why me?" Indignation may surface when denial subsides. Patients are irritable, demanding, and critical; anger may be directed at themselves, caretakers, family and friends, or God.

3. ***Bargaining.*** "Yes me, but . . ." This stage entails promises to buy additional time. Patients may promise to donate their organs to research or they may reaffirm an earlier faith in God.

4. ***Depression.***"Yes, me." The patient comes to a full realization of what is going to happen and to whom. With the impending loss of life, a pervasive despondency may set in.

5. ***Acceptance.*** The patient begins to accept the inevitable. This stage need not constitute defeat or total surrender. "Yes, me, and I'm ready."

These five stages are not all encompassing or prescriptive. Not everyone will reach these stages; perhaps only a few will reach acceptance. A patient may demonstrate aspects of all five stages in one interview or may fluctuate between stages. Moreover, patients may exhibit other coping methods—such as terror, humor, or compassion—to offset each stage.

STRATEGIES OF BREAKING BAD NEWS

Breaking the News After diagnosis and prognosis have been made, physicians need to tell the patient and families. Formerly, doctors subscribed to a conspiracy of silence, believing that the less patients knew, the better their chance for recovery; it was believed that news of impending death brought despair, so the truth was withheld. The current policy is honesty and openness; the question is not whether to tell the patient, but when and how. The American Hospital Association in 1973 drafted the "Patients' Bill of Rights,"declaring that patients have a "right to obtain complete current information regarding diagnosis, treatment and prognosis in terms the patient can be reasonably expected to understand." Full disclosure is mandated whether the patient requests it or not.

In breaking the news of impending death to the patient, diplomacy dictates the following guidelines: (1) The health team physician should participate. (2) The spouse or partner should be present if possible. (3) Relatives need comfort, as does the patient. (4) Use simple words, even with educated patients. (5) Show compassion and emotional support, avoiding bluntness or abruptness. (6) Guessing how long a patient has to live may be inaccurate and inadvisable. (7) Encourage and answer questions, signaling one's availability for honest communication. (8)

Truth is not the enemy of hope. (9) Communicate a willingness to see the patient through until death occurs. (10) Explain the situation and introduce the next step. A gentle, sensible approach helps to modulate the patient's own denial and acceptance. The physician must choose how much information to give and when, on the basis of the patient's needs and capacities.

BEREAVEMENT AND GRIEF COUNSELLING

When a person's grief-related thoughts, behaviors, or feelings are extremely distressing, unrelenting, or incite concern, a qualified mental health professional may be able to help. Therapy is an effective way to learn to cope with the stressors associated with the loss and to manage symptoms with techniques such as relaxation or meditation.

Each experience of grief is unique, complex, and personal, and therapists will tailor treatment to meet the specific needs of each person. For example, a therapist might help the bereaved find different ways to maintain healthy connections with the deceased through memory, reflection, ritual, or dialogue about the deceased and with the deceased.

In addition to individual therapy, group therapy can be helpful for those who find solace in the reciprocal sharing of thoughts and feelings, and recovery results are often rapid in this setting. Similarly, family therapy may be suitable for a family whose members are struggling to adapt to the loss of a family member.

MANAGEMENT

Management of physical symptoms

Physical symptoms other than pain often contribute to suffering near the end of life. In addition to pain, the most common symptoms in the terminal stages of an illness such as cancer or acquired immunodeficiency syndrome are fatigue, anorexia, cachexia, nausea, vomiting, constipation, delirium and dyspnea. Management involves a diagnostic evaluation for the cause of each symptom when possible, treatment of the identified cause when reasonable, and concomitant treatment of the symptom using non pharmacologic and adjunctive pharmacologic measures.

Management of psychological symptoms.

Psychotherapy with dying patients shares many features with all other psychotherapy. However, the unique status of the dying person presents special problems for the mental health professional. Clearly, everyone will die, and in this sense all therapy is done with patients of a limited life span. The labeling of a person as a "dying patient", identifies that person as belonging to a special category of humanity, and creates profound changes in the emotional, social, and spiritual climate of therapy. The dying person is one who is seen to be in a life- threatening condition with relatively little remaining time rind little or no hope of recovery. This unique existential position of the dying person necessitates some adaptations of the typical psychotherapeutic attitudes and strategies. The goals, structure, and process of therapy must change to meet the special needs and circumstances of the dying patient.

How does therapy with a dying person differ from "typical" therapy? There are several features which distinguish it.

First, therapy is more time-limited and time-focused. The dimension of time takes on special urgency with the dying patient. While many therapies are time-limited, often they proceed as if time were an inexhaustible resource. The brief remaining time for the dying patient intensifies the therapy process, and accelerates it.

Second, the goals of therapy with dying patients are often more modest. Recognizing the limits of possible change is an essential feature of therapy with the dying. What can be accomplished is quite restricted by time, disability, and other aspects of the patient's condition.

Third, the treatment of the dying patient often requires careful coordination with a variety of medical, nursing, and pastoral professionals. The physical condition, medical treatments, and institutional settings of the patient complicate the practical and psychological context of therapy.

THERAPEUTIC APPROACHES

Reflecting the increased maturity of the field, there are presently many therapists and researchers focusing on this population, and in addition several scholarly journals which devote some attention to the care of the dying person. Psychotherapy is beginning to be incorporated into the more general and growing field of clinical thanatology, which is concerned with the overall care and treatment of the dying person - mind, body, and spirit.

Modern psychotherapies are divided into four main groups - psychodynamic, humanistic, behavioral, and family therapy.

The main features of these therapies as used with all patients are preserved in the treatment of the dying, but each has been modified somewhat to fit the unique needs of dying persons.

PSYCHODYNAMIC THERAPIES

The psychodynamic approaches are primarily concerned with the emotional conflicts and defense mechanisms of the individual. Special issues of conflict and defense arise in the dying person, and this approach addresses them in the hope of resolving the psychic crisis to the fullest extent possible. Dying is the ultimate crisis of ego development, and as such is associated with intense infra-psychic turmoil. Psychoanalyst Erik Erickson labels the last stage of ego development, "ego integrity versus despair", and identifies it with the crisis provoked by the confrontation with one's mortality. The fear of death may precipitate a breakdown of previously integrated ego functioning, and result in an attitude of despair and disgust.

In most people the threat of death generates powerful defensive reactions, and although these defenses provide some limited relief of emotional distress, in the end they prohibit the person from effectively coping with the death crisis. Common defenses which are found in the dying person include denial, displacement, projection, and regression. As Kubler-Ross pointed out, denial is a very typical reaction of the dying person. The refusal to accept the reality of death makes it impossible for people to prepare themselves and their families adequately for it.

Through the displacement defense the fear of dying is channeled into other, "substitute" fears. For example, one may become preoccupied with anxiety about family members, personal business, household jobs, or other matters, and, thus, obtain partial release of one's death anxiety. The dying person's projection defense typically expresses itself in hostility and resentment toward others, e.g., doctors, nurses, and family. The person may irrationally blame others for the illness, or accuse them of not doing enough to cure or help. Regression in the dying person is often manifested in increasingly immature, dependent, and occasionally self-threatening behaviors and attitudes. An example is the extremely helpless, "infantilized" position of the person who has completely given up and merely waits for death.

A major goal of dynamic therapy with the dying is to help the person recognize, confront, and replace the defenses which run counter to an emotionally healthy attitude toward death. In the process it may be necessary to try to work through some long-standing problems and fixations which are intensified by the death crisis.. Dynamic therapy with dying patients is not directed as much toward the goal of insight, as it is with others. Time limits the course of therapy with the dying, and the goals are therefore more short term changes; rather than long-term personality change.

THE HUMANISTIC THERAPY

More than other approaches the humanistic view of therapy clearly integrates a philosophy of human nature in which death plays an essential role. Existentialism is a philosophy which has had a significant effect on the humanistic approach, and in this philosophy living the "good life" demand a confrontation with the reality of death. Death awareness helps us to clarify our values and purpose in life, and motivates us to live our lives with fullness and meaning. Death is the absolute existential threat, and it forces us to acknowledge the limit of our life plans and face "nothingne".

Humanistic therapy aims to help the dying patient live as full a life as possible in the face of death. Without giving false hope or optimism, the therapist attempts to mobilize the patient's will to live, to encourage the expression and growth of the self, and to facilitate the patient's self- actualization .

THE BEHAVIORAL TECHNIQUES

The behavioral approach to therapy relies on educating patients about more adequate coping skills to help deal better with the death crisis. Impending death is a terribly stressful situation, and it produces extreme emotional reactions like anxiety and depression, which inhibit patients from living out the remainder of their lives in a satisfactory way. The symptoms of the dying patient are partially manageable through some standard behavioral techniques. For example, relaxation training and desensitization can help to alleviate excessive fear and tension. Other self-management skills, like biofeedback and self-hypnosis, are also useful in controlling the distressing emotions of the patient.

One example of a valuable behavior therapy technique is "stress inoculation training". With the dying patient this strategy may be used to help cope with the physical and emotional aspects of pain. In this approach the patient is taught how to employ cognitive and behavioral skills in preparing for pain and managing pain.

FAMILY THERAPY

The impending death of a family member places the entire family in a state of crisis. Death presents a threatening situation for each member of the dying person's family. The degree of disturbance in the family depends on many factors such as the role of the dying member, the stage of development of the family, and the quality of relationships among family members. A family systems approach conceives of the entire family, not just the dying person, as the recipient of therapy. This approach seeks to provide the family unit the opportunity to learn to deal with the tragedy. Some therapists will continue treatment beyond the death, offering grief counseling for the survivors.

Though family therapy may be integrated into therapies of various types, there are several issues on which family therapists are more likely to focus. Lying patients often experience a need to feel the closeness and support of their families in facing the death crisis. In families where past conflicts have interfered with relationships between the patient and others, family therapy can facilitate more open and productive communication. This can benefit all members concerned in terms of finding closure for "unfinished business". The defenses of family members can make it very difficult for the dying patient to confront death. It often happens that family members share the defensive reactions of the dying person, such as denial of the facts and displaced anger.

An advantage of the family approach to therapy is that it offers an experience that may enable everyone to accept the facts and to work together to enhance the quality of life for the dying person. Families generally experience a range of intense emotions regarding the dying patient, including anger, guilt, fear, and depression. In family therapy members are encouraged to understand and express these feelings in anticipation of the death of their loved one.

EXISTENTIAL PSYCHOTHERAPY.

Existential Psychological Theory and Existential Psychotherapy have not been directly cited in the literature with the terminally ill; however, several authors have reported efficacious results when the focus on principles of Existential Psychological Theory. Levitan (1985) developed a hypnotic procedure called "hypnotic death rehearsal" that was designed to address and resolve the existential principle of death anxiety. Rosenberg (1983) employed revivification of past accomplishments in the lives of terminally ill cancer patients as a vehicle with which to approach and resolve the existential sense of meaninglessness. Kaye (1984) demonstrated the value of using hypnosis and family therapy to effectively deal with existential isolation in a cancer patient.

Spiegel, Bloom, and Yalom (1981) provided empirical evidence supportive of existential therapy (1977) prognostications regarding the benefits of meeting the emotional needs of the terminally ill. Their study shows the effects of existential therapy for terminally ill cancer patients and demonstrated benefits in patients' self-esteem, mood and affect, overall efficacy of coping capacities and enhancement of the patients' quality of life.

HOSPICE CARE

Dying patients may choose hospice care. A holistic and philosophical approach to end of life care, hospice brings doctors, nurses, social workers and other professionals together as a care team. The hospice team's goal is to make the patient as comfortable as possible during his or her final days. Hospice emphasizes pain control, symptom management, natural death, and quality of life to comfort the patient's physical body. Nearly all definitions of a "good death" respect the principle of autonomy and encourage helping an individual choose and participate in decisions about medical options at the end of life. (Autonomy is an individual's ability to control situations and circumstances). Part of the philosophy of hospice involves restoring and supporting both the patient and his or her family's control over the circumstances of death.

The hospice team cares for the dying patient wherever that patient is at home, in a nursing home, in a hospital, or in a separate hospice facility. In addition to medical care, the hospice team may provide emotional and spiritual support, social services, nutrition counseling, and grief counseling for both the patient and loved ones.

Key to a hospice professional's self-care is the ability to fully enter into relationships with patients while maintaining one's personal life and well-being. Challenges to this balancing act include preserving the professional relationship framework and managing powerful emotions evoked in hospice work.

Several aspects of hospice work with older adults may result in making exceptions to usual professional limits, making the relationship more personal. These include the sense of urgency and finality of death or being with patients in their homes during this significant life juncture. Patients' expectations, desire for a mutual relationship, or quest for companionship may also result in extending the usual limits.

Professionals can feel pulled to give or receive gifts, extend the time of visits, or share more personal information than usual. Training that includes practice handling such situations is particularly helpful.

Hospice work with older adults sometimes taps into feelings and unresolved issues from many sources. Emotions may be evoked regarding parents, grandparents, or other older adults in the lives of professionals. Some professionals may unconsciously enter this field partly to fulfill unmet childhood desires for approval, love, or recognition or to access someone to admire and emulate. While hospice work sometimes results in feeling loved and appreciated by clients, this unconscious motivation can also lead to over involvement in an attempt to fill a void. Emotions evoked in hospice work hold the potential to enhance helpers' skills or, if kept outside awareness, interfere with a clear view of patients and their needs. Therefore, hospice professionals must acknowledge their own vulnerability and the need to process their feelings, particularly grief, along with the associated pain and enrichment it includes. This improves professionals' self-care because they have reservoirs of resources with which to respond empathetically and clearly to clients' needs rather than distancing themselves from clients or overinvesting to meet their personal needs.

Self-Care Plan

The challenges of hospice work make self-care planning a wise choice and another fringe benefit. It involves mapping out a plan that addresses individual physical, emotional, cognitive, relational, and spiritual strengths and challenges (Jones, 2005), serving as a guide through the ups and downs of a hospice career to prevent burnout, maintain motivation, and address obstacles.

Physical Self-Care — Listening to the Body

Since stress is experienced physically, it is important to identify where stress manifests itself in the body, routinely check vulnerable areas, and find effective ways to counteract physical stress with relaxation. A variety of methods exist, including simple breathing techniques (Weil, 1990), progressive muscle relaxation, acupressure, massage, exercise, yoga, and meditation (Benson,1995; Davis, Eshelman, & McKay, 2000; Kabat-Zinn, 1995; Keating, 2002). Attending to ongoing difficulties, such as depression or insomnia, is included. New hospice professionals are susceptible to anxiety that they or loved ones have a terminal disease (Larson, 1993). Professionals need to recognize this as a common attempt to integrate heavy exposure to terminal illness and channel these worries into preventive action based on their own or loved ones' specific disease predispositions.

Emotional and Cognitive Self-Care — Express, Soothe, Release

Emotional self-care includes maximizing energizing emotions and processing grief, routinely letting it in and out of one's life. Identifying individual emotional stress indicators, such as increased crying, irritability, anxiety, numbness, self-doubt, or addictive behaviors, is important. Key to emotional self-care is routinely expressing, soothing, and releasing emotions. Allowing for more frequent crying may be appropriate for hospice professionals, even if a movie or music is needed to "jump-start" a good cry. Other methods include writing, creating, listening to music, talking with confidants, enjoying hot baths, being held, or cuddling a pet. Aromatherapy, massage, meditation, mindfulness, prayer, gardening, and cleaning offer other emotionally soothing outlets. Allowing time to soak up joyful times and successes or engaging in pleasurable activities and humor is energizing.

I recommend a simple, brief, daily release ritual to intentionally let go of emotions that professionals often carry home from clients, particularly the heavy emotion of grief. The ritual includes acknowledging the detriment of carrying others' emotions, reviewing the day's situations, and letting them go. This can be done while listening to music on the drive home or before sleep, changing clothes after work, meditating or praying, visualizing the day's concerns going down the drain while showering, or getting farther away while running or walking.

Since thoughts affect emotions, self-care includes healthy internal dialogue. Keeping a log of thoughts for one week identifies harmful patterns that, for example, polarize, self-denigrate, blame, or expect perfection, especially related to challenging hospice situations. Distorted thought patterns are then replaced with reasonable alternatives or at least with challenges to the veracity of destructive thoughts. Supervisors and peers can offer valuable feedback. Professionals may also model their internal dialogue on how they talk to loved ones or valued colleagues.

Relational Self-Care — Support, Support, Support

The emotionally demanding work of hospice care makes a strong support system essential. Stress responses include increased irritability, distance, or dependence. Finding those able to listen and support is crucial. It is helpful to educate significant others about work stresses, when "it's about work, not about you," and ways they can offer meaningful support. This means knowing what you need and being able to ask for it, which is often difficult for professionals. In addition, self-care requires setting healthy limits in personal and professional relationships. Helpful tools include identifying warning signals of overextending, practicing setting limits, and handling conflicts by dealing directly with the person when an issue first arises, while remaining focused on solutions without blaming or personalizing.

Regularly scheduled supervisory and peer sessions are vital to preventing burnout and compassion fatigue, to the extent they provide positive, constructive feedback that assists in managing emotions, maintaining confidence and self-esteem, normalizing experiences, and developing new resources and coping methods (Leon, Atholz, & Dziegielewski, 1999; Keidel,
2002; Poulin & Walter, 1993). In addition, participation in political advocacy to address gaps in care is an outlet for frustration over inadequate resources.

Spiritual Self-Care — Tuning In to the Bigger Picture

End-of-life work is often spiritually rejuvenating, since it involves clients' big-picture concerns. Sometimes, the big picture gets lost in the details of paperwork and finding resources, requiring renewed attention to one's connection to the meaning of life and hospice work. Staying attuned spiritually includes reading sacred texts, praying, attending services, connecting to nature, listening to music, meditating, and engaging in creative endeavors. Since hospice work with older adults involves a heavy focus on the end of life, it is important to balance this with involvement in other aspects of life, such as being with children and healthy older adults. Opportunities to hold babies are thoroughly relished at hospices.

Self-Care Is Not Optional

Professionals often say that although they know self-care is important, they feel selfish when setting a limit or caring for themselves. I ask hospice professionals to think about an older client's caregiver whose self-neglect has reached the point where she will soon need care herself, a common problem. Then I suggest that they will be unable to help that caregiver until they do what they are asking her to do. I propose starting with one small step and considering an accountability partner for support. Since the professional's self is the vehicle for serving clients, self-care is similar to musicians caring for their instruments, an occupational responsibility. Tending to the source of one's gifts results in a long career of privilege as a compassionate sojourner in many clients' unique lives as they approach their final passage.

DEATH ANXIETY

Death anxiety is the morbid, abnormal or persistent fear of death or dying. The British National Health Service defines death anxiety as a feeling of dread, apprehension or solicitude (anxiety) when one thinks of the process of dying, or ceasing to be or what happens after death. It is also referred to as thanatophobia (fear of death) and necrophobia (fear of death or the dead). Lower ego integrity, more physical problems, and more psychological problems are predictive of higher levels of death anxiety in elderly people.

TYPES DEATH ANXIETY

1. Predatory death anxiety

Predatory death anxiety is the most basic form of death anxiety, with its origins stemming from the first unicellular organisms' set of adaptive resources.

2. Predation death anxiety.

Predation death anxiety is a form of death anxiety that arises, often occurring unconsciously rather than consciously, when an individual physically and/or mentally harms another.

3. Existential death anxiety;

Existential death anxiety is the basic knowledge and awareness that natural life is short. It is said that existential death anxiety directly correlates to language; that is, language has created the basis for this type of death anxiety through communicative and behavioural changes. However, existential death anxiety, unlike predatory death anxiety, does not involve episodes of psychological or physical harm.

PERSONAL MEANINGS OF DEATH

Humans develop meanings and associate them with objects and events in their environment. These meanings that we associate to certain things are what provoke certain emotions within an individual. People tend to develop personal meanings of death and those meanings could accordingly be negative or positive for the certain individual. If they are positive, then the consequences of those meanings can be comforting to the individual. If negative they can cause emotional turmoil in the individual when faced with the death of someone or when faced with death itself. Depending on the certain meaning one has associated with death, the consequences will vary accordingly whether they are negative or positive meanings.(Cicirelli, V. G.,1998).

EUTHANASIA

History

According to the historian N. D. A. Kemp, the origin of the contemporary debate on euthanasia started in 1870.(Nick Kemp ,7 September 2002). Nevertheless, euthanasia was debated and practiced long before that date. Euthanasia was practiced in Ancient Greece and

Rome: for example, hemlock was employed as a means of hastening death on the island of Kea, a technique also employed in Marseilles and by Socrates in Athens. Euthanasia, in the sense of the deliberate hastening of a person's death, was supported by Socrates, Plato and Seneca the Elder in the ancient world, although Hippocrates appears to have spoken against the practice, writing "I will not prescribe a deadly drug to please someone, nor give advice that may cause his death" (noting there is some debate in the literature about whether or not this was intended to encompass euthanasia). (Mystakidou, et. Al,2005) Stolberg, Michael (2007), Gesundheit, et.al (2006).

Definition

The word "euthanasia" was first used in a medical context by Francis Bacon in the 17th century, to refer to an easy, painless, happy death, during which it was a "physician's responsibility to alleviate the 'physical sufferings' of the body." Bacon referred to an "outward euthanasia"—the term "outward" he used to distinguish from a spiritual concept—the euthanasia "which regards the preparation of the soul." (Francis Bacon: the major works By Francis Bacon, Brian Vickers) . "Euthanasia is the "painless inducement of a quick death".(Kohl, Marvin ,1974). "The painless killing of a patient suffering from an incurable and painful disease or in an irreversible coma".(The definition offered by the Oxford EnglishDictionary) "A mode or act of inducing or permitting death painlessly as a relief from suffering" (Marvin Khol and PaulKurtz's)

Classification of euthanasia

Euthanasia may be classified according to whether a person gives informed consent into three types: voluntary, non-voluntary and involuntary.

There is a debate within the medical and bioethics literature about whether or not the non- voluntary (and by extension, involuntary) killing of patients can be regarded as euthanasia, irrespective of intent or the patient's circumstances. In the definitions offered by Beauchamp & Davidson and, later, by Wreen1998 , consent on the part of the patient was not considered to be one of their criteria, although it may have been required to justify euthanasia. (Wreen, Michael ,1988) However, others see consent as essential.

Voluntary euthanasia- Euthanasia conducted with the consent of the patient is termed voluntary euthanasia. Active voluntary euthanasia is legal in Belgium, Luxembourg and the Netherlands. Passive voluntary euthanasia is legal throughout the U.S. per Cruzan v. Director, Missouri Department of Health. When the patient brings about his or her own death with the assistance of a physician, the term assisted suicide is often used instead. Assisted suicide is legal in Switzerland and the U.S. states of Oregon, Washington and Montana.

Non-voluntary euthanasia- Euthanasia conducted where the consent of the patient is unavailable is termed non-voluntary euthanasia. Examples include child euthanasia, which is illegal worldwide but decriminalised under certain specific circumstances in the Netherlands under the Groningen Protocol.

Involuntary euthanasia- Euthanasia conducted against the will of the patient is termed involuntary euthanasia.

Passive and active euthanasia- Voluntary, non-voluntary and involuntary euthanasia can all be further divided into passive or active variants.

Passive euthanasia entails the withholding of common treatments, such as antibiotics, necessary for the continuance of life.

Active euthanasia entails the use of lethal substances or forces, such as administering a lethal injection, to kill and is the most controversial means. A number of authors consider these terms to be misleading and unhelpful.(Harris, NM. Oct 2001).

Beginnings of the contemporary euthanasia debate

In the mid-1800s, the use of morphine to treat "the pains of death" emerged, with John Warren recommending its use in 1848. A similar use of chloroform was revealed by Joseph Bullar in 1866. However, in neither case was it recommended that the use should be to hasten death.

In 1870 Samuel Williams, a schoolteacher, initiated the contemporary euthanasia debate through a speech given at the Birmingham Speculative Club in England, which was subsequently published in a one-off publication entitled Essays of the Birmingham Speculative Club, the collected works of a number of members of an amateur philosophical society.(Rachels(January

1975)Williams' proposal was to use chloroform to deliberately hasten the death of terminally ill patients: Euthanasia may also lead to counterexamples: such definitions may encompass killing a person suffering from an incurable disease for personal gain (such as to claim an inheritance), and commentators such as Tom Beauchamp & Arnold Davidson have argued that doing such would constitute "murder simpliciter" rather than euthanasia.

Some element incorporated into many definitions is that of intentionality – the death must be intended, rather than being accidental, and the intent of the action must be a "merciful death".

Michael Wreen argued that "the principal thing that distinguishes euthanasia from intentional killing simpliciter is the agent's motive: it must be a good motive insofar as the good of the person killed is concerned"

A view mirrored by Heather Draper, who also spoke to the importance of motive, arguing that "the motive forms a crucial part of arguments for euthanasia, because it must be in the best interests of the person on the receiving end."

Definitions such as that offered by the House of Lords Select Committee on Medical Ethics take this path, where euthanasia is defined as "a deliberate intervention undertaken with the express intention of ending a life, to relieve intractable suffering."

Beauchamp & Davidson also highlight Baruch Brody's "an act of euthanasia is one in which one person ... Kills another person for the benefit of the second person, who actually does benefit from being killed".

Draper argued that any definition of euthanasia must incorporate four elements: an agent and a subject; an intention; a causal proximity, such that the actions of the agent lead to the outcome; and an outcome. Based on this, she offered a definition incorporating those elements, stating that euthanasia "must be defined as death that results from the intention of one person to kill another person, using the most gentle and painless means possible, that is motivated solely by the best interests of the person who dies." Prior to Draper, Beauchamp & Davidson had also offered a definition which includes these elements, although they offered a somewhat longer account, and one that specifically discounts fetuses in order to distinguish between abortions and euthanasia.

In discussing his definition, Wreen noted the difficulty of justifying euthanasia when faced with the notion of the subject's "right to life". In response, Wreen argued that euthanasia has to be voluntary, and that "involuntary euthanasia is, as such, a great wrong". Other commentators incorporate consent more directly into their definitions. For example, in a discussion of euthanasia presented in 2003 by the European Association of Palliative Care (EPAC) Ethics Task Force, the authors offered: "Medicalized killing of a person without the person's consent, whether nonvoluntary (where the person in unable to consent) or involuntary (against the person's will) is not euthanasia: it is murder. Hence, euthanasia can be voluntary only." Although the EPAC Ethics Task Force argued that both non-voluntary and involuntary euthanasia could not be included in the definition of euthanasia, there is discussion in the literature about excluding one but not the other.

ARGUMENTS AGAINST EUTHANASIA

It's possible to argue about the way we've divided up the arguments, and many arguments could fall into more categories than we've used.

Ethical arguments

- Euthanasia weakens society's respect for the sanctity of life.
- Accepting euthanasia accepts that some lives (those of the disabled or sick) are worth less than others.
- Voluntary euthanasia is the start of a slippery slope that leads to involuntary euthanasia and the killing of people who are thought undesirable.
- Euthanasia might not be in a person's best interests..
- Euthanasia affects other people's rights, not just those of the patient

Practical arguments

- Proper palliative care makes euthanasia unnecessary
- There's no way of properly regulating euthanasia
- Allowing euthanasia will lead to less good care for the terminally ill
- Allowing euthanasia undermines the commitment of doctors and nurses to saving lives
- Euthanasia may become a cost-effective way to treat the terminally ill
- Allowing euthanasia will discourage the search for new cures and treatments for the terminally ill

- Euthanasia undermines the motivation to provide good care for the dying, and good pain relief
- Euthanasia gives too much power to doctors
- Euthanasia exposes vulnerable people to pressure to end their lives
- Moral pressure on elderly relatives by selfish families
- Moral pressure to free up medical resources
- Patients who are abandoned by their families may feel euthanasia is the only solution.

Historical arguments

Voluntary euthanasia is the start of a slippery slope that leads to involuntary euthanasia and the killing of people who are thought undesirable

Religious arguments

- Euthanasia is against the word and will of God
- Euthanasia weakens society's respect for the sanctity of life
- Suffering may have value
- Voluntary euthanasia is the start of a slippery slope that leads to involuntary euthanasia and the killing of people who are thought undesirable.

ARGUMENTS FOR EUTHANASIA

Supporters of euthanasia would respond that this argument includes a number of completely misleading suggestions, and refute them:

- Dying is not the same as never having been born
- The debate is nothing to do with preventing disabled babies being born, or preventing
- people with disabilities from becoming parents
- Nobody is asking for patients to be killed against their wishes - whether or not those
- patients are disabled
- The euthanasia procedure is intended for use by patients who are dying, or in a condition

- that will get worse - most disabilities don't come under that category
- The normal procedure for euthanasia would have to be initiated at the patient's request
- Disabled people who are not mentally impaired are just as capable as able-bodied people of deciding what they want
- Protections will be in place for patients who are mentally impaired, whether through disability or some other reason
- It is possible that someone who has just become disabled may feel depressed enough to ask for death, which is why any proposed system of euthanasia must include psychological support and assessment before the patient's wish is granted
- All people should have equal rights and opportunities to live, or to choose not to go on Living.

UNIT- XII

OTHER GENERAL CLINICAL CONDITIONS

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OTHER GENERAL CLINICAL CONDITIONS

Introduction: The importance of the mental health expert in the management of medically ill individuals has grown with the proliferation of information linking health and behaviour. The mental health professional intervenes to develop healthful behaviours and remove unhealthful behaviours in all medically ill patients and to relieve emotional distress in the subset of patients with evident psychopathology (Cohen & Rodriguez, 1995).

There are several general medical conditions where applications of psychological techniques or services appear to affect the outcome of medical management positively. In this unit we are going to discuss some of them. These conditions are as follows:

1. Diabetes:

Diabetes is a chronic condition of impaired carbohydrates, protein and fat metabolism that results from insufficient secretion of insulin or from resistance. According to the Centers for Disease Control and Prevention, 20.9 million Americans were living with diagnosed diabetes in 2011, up from just 5.6 million in 1980. And an estimated seven million more people had diabetes but were undiagnosed. Diabetes goes hand in hand with an increase in obesity and inactivity.

Type 1 and Type 2 Diabetes:

Type 1 diabetes or insulin-dependent diabetes, is usually diagnosed in children and young adults. There are few known risk factors, though family history may play a role. Brittle diabetes, also known as unstable diabetes or labile diabetes is a term that was traditionally used to describe the dramatic and recurrent swings in glucose levels, often occurring for no apparent reason in insulin-dependent diabetes.

Type 2diabetes or insulin-resistant diabetes, is most often diagnosed in adults, though the rate in youth is on the rise.

In contrast to type 1diabetes, type 2 may not require insulin injections and can often be controlled with lifestyle changes, oral medications or both. Obesity, inactivity, family history and poor diet are risk factors for type 2 diabetes.

Evidence demonstrating the importance of psychological, behavioural, and social factors in diabetes has been accumulating for more than 20 years (Rubin & Peyrot, 1992) and the role of the mental health expert in diabetes care has expanded accordingly (Anderson & Rubin, 1996; Lorenz, Bubb, Davis, Jacobson, Jannasch, Kramer, Lipps, Schlundt,1996).

<u>Neurocognitive functioning and Diabetes:</u>

Studies indicate that children who develop diabetes before 5 years of age and/or who have frequent episodes of hypoglycemia are at risk for neurocognitive deficits, particularly in visual-spatial functioning (Holmes & Richman, *1985*; Rovet, Ehrlich & Hoppe, *1988;* Ryan, Vega & Drash, *1985*). Studies have also shown that diabetic children, especially boys, are more likely to have learning problems (Holmes, Dunlap, Chen & Cornwell, 1992). Other research has found poorer attentional functioning and lower verbal intelligence in children with a history of significant hypoglycemia (Rovet & Alvarez, 1997). A longitudinal study of newly diagnosed children revealed declines in verbal intelligence and school grades, predicted in part by memory dysfunction (Kovacs, Goldston and Iyengar, 1992; Kovacs, Ryan& Obrosky, 1994). Another study showed that 2 years after diagnosis, children exhibited mild neuropsychological deficits, including reduced speed of information processing, and decrements in conceptual reasoning and acquisition of new knowledge (Northam, Anderson, Werther, Warne, Adler & Andrewes, 1998), which were predicted by both recurrent hypoglycemia and hyperglycemia, as well as early onset of diabetes (before 5 years of age) (Northam, Anderson, Werther, Warne & Andrewes, 1999). Neurocognitive deficits have been observed in adults with type 1diabetes, particularly those with at least five episodes of severe hypoglycemia (Langan, Deary, Hepburn & Frier, 1991), and in patients with peripheral neuropathy (Ryan, Williams, Orchard & Finegold, 1992). Among older adults with type 2 diabetes, cognitive deficits have been reported in association with poor glycemic control (Reaven, Thompson, Nahum & Haskins, 1990).

<u>Psychopathology and Diabetes</u>: Approximately one-third of patients with diabetes have diagnosable psychological problems at some point during their lifetime.

<u>Stress:</u> Stress is common in diabetes and is brought on by ordinary daily hassles (e.g., driving in traffic, conflict with family members, work deadlines), negative life events (e.g., death of a loved one, financial problems, divorce), and the additional burdens of coping with diabetes.

Stress may have direct effects on health via elevated blood glucose values (Surwitt & Schneider, 1992; Auslander, Bubb, Rogge & Santiago, 1993) and indirect effects on health via disruption in behavioural patterns and routines (e.g., eating and sleeping). Stress has been associated with an increased risk of type 2 diabetes. When one is under stress, his body signals its nervous system and pituitary gland to produce epinephrine and cortisol, known as "stress hormones." When cortisol and epinephrine are released, the liver produces more glucose, a blood sugar.

Affective disorder and Anxiety disorder: Affective and anxiety disorders are the most common diagnoses and occur significantly more often in patients with diabetes than in the general U.S. population (Gavard, Lustman & Clouse, 1993). These disorders can lead to poor glycemic control through alterations in neurohormonal and neurotransmitter functioning and through disruption in diabetes self-care.

Depressive disorder: Major depression affects approximately one of every five patients with diabetes and severely impairs quality of life and all aspects of functioning (Gavard, Lustman & Clouse, 1993). It has added importance in diabetes because of its association with treatment nonadherence, poor glycemic control, and increased risk for micro- and macrovascular disease complications (Lustman, Griffith & Clouse, 1997). Depression remains unrecognized and untreated in the majority of cases despite its specific relevance to diabetes (Lustman & Harper, 1987).

Eating disorder: Eating disorders are clinically important because of their association with poor glycemic control and an increased risk for retinopathy (Rydall, Rodin, Olmsted, Devenyi & Daneman, 1997). Eating disorders can be effectively treated with psychotherapy.

Psychological problem in children and adolescents: Pediatric and adult patients with diabetes may manifest psychological problems in different ways. Evidence of psychosocial problems related to diabetes in children is often observed in poor school performance, impaired peer relations, or behavioral changes at home, at school, or with friends (Eiser, 1990). In addition, conflictual family relations are often clues to psychosocial problems for children with diabetes.

Treatment:

Healthy eating: Contrary to popular perception, there's no specific diabetes diet. However, it's important to center patient's diet on high-fiber, low-fat foods: Fruits, Vegetables, Whole grains. The patient needs to eat fewer animal products, refined carbohydrates and sweets. Low glycemic index foods also may be helpful. The glycemic index is a measure of how quickly a food causes a rise in your blood sugar. Foods with a high glycemic index raise blood sugar quickly. Low glycemic foods may help achieve a more stable blood sugar. Foods with a low glycemic index typically are foods that are higher in fiber.

Physical activity: Everyone needs regular aerobic exercise, such as walking, swimming and biking. What's most important is making physical activity part of the patient's daily routine.

Monitoring blood sugar: Depending on the treatment plan, the patient may check and record his blood sugar level every now and then or, if he is on insulin, multiple times a day. Sometimes, blood sugar levels can be unpredictable. With help from the diabetes treatment team, the patient will learn how blood sugar level changes in response to food, exercise, alcohol, illness and medication.

Medication: Most commonly used medication in Type 2 diabetes are Metformin, Sulfonylureas, Meglitinides, Thiazolidinediones, DPP-4 inhibitors**,** GLP-1 receptor agonists etc.

Insulin therapy: Some people who have type 2 diabetes need insulin therapy as well. In the past, insulin therapy was used as last resort, but today it's often prescribed sooner because of its benefits. Because normal digestion interferes with insulin taken by mouth, insulin must be injected. Depending on the patient's needs, doctor may prescribe a mixture of insulin types to use throughout the day and night. Often, people with type 2 diabetes start insulin use with one long-acting shot at night.

Role of psychologist in diabetic care: Ideally, diabetes treatment is provided by a team of health care professionals that consists of a physician, diabetes nurse educator, dietician, and psychologist (Lorenz, Bubb, Davis, Jacobson, Jannasch, Kramer, Lipps, Schlundt, 1996). The psychologist provides direct services to the patient via promotion of health behaviours and treatment of psychological problems, and also provides consultation to the medical team on how to incorporate psychological principles into patient care to enhance clinical outcomes.

Diagnoses and Assessment: Clinical psychologists are an appropriate resource to the diabetes treatment team for the diagnosis, assessment, and treatment of mental health problems of patients with diabetes. Psychologists with expertise in reinforcement strategies, learning principles, and behaviour modification are highly desirable given the usefulness of these skills for developing health behaviours. While not all psychologists are trained in diabetes, it is recommended that clinical psychologists working with diabetic patients have training in health and/or pediatric psychology and be licensed by the state in which they work. This will provide some degree of quality assurance and will increase the chances that services rendered by a psychologist will be reimbursed by insurance companies.

Non-adherence: Non-adherence to the diabetes regimen is the most common reason for psychological referral. Non-adherence is a central focus of psychological treatment because of its recognized association with poor metabolic control (Kutz, 990; Johnson, 1992) and increased risk for diabetes complications. Non-adherence is often mistakenly attributed to inadequate knowledge about proper diabetes care. Many other psychosocial factors contribute significantly to this problem, such as inadequate social support, time pressures, stress, and health beliefs that are incompatible with the regimen (Weissberg-Benchell, Glasgow, Tynan, Wirtz, Turek & Ward,1995; Boehm, Schlenk, Funnell, Powers & Ronis,1997; Schlundt, Rea, Kline & Pichert, 1994). The psychologist can provide treatment to develop new healthful behaviours, enhance existing healthful behaviours, and extinguish unhealthy behaviours as they relate to improved glycemic control. Nonadherence may also be a manifestation of more serious psychological problems, such as depression, anxiety, or eating disorders. These problems may be treated effectively with psychotherapy or psychotropic medication (Lustman, Clouse, Alrakawi, Rubin & Gelenberg, 1997; Mendez & Belendez, 1997; Wysocki, White, Bubb, Harris & Greco, 1995).

Providing information: With both children and adolescents, the psychologist can provide information to the diabetes treatment team about psychosocial development and how best to resolve some of the struggles between parents and youth that may be an impediment to proper diabetes management. In addition, the psychologist can assist parents in negotiating control with their child over the diabetes treatment regimen.

Psychotherapies:

Supportive therapy focuses on increasing the individual's self-esteem and adaptive skills and identifies unhelpful behaviours by exploring the person's interpersonal patterns (Woller et al, 1996).

Motivational interviewing (MI) involves helping the individual identify and resolve discrepancies between their goals, values and current behaviour (Williams et al, 1998).

Psychodynamic (or psychoanalytic) therapy focuses more on understanding current behaviour through past experiences (Pusch et al, 2012).

Behavioural therapy uses techniques such as goal-setting and reinforcement (Pusch et al, 2012).

Interpersonal therapy focuses on coping strategies and interpersonal relationships (Wilfley, 2001).

Cognitive Behaviour Therapy examines the relationship between a person's thoughts, feelings and behaviours, and focuses on practical solutions to current problems. The cognitive element involves identifying and modifying dysfunctional cognitions, or thoughts, into more realistic and helpful ones; this is called "cognitive restructuring". The behavioural element refers to the use of behaviour change strategies such as cueing, goal-setting and positive reinforcement, as well as activity scheduling and stress management techniques

Six steps to living with diabetes:

1. Get the facts.
2. Accept your feelings.
3. Maintain a balanced perspective.
4. Be realistic.
5. Try new things.
6. Develop a strong support network.

2. Sleep Disorders:

Sleep disorders or somnipathy are changes in sleeping patterns or habits. Signs and symptoms of sleep disorders include excessive daytime sleepiness, irregular breathing or increased movement during sleep, difficulty sleeping, and abnormal sleep behaviours. A sleep disorder can affect one's overall health, safety and quality of life.

Sleep problems are highly prevalent, affecting approximately 75% of adults (National Sleep Foundation, 2005) and 25–40% of children (Owens, 2005). A complex bidirectional relationship exists between sleep problems and mental health problems, including depression, anxiety, and traumatic stress disorders (American Psychiatric Association [APA], 1994).

Major categories:

Dyssomnias: Dyssomnias are a broad classification of sleeping disorders that make it difficult to get to sleep, or to remain sleeping. Dyssomnias are primary disorders of initiating or maintaining sleep or of excessive sleepiness and are characterized by a disturbance in the amount, quality, or timing of sleep. It can be intrinsic or extrinsic.

Insomnia: Insomnia is characterized by an extended period of symptoms including trouble with retaining sleep, fatigue, decreased attentiveness, and dysphoria. To diagnose insomnia, these symptoms must persist for a minimum of 4 weeks. The DSM-IV categorizes insomnias into primary insomnia, insomnia associated with medical or mental illness, and insomnia associated with the consumption or abuse of substances. Individuals with insomnia often worry about the negative health consequences, which can lead to the development of anxiety and depression.

Hypersomnia: The main symptom of hypersomnia is excessive daytime sleepiness (EDS), or prolonged nighttime sleep, which has occurred for at least 3 months prior to diagnosis. Hypersomnia affects approximately 5% of the general population, with a higher prevalence for men due to the sleep apnea syndromes.

Parasomnias: A category of sleep disorders that involve abnormal and unnatural movements, behaviours, emotions, perceptions, and dreams in connection with sleep.

- REM sleep behaviour disorder (abnormal behaviour during sleep phase, eg. Loss of muscle atonia)
- Sleep terror (or Pavor nocturnus)
- Sleepwalking (or somnambulism)
- Bruxism (Tooth-grinding)
- Bedwetting or sleep enuresis.
- Sleep talking (or somniloquy)
- Sleep sex (or sexsomnia)
- Exploding head syndrome (Waking up in the night hearing loud noises).

Other common sleep disorders include sleep apnea (stops in breathing during sleep), narcolepsy and hypersomnia (excessive sleepiness at inappropriate

times), cataplexy (sudden and transient loss of muscle tone while awake), and sleeping sickness (disruption of sleep cycle due to infection).

Medical or psychiatric conditions:

- Psychosis (such as Schizophrenia)
- Depression
- Anxiety
- Panic
- Alcoholism
- Sleeping sickness (a parasitic disease which can be transmitted by the Tsetse fly).

Assessment:

- **Sleep diary:** Tracking sleep patterns may help a doctor reach a diagnosis.
- **Epworth Sleepiness Scale:** A validated questionnaire that is used to assess daytime sleepiness.
- **Polysomnogram:** A test measuring brain and muscle activity including breathing during sleep
- **Multiple Sleep Latency Test:** A test for daytime sleepiness, usually administered the day after overnight polysomnography.
- **Actigraphy:** a test to assess sleep-wake patterns, usually for a week or more. Actigraphs are wrist-worn devices, about the size of a wristwatch, that measure movement.
- **Mental health exam:** Because insomnia may be a symptom of depression, anxiety, or another mental health disorder, a mental status exam, mental health history, and basic mental evaluations may be part of the assessment for a person complaining of insomnia.

Treatment options:

Medication: Certain disorders like narcolepsy, are best treated with prescription drugs such as Modafinil. Temazepam is a benzodiazepine group of drug used for insomnia.

Sleep Education: In sleep education the patient is provided information regarding normal sleep hours, sleep disorders, misconception about sleep, major concepts related to biology of sleep, its physiological pathway, how it can be affected etc.

Sleep Hygiene: Sleep-enhancing directives include maintaining a regular sleep-wake schedule; keeping a steady programme of daily exercise; protecting bedroom against excessive noise, light, cold and heat; eating light food before retiring if hungry; and setting time aside to relax before getting into bed. Finally to keep the person from developing conditional arousal associated with the bed and bedroom. There are certain don’ts of sleep hygiene. They include avoiding strenuous exercise immediately before bedtime; abstaining from alcohol, tobacco, coffee, tea in the evening; not watching television in bed; and not chronically taking sleeping pills.

Cognitive Behavioral Treatment for insomnia (CBT-I) has been found to be as effective as hypnotic medications for short-term treatment of insomnia (Morgenthaler, Kramer et al., 2006; Smith et al., 2002), and more effective than medications for the longer term management of insomnia (Morin, Colecchi, Stone, Sood, & Brink, 1999). A recent National Institute of Health (NIH) state-of-the-science conference statement on insomnia also concluded that cognitive–behavioural treatments of insomnia are as effective in the short-term and more effective in the long-term for the treatment of insomnia (NIH, 2005).

Behaviour Therapies (e.g., extinction, graduated extinction, preventive education) have also been empirically validated in the treatment of bedtime problems and night waking in young children, resulting in a recent standards of practice statement for their efficacy by the American Academy of Sleep Medicine (AASM; Morgenthaler, Owens et al., 2006).

Evidence Based Therapy: Sleep restriction–sleep compression therapy and multicomponent cognitive– behavioral therapy are found to be most effective in treating insomnia, sleep apnea etc.

Hypnosis: Research suggests that hypnosis may be helpful in alleviating some types and manifestations of sleep disorders in some patients. "Acute and chronic insomnia often respond to relaxation and hypnotherapy approaches, along with sleep hygiene instructions."Hypnotherapy has also helped with nightmares and sleep terrors. There are several reports of successful use of hypnotherapy for parasomnias specifically for head and body rocking, bedwetting and sleepwalking.

3. OBESITY:

Obesity is a medical condition in which excess body fat has accumulated to the extent that it may have a negative effect on health, leading to reduced life expectancy and/or increased health problems. In Western countries, people are considered obese when their body mass index (BMI), a measurement obtained by dividing a person's weight by the square of the person's height, exceeds30 kg/m^2, with the range 25-30 kg/m^2 defined as overweight. Obesity increases the likelihood of various diseases, particularly heart disease, type 2 diabetes, obstructive sleep apnea, certain types of cancer, and osteoarthritis. Obesity is most commonly caused by a combination of excessive food energy intake, lack of physical activity, and genetic susceptibility, although a few cases are caused primarily by genes, endocrine disorders, medications, or psychiatric illness.

Psychological distress in obese people: Studies do indicate that subgroups within the obese population, such as obese individuals presenting for clinical weight-loss treatment and obese binge-eaters, show elevated psychopathology (Fitzgibbon, Stolley & Kirschenbaum, 1993; Black, Goldstein & Mason, 1992; Telch & Agras, 1994). Individuals seeking treatment for weight loss have consistently demonstrated a higher prevalence of distress than their nontreatment-seeking counterparts. Goldsmith et al. (1992) reported that 55.6% of their participants who were seeking weight-loss treatment met criteria for current or past psychiatric illness, especially major depression and dysthymia. Binge eating, which occurs in 30% of obese individuals seeking weight loss treatment (Spitzer et al., 1992), also has been linked to elevated levels of depression. Marcus et al. (1992) found that obese binge-eaters had substantial depressive symptomatology compared with their nonbinging counterparts. Additionally, Kuehnel and Wadden (1994) reported that individuals diagnosed with binge eating disorder had significantly higher levels of depression than both nonbingers and problem eaters. This binge eating mediated pattern of results has been confirmed in numerous additional studies (Costanzo et al., 1999; Musante, 1998). The results of these studies suggest that when attempting to understand the relationship between obesity and symptoms of psychological distress, it seems useful to consider mediating variables, such as treatment-seeking status or binge eating, to explain the presence of psychological symptomatology. Brownell (1995) have proposed several additional risk factors that may determine which obese individuals will suffer negative psychological consequences. These include social class, degree of obesity, and body-image dissatisfaction. In this study one of the proposed risk factors, body-image dissatisfaction, is investigated as a potential mediator of the relationship between dysphoric psychological states (e.g., depression and low self-esteem) and obesity in a treatment seeking population.

Body image and obesity: Body image is an individual’s psychological experience of the appearance and function of his/her body and is one aspect of an individual’s mental representation of him/ herself (Cash & Prunzinsky, 1997; Garner, 1997). Large-sample survey research suggests that 52% of men and 66% of women in America are dissatisfied with their weight (Garner, 1997). While the overwhelming majority of women desire to lose weight, a somewhat different picture emerges for men (Garner, 1997). Although a substantial percentage of men (88%) who are dissatisfied with their weight do desire to lose weight, 22% of men who express dissatisfaction with their bodies actually wish to gain weight (Garner, 1997), most likely to meet the muscular ideals that are portrayed for men's physiques (Drewnowski &Yee, 1987).

Role of Psychologist: Psychologists should obtain necessary information from assessments and ask patients any additional follow up questions necessary. Obese patients and their families can have significant socioeconomic and cultural differences, and psychologists should be active in seeking an understanding of these factors of diversity.

Assessment: The assessments including medical and developmental history, family history, educational history, patient's emotional history and body image, behavioural, social and psychiatric history are important. In addition to it patient's sleep pattern, kind of nutr ition he takes, his motivation or readiness to change must be assessed by the psychologist.

Cognitive Behaviour Therapy and Behaviour Therapy: Combination of both CBT and BT has been found to be effective in people who are overweight and are being in the process of weight loss. There are 5 cognitive behavioural strategies which helps the patient to gain a healthy life style:

1. Goal setting: specific goal, ambitious goal and regular feedback.

2. Self monitoring: pay attention to physical cues and identify challenges to changing behaviour.

3. Feedback and reinforcement: Feedback about diet or exercise routine can provide motivation or help to adjust one's behaviour.

4. Boosting the belief: Setting concrete and achievable goals.

 Incentives: Offering lower-priced onsite fitness facilities as an incentive to exercise, offering cash incentives and gift cards, providing free health coaching and offering insurance premium discounts to those who meet certain standards.

Group and Family Therapy: When youth with complex medical conditions such as obesity and its comorbid conditions are treated, parent training and parent consultation is often needed and would be most beneficial without the patient present. An example is: When a pediatric psychologist works to coach parents in behavioural principles with a disruptive child, the ideal treatment plan includes at least some sessions without the child present. A CPT (current procedural terminology) code for this model of therapy exists under both mental health and health and behaviour CPT codes, but family therapy without the patient present is rarely covered by insurance companies. Group intervention represents another challenge; many obesity group interventions are multidisciplinary in nature and are not focused on psychiatric diagnoses. Thus, mental health CPT codes for group would be inappropriate. Health and behaviour codes represent a mechanism for billing for these multidisciplinary groups; however, a psychologist must be in a lead role as the codes do not cover licensed social workers, dieticians, or exercise physiologists.

Recommendations:

1. Use of a hunger scale (a patient-friendly scale to assist in determining level of hunger at any given time, typically 10-point scale with faces reflecting different levels of hunger/discomfort).
2. Provide education about metabolism and factors affecting metabolic rate such as the negative effects of skipping breakfast.
3. Review the importance of sleep hygiene and impact on hunger, fatigue, health, and behaviour.
4. Review the impact of teasing/bullying on patient; provide education on school policies related to bullying; support caregivers in making changes in the home and school environment to reduce or eliminate teasing/bullying.
5. Eating slowly, shaping toward appropriate portion sizes.
6. 5 small meals/snacks a day, shaping toward that goal.
7. Review benefits of regular exercise, shaping toward moderate/vigorous exercise
8. Take variety of fruits/vegetables through tasting opportunities, pairing with liked foods (dips), repeated presentations, shaping serving size over time, caregiver modelling.
9. Provide psychotherapy to address coping with chronic illness, functional impairment, body image.
10. Discuss healthy snack options and how to reinforce choosing those over unhealthy options.

4. Dental anxiety:

Dental anxiety (odontophobia or dental fear) is the fear of dentistry and of receiving dental care. Dental anxiety can prevent patients from getting the treatment they need. Postponing treatment can result in additional pain, helping to reinforce fear of the dentist as the pain becomes associated with the visit. Direct or indirect (eg, vicarious learning, stimulus generalization, helplessness and perceived lack of control) experiences can cause it.

Treatment: Treatments for dental fear often include a combination of behavioural and pharmacological techniques. Specialized dental fear clinics use both psychologists and dentists to help people learn to manage and decrease their fear of dental treatment. The goal of these clinics is to provide individuals with the fear management skills necessary for them to receive regular dental care with a minimum of fear or anxiety.

Pharmacological techniques to manage dental fear range from mild sedation to general anesthesia, and are often used by dentists in conjunction with behavioural techniques (Milgrom & Heaton, 2007). One common anxiety-reducing medication used in dentistry is nitrous oxide (also known as "laughing gas"), which is inhaled through a mask worn on the nose and causes feelings of relaxation and dissociation. Dentists may also prescribe an oral sedative, such as a benzodiazepine like temazepam, alprazolam, diazepam, ortriazolam.

Behavioural techniques include teaching individuals relaxation techniques, such as diaphragmatic breathing and progressive muscle relaxation, as well as cognitive, or thought-based techniques, such as cognitive restructuring and guided imagery (Milgrom, Weinstein & Getz, 1995). Both relaxation and cognitive strategies have been shown to significantly reduce dental fear (Lundgren, Carlsson & Berggren, 2006). One example of a behavioural technique is systematic desensitization, a method used in psychology to overcome phobias and other anxiety disorders (Wolpe, 1958). This is also sometimes called graduated exposure therapy or gradual exposure. For example, for a patient who is fearful of dental injections, the therapist first teaches relaxation skills to the patient, then gradually introduces the feared object (in this case, the needle and/or syringe) to the patient, encouraging the patient to manage his/her fear using the relaxation skills previously taught. The patient progresses through the steps of receiving a dental injection while using the relaxation skills, until the patient is able to successfully receive a dental injection while experiencing little to no fear. This method has been shown to be effective in treating fear of dental injections (Coldwell et al., 2007). Cognitive restructuring, if applied in a non-threatening situation, might be a useful alternative as a first step after years of avoidance of dental care and less threatening than immediate exposure to the feared stimuli (de Jongh, Muris, ter Horst, van Zuuren, Schoenmakers & Makkes, 1995).

Self-help and peer support, recent research has focused on the role of online communities in helping people to confront their anxiety or phobia and successfully receive dental care. The findings suggest that certain individuals do appear to benefit from their involvement in dental anxiety online support groups (Buchanan & Coulson, 2007; Coulson & Buchanan, 2008).

5. burns injury:

A burn is a type of injury to flesh or skin. Burns that affect only the superficial skin are known as superficial or first-degree burns. When damage penetrates into some of the underlying layers, it is a partial-thickness or second-degree burn. In a full-thickness or third-degree burn, the injury extends to all layers of the skin. A fourth-degree burn additionally involves injury to deeper tissues, such as muscle or bone. 4 types of sources can cause burns: thermal (flash, flame, scalds, and contacts to hot metal or plastics), chemical (strong acids or alkali substances), electrical (Ac or DC current) and radiological (alpha, beta or gamma radiation).

Individuals who sustain burns often experience substantial biological, psychological and psychosocial stressors (Patterson, Everett, Bormbadier, Questad, Lee & Marvin, 1993). Burns may result in traumatic stress, pain, shock, sepsis and altered functioning of the hypothalamic adreno-cortical system or the immune system (Madianos, Papaghelis, Ioannovich & Dafni, 2001). A major burn injury can impair skin integrity, sensation and may lead to hypertrophic scarring. In addition to changes in appearance and function brought about by scarring, deeper burns may result in damage to, or complete loss of functionally or cosmetically important body parts (Rose & Herdon, 1997). The physical and psychological consequences of a major burn injury can interfere significantly with social and occupational performance, which may be exacerbated by environmental barriers or lack of social support (Druery, Brown & Muller, 2005).

Biological complication after burn injury:

- Post injury infections (Bacteremia, Septicemia).
- Acute gastrointestinal ulcers.
- Circumscribed lesions within the lining of the stomach and duodenum.

Psychological distress in burn injury:

- Feeling sad, anxious or irritable.
- Feeling helpless and hopeless.
- Feeling upset about depending on other people for assistance.
- Feeling distant from family, friends or the general public.
- Feeling alone.
- Difficulty falling asleep.
- Difficulty staying asleep.
- Difficulty relaxing mind and body.
- Difficulty concentrating.
- Low energy or feeling tired all the time.

While in the hospital, survivors may find they have a lot of time to focus on their burn injury. Many people report having psychological distress several days or a few weeks after they were injured. For most, periods of distress become less frequent and less upsetting after a couple of weeks to a couple of months. However, about one third of people with major burn injuries continue to feel very distressed for up to 2 years.

Causes that lead to distress:

- Thinking about the event itself.
- Worries about the future.
- Appearance of the injury (public reaction, staring, curiosity).
- Remembering the way the wounds looked both at the scene and in the hospital.
- Changes in appearance because of scars and contractures.
- Physical discomfort.
- Pain while the wound is still healing (especially during the repeated dressing changes) and pain that continues for months or years afterward.

Changes in lifestyle and circumstances:

- Limitations in physical abilities.
- Loss of independence.
- Difficulty in returning to work or school.
- Loss of property, residence, pets, etc.
- Interruption of daily life activities and roles.
- Stress on intimate relationships.
- Challenges with sexual interests and intimacy.
- Extensive medical needs and new financial burdens.

Hospital care after burn injury:

Burn wound healing: Factors that will enable healing to occur include wound care, good nutrition, and maintenance of function, positive attitude and co-operation from the patient. Oedema reduction, prevention of burn wound infection and adequate analgesia will also contribute to optimal patient outcome.

Wound cleansing: It is interesting to note that in the general wound care arena; there has been much discussion on the choice of wound cleansing solutions. Most studies have not looked at burn wounds.

Mechanisms of wound debridement: There are five methods of wound debridement that may be utilised in wound management.

1. **Autolytic** - the use of moist dressing such as hydrogels or hydrocolloids may facilitate this.

2. **Surgical** – fascial, tangential or sharp debridement with scissors or scalpel.

3. **Enzymatic** – e.g. fibrinolysins, (mashed papaya is used in Africa).

4. **Mechanical** – wet to dry dressings (danger of damage to new epithelium), pulse lavage, gentle washing.

5. **Biological** – larvae of Lucilia sericata.

Burn wound dressing: Various biologic, biosynthetic and synthetic wound dressings are used in burn care. Selection and use of these products depends on the condition of the wound bed, the inherent properties of the dressing and the goals of therapy (Carrougher, 1998). In practice, different products seem to work for some centres and not for others. Patient population demographics and the local environment may all impact on the success or failure of certain dressing products.

Management of minor burns

- Clean burns with soap and water, or a dilute water-based disinfectant to remove loose skin.
- All blisters should be left intact to minimise the risk of infection.
- Larger blisters or those in an awkward position (in danger of bursting) should be aspirated under aseptic technique.
- Non-adhesive dressing, with gauze padding is usually effective, but biological dressings are better, especially for children.
- Ensure adequate analgesia and assess the need for tetanus prophylaxis.

Management of major burns

- Direct thermal injury producing upper airway oedema and/or obstruction.
- Inhalation of products of combustion (carbon particles) and toxic fumes, leading to chemical tracheobronchitis, oedema, and pneumonia.
- Carbon monoxide (CO) poisoning.

Psychological Intervention:

Mental health professionals are trained in methods for assessing and treating psychological distress. Professional help is particularly important if the distress is severe and interferes with things that are important to the patient. There are many health care practitioners such as psychiatrists, psychologists, social workers, and pastoral counsellors that can help. It is best to work with a mental health professional who has experience in treating people with severe injuries and expertise in treating the problems the patient may be experiencing (e.g., body image, social discomfort, post-traumatic stress disorder or PTSD).

Burn specific health scale (Munster, Fauerbach & Lawrence, 1996): The scale is composed of 114 items, based on the selection of items from a much larger pool by a group of professional and patient judges. The scale is used to quantify dysfunction and distress across six major domains of health.

Measurement of Quality of Life: A significant measure of the degree of recovery from major burn injury is health related quality of life (HRQL, Osaba, Rodrigues, Myles, Zee & Pater, 1998). This has been defined as a multifactorial construct that involves an individual's degree of satisfaction and level of functioning in several core domains, including physical-behavioural function, psychological well being, social and work role performance, and personal perception of health (Guilliemin, Bombardier & Beaton, 1998).

At all stages of patient care, it is essential to closely monitor and effectively treat the patient's pain symptoms. When intervening to help reduce patient distress, it is essential to tailor make approaches to stress reduction according to where the patient is in their burn care taking into consideration various social, individual and family determinants of the patient. Patient can receive education and interventions for problems such as acute stress disorder or PTSD, preparation for school return, and community reintegration (Burns, 1997). The next step is rehabilitation. Care is typically provided on an outpatient and initial reconstruction evaluations will often occur at this phase. Pain associated with open wounds and anxiety about survival becomes less of an issue. However, patients are faced with new issues that might include itching, poor sleep and integration into the community, including a return to work (Edgar & Brerton, 2004). Depression and PTSD often become more solidified at this time (Winter & Irle, 2004). Relationship issues and with sexual functioning may come to the forefront (Tucker, 1986). It is at this phase that long term psychotherapy with a mental health professional may become particularly important (Frank & Elliot, 2000). An important role of the burn team is to recognize when distress reaches the level where there is a need for psychological or psychiatric assessment (Pruzinsky, Rice, Himel, Morgan & Edlich, 1992). However, when a patient develops PTSD or major depressive disorder, then SSRI antidepressant agents are often warranted either alone or in combination with cognitive behaviour therapy (CBT). CBT enables the burn survivor to understand the thoughts, feelings and behaviours of others when they encounter them. It helps to develop new cognitive and social skills that can enhance confidence during challenging social interactions (Thompson & Kent, 2001). CBT also provides a set of skills and strategies known as REACH OUT.

R- Reassurance
E- Energy/ Effort
A- Assertiveness
C- Courage
H- Humor
O- Over there
U- Understanding
T- Tenacity

The skills and strategies are taught in various formats, including face to face sessions, group activities, and workshops and weekends, self help guidelines and social skill videos. Some burn centers have programs that encourage peer support visits. Others use professionally run support groups (Davison, Pennebaker & Dickerson, 2000). The Phoenix Society provides a means for burn survivors to get in touch with other survivors to get in touch with other survivors and available programs (Maslow & Lobato, 2010).

6. Pre and post surgery:

a. <u>Pre-surgery or pre-operative care:</u> Preoperative care is the preparation and management of a patient prior to surgery. It includes both physical and psychological preparation. Preoperative teaching includes instruction about the preoperative period; the surgery itself and the postoperative period. Preoperative teaching must be individualized for each patient. Some people want as much information as possible, while others prefer only minimal information because too much knowledge may increase their anxiety. Patients have different abilities to comprehend medical procedures; some prefer printed information, while others learn more from oral presentations. Preoperative care involves many components, and may be done the day before surgery in the hospital, or during the weeks before surgery on an outpatient basis.

Physical preparation: Physical preparation may consist of a complete medical history and physical exam, including the patient's surgical and anaesthesia background. The patient should inform the physician and hospital staff if he or she has ever had an adverse reaction to anaesthesia (such as anaphylactic shock), or if there is a family history of malignant hyperthermia. Laboratory tests may include complete blood count, electrolytes, prothrombin time, activated partial thromboplastin time and urinalysis. The patient will most likely have an electrocardiogram (ECG) if she or he has a history of cardiac disease, or is over 50 years of age. A chest X-ray is done if the patient has a history of respiratory disease. Part of the preparation includes assessment for risk factors that might impair healing, such as nutritional deficiencies, steroid use, radiation or chemotherapy drug or alcohol abuse, or metabolic disease such as diabetes. The patient should also provide a list of all medications, vitamins, and herbal or food supplements that he or she uses. Bowel clearance may be ordered if the patient is having surgery of the lower gastrointestinal tract. The patient should start the bowel preparation early in the evening before surgery to prevent interrupted sleep during the night. Some patients may benefit from a sleeping pill the night before surgery. The night before surgery, skin preparation is often ordered, which can take the form of scrubbing with a special soap (i.e., Hibiclens), or possibly hair removal from the surgical area.

Psychological preparation: Patients are often fearful or anxious about having surgery. It is often helpful for them to express their concerns to health care workers. This can be especially beneficial for the patients who are critically ill, or who are having a high risk procedure. The family needs to be included in psychological preoperative care. Pastoral care is usually offered in the hospital. If the patient has a fear of dying during surgery, this concern should be expressed. Patients and families who are prepared psychologically tend to cope better with the patient's postoperative course. Preparation leads to superior outcomes since the goals of recovery are known ahead of time, and the patient is able to manage postoperative pain more effectively.

b. Postoperative care: Surgery can be tough on a patient's body, and physicians do all that they can to aid their patients' recovery. However, it is important for all those involved in the patient's recovery to be concerned with their mental health as well as their physical health.
The body is a complicated and integrated organism; a physical surgery can have a resounding effect on a patient's mental state, but in the same way, a patient's mental state will have a significant impact on their body's physical state. Being prepared to care for the patient's postoperative psychological condition will greatly aid in the overall well-being of a patient and speed them on their way back to full health. The psychologists must take care of certain issues.

Short Term Memory Impairment: As patients return to consciousness, they will experience varying states of awareness. During this time, patients may struggle to remember things that happen around them or that are said to them. The psychologist should be patient and be prepared to repeat explanations and updates to the patient as they recover.

Loss of Concentration: In addition to recovering from the anaesthesia, patients will most likely be on medication to manage their pain. These two factors can lead to patients struggling to focus or concentrate on simple tasks. As the patients recover, the psychologist should make them understand that they should see an improvement in their attention span, but do not be surprised if they cannot perform basic arithmetic or if they get distracted easily.

Confusion: Another side effect of the pain medication can be confusion in the patient. Depending on the type and strength of the pain medication they are receiving, patients may experience anything from drowsiness to hallucinations. As their pain lessens, the patient will be able to decrease their pain medication and these side effects should disappear. The psychologist along with the nurse should redirect them, and make it sure to prevent them from performing tasks in which they could injure themselves.

Fatigue and Sleepiness: The more extensive the surgery, the more trauma the body experiences. After coming out of surgery, the patient will be fatigued and will spend most of their day sleeping. It is essential to allow them to sleep as much as possible to help them recuperate.

Self-Esteem Issues: Depending on the area of the surgery, patients may be self-conscious about the aesthetic effect of the surgery. This is especially true if the surgery occurred in the head and neck region. The knowledge of a change in their body can make a patient feel uncomfortable. It is important that good care is given to the incision to prevent infection and to aid in correct healing.

Post-Operative Depression: While most surgeons work with patients to reduce pre-operative stress, few doctors address the possibility of post-operative depression with patients. However, it is becoming more recognized in the medical world that patients frequently experience a post-operative depression that can last anywhere from a few weeks to six months. While the cause of this post-operative depression is still being researched, the symptoms have been shown to include excessive fatigue, apathy, irritability, shifts in appetite, and feelings of helplessness or despair. For some this depression may resolve itself, but it is important for patients and their caretakers to monitor their mental state and help them receive professional help if it becomes necessary.

7. Amputation:

Amputation is the intentional surgical removal of a limb or body part. It is performed to remove diseased tissue or relieve pain. Arms, legs, hands, feet, fingers and toes can be amputated. Most amputations involve small body parts such as a finger, rather than the entire limb. Amputation is performed to remove tissue that no longer has an adequate blood supply, to remove malignant tumours and because of severe trauma to the body part. Amputations can be either planned or emergency procedures. Details of operation vary slightly depending on what part is to be removed. The goal of amputations is twofold: to remove diseased tissue so that the wound will heal cleanly and to construct a stump that will allow the attachment of a prosthesis or artificial replacement part. The surgeon makes an incision around the part to be amputated. The part is removed and the one is smoothed. A flap is constructed of the muscle, connective tissue and skin to cover the raw end of the bone. A flap is closed over the bone with sutures (surgical stitches) that remain in place for about one month. Often a rigid dressing or cast is applied that stays in place for about two weeks.

Preparation: Before an amputation is performed, extensive testing is done to determine the proper level of amputation. The goal of the surgeon is to find the place where healing is most likely to be complete, while allowing the maximum amount of limb to remain for effective rehabilitation. The greater the blood flow through an area, the more likely healing is to occur. Several tests are designed to measure the blood flow and to choose the proper level of amputation.

- Measurement of blood pressure in different parts of the limb.
- Xenon 133 studies, which use a radiopharmaceutical to measure blood flow.
- An oxygen electrode is used in Oxygen tension measurement to measure oxygen pressure under the skin. If the pressure is 0, the healing will not occur. If the pressure reads higher than 40 mm Hg. Healing of the area is likely to be satisfactory.
- Laser Doppler measurements of the microcirculation of the skin
- Skin fluorescent studies that also measure skin microcirculation
- Skin perfusion measurements used to measure a blood pressure cuff and photoelectric detector.
- Infrared measurement of skin temperature.

The psychological aspect before Amputation: Immediate reactions to the prospect of amputation vary; they depend on whether the amputation was planned, occurred within the context of a chronic medical illness, or was necessitated by the sudden onset of infection or trauma. The context for amputation affects the psychological sequelae during the rehabilitation phase as well. When there is time to think about impending loss, classic stages of grief may be experienced (Parkes, 1975). Among these stages are denial (often manifest as a refusal to engage in discussion or to ask basic questions about the planned procedure), anger (which may be directed toward the medical team, with expressions of being “cheated” or “tricked” into agreeing to an amputation), bargaining (by attempting to forestall the surgery or to delay it indefinitely for a myriad of reasons such as “I'm too tired, I don't want to go through with any major surgery”), depression (taking the form of “learned helplessness,” feelings of passivity, and being overwhelmed), and acceptance (which may not be reached until the patient is well into the rehabilitation process) (Kubler Ross, 1969).

After learning that amputation may be required, anxiety often alternates with depression. This anxiety may be generalized (e.g., manifest by jitteriness, a decreased ability to sleep, silent rumination, and social withdrawal) or result in disturbed sleep and irritability. Not surprisingly, anxiety may be directed toward the fate of the limb that will be removed (Noble, Price, Gilder, 1954) as well as about the prospect of phantom limb pain, which many patients (who know of other amputees) may be familiar with. Intense sensitivity to the perceived negative attitudes of others toward people with disabilities may also be present, and this may initially be revealed by help-rejecting behaviour or expressions of indifference to questions related to what level of function to expect (Noble, Price, Gilder, 1954).

Psychological adaptation: The manner in which the surgery is presented by the surgeon can have much bearing on the magnitude and kind of affective response. Mendelson and co-workers recommend that the surgeon paint a realistic picture of the immediate and long-term goals for the patient and his family (Mendelson, 1986). Labeling the amputation as a reconstructive prelude to an improved life is a much different matter from implying that it is a mutilation and a failure. Furthermore, a hopeful attitude, detailed explanation of all aspects of the surgery and the rehabilitative process, and full response to all questions (especially those that seem trivial) appear to diminish anxiety, anger, and despair.

Psychological aspects after Amputation: According to Kolb (1975), an alteration in an individual’s body image sets up a series of emotional, perceptual and psychological reactions. Studies have reported depression being prevalent after amputation (Newell, 1991; Kashani, Frank, Kashani, Wonderlich & Reid, 1983). The most common psychopathogical symptoms included reaction to stressful events and adjustment disorder (predominantly affecting other emotions; mixed disorder of conduct and emotions; prolonged depressive reaction) and dysthymia (Friedman, 1987; Plastica & Deveceski, 2006).

Rehabilitation: There are many aspects should be taken care of after amputation like pain management, dressing, skin care and exercises.

Psychological intervention: Psychological treatment interventions should address both the person with an amputation and the family. It is important to provide a supportive environment where the person can discuss his feelings of loss and fears for the future. Some attempt should be made to support the person as he re-enters society and to continue to discuss with him his changed body image and how people may react poorly to him in public. An excellent way to give persons with amputations psychological support is through peer counselling or support groups. Every attempt should be made to have a person of similar age and type or level of amputation speaks with a person with recent amputation. Whenever possible the person with an amputation should be encouraged to return to work or previous life roles. If this is not possible, finding new roles for the person with an amputation will also help him to see that he is useful and that he does still contribute to society. Most people who have had an amputation experience phantom sensation, described as the feeling of the missing part of the limb. This requires no treatment but some people will also feel pain coming from the missing limb which is known as phantom limb pain. The pain is often described as a cramping or twisted posture of the missing limb. If a painful wound was present prior to the amputation, the phantom pain may mimic the pain of that lesion. Phantom limb pain is difficult to treat, causing frustration for the person with an amputations and care givers alike. Treatment for phantom pain should begin with touch (tactile stimulation) and biofeedback.

8. Evaluation of Organ donor and recipient:

Organ Donor: Maximizing the psychosocial status and well being of donors, both before and after the transplant, is among the foremost goals of transplant centres that have living organ programs for kidney, liver, lung, intestine, or pancreas transplantation. The psychosocial issues that are currently of greatest concern in the context of living organ donation, e.g. prevention of psychological harm, ensuring that donors are fully informed and decide to donate without coercion, monitoring donor's psychosocial outcomes are intimately linked to the factors that historically served as barriers to use of organs from living donors.

Medical evaluation: There are some of the syndromes that must be examined before organ transplantation. Tests should be done to examine virus of the diseases like, HIV *(*Simmonds, 1993*),* Cytomegalovirus (CMV), *(*Ho, 1994; Dummer, Hardy,Poorsattar, 1983*)* Hepatitis C virus (HCV)*(*Bouthot, 1997; Pereira, 1992; 1994; Pfau, 2000; Wreghitt, 1994*).*

Psychosocial evaluation: Although there is uniform recognition that psychosocial evaluation if a potential donor is critical (Ethics Committee of the Transplantation Society, 2004, Abecassis, 2000, New York State Committee on Quality Improvement in Liver Donation, 2002), there are no widely adopted standards for the content of evaluation. In many ways, the depth, value and purpose of the complete psychosocial evaluation of donors are analogous to those of the similarly extensive evaluation of candidates to receive organ transplants.

Core components of pre-donation psychosocial evaluation of living organ donors (Dew et al., 2005):

components	**Areas to be assessed**
Motivation for donation	Reasons for donation; how decision to donate was made; evidence of coercion/ inducement; expectations; ambivalence about donation.
Relationship between donor and recipient	Nature of relationship (biological, emotional, unrelated directed or unrelated nondirected); if related, quality of the relationship.

Attitude of significant others toward the donation	Support, pressure, and /or opposition by family, friends; availability of emotional and practical assistance during recovery.
Knowledge about the surgery and recovery	Understanding of risks of surgery, possible complications, expected recovery and recuperation time; understanding of basic issues.
Work and school related issues	Arrangements made with employer or school; financial resources.
Mental health history and current status	Psychiatric disorders (mood disorders, anxiety disorders, psychosis, suicidal ideation and/or attempts); personality disorders; substance use history (symptoms of abuse and/or dependence; quality and frequency of current use of alcohol and other substances); cognitive ability, and competence and capability to make informed decisions.
Psychosocial history and current status	Marital status and relationship stability, living arrangements; religious beliefs and orientations; community or religious activities; concurrent stressors (work related, home related, other); strategies used to cope with health related and other life stressors

Organ recipients: The aims of screening the potential recipient of a solid-organ transplant are 4-fold:

1. To determine the immune status of the recipient against common pathogens that can be transmitted by transplants. This is because established immunity against pathogens such as CMV, T. gondii, and possibly HBV protects the recipient from severe sequelae of a primoinfection with these agents.
2. To permit the allocation of organs from donors infected with a certain pathogen to recipients who are already carriers of this agent, such as HCV infection.
3. To recognize and possibly treat infections that can be expected to exacerbate or reactivate after immunosuppression such as tuberculosis, the endemic dimorphic mycoses such as coccidioidomycosis and histoplasmosis, or strongyloidiasis.

4. To avoid transplantation in patients with a poor prognosis after transplantation, such as HIV infection or colonization with certain panresistant bacteria.

A clinical and radiological workup for the detection of occult or latent infection should, in addition to a thorough history and physical examination, include in all patients a chest film in 2 planes for detection of infiltrates and residues of chronic infections such as tuberculosis, coccidioidomycosis, or histoplasmosis (B-III), a tuberculin skin test (C-III), and stool examinations for parasites (C-III).

Psychological evaluation: The main purpose of the psychosocial evaluation is to identify potential risk factors that could complicate surgical outcome. Risk factors such as substance abuse, compliance issues, and serious psychopathology can increase postoperative noncompliance and morbidity (Olbrisch et al., 2002). Evaluating such risk factors has two objectives: determining who should get an organ and identifying what resources the patient might need or what rehabilitation is needed before proceeding with the surgery. For example, learning that a patient has poor coping skills and a history of depression does not mean that patient is ineligible for a transplant from a psychosocial perspective. Rather, it tells the clinician that the patient may need services, such as therapy, before proceeding with the transplant procedure to ensure that the patient can adequately care for him/herself and the new organ.

The method used to obtain a psychosocial evaluation of transplant candidates can vary significantly according to medical facility, third-party payer requirements, and specific organ of transplant. The evaluation can be conducted by a psychiatrist, psychologist, and/or social worker, depending on a site's protocol. Some programs have a social worker conduct the psychosocial evaluation and only make referrals to psychologists or psychiatrists if indicated (Levenson & Olbrisch, 1993).

Psychiatric disorders are another focus of psychological evaluations. While some psychiatric disorders are linked to poor post transplant outcomes, psychiatric history itself do not necessarily predict a poor outcome (Olbrisch et al., 2002). A survey of cardiac, liver, and renal transplant programs indicates that the psychopathology most often considered absolute contraindications include active schizophrenia, current suicidal ideation, and dementia (Levenson & Olbrich, 1993). Other factors considered a relative contraindication include mental retardation, recent or distant suicide attempts, current or past affective disorders, controlled schizophrenia, and personality disorders.

Another major area of focus for the psychosocial evaluation is compliance. Dietary and medical noncompliance is most strongly linked to poor transplant outcome or death, so it is critical to assess a patient's ability to comply with the stringent post-transplant medical regimen. If the patient shows a pattern of medical noncompliance, the evaluator may suggest that the patient improve compliance before being considered an appropriate candidate.

The psychosocial evaluation provides the transplant team an opportunity to assess psychological factors that may inhibit successful transplant outcomes. If found, there are recommendations (e.g. develop better coping skills, demonstrate compliance) so that the patient may be considered a more appropriate transplant candidate in the future.

9. Pre and post transplantation:

Organ transplantation is the moving of an organ from one body to another or from a donor site to another location on the person's own body, to replace the recipient's damaged or absent organ. The emerging field of regenerative medicine is allowing scientists and engineers to create organs to be re-grown from the person's own cells (stem cells, or cells extracted from the failing organs).Organs and/or tissues that are transplanted within the same person's body are called autografts.

Psychosocial issues are present before, during, and after transplantation. Patients who hope to receive a donor organ typically try to increase their medical knowledge before the transplantation process takes place. With the availability of increased information on the Internet and through support groups, most organ transplant recipients become well informed. With this knowledge, however, come additional sources of anxiety, including transplant evaluation outcome, the shortage of organs, the uncertainty of donation, and increased understanding that having a transplant may not offer a cure. This overall uncertainty is a large precipitant to psychosocial problems for individuals prior to transplant (Engle, 2001).

Psychosocial issues for individuals in beginning stages of the transplant process include pre-transplant evaluation stress; family role adjustment as the recipients' physical capabilities change; patients' fear and anxiety about futures; loss of comfort, independence, autonomy, and privacy; and increased strain in relationships with friends and work associates. It is essential that patients recognize and consider these issues prior to transplantation because accompanying psychological characteristics such as anxiety and depression are associated with poor health practices throughout the transplant process and impaired post transplant health outcomes.

Psychosocial stressors change during the transplant process. These changes result from relocation to the transplant center, guilt over knowing that a donor death occurred in order to procure a donor organ, coping with the medical regimen, body image concerns, and exposure to loss. Diagnosable disorders among patients during hospitalization have been identified, including anxiety disorders, cognitive impairment, depression, and even posttraumatic stress disorders (e.g., Bunzel, Laederach-Hofmann, Wieselthaler, Roethy, & Drees, 2005).

After the transplant procedure, psychosocial concerns tend to revolve around readjustment to a new lifestyle. These concerns include anxiety at discharge, perhaps due to loss of security or fear of adjustment; sadness or guilt from leaving relationships formed with other transplant patients; estrangement from family and community during readjustment to family and work roles; and increased physical and functional impairment (Dew et al., 2000; Olbrisch, Benedict, Ashe, & Levenson, 2002).

Psychosocial intervention: Clinicians can facilitate support groups and bibliotherapy for patients and families at the stages of the transplant process to normalize and respond to psychosocial issues (Hodges, Craven, & Littlefield, 1995). Intervention programs for enhancing positive adaptation among chronic illness populations (e.g., cancer) are developed and supported by research (Antoni, Carver, & Lechner, 2009). Such programs use cognitive behavioural stress management (CBSM) techniques - changing faulty cognitive appraisals of stressors, enhancing emotional processing, improving relaxation skills, and bolstering social support—to increase psychological and physiological outcomes. Such programs offer psychosocial interventions for enhancing positive adjustment among transplant populations. Caring for donor families involves interventions targeted to issues of grief and loss. Maloney and Wolfelt (2001) note particular challenges face donor families. Initially, families must face the decision for or against donation. In spite of the importance of this decision for donation, there are many barriers to its implementation. Families are usually shocked and disbelieving; donation requests often occur after automobile or other accidents, and family members may be summoned to the hospital without even knowing their loved one is on life-support. Procurement coordinators may be more focused on the facts behind the decision, for example, being sure families understand the definition of brain death and the steps of donation, and less emotionally available for support. Psychologists can serve a critical role by giving family members support for anger and grieving, and promoting rituals surrounding death, including viewing the body.

10. Organ replacement:

Various new technologies, including stem cells, tissue engineering, xenotransplantation, and organogenesis, all have potential for replacing or augmenting organ function. According to Jeffrey L. Platt (MD, Director of Transplantation Biology at the Mayo Clinic in Rochester, Minnesota), these technologies have significant overlap. Organogenesis may use xenotransplantation as a way of producing organs in the future, that is, human organs developed in animals. However, biological and immunologic obstacles may delay application of these technologies (Cascalho & Platt, 2001). In addition, the government and the public have voiced ethical objections to studying stem cells, because of their capacity to become embryos.

Tissue Engineering is the science of growing replacement organs and tissue in the lab to replace damaged or diseased tissue. The process usually starts with a three-dimensional structure called a scaffold that is used to support cells as they grow and develop. Skin, blood vessels, bladders, trachea, oesophagus, muscle and other types of tissue have been successfully engineered; and some of these tissues have already been used in treating human disease. Solid organs such as the liver, kidney, heart and pancreas are especially challenging and are considered the "Holy Grail" of tissue engineering.

According to WHO **Xenotransplantation** is the transplantation of living cells, tissues or organs from one species to another. Such cells, tissues or organs are called xenografts or xenotransplant. In contrast, the term allotransplantation refers to a same species transplant. Human xenotransplantation offers a potential treatment for end-stage organ failure, a significant health problem in parts of the industrialized world. A worldwide shortage of organs for clinical implantation causes about 20–35% of patients who need replacement organs to die on the waiting list (Heally et al., 2005). Certain procedures, some of which are being investigated in early clinical trials, aim to use cells or tissues from other species to treat life-threatening and debilitating illnesses such as cancer, diabetes, liver failure and Parkinson’s disease. If vitrification can be perfected, it could allow for long-term storage of xenogenic cells, tissues and organs so that they would be more readily available for transplant.

Organogenesis in embryology is the series of organized integrated processes that transforms an amorphous mass of cells into a complete organ in the developing embryo. The cells of an organ-forming region undergo differential development and movement to form an organ primordium, or anlage. Organogenesis continues until the definitive characteristics of the organ are achieved. Concurrent with this process is histogenesis; the result of both processes is a structurally and functionally complete organ. The accomplishment of organogenesis ends the period during which the developing organism is called an embryo and begins the period in which the organism is called a foetus (Encyclopaedia Britannica).

A **stem cell transplant** (also called a blood or marrow transplant) is the injection or infusion of healthy stem cells into the body to replace damaged or diseased stem cells. A stem cell transplant may be necessary if one's bone marrow stops working and doesn't produce enough healthy stem cells. This procedure also may be performed if high-dose chemotherapy or radiation therapy is given in the treatment of blood disorders such as leukemia, lymphoma, multiple myeloma or sickle cell anaemia.

Cell therapy (also called **cellular therapy** or **cytotherapy**) is therapy in which cellular material is injected into a patient (American Cancer Society, 2013) this generally means intact, living cells. For example, T cells capable of fighting cancer cells via cell-mediated immunity may be injected in the course of immunotherapy.

The psychological distress in patients and psychological intervention are same as described in organ transplantation.

11. Haemophilia:

Haemophilia is characterized by a tendency to have lengthy bleedings. The bleedings can appear to occur spontaneously or after minimal trauma. It is typical that bleeding occurs in the joints and muscles. Untreated bleedings cause gradual degeneration of the body's motility apparatus. Sometimes bleeding in the mucous membrane happens. Life threatening bleedings can occur after a slight trauma to the skull or internal organs. It is caused by a congenital deficiency of/or complete absence of the functioning coagulation factor VIII (haemophilia A) or coagulation factor IX (haemophilia B). These are hereditary and transmitted recessively. The disorder affects almost exclusively men, while women are the carriers.

Psychological aspects:

Acceptance of initial diagnosis: The disease can be frightening for those who haven't previously encountered it in the family, and initial diagnosis tends to have a huge financial and emotional impact on the whole family. The literature highlights that diagnosis of the disease has a life changing effect on parents who feel shock and guilt, resulting in a lack of self esteem (Beeton *et al*, 2007). Studies reveal that the quality of life is most impaired at the initial stages of the diagnosis (Bullinger *et al*, 2003). The majority of the literature suggests that mothers have a key role in the treatment of their sons and some papers refer to the 'special guilt' some mothers feel as carriers, who feel they are ultimately responsible for their son's haemophilia (Ross, 2004). This guilt can be overwhelming and can give way to passiveness and despair at a time when parents need to focus on getting the best medical treatment for the child (Ross, 2004). Guilt can result in depression and a rejection of their child (Cassis, 2007) or by contrast overprotection and overcompensation which can ultimately lead to the child using their haemophilia as a means to avoid challenges, responsibility and adulthood (Ross, 2004). Parents can feel overwhelmed by responsibility in the management of the life of their child with haemophilia. While most of the time a normal life is possible, the unpredictable nature of the disease means that setbacks can occur. After long periods of normality, these can make the patient and family feel vulnerable once again. (Beeton *et al*, 2007).

Fear of pain: One of the major challenges for the patient themselves is learning to manage the stress of chronic pain and dysfunctional joints (Manco-Johnson, 2003; Valentino, 2007).

Anxiety and depression, anger and frustration: Depression is most commonly found in individuals with haemophilia who have contracted HIV or Hepatitis C through contaminated blood products. The social stigma that is attached to these diseases along with the side effects of treating these co-morbid conditions, can lead to depression and anxiety (Barlow *et al*, 2007). Anger and frustration, anxiety and depression are common themes in the literature along with a feeling of uncertainty about the future both for themselves and for their families (Barlow *et al*, 2007; Beeton *et al*, 2007; Bottos *et al*, 2007).

<u>Treatment approach:</u>

Multi-disciplinary team: Multidisciplinary teams are now a common feature in some areas of healthcare and bring a range of professionals and their complementary skills to bear on treatment and support. Such a team could include physicians (often with different specialist skills), specialist nurses, psychologists, physiotherapists, social workers and occupational therapists. They would therefore hold expertise in several areas identified in the literature as key to psychosocial support.

Medical intervention: The aim of the treatment of haemophiliacs is to counteract and prevent bleedings and the reduction of the body's motility capacity and other complications of the disorder. Preventive treatment, so-called "prophylaxis", with factor concentrate has proved to be the key to a life without the motility apparatus being destroyed. Thoroughly purified and virus inactivated and recombinant preparations have meant less and less risk of side-effects lately. Progress within gene therapy research provides new hope for haemophiliacs but can only primarily be seen as a cure for the disorder in a longer time perspective.

Regular check up: Individuals with haemophilia are monitored regularly at a haemophilia centre to evaluate the effects of the treatment. The check-ups comprise a medical examination, an assessment of the status of the joints, hepatitis and HIV infection as well as the presence of antibodies to factor concentrate, a decision on continued dosage, the need for pain relief, physiotherapy and orthopaedic surgery and the need for dental care. When necessary, genetic advice is also given. The check-ups are performed every six months or every other year depending on the severity of the disorder and possible complications. Each check-up is followed by a written result including treatment recommendations, which are sent to the patient's local doctor and to the haemophiliac.

Education and information: The haemophilia centres provide education and information for health-care staff as well as for the haemophiliac and his/her relatives. Individuals with severe or moderate haemophilia need continuous education about prophylactic treatment, about dosages in different types of bleedings and about possible complications.

Psychological Counselling: Bottos *et al* (2007), have mentioned the positive change to or strengthening of the use of adaptive, problem solving coping strategies resulting from regular counselling programmes, where families are given information and the chance to share their experiences and feelings with other families with the disease. Counselling adults allows them to express fully their emotions, fears and frustrations so as gradually to move them to a practical mindset that allows them to cope with the disease and parent the child in a way that it develops good mental and physical health (Cassis, 2007; Beeton *et al*, 2007). Also, counselling young children to understand the benefits of their medical treatment so as to cope with their frustration will enable them to focus on their possibilities. Cassis (2007) stressed the importance of careers counselling regarding training options and work for young adults in order to find work that is suitable for them. The need to make a contribution to society by working is essential for adult individuals with haemophilia and has a hugely positive impact on their self-esteem.

Psychological care should be provided in every developmental stages of the patient since his childhood. To provide optimal care, healthcare professionals need to be able to identify issues and challenges related to having a bleeding disorder that may be affecting their patients' cognitive and emotional development. Some people with haemophilia may focus on the emotional challenges, while others find ways to better cope with their situation. Short-term psychotherapy, alternative therapies, and social services can help individuals with haemophilia cope with symptoms and limitations and develop a healthy sense of self.

12. Sensory impairment:

Sensory impairment is when one of the five senses; sight, hearing, smell, touch, taste and spatial awareness, is no longer normal. Examples - If you wear glasses you have sight impairment, if you find it hard to hear or have a hearing aid then you have a hearing impairment or it can be said that The term sensory impairment encompasses visual loss (including blindness and partial sight), hearing loss (including the whole range) and multisensory impairment (which means having a diagnosed visual and hearing impairment with at least a mild loss in each modality or deaf-blindness).

Vision impairment (VI): This term covers varying degrees of vision loss including those who are registered severely sight impaired (blind). Even the latter may have some vision, such as being able to tell the difference between light and dark. There are many conditions that cause different kinds of vision loss; the main distinction between conditions is whether the impairment is ocular (eye) or cerebral (brain). Cerebral VI (also known as cortical VI) is common in children with CLDD/PMLD. Functional vision refers to the interaction between the environment and how the visual information is processed. Knowing a student's condition and degree of functional vision may help staff to understand what they can see.

Hearing impairment (HI): Two main types of hearing impairment are:

Conductive hearing loss: The most common type and results from interference in the conduction pathways through which sound reaches the inner ear. This hearing loss usually affects the volume of sound reaching the inner ear. People with conductive hearing loss may benefit from the surgical insertion of grommets or from hearing aids. It is commonly a temporary hearing loss.

Sensorineural hearing loss: It is caused by damage to the hair cells lining the inner ear, or the nerves that supply them. This hearing loss can range from mild to profound, and affects certain frequencies more than others. Consequently, people with sensorineural hearing loss need high quality hearing aids or cochlear implants to gain access to the spoken word and sound in the environment.

Multisensory impairment (MSI): This is a term used to describe students who have a combination of visual and hearing loss. They are sometimes referred to as deaf-blind, although many have some residual sight and/or hearing. The combination of the two sensory losses intensifies the impact of each. Students with multisensory impairment have much greater difficulty in accessing the environment and the curriculum, than those with a single sensory impairment.

Locomotor Disability: Locomotor disability is defined as a person's inability to execute distinctive activities associated with moving both himself and objects, from place to place and such inability resulting from affliction of musculoskeletal and/or nervous system.

<u>Psychological distress related to sensory impairment:</u>

There is a study was done by Bodsworth, Clare, Simblett, and Deaf-blind UK (2011) on deaf-blindness done in the leading organizations of UK (England, Wales and Scotland). They found that almost half the men and women in their sample (45.8% of 439 respondents) reported high levels of anxiety, depression, physical symptoms and/or social impairment. The findings showed that mental distress is three times more common among people with deaf-blindness than among the general adult population. It is more than twice as common as among other older people. Respondents reported experiences of social isolation, a loss of independence, and the impact of other people's negative attitudes.

Psychological and community services: The community psychologists provide services including: therapy for mental health crisis like emotional and behavioural issues, personality difficulties, drug and alcohol abuse, communication and language disorders. Vocational skill training, occupational therapies are also provided to sensory impaired people.

13. Rheumatic diseases:

Rheumatic diseases are painful conditions usually caused by inflammation, swelling, and pain in the joints or muscles. Almost any joint can be affected in rheumatic disease. There are more than 100 rheumatic diseases.

Osteoarthritis: About 27 million Americans have osteoarthritis (OA), the "wear-and-tear" arthritis. OA causes damage to the cartilage over time. Cartilage is a material that cushions the end of bones and allows joints to move smoothly. As cartilage of a joint wears down, this joint movement becomes painful or limited. OA can be a normal part of aging that can affect many different joints. However, it usually affects the knees, hips, lower back, neck, fingers, and feet. The signs and symptoms of OA, depending on the joints involved, include:

- Pain in joint
- Joint swelling
- Joint may be warm to touch
- Joint stiffness
- Muscle weakness and joint instability
- Pain when walking
- Difficulty gripping objects
- Difficulty dressing or combing hair
- Difficulty sitting or bending over

Rheumatoid Arthritis: In Rheumatoid Arthritis (RA), the body's immune system attacks its own tissues, causing joint pain, swelling, and stiffness that can be severe. The condition can result in permanent joint damage and deformity. RA signs and symptoms include:

- Joint pain and swelling
- Involvement of multiple joints (usually in a symmetrical pattern)
- Other organ involvement such as eyes and lungs
- Joint stiffness, especially in the morning
- Fatigue
- Lumps called rheumatoid nodules

Lupus: SLE or systemic lupus erythematosus is another autoimmune disease; the cause of SLE is unknown. Lupus signs and symptoms include:

- Joint pain
- Fatigue
- Joint stiffness
- Rashes, including the "butterfly rash" across the cheeks
- Sun sensitivity
- Hair loss
- Discoloration of the fingers or toes when exposed to cold (calledRaynaud's phenomenon)
- Internal organ involvement, such as the kidneys
- Blood disorders, such as anemia and low whit blood cell or low platelet counts
- Chest pain from inflammation of the lining of the heart or lungs
- Seizures or stroke.

Ankylosing Spondylitis: Ankylosing Spondylitis **(AS)** usually starts gradually as lower back pain. The hallmark feature of AS is the involvement of the joints at the base of the spine. This is where the spine attaches to the pelvis, also known as the sacroiliac joints. Ankylosing spondylitis is more common in young men, especially from the teenage years to age 30. With progression of AS, the spine may become stiffer. It may become difficult to bend for common everyday activities. AS symptoms include:

- Lower back pain that worsens and works its way up the spine
- Pain felt between the shoulder blades and in the neck
- Pain and stiffness in the back, especially at rest and on arising
- Pain and stiffness get better after activity
- Pain in the middle back and then upper back and neck (after 5-10 years)

Sjogren's Syndrome: Sjogren's syndrome **is** an inflammatory, autoimmune disease. It can occur with other autoimmune diseases such as RA and lupus, but also on its own. Although the cause of Sjogren's is unknown, it is more common in women. Sjogren's signs and symptoms include:

- Dry eyes (the glands in eyes do not make adequate tears)
- Eye irritation and burning
- Dry mouth (the glands in mouth do not give adequate saliva)
- Dental decay, gum disease, thrush
- Swelling of the parotid glands on the sides of the face
- Joint pain and stiffness (rarely)
- Internal organ diseases (rarely)

Psychological aspects of rheumatic disease: Rheumatic diseases have a great impact on the quality of life. They affect not only physical functioning but also psychological and social aspects (Anderson et al., 1985). Rheumatic diseases have on the one hand behavioural, psychological and social consequences, but on the other hand behavioural, psychological and social variables may be determinants of disease development and of patients' ability to adapt to their condition. The rheumatic disease will have physical as well as behavioural and psychosocial implications for the patient and his or her family. The quality of life of rheumatic patients can be characterized by functional disability (e.g. restrictions in mobility and daily activities), pain, loss of independence, psychological problems like depression and anxiety, changes in family functioning and social activities, work disability and financial problems (Anderson et al., 1985; Cornelissen et al., 1988). McFarlane and Brooks (1988) found that psychological factors, like denial of emotional distress or depression, predicted more of the variance in disability, over a 3-year period, than disease activity in 30 RA patients. Several studies have demonstrated relationships between the use of certain coping strategies and adjustment (Bradley, 1989b). RA patients' use of negative coping strategies such as catastrophizing, escapist fantasies and engaging in passive pain management strategies is associated with poor psychological status and high levels of functional impairment (Brown et al., 1989; Keefe et al., 1989; Revenson and Felton, 1989). Various studies have indicated that cognitive factors like self-efficacy or learned helplessness are related to functional and psychological impairment (Lorig et al., 1989; Nicassio et al., 1985). The unpredictable nature of RA and the varying disease activity may cause patients to view their disease as uncontrollable. This often leads to low self efficacy expectations about needed causes of action to cope with the consequences of the disease or to more generalized feelings of learned helplessness; that is the experience of having no control over one's life, in general, and across different situations.

Treatments for Rheumatic Diseases:

Treatments for rheumatic diseases include medications to improve symptoms and control disease. Along with drugs, other parts of a treatment plan include: regular exercise, balanced diet, stress reduction and rest. Working with a rheumatologist can help to find the best ways to manage the condition.

Role of Clinical Psychologist:

The psychologist assesses the individual's and family's psychological status and ability to cope with the unpredictable nature and changing health status associated with rheumatic diseases. The psychologist conducts psychological tests and interviews that may be used to assess an individual's psychosocial status including;

• Adjustment to disability

- Adherence to treatment
- Coping style
- Family interaction/communication
- Mood, such as levels of anxiety and depression
- Cognitive functioning
- Transition planning (adolescents)

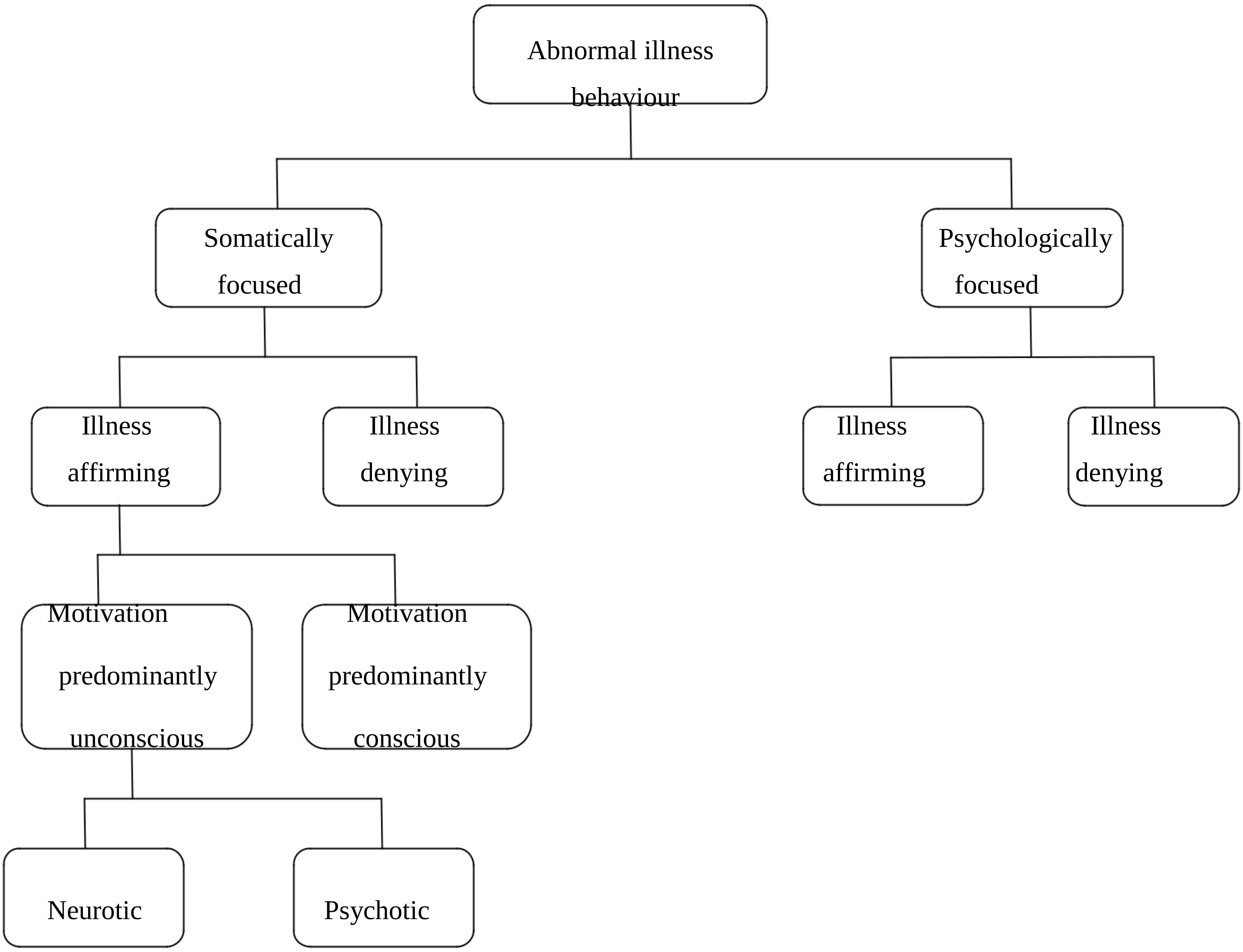
Based on an evaluation, the psychologist tailors a treatment plan to meet the needs of the patient. The psychologist provides a wide range of interventions designed to enhance coping and overall psychological well-being, including cognitive behavioural therapy, pain, sleep and stress management, sexual and relationship counselling, family or couples counselling, and psychotherapy.

Psychological therapy ranges from short-term interventions for dealing with normal adjustment issues and crisis management and interventions (involving persons high-risk for harm to self or others) to long-term psychotherapy for the treatment of more chronic psychological disorders, which are typically provided in a private practice setting, in a hospital or outpatient clinic. In the rehabilitation setting, the licensed psychologist may be called upon for consultation in matters related to behavioural management, treatment adherence, and cognitive dysfunction.

14. Abnormal illness behaviour:

Abnormal Illness Behaviour (also named "dysnosognosia") (Pilowsky, 1986) is defined as "the persistence of an inappropriate or maladaptive mode of perceiving, evaluating or acting in relation to one's own state of health, despite the fact that a doctor (or other appropriate social agent) has offered an accurate and reasonably lucid explanation of the nature of the illness and the appropriate course of management to be followed (if any) with opportunities for discussion, negotiation and clarification based on an adequate assessment of all biological, psychological, social and cultural factors".

In 1969, the concept of Abnormal Illness Behaviour (AIB) was introduced (Pilowsky, 1986) in an attempt to clarify the confusion over such terms as "hysteria" and "hypochondriasis". The concept and some related terms were further refined in a paper entitled "A general classification of abnormal illness behaviors" which also described a way of classifying such behaviours (Pilowsky, 1978).



The forms of illness behaviour which will be seen by a clinician will vary, depending on whether patients are seen in the community, in a general hospital or a psychiatric service. They may be usefully subdivided as follows:

Somatic Misattribution: Misattribution is frequently encountered in the community general practitioners and physicians. The overwhelming majority of patients attending general practitioner's present with a somatic symptom. Of these, up to a half may have a somatic cause. In the remaining patients the symptoms will be part of an emotional response to a stress which will be readily identifiable e.g. bereavement, loss of employment etc. The stress may be ongoing or there may have been a recent crisis. After a careful history and physical examination (i.e. complete clinical evaluation) has been carried out, the doctor is in a position to explore psychosocial issues and then explain to the patient how somatic symptoms may be the most prominent features of a response to a stress. Many of these patients are seen by specialist physicians, neurologists, gastroenterologists, thoracic physicians and cardiologists, depending on the localization of symptoms e.g. headaches, dyspepsia, diarrhoea, palpitations, hyperventilation or combinations of these. It has been suggested that these syndromes be labelled "Autonomic Arousal Disorders" [DSM-IV Options book: Work in Progress 9/1/91 Task Force on DSM-IV, American Psychiatric Association]. However diagnoses such as Adjustment disorder, Anxiety disorder, Phobic disorder and Panic disorder may apply.

Abnormal Illness Behaviour- Neurotic: Abnormal Illness Behaviour of the neurotic type encompasses a number of diagnoses which have been grouped as somatoform disorders, e.g. hypochondriasis, conversion disorder, somatoform pain disorder etc. In these conditions the sick role is adopted as a means of achieving psychological equilibrium on the basis of current stress and predisposing personality factors arising from childhood developmental experiences such as hospitalization, parental models, being labelled as "sickly" or "almost died at birth," over protective parents etc. Eventually many of these patients will reach psychiatrists, especially those working in general hospitals on consultation-liaison services or in pain clinics.

Abnormal Illness Behaviour- Psychotic: In their most obvious forms such conditions present with delusions, of which the most commonly described are hypochondriacal in nature. These syndromes are usually encountered as part of affective (especially depressive) psychoses or schizophrenia. In some cases the delusion is the most prominent, if not the only feature, such that it is considered to represent a form of paranoia. Monosymptomatic hypochondriasis and body dysmorphic disorders may be grouped with these conditions. These patients are seen most commonly by psychiatrists. They may be treated as outpatients, in-patients on general hospital psychiatric wards, or in psychiatric hospitals. Studies of hypochondriasis in these settings are likely to conclude that the condition is secondary, usually to a depressive illness.

Treatment option:

1. Pharmacological treatments: The tricyclic antidepressants have been widely used in the treatment of chronic pain and they are very effective when a depressive syndrome is present. However, it has also been proposed that the tricyclics may reduce pain intensity by enhancing the activity of the endogenous pain suppression system (which is dependent on both serotonin and noradrenaline). Pilowsky et al. (1990) conducted a double-blind placebo controlled cross-over trial of amitriptyline in patients referred to a pain clinic.

2. Somatic Treatments: Somatic treatments such as relaxation training, physiotherapeutic massage and transcutaneous nerve stimulation to be helpful in providing some patients with a degree of relief. In addition, these treatments help to convince the patient that the somatic dimension of the pain experience is being taken seriously and thus helps to establish rapport.

3. cognitive Behaviour Therapy: The cognitive-behavioural approach, typically requires a 3-6 weeks inpatient stay and participation in a highly structured program involving graduated exercises and activities, cognitive restructuring, individual psychotherapy and family therapy. It is most important that patients are well prepared for the program and show motivation for change. Fordyce who pioneered this approach has described the methods well (Fordyce, 1976). Barsky et al. (1988) describe a cognitive-educational treatment for hypochondriasis based on the idea of somatosensory amplification. Four factors are considered to amplify somatic symptoms:

(1) Attention expectation;
(2) Symptom attribution and appraisal;
(3) The context used for interpreting the symptoms, and
(4) Disturbing affect and dependency needs.

Groups of six to eight patients meet for six consecutive weeks for a 'course' on the perception of physical symptoms. The educational component is stressed in order to reduce patient resistance and the 'stigma attached to psychiatric treatment'. Patients are introduced to techniques for reducing somatic 'hypervigilance' such as attention and relaxation exercises and distraction techniques. A good deal of didactic material is presented, and this approach to hypochondriasis seems to be acceptable to patients and to help by offering a logical, internally consistent model within which to gain sense of mastery over their abnormal illness behaviours. The cognitive-behavioural approach has been used widely in the treatment of somatoform pain disorders, especially in North America. Benjamin (1989) has reviewed the psychological treatment of chronic pain and concluded that the various approaches may well be complementary. He urges that approaches should be eclectic, be free of dogmatism and employ methods which have been shown to work.

4. Psychotherapeutic approaches: A variety of psychotherapeutic approaches may be taken, including individual, group, marital and family. In their work, Pilowsky & Bassett (1982) have focussed particularly on the brief individual psychotherapeutic approach. In a pilot study on a small group of chronic pain patients, they found that dynamically orientated psychotherapy consisting of 12 weekly 45 minute sessions produced better results as regards global functioning, when compared to 6 fortnightly 15 minute supportive sessions (Basset & Pilowsky, 1985). In the course of their experience with the use of individual psychotherapy, they have found that reduction in pain complaints cannot be regarded as the only index of improvement.

The use of **group psychotherapy** has been well described by Pinsky working at the City of Hope Medical Centre (Pinsky, 1978). In any program where patients are admitted in cohorts for a fixed period, groups have a part to play in facilitating information exchange and therapeutic modelling. It is crucial that the role of the spouse and family not be overlooked in the management of abnormal or discordant illness behaviours.

Hypnotherapy is an approach which has been poorly evaluated for its contribution to the treatment of somatoform disorders in general, but it would appear to have a role in pain control when used for certain individuals in appropriate contexts (Turner & Chapman, 1982; Barber & Adrian, 1982; Chapman, 1985).

5. **Combination Therapy:** In practice most patients are treated with combinations of therapies. Pilowsky and Barrow (1990) have reported on a controlled evaluation of brief psychotherapy and amitriptyline (AMI). Outcome was independently assessed in terms of 'categorical' variables (pain, well-being and activity) and a number of 'continuous' variables (intensity of pain, amount of time in pain and 'productivity', i.e. ability to carry out usual tasks and duties). Analysis of the categorical data showed significant findings only for 'activity' in that patients receiving supportive psychotherapy (i.e. 6,15 minute fortnightly sessions) did better with AMI than with placebo. Further, those on AMI did better without psychotherapy (12,45 minute, weekly sessions). Overall, those on AMI showed improved activity levels. An interesting finding which emerged was that patients on psychotherapy and placebo reported a significant increase in pain intensity, but also a significant increase in productivity.

15. Health anxiety:

Health anxiety (Hypochondria) is excessive worrying about health, to the point where it causes great distress and affects everyday life. Some people with health anxiety have a medical condition, which they worry about excessively. Others have medically unexplained symptoms, such as chest pain or headaches, which they are concerned may be a sign of a serious illness, despite the doctor's reassurance. Health anxiety can be a vicious circle. If a person constantly checks his body for signs of illness, such as a rash or bump, he will eventually find something. It often won't be anything serious – it could be a natural body change, or he could be misinterpreting signs of anxiety (such as increased heart rate and sweating) as signs of a serious condition. However, the discovery tends to cause great anxiety and make him self-check even more. He may find himself needing more reassurance from doctors, friends and family. The comfort he gets from this reassurance may be short-lived, or he may stop believing it, which only means he needs more of it to feel better. Seeking reassurance just keeps the symptoms in his head, and usually makes him feel worse. When physical symptoms are triggered or made worse by worrying, it causes even more anxiety, which just worsens the symptoms. Excessive worrying can also lead to panic attacks or even depression.

<u>Types of health anxiety:</u> People with health anxiety can fall into one of two extremes

Constantly seeking information and reassurance – for example, obsessively researching illnesses from the internet, booking frequent appointments to general practitioners, and having frequent tests that don't find any problems.

Avoidant behaviour – avoiding medical TV programmes, general practitioner appointments and anything else that might trigger the anxiety, and avoiding activities such as exercise that are perceived to make the condition worse.

<u>Treatment:</u>

Role of General Practitioners:

Once the general practitioner has established that his patient does suffer from health anxiety, and there is no serious underlying physical cause for any symptoms he might have, the practitioner should investigate whether the patient might have a problem, such as depression or anxiety disorder that may be causing or worsening his symptoms. If this is the case, the person may be referred for psychological therapy and he may benefit from antidepressants.
Antidepressants may be helpful if the person has a mental health condition such as depression. For some people, these may work better than CBT. The physician can directly prescribe antidepressants or refer to a mental health specialist for treatment.

Role of Psychologists:

The psychologists assess first the symptoms of health anxiety through interview and questionnaires. Cognitive behavioural therapy (CBT) is an effective treatment for many people with health anxiety. It involves working with a trained CBT therapist to identify the thoughts and emotions the person experiences and the things he does to cope with them, with the aim of changing unhealthy thoughts and behaviours that maintain health anxiety. CBT looks at how to challenge the way the person interprets symptoms, to encourage a more balanced and realistic view. It should help to:

- learn what seems to make the symptoms worse
- develop methods of coping with the symptoms
- keep yourself more active, even if you still have symptoms

However, CBT is not the best treatment for everyone with health anxiety. Some people may benefit more from a different psychological therapy, such as trauma-focused therapy or a psychotherapy that will help a particular psychological condition.

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UNIT-XIII

EFFECTS OF PSYCHOTHERAPY ON THE BIOLOGY OF THE BRAIN

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"Effects of Psychotherapy on the Biology of the Brain"

Introduction

Sometime in the 1980s, the Dalai Lama was visiting a university hospital in the United States (Begley, 2007) observing a surgery to remove a tumor from the brain of a patient. At the conclusion of the operation, the surgeon announced that the patient would be fully functioning when he recovered. The Dalai Lama voiced his understanding that a tumor in the brain would impair mental functions. However, he asked the surgeon whether or not a mental activity, a thought, could affect the physical structure of the brain. The prompt response was a definite "No! There is no way mental activity can have physical effects in the brain." Some 20 years later, that "no" has been shown to be wrong. Years of neuroscience research support what H. H. the Dalai Lama was implicitly suggesting: Mental activity effects physical, neuroplastic changes in the brain.

In just over two decades, developments in a branch of biology—the discipline of Neuroscience—promise to enhance our understanding of mental illness and mental health, and expand our treatment strategies and tools. They are leading to a major shift in how we understand the relationships between brain biology and effects of experience.

Three major developments

One is the advance of neuroimaging technologies allowing us to see electrical and metabolic activities in a living working brain.

The second development involves the concept of neuroplasticity; imaging shows that the brain exhibits durable physical, biological changes as a result of learning. When we observe behaviors that change and persist as a result of experience that is what we refer to as learning.

What we know now, is that what we observe as learning is accompanied by predictable biological changes that have occurred in the brain.

The third discovery is what can be called self-directed neuro-plasticity. Conscious voluntary directing of attention and practice with voluntary behaviors can produce durable biological changes in the brain. The processes of psychotherapy always involve this self-directed neuroplasticity.

Later research in neuroplasticity supports the idea that humans have the power to correct biological brain glitches by retraining the brain with more positive learning experiences. An important note is that brain activity and organization are here-and-now phenomena. Although these are partly results of history, in the form of genetic blueprints and experiences from the past, neuroplasticity takes place in the present. We should be cautious, at a minimum, about deciding if an impairment is correctible, based on origin—e.g. genetic source, or brain injury source, or early life experience source, or long-term chronic source. We do better with prognosis by looking at evidence for if and how a particular impairment can be corrected.

Assumptions about the origins of malfunctions don't lead to answers about how to correct the condition. Issues framed in terms of genetics versus experience, for instance, will not lead to solutions for correcting a condition. The empirical question is whether or not the impaired biology can be corrected. The science of genetics has, of necessity, spawned an emergent field of epigenetic, with discoveries of how other life factors influence how a specific genetic blueprint will express itself (Jaenisch & Bird, 2003). Illness, medication, nutrition, and other life experiences can alter outcomes for identical genes. Obviously this does not hold for all genes. However, many genetic programs are not immutable predetermined fate; they are tendencies with more than one possible outcome.

An interesting example is the work by Kagan (1992) with his studies of differences between timid and bold children. Mothers had been bringing their toddlers to his laboratory at Harvard University for many years, for studies of developmental changes overtime. Studies of temperament differences followed children about age two, and in later years. Mothers brought their toddlers of about 21 months of age to the new experience of the lab. Observers noted sharp differences in the way different toddlers responded.

Most of these children exhibited a robust attraction to the play materials and to the other children. They were outgoing and began readily playing with other children. However, about a fifth of these children reacted very differently, showing signs of fearfulness in these new surroundings. They would cling to their mothers and resisted engaging with other children, showing obvious signs of discomfort. These children had been timid about anything unfamiliar from the time they were infants. A most intriguing observation however, was noted several years later when the children entered kindergarten. About a third of the previously timid children no longer exhibited the earlier timid, fearful behaviors. They acted just like the other bold children. Research staff had also observed interactions in the homes of these children, and described differences in parent-child interactions in the homes of those children whose timidity had faded, from those of children whose timidity persisted.

It appears that learning experiences can modify how genes will express. The altering of early life temperament tendencies involved biological changes in the brain. None of the bold children had become timid by kindergarten age, suggesting some genetic tendencies are more durable than others.

Knowing what caused the observed behaviors initially does not tell you whether or not that behavior can be changed. Whether the source is genetics or experience, observed behavior is always an expression of how the brain is working now. We don't inherit a behavior control gene. We inherit a brain structure with certain propensities; at least some of these tendencies can be altered with neuroplastic changes in that structure. Whether the brainwork can be changed, or not, is an empirical question. It cannot be deduced from the initial source.

Neurobiology and psychotherapy: an emerging dialogue

Following a long period of mutual neglect, contemporary neuroscience and psychotherapy have entered a new stage of their relationship. With growing sophistication in its methods, neuroscience has started to identify neural correlates not only of mental disorders but also of therapeutic changes. The traditional dualism of psychological and somatic psychiatry seems no longer tenable, since even fleeting emotions or thoughts have been shown to leave their traces in the brain. This offers two diverging paths for psychiatry. On the one hand, neurobiology will claim hegemony over psychological approaches, implying some kind of reductionism which holds that all psychological states are 'really' brain states.

Mental disorders should then be regarded as nothing other than chemical imbalances, and psychiatrists should not treat individuals, but their brains. On the other hand, psychological psychiatry could turn the tables and demonstrate that the formation of the brain is inseparably connected to a person's environment and life history. In this view, the brain may only be properly understood as a social and historical organ, along the lines of a 'social neuroscience' [Cacioppo JT, 2002] or 'neurophenomenology' [Fuchs, 2003; Varela FJ, 1996]. This would potentially integrate analyses of biological, psychological and social determinants of mental disorders into a coherent framework that would even stimulate psychotherapeutic theory and practice.

Neuroplasticity and memory

The investigation of neuroplasticity has added new insight to our understanding of therapeutic change. As we currently know, growth and differentiation of the brain are not only determined genetically, but also by its continuous interaction with the environment. This epigenetic formation of the brain does not end in early childhood: There is a life-long re-mapping of cortical networks according to the individual experience, including the de-novo generation of neurones in the adult hippocampus, as proven recently [Bjo¨ rklund A, 2000]. Neuroplasticity is a prerequisite for any enduring change in behavior cognition, and emotion, which is the focus of psychotherapy.

In order to produce lasting effects, psychotherapy should arrive at restructuring neural networks, particularly in the subcortical-limbic system which is responsible for unconscious emotional motivations and dispositions. 'Insight' or 'appeal' reach only corticohippocampal structures, which correspond to conscious memory and cognition, but have only very limited effects on the motivational system [Panksepp J, 1998; Davidson RJ, 1999].

Memory research is directly relevant for the processes of learning and change that are dealt with in psychotherapy. Of particular importance is the distinction established by cognitive neuroscience between two memory systems [Schacter DL, 1987; Squire LR, 1993]: procedural (implicit) memory encompasses all automatic performances, unconscious dispositions and nonverbal habits of behavior, whereas declarative (explicit) memory records single experiences for later recall. Both are based on different sets of neural structures: procedural or implicit memory involves, among others, the basal ganglia, cerebellum and amygdala; declarative memory is mainly located in the temporal lobe, especially in the hippocampus and connected cortical structures [Kandel ER, 1999]. Since the implicit memory system also contains stored patterns of bodily and emotional interaction which are pre reflectively activated by subtle situational cues (e.g. facial expressions, gestures, undertones, atmospheres), it is crucial for the patient's relationships as well as for the therapeutic process.

Developmental research, attachment, and the roots of empathy

In this context, further important results come from developmental research into memory and learning. Mother–infant interaction research has shown that procedural learning and the cerebral mapping of interaction patterns are fully developed in infants aged 3–4 months [Beebe B, 1997]. By unconscious processing of affective information, their implicit memory system is already capable of extracting prototypes and rules from repeated experiences. Thus procedural 'schemes-of-being-with' [Stern D, 1985] are acquired that organize the child's interpersonal behavior, and which will later be transferred to other environments whether these are congruent with the early experiences or not. Thus implicit memory also serves as the link between deficient early interaction experiences, dysfunctional bonding patterns and disturbed affect regulations, which play a decisive role in most mental disorders. These convergent results from developmental and neurobiological research have confirmed, on the one hand, the role of the unconscious as emphasized by psychoanalysis.

On the other hand, this implicit unconscious is quite different from the dynamic unconscious due to repression, defence, anxiety, or conflict, which Freud conceived as the predominant form [Kandel ER, 1999]. Neuro-developmental research has also shown that childhood amnesia is not the result of a repression during resolution of the oedipal complex, but corresponds to late maturation of the declarative memory system [Clyman R, 1991]. Thus neurobiological findings have in part contradicted central assumptions of psychoanalytic metapsychology. The 'biological turn' of psychology has also drawn renewed attention to John Bowlby's [Bowlby J, 1982] attachment theory of social bonding. Supported and expanded by animal research on disturbed neurophysiologic homeostasis following early deprivation [Amini F, 1996; Hofer MA, 1984], Bowlby's theory may serve as a psychobiological model for the social development of the brain. There is growing evidence that the attachment system is a central organizing system in the brain of higher social mammals, allowing infants to use their parents for regulating their inner states until their own psycho neurobiologic functions become mature and autonomous.

As Amini and colleagues have pointed out, the developing nervous system consists of 'open homeostatic loops' which require external regulation or 'tuning' from others [Amini F, 1996; Insel TR, 2001]. On the phenomenological level, this corresponds to the shared states of 'affect attunement' [Stern D, 1985] or 'dyadic states of consciousness' [Tronick EZ, 2003] of mother and infant. These early attachment experiences are internalized and encoded as procedural memory, thus establishing secure and stable bonds to others. Conversely, attachment deficits may result in disorganized behavioral repertoires and deficient 'body micro-practices' [Downing G, 1994], as well as an impaired physiologic capacity for self-regulation of stress and affects [Amini F, 1994]. These findings highlight how deeply human sociality is weaved into the physiological structures of the body.

The intersubjective nature of the human brain is underscored by the discovery of a neural mirroring system in the premotor cortex and other areas of the brain, obviously serving as the neurobiological correlate of action understanding, nonverbal communication and empathy [Rizzolatti G, 1996; Gallese V, 2003]. Mirror neurons discharge both when an action is performed and when a similar action is observed in another individual.

They seem to represent a system that matches intentional behavior of others to one's own action experience, and in this way they form a link of mutual understanding through bodily simulation or resonance. Recently, 'pain neurons' activated by pain observed in others have also been found in the cingulate cortex [Hutchison WD, 1999]. Though not having direct applicability for psychotherapy, the concept of a mirror matching network supplies strong evidence for what the phenomenologist

Merleau-Ponty [Merleau-Ponty M, 1967] has called 'intercorporality': there is a sphere of bodily sensibility and mutual resonance which we share from the beginning with others as embodied subjects. To become aware of these prereflective processes going on during verbal exchange may enhance therapeutic effectiveness.

Consequences for psychotherapeutic concepts

These convergent influences of various research results, though still at an early stage, have already changed the overall framework of psychodynamic approaches and other psychotherapies considerably. The established role of procedural memory and emotional learning, the implicit nature of early acquired relational patterns, the crucial importance of attachment, intercorporality and empathy, in contrast to a decreasing role of repression and declarative memory, have shifted the emphasis from insight-oriented, interpretative or cognitive techniques towards procedural and emotional learning. Alteration of implicit memory patterns presupposes their activation as 'enactments' in the therapeutic process. Accordingly, Stern and other members of the Boston Process of Change Study Group [Stern D, 1998; Lyons-Ruth K, 1998] have developed a therapeutic model centered on ‘now-moments’ in the interaction which represent a particularly striking convergence of procedural relearning and insight. Psychotherapy may thus be regarded as a new attachment relationship which is able to regulate affective homeostasis and restructure attachment-related implicit memory [Gabbard GO, 2000]. In this view, the core of therapeutic interaction lies in the affective communication mediated by bodily resonance, undertones and atmosphere much more than by symbolic language. Thus it is not so much the explicit past that is in the focus of the therapeutic process but rather the implicit past which unconsciously organizes and structures the patient’s ‘procedural field’ of relating to others. In sum, the growing emphasis on implicit relearning in psychotherapy supports the present, experiential aspects of the therapeutic relationship as agents of change. Neurobiological effects of psychotherapy with increasing influence of neurobiological paradigms on psychotherapy, the question arises whether psychotherapeutic effects may also be demonstrated on the neurobiological level. Procedural relearning in psychotherapy should be expected to influence the structure and functions of the brain by altering synaptic plasticity and gene expression [Kandel ER, 2001; Liggan DY, 1999]. There is growing evidence for a modification of gene expression by emotional experiences [26]; for example, tender touch activates the expression of an ‘immediate early gene’ which promotes cellular processes of growth and maturation (Rossi EL, 2002). Of course, psychotherapeutic effects will be more dependent on long-term changes in pathophysiological patterns of the brain which may, for example, be shown by neuroimaging studies.

Psychotherapy seems to be mainly based on cortical ‘top-down’ mechanisms, and pharmacotherapy on subcortical ‘bottom-up’ mechanisms [Goldapple K, 2004]. This would correspond to a concept of the brain as an organ of transformation [Fuchs T, 2002;], which may be addressed by input on different hierarchical levels and translates it in both directions. The transformation runs ‘top-down’ in the one case, that is from subjective experience of meaning and interaction to the neurochemical level mental acts change the brain; and it runs ‘bottom-up’ in the other case, that is from pharmacological effects on subcortical transmitter metabolism to a change in subjective mood and cognition. Such a bidirectional concept is also supported by the results of the Mayberg group [Mayberg HS, 2002], showing mainly cortical (‘subjective’) effects of placebo in contrast to subcortical-limbic and brainstem effects of fluoxetine in major depression.

Psychological treatments and Brain biology

This talk will attempt to answer the question 'Is there biological basis for psychological treatments?' It will attempt to identify biological substrates for psychological treatments, and to assess the research evidence. There are three major forms of psychotherapies/psychological treatments; Behavioral, Cognitive and Psychodynamic. They reflect on interventions at different levels of psychological organization. Behavioral psychotherapy focuses on dysfunction in simple forms of learning and memory (operant and associative conditioning) and related motor behaviour. The brain structures which it involves are the amygdala, basal ganglia and the hippocampus. Cognitive psychotherapy focuses on patterns of information processing thinking patterns in particular disorder. It addresses negative cognitions that play a role in the development and maintenance of the psychopathological state patient learns to evaluate and modify such thinking patterns. The brain areas which it involves include neocortex, specifically the frontal cortex. Psychodynamic psychotherapy focuses on interpersonal representation of a set of expectations about self, others, and their relationship that organizes related affect, thought, and behaviour and the neuropsychological underpinnings of interpersonal representations. The brain areas probably involved include Neurocircuitory incorporating, lateralized cerebral hemispheres and subcortical areas.

Biological treatments include Medications, ECT, Light therapy, and repetitive Transcranial Magnetic stimulation (rTMS). They work by modifying gene expression, neurotransmitter modulation, neuronal firing, and modification of neural circuitories, causing pervasive changes and hence causing behavioral and symptom changes.

If Biological treatments work by affecting the Brain's structure and functioning, then psychological treatments should surely work in similar way in order to produce change. How else could they work?

The Memory model of psychotherapy shows that there are two distinct memory systems. PET and EEG studies have shown that these two types of memory are separate brain functions and rely on different sets of neural structures and physiologic properties which result in distinct patterns of neural activity.

The Implicit memory system processes information regarding affect and forms large amounts of complex information extracts and stores rules. These rules are learned implicitly, which causes self-perpetuating bias for interpreting later experience whether appropriate or not, and which guides behaviour which is not available for conscious processing and reflection.

"In psychotherapy, these patterns of implicit rules are revealed and reflected upon, and change occurs through the learning of new patterns explicitly repeated until the new habit-based manner is engrained in the implicit memory system". Effective and successful psychotherapy should lead to long-lasting change in behavior, cognition, and emotions. When using the memory model to describe the changes listed above, there is a need to postulate plasticity or adaptability of the brain.

The Biological basis of Memory & Plasticity may be summarized as follows:

Cajal has suggested that information can be stored by modifying the connections between communicating nerve cells in order to form associations.

Hebb has observed that modifications only takes place between the connected cells, if both neurons were simultaneously active.

Information is encoded by strengthening the connections between neurons that are simultaneously activated "Neurons that fire together will wire together"; this is known as Hebb's rule, and it leads to 'Hebb - like synaptic plasticity'.

Cortical maps are dynamic and are remodeled as result of important experiences throughout life. One interesting study describing this was that of Structural MRIs of the brains of 16 London taxi drivers with extensive navigation experience compared with those of control subjects who did not drive taxis. The Posterior hippocampi of taxi drivers were significantly larger relative to those of control subjects. This was reported in the popular press as follows; "The scientists also found part of the hippocampus grew larger as the taxi drivers spent more time in the job". She (Maguire) said "The hippocampus has changed its structure to accommodate their huge amount of navigating experience."

David Cohen, one of the taxi drivers commented 'I never noticed part of my brain growing - it makes you wonder what happened to the rest of it.' (Maguire 2000).

The Posterior hippocampus stores a spatial representation of the environment and can expand regionally to accommodate elaboration of this representation in people with a high dependence on navigational skills. The volume of gray matter in the right hippocampus was found to correlate significantly with the amount of time spent learning to be and practicing as a licensed London taxi driver. Thus, it was stated that 'Local plastic change in the structure of the healthy adult human brain in response to environmental demands'.

Aydin et al. (2005) described MR spectroscopy of auditory cortex 10 musicians and showed that, Long-term, professional musical activity caused significant changes in the neurometabolite concentrations, possibly reflecting use-dependent adaptation in the brains of musicians.

Psychotherapy facilitates changes in the permanent storage of information acquired throughout the individual's life. The mechanisms involved in neuronal learning and memory, such as LTP and LTD, are used and reused in the moulding of personality and behavior based on experience (Post 1998).

Attachment memories may be implicit. Attachment research suggests that early patterns of responsiveness exhibited by attachment figures has consequences during neural development. Affective self-regulation appears to be minimal at birth and enhanced by exposure to experiences of appropriate attachment relationships. Inability to self-regulate leads to inability to self-soothe or to modulate anger. In personality disorders, the consequence of early attachment failure possibly results in exaggerated and prolonged reliance on external sources of regulation. Rosenblum et al showed that exposure of monkeys in early life to an inadequate attachment figure engendered permanent vulnerability to anxious and depressed states and to poor social functioning.

Psychotherapy is a form of attachment relationship. A physiological process capable ofregulating neurophysiology and altering underlying neural structure. When patients participate in psychotherapy, they first of all activate the implicit memory system and then engage the mechanism whereby implicitly stored material can be modified (Amini 1996).

<u>Like Drugs, Talk Therapy Can Change Brain Chemistry</u>

Pharmacotherapy and psychotherapy can produce remarkably similar effects on functional brain activity. But does that mean that antidepressants and psychotherapy are really equivalent? In a word, no. Psychotherapy alone has so far been largely ineffective for diseases like schizophrenia, where there is strong evidence of structural, as well as functional, brain abnormalities. So it seems that if the brain is severely disordered, then talk therapy cannot alter it. But it is clear that talk therapy can alter brain function. The reason may come from the elegant work of a Nobel Prize-winning psychiatrist and neurobiologist, Dr. Eric Kandel. Studying the simple and well-mapped nervous system of a sea slug, Aplysia, Dr. Kandel showed that learning leads to the production of new proteins and, in turn, to the remodeling of neurons. Sea slugs exposed to the controlled-learning condition that produced long-term memory ended up with double the number of neuronal connections as the untrained animals. In essence, Dr. Kandel has proved that learning involves the creation of new neuronal connections. The clear implication for humans is that learning literally changes the structure and function of the brain. Now it may seem a big leap from a snail to a human. But if psychotherapy is thought of as a form of learning, then when therapists talk to patients, they cause them to learn, perhaps changing their brain function and, perhaps, for the long run. In the end, Eric chose cognitive behavior therapy and improved drastically. Through exposure to those situations that he feared like messy dirty places, he became desensitized to them and lost his compulsion to wash.

Information transduction

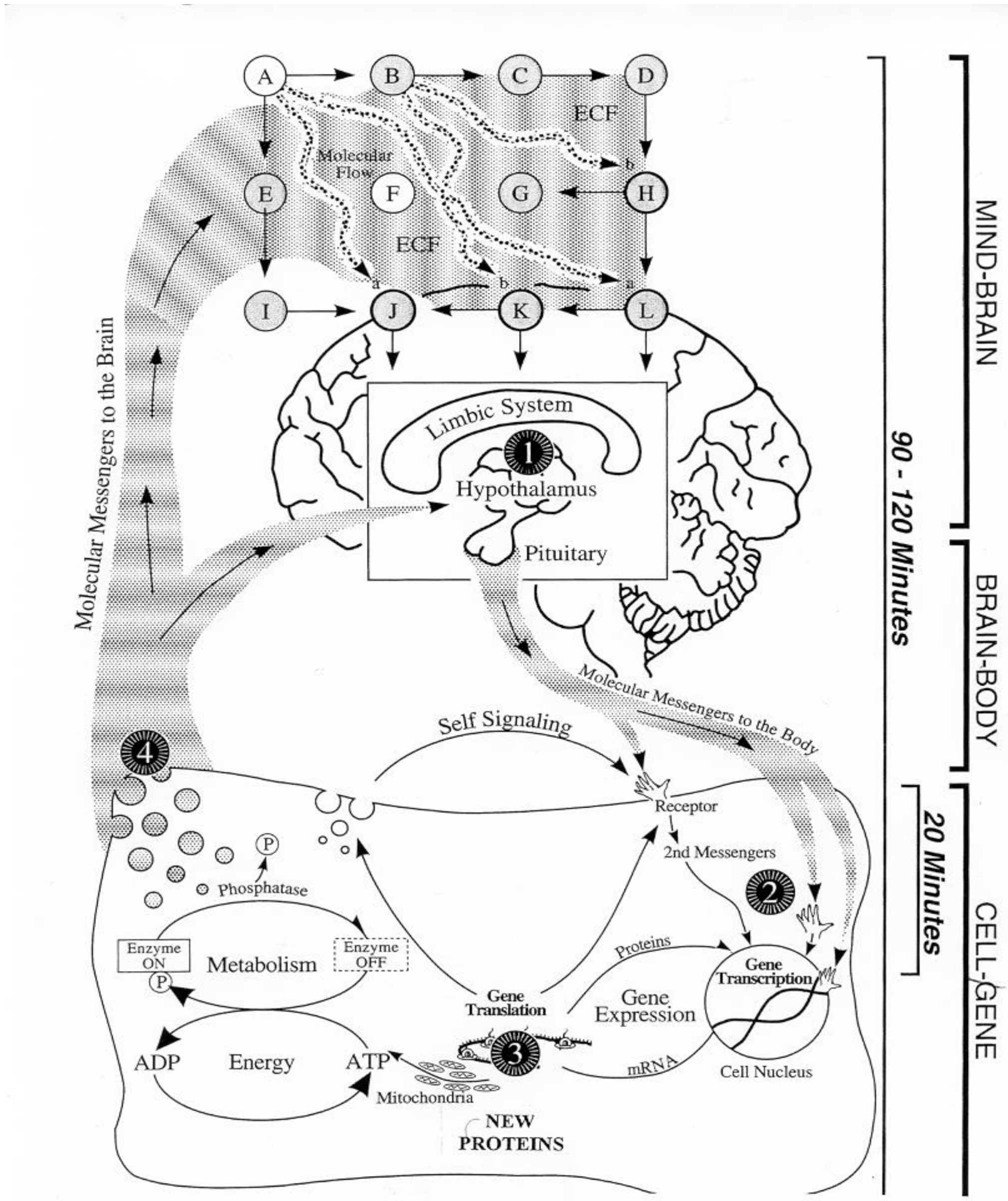
The fundamental question for a deep psychobiology of psychotherapy is, “How do we integrate the many levels of mindbody communication and healing from the psychosocial to the cellular-genetic?” Is it possible to use the concept of “information” and the “transformations of information” to do this? Is it possible to create a new science of “information transduction” that explores how information experienced as human cognitive behavior (thoughts, words, images, emotions, meaning etc.) is transformed into other forms of information expressed as the physical structure of our genes and proteins and visa versa? A major mission of this new science of information transduction is to trace the pathways by which human cognition and emotional experience modulates biological processes in health, sickness and the dynamics of mindbody healing. An interesting step in the creation of this new science of information transduction was pioneered the biologist, (Thomas Stonier, 1990). A visual summary of his ideas was illustrated by the author (Rossi, 1986/1993)

Stage One: Mind Brain Information Transduction

The Limbic-Hypothalamic-Pituitary System is currently recognized as a major information transducer between the brain and the body. Cells within the hypothalamus transduce (transformation information from one form into another) the essentially electrochemical neural impulses of the brain that encode the phenomenological experience of “mind” and emotions into the hormonal “messenger molecules” of the endocrine system. These messenger molecules then travel through the blood stream in a cybernetic loop of information transduction where the four major stages of interest for psychotherapy are numbered.

The complex loop of mindbody communication illustrated in the Figure modulates the action of neurons and cells of the body at all levels from the basic pathways of sensation and perception in the brain to the intracellular dynamics of gene transcription and translation. It has been proposed that the molecular messengers (also called “informational substances”) of the endocrine, autonomic and immune systems mediate stress, emotions, memory, learning, personality, behavior and symptoms (Rossi, 1986/1993, 1996). This communication loop is a two way street by which (1) mind can modulate physiology of the brain and body and (2) biology can modulate mind, emotions, learning and behavior.

The Mindbody Communication Loop

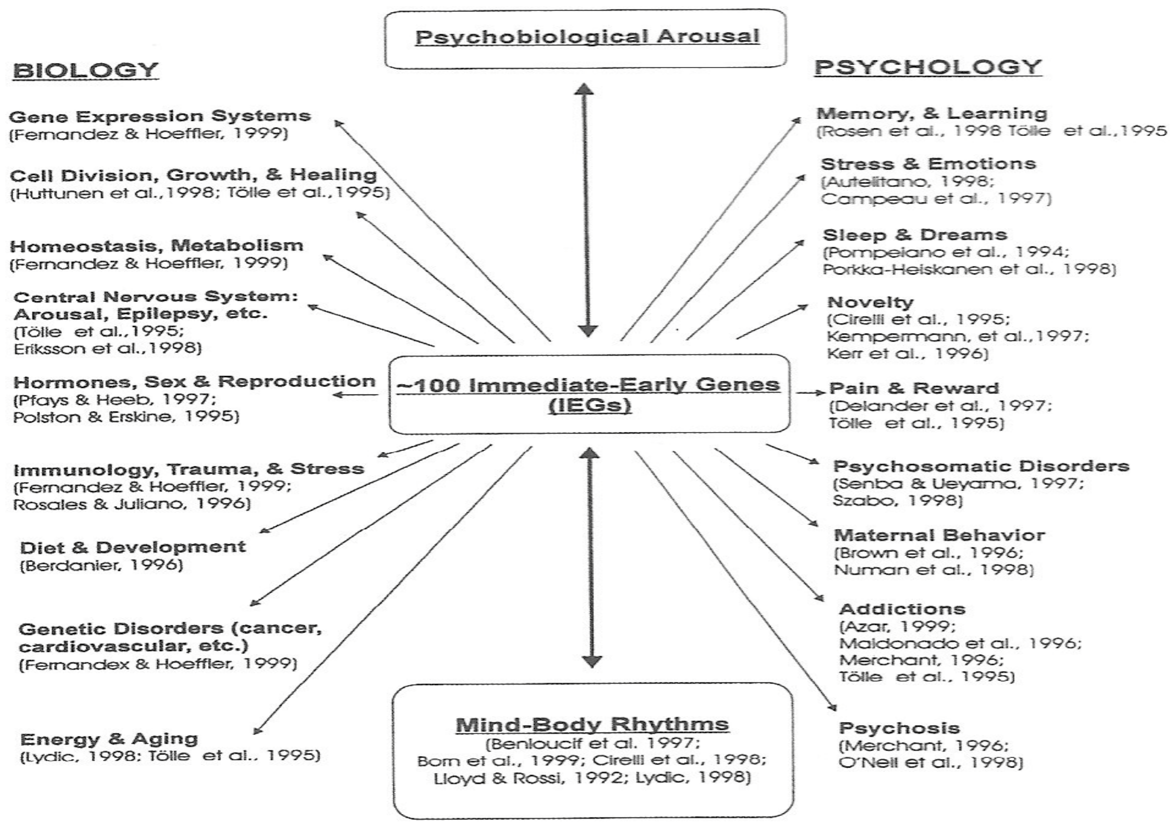


Stage Two: Immediate-Early Genes in Creative Adaptation

A generation ago it was believed that genes were simply the units of physical heredity that were transmitted from one generation to another through sexual reproduction. Today we know that genes have a second major function: a major class of genes, sometimes called, “Immediate-Early Genes,” (also called “Primary Response Genes” or “Third Messengers”) are continuously active in responding to the hormonal messenger molecules signaling the need for creative adaptation to important changes in the environment. Everything from novelty, shock, surprise, touch, sexual stimuli, maternal behavior and psychosocial stress to temperature, food, physical trauma and toxins in the environment can be signaled to the genes via the hormonal messenger molecules that arrive from the limbic-hypothalamic-pituitary system (Merchant, 1965). Immediate-early genes (IEGs) are the newly discovered mediators between nature and nurture. Immediate-early genes act as information transducers allowing signals from the external environment to regulate genes within the internal matrix of the nucleus of life itself. Many researchers now believe that memories along with new experiences are encoded in the central nervous system by changes in the structure and formation of new proteins within the synapses between neurons (Eriksson et al., 1998). IEGs function as transcription factors regulating the “housekeeping genes” that make the proteins within the neuron that encode new memory and learning (Tölle et al., 1995). More than 100 immediate-early genes have been reported at this time. While many of their functions still remain unknown, neuroscientists are exploring the interrelated biological and psychological functions that immediate-early genes such as *c-Fos* and *c-Jun* are already known to serve as illustrated in Figure. Most arousing psychosocial stimuli can induce immediate-early genes within minutes and their main effects are mediated within 20 minutes to an hour or two.

Stage Three: Protein Synthesis in Memory, Stress, and Healing

The third stage in the mind-gene communication loop is the process of gene translation leading to the production of new proteins in the Figure. The time required making new proteins in response to psychological arousal and physical stress as illustrated on the right side of Figure provides an important window into the informational dynamics of new approaches to psychotherapy. The domain of psychological time in minutes and hours as illustrated in Figure relates the Basic Rest-Activity Cycle of human behavior to the processes of mindbody communication. This is in sharp contrast to the more recent evolutionary form of more rapid mindbody communication mediated by the central nervous system in small fractions of a second that are briefer than the usual phenomenological span of consciousness.

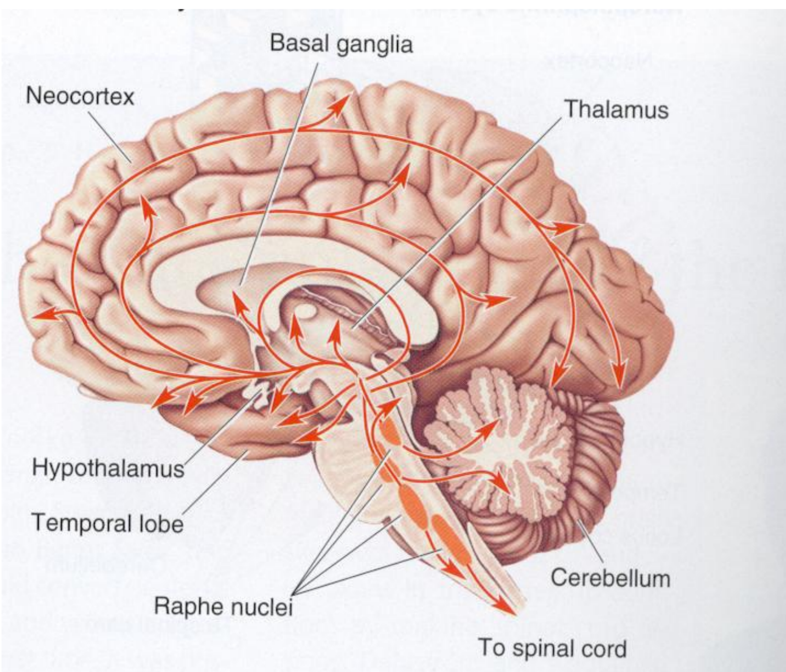


The central role of immediate-early genes (IEGs) in the deep psychobiology of psychotherapy.

Last stage: Messenger Molecules and State Dependent Memory

Stage four of Figure 3 illustrates how messenger molecules (such as hormones, neurotransmitters, growth factors, etc.) that have their origin in the processing of the larger protein “mother-molecules” in stage three may be stored within the cells of the brain and body as a kind of “molecular memory.” These molecular messengers are released into the blood stream where they can complete the complex cybernetic loop of information transduction by passing through the “blood-brain barrier” to modulate the brain’s neural networks and psychological experience as illustrated by the block of letters A through L at the top of the Figure. Such localized neuronal networks of the brain are modulated by a complex field of messenger molecules that can reach the limbic-hypothalamic-pituitary system as well as certain areas of the cerebral cortex. This is a new model of how the sexual hormones, stress hormones, immune system cytokines (messenger molecules like IL-1 and Il-2) and growth factors of the body can modulate mind and emotions and vise versa. If we are willing to grant that communication within the neuronal networks of the brain is modulated changes in the strengths of synaptic connections, then we could say that *meaning* is to be found in the complex dynamic field of messenger molecules that continually bath and contextualize the information of the neuronal networks in ever changing patterns. It is truly amazing to learn, for example, that most of the sexual and stress hormones that have been adequately tested have state dependent effects on our mental and emotional states as well as memory and learning. Recent research indicates that most forms of learning (Pavlovian, Skinnerian, imprinting, sensitization, etc.) are now known to involve these hormonal messenger molecules from the body that can reach the brain to modulate the neural networks that encode mind, memory, learning and behavior. Insofar as these classical forms of learning use messenger molecules, they *ipso facto* have a "state-dependent component" (Rossi, 1986/1993, 1996).

Psychotherapy and serotonin



The first studies on serotonin and psychotherapy were conducted at the Department of Psychiatry, University of Eastern Finland (Kuopio) in the late 90's and early this millennium

A male depressed subject, with traits of personality disorder, showed normalization of SERT during one year of psychotherapy whereas an untreated control case with similar symptoms had no follow-up SERT changes (*Viinamäki et al. Nord J Psychiat 1998*)

In a female patient with anxiety and depression the reduced SERT level increased to a normal range during one year of psychotherapy. Her clinical recovery was delayed, however, with six months from SERT normalization (*Saarinen et al. Am J Psychotherapy 2005*)

Naturalistic sample of depressed subjects (n=18) with six months of dynamically oriented, supportive psychotherapy (1x week)

A trend of SERT improvement as a function of clinical improvement, however, with an inverted U-shape curve. Some subjects seem to improve without a SERT change. In some subjects, SERT and clinical symptoms deteriorated

Changes Related To CBT in Anxiety Disorders

Studies with Spider Phobia

Cognitive behavioral therapy has been shown to be effective to reduce symptoms of specific phobia. The neuroimaging studies in CBT were conducted by Paquette et al.10 and Straube et al.

In their 2003 study, Paquette et al.10 evaluated the subjects with fMRI before and after CBT treatment. The participants of the study comprised of 12 women with spider phobia and a comparison group of 13 women without history of neurological or psychiatric disease and absence of anxiety response to spider exposure. The treatment with CBT consisted of the gradual exposure to spiders and cognitive restructuring. All subjects responded successfully to the therapy.

The neuroimaging findings showed that before the treatment, phobic patients presented significantly activated dorsolateral prefrontal cortex and parahippocampal gyrus. The findings after CBT treatment showed that there was no significant activation of these structures in the phobic subjects. The absence of activation of the dorsolateral prefrontal cortex and parahippocampal gyrus after CBT demonstrated for the authors strong support to the hypothesis that CBT reduces phobic avoidance through the extinction of the contextual fear learned in the hippocampal/parahippocampal region and reduces the dysfunctional and catastrophic thoughts in the prefrontal cortex. Therefore, the process of extinction would be able to prevent reactivation of traumatic memories, allowing the phobic subject to modify his or her perception of stimuli which evoked fear before the treatment. With the modification of stimulus perception, it ceases to be a threat, and this cognitive restructuring could inhibit the activation of brain regions previously associated with a phobic reaction.10

Straube et al.11 performed another study of symptom provocation using fMRI. They also investigated the neurobiological effect of a successful therapeutic intervention with CBT. The study included healthy as well as phobic individuals on a waiting list. The phobic subjects were scanned before and after CBT treatment.

Twenty-eight women with spider phobia and 14 healthy women took part in the study. The subjects with spider phobia were randomly assigned to the therapy group and comparison group on a waiting list. The groups did not differ in phobia severity, age, or educational level. The CBT treatment consisted of gradual exposure to spiders and cognitive restructuring. All of the subjects in the therapy group responded successfully to the treatment.

The neuroimaging findings before the treatment showed that only the phobic subjects displayed activation of the insula and anterior cingulate cortex, while the activation of the amygdala was restricted to the healthy control subjects. No activation of other areas was found among the phobic groups.

The neuroimaging findings after the treatment showed significant differences between the waiting list group and the therapy group. The therapy group showed absence of activation during symptom elicitation in the anterior cingulate cortex and only a small area of activation of the anterior ventral insula, while the waiting group displayed marked responses bilaterally in the insula and anterior cingulate cortex. The therapy group did not show any significant difference from the healthy comparison subjects, while the waiting list patients showed more activation of the right insula and anterior cingulate cortex.

Straube et al.11 showed that the processing of phobic threat is associated with increased activation of the insula and anterior cingulate cortex in subjects with specific phobia. It is important to highlight that successful CBT led to reduction of hyperactivity in these regions.

Studies with Social Phobia

The neurofunctional changes associated with the reduction of social anxiety in patients submitted to CBT treatment were investigated by Furmark et al.12 through PET. The research also had the aim of exploring whether the brain change was associated with the long-term results of the treatment.

Eighteen individuals who fulfilled the criteria for diagnosis of social phobia according to DSM-IV participated in the study. The participants were sorted according to symptom severity, age, and gender, and then randomized for treatment with citalopram, CBT; each group comprised six participants. The CBT group used techniques of exposure, cognitive restructuring, and homework.

The therapeutic effect on regional blood flow was assessed by contrasting the task of speaking in public before and after the treatment in each group. Thus, the improvement of social anxiety was associated with significant reduction of the response of regional blood flow bilaterally in the amygdala, hippocampus, and median and anterior temporal cortex, including the entorhinal, perirhinal, parahippocampal, and periamygdaloid areas both for the CBT and citalopram groups. There was no significant alteration in the regional blood flow in the waiting-list group.

The study showed that the degree of reduction of the limbic response with the treatment is associated with the long-term clinical outcome. The decrease in the response of the brain blood flow in the amygdala, periaqueductal gray matter, and left thalamus can indicate which patients show greater improvement in an interval of 1 year. Thus, favorable results at 1-year follow-up were associated with greater attenuation of the subcortical blood flow response while speaking in public.

The study concluded that the neural sites of activation for the treatment with citalopram and CBT in social anxiety converge to the amygdala, hippocampus, and adjacent cortical areas, possible representing a common way in the successful treatment of anxiety. The attenuation of the activity in the amygdale and limbic region with the treatment was associated with a favorable long-term result and can be a prerequisite for clinical improvement.

Studies with Posttraumatic Stress Disorder

The main findings of Farrow et al in 2005 were that there was an increase in the activation of the left medial temporal gyrus in response to the paradigm of empathy. The same process occurred with the posterior cingulate gyrus, which had its activation increased in response to the condition of forgiveness after the treatment. From this study Farrow et al concluded that CBT can promote changes in the brain area.

Studies with Obsessive-Compulsive Disorder

Three studies were identified evaluating the neurobiological effects of CBT in patients with OCD. Baxter et al. and Schwartz et al. used PET, and Nakao et al. examined the patients through fMRI.

The participants presented improvement of symptoms both in the fluoxetine and in the behavioral therapy group. The neuroimaging findings after treatment showed decrease of the right anterior cingulate and left thalamus in the fluoxetine group who responded to the treatment. The head of the right caudate nucleus presented a significant decrease in both treatments.

Baxter et al. concluded that the glucose metabolism of the head of the right caudate nucleus was changed in the patients treated successfully with both behavioral therapy and fluoxetine. There was a significant correlation of activity of the orbital cortex with the caudate nucleus and the thalamus before the treatment in patients who responded. This correlation disappeared after the success of the treatment.

Schwartz et al. concluded that the results of this study replicated those of the first study, presenting a significant change in the metabolic activity of the right caudate, which was normalized after effective behavioral therapy. This change was not observed in the patients who did not respond to treatment. When the participant data from the first study were combined with the participant data from the subsequent study, it was possible to demonstrate a statistically significant pretreatment correlation between the right orbital gyrus, caudate nucleus, and the thalamus, which decreases after effective treatment. In the previous study, similar results were found with a sample treated with behavioral therapy or fluoxetine. The finding that these effects can be demonstrated following effective treatment only with behavioral therapy (without using medication) and that the correlation between regions is not observed in healthy control subjects, suggests that the association of activity between elements of the cortical-striate-thalamic circuit may be related to expression of OCD symptoms.

Nakao et al, Ten patients were randomly assigned to receive fluvoxamine (n=4) or behavioral therapy (n=6). After treatment the clinical symptoms in both groups were significantly reduced; two patients in the medication group did not show any improvement.

Concerning the neuroimaging findings, the patients presented activation of the left orbital frontal cortex, temporal cortex, and parietal cortex during the task of symptom provocation before the treatment. After the treatment, the patients showed decrease of the activation in the orbital frontal cortex.

The study concluded that the hyperactivation of the circuits involved in the symptomatic expression of OCD, namely, orbital frontal cortex, anterior cingulate gyrus and basal ganglia, can decrease with symptom improvement.

Studies with Panic Disorder

Two studies were performed investigating the neurobiological substrates of CBT in patients with panic disorder through PET.

Prasko et al. submitted resting subjects to PET scans before and after treatments with CBT and medication. All of the patients were without medication for at least 2 weeks.

Twelve patients fulfilling the criterion for panic disorder with or without agoraphobia according to the DSM-IV participated in the study. Ten of the 12 patients suffered from agoraphobia. The patients were randomly distributed into the two treatment groups. The patients of both groups did not show any significant difference in symptom severity at the beginning of the study. The CBT and medication groups were composed of six individuals.

Prasko et al. concluded that both treatments were effective regarding panic symptoms. The changes of brain metabolism in the cortical regions were similar for both treatments. The increased activity of the brain metabolism in the left hemisphere was mainly at the prefrontal, temporoparietal, and occipital regions and posterior cingulate. The decrease was predominantly at the left hemisphere of the frontal region, and at the right hemisphere of the frontal, temporal, and parietal region. No changes were observed in the metabolic activity of subcortical areas. The results of the study indicate that both CBT and antidepressant treatments can activate the temporal cortical processing.

Sakai et al.19 also used FDG-PET to investigate the changes in the use of regional brain glucose associated with anxiety reduction after CBT treatment. The authors worked on the hypothesis that regions above the amygdala such as the medial prefrontal cortex, anterior cingulate cortex, and hippocampus could be modulated in the patients who responded to CBT. Also, the amygdala bilaterally, hippocampus, thalamus, midbrain, caudal pons, medulla, and cerebellum would present an increase in glucose uptake at the baseline condition before the treatment and would have a reduction of this activation after treatment. According to the authors, these regions would be part of the "neurocircuit of panic."

Twelve patients who fulfilled the criteria for panic disorder of the DSM-IV and who had not used fluoxetine and CBT prior to the study participated in the study. The first PET scan was performed before the treatment with CBT and the second was performed after the treatment. The individuals remained at rest during the performance of the procedure.

The neuroimaging findings after CBT showed decreased metabolism in the right hippocampus, left ventral anterior cingulate cortex, left cerebellum, and pons. The increased regional brain glucose metabolism was found at the prefrontal medial region bilaterally.

The PET results following successful CBT treatment evidenced that the level of glucose uptake in the right hippocampus, medial prefrontal cortex, and left ventral cingulate cortex was modulated by the treatment. The findings are consistent with the hypothesis that regions above the amygdala can be adaptively modulated in patients who respond to CBT. Thus, Sakai et al concluded that improvement in panic symptoms through CBT can promote brain effects.

Cognitive Behavioral Psychotherapy and Neurobiology of Depression

Cognitive behavioral psychotherapy (CBT) received most attention in the studies that deal with neurobiological effects of psychotherapies in MDD. CBT is an effective psychotherapy method commonly used for treatment of depression. According to CBT, maladaptive information processing processes, such as misleading beliefs and recurrent negative thoughts, over generalization, over-personalization, selective abstraction, thought of all or nothing, lie under depression (Beck, 2005). In MDD, it is found that maladaptive cognitive processes such as ruminations and strongly negative self focus are connected to hyperactivity of PFC and rostral anterior cingulate cortex (ACC) (Yo- shimura, 2010). It is detected that while emotional decisions are made by depression patients, there is hypoactivity in the left and hyperactivity in the right of the dorsolateral prefrontal cortex (DLPFC) (Grimm, 2008). It is found that effective connections between PFC, ACC and amygdala got disturbed in patients with MDD. Regulative effect of PFC on amygdala becomes insufficient and bottom to up activity increases in the connections between amygdala and ACC, and between ACC and PFC. It is claimed that in MDD, reason behind the difficulty in coping with emotional stimuli is the breach of the connection between the prefrontal cortex and amygdala (Siegla 2007; Moses-Kolko, 2010; Carballedo, 2011; Lu, 2012).

Neurobiological effects of CBT were observed using functional imaging techniques (positron emission tomography scanning/PET and functional magnetic resonance imaging/fMRI). Goldapple and colleagues (2004) studies all brain metabolisms before and after the treatment with PET and compared psychotherapy and pharmacotherapy. In both treatment groups, similar levels of recovery were observed in the severity of the depression. In the psychotherapy group, hippocampus and dorsal cingulate metabolisms increased and PFC metabolism decreased in the direction of normalization. In the pharmacotherapy group, exactly opposite changes were observed as prefrontal activity increased and hippocampus and cingulate activity decreased. It was commented that there may be a relationship between the decline in ruminative thoughts and other dysfunctional thoughts achieved by psychotherapy and the decline in the PFC activity;

In a similar study, 12 patients diagnosed with MDD were applied CBT and 12 patients were applied venlafaxine treatment. After a treatment period of sixteen weeks, changes in brain metabolism were investigated by PET. In both treatment groups, it was found that orbitofrontal cortex and left medial PFC metabolisms decreased and right occipital temporal cortex metabolism increased. Difference was detected in terms of effects on subgenual cingulate and caudate in the two treatment groups. It was concluded that subgenual cingulate might have an important role in response to the treatment (Kennedy et al., 2007).

Cingulate cortex plays an important role in depression therapy. In a study, hypermetabolism was detected on the interface between pre-treatment pregenual and subgenual cingulate cortex in the patients who did not respond to psychotherapy and venlafaxine treatment. However, hypometabolism was detected in the same region in the patients who responded to the treatment. To a lesser extent, changes in other limbic and subcortical regions (hippocampus, amygdala, posterior cingulate, striatum and thalamus) were reported. It was found that pharmacological, cognitive and somatic therapies cause different changes in different regions of the prefrontal and cingulate cortex (Seminowicz et al., 2004).

In some studies, neurobiological effects of psychotherapies were investigated using functional Magnetic Resonance Imaging (fMRI) method. It has been determined that in major depressive disorder there is a general decrease of activity in the ventromedial prefrontal cortex (PFC); that the distinction of emotional and neutral stimuli in amygdala, caudate, hippocampus decreases and that the response to negative stimuli in left anterior temporal lobe and right dorsolateral PFC in comparison to positive stimuli. Following CBT, MDD patients exhibited overall increases in ventromedial PFC activation, enhanced arousal responses in the amygdala, caudate, and hippocampus, and a reversal of valence effects in the left anterior temporal lobe (Ritchey et al., 2011).

Fourteen patients with MDD and 21 healthy control subjects were compared in terms of activity during emotional response. It was found that in patients with MDD, both subgenual cingulate cortex and amygdala showed abnormal activity. Strong relationship was found between low reactivity in subgenual cingulate cortex, high reactivity in amygdala and the response to CBT. Decline in subgenual cingulate cortex activity before the therapy is a predictor for response to CBT and shows a deficit in regulation of this area. It is argued that CBT might be providing recovery by fixing the disturbances in the emotional regulation (Siegle et al., 2006). ACC is the first piece of the Papez circuit that is known to be associated with regulation of emotions. It is known that in depression patients, grey matter volume, blood flow and glucose metabolism decrease in the subgenual cingulate cortex (Drevets, 2008). Data at hand shows that cingulate cortex functions that get disturbed in MDD might be recovered by CBT.

Few studies supports that in depression neural network that contains ACC is important for response to the CBT, and that changes in the ACC activity may be predictor of the response to pharmacotherapy and CBT (Costafreda et al., 2009). CBT possibly shows its effect more controlled information processing processes replace emotional responses, automatic limbic reactions are prevented and role of inhibitor driving mechanisms increases (Derubeus, 2008; Holtzheimer &Mayberg, 2008).

Findings derived from these studies show that CBT ensures recovery in the maladaptive information processing processes in MDD by causing changes in the PFC, ACC, and amygdala metabolisms. There are clues that metabolism and activity changes in the cingulate cortex (especially in the subgenual area) might be decisive on response to the therapy.

Interpersonal Psychotherapy and Neurobiology

Interpersonal psychotherapy (IPT) aims to alleviate depression by raising quality of interpersonal world of the patient. Phenomenological formulation of this method is based on social environment of the individual. Depression is associated with current interpersonal relationship problems. The patient is assisted to gain social skills so that she/he can solve his/her interpersonal relationships. In IPT, 12 to16-week sessions are applied and 4 main topics are focused on: 1) Unresolved Grief: The patient is encouraged to mourn after the relative lost and establish new relationships; 2) Role conflict: In case of social role conflict, the individual is encouraged to reevaluate the difficulties and search for ways of solution; 3) Role transitions: Possible gains for the patient from role transitions are worked on; 4) Interpersonal deficiencies: When interpersonal conflicts are evident, psychotherapy aims at overcoming social isolation. Another characteristic of this method is supporting of strong aspects of the patient (Alkan, 2007; Friedman & Thase, 2009).

Three studies done in this area has been reached. A study conducted by Brodly and colleagues (2001a) compared brain metabolism of 24 patients suffering major depressive disorder with a healthy control group of 16 people (using PET). Ten patients with depression were applied paroxetine and 14 patients were applied IPT. Patients who had MDD before the treatment had PFC, caudate and thalamus metabolisms higher than and temporal lobe metabolism lower than the control group. In both group, PFC (bilateral in the group taking paroxetine and right in the group receiving IPT) and left anterior cingulate gyrus metabolisms decreased and left temporal lobe metabolism increased to return to normal. In the other study of the same author, metabolic values of 14 patients who were applied IPT for 12 weeks were compared to 24 patients who took paroxetine. Correlation was found between recovery from anxiety, psychomotor retardation, tension and fatigue symptoms and the decline in the ventral frontal lobe metabolism; between recovery from anxiety and tension symptoms and the decline in ventral anterior cingulate gyrus and anterior insula activity; between psychomotor retardation and the increase of dorsal anterior cingulate activity; and between cognitive disturbance and the increase of DLPFC metabolism. It was concluded that findings supported limbic-cortical dysregulation theory in MDD, metabolic changes occurred in the frontal zone in depression and IPT fixed these (Brody et al., 2001b).

In another study, 13 patients diagnosed with MDD and suffer medium to severe depression were applied IPT and 15 patients were applied 75 mg/g venlafaxine. After 6 weeks, single photon emission computed tomography (SPECT) scan was taken to eva- luate brain blood flow. In both groups, severity of depression decreased. In the venlafaxine group, activity was detected in the right posterior temporal and right basal ganglia; and in the IPT group activity was detected in the right posterior cingulate and right basal ganglia. Increase in limbic blood flow was detected in the IPT group, and increase in basal ganglion blood flow was detected in both treatment groups (Martin et al., 2001).

Although there are lower number of studies in this field, the findings support that IPT ensure recovery of the metabolism and blood flow in the prefrontal cortex, cingulate cortex and basal ganglia.

Psychodynamic Therapies and Neurobiology

Psychodynamic psychotherapy model is based on the idea that depression is related to unconscious conflicts and desires. According to this, interpersonal relationships are closely related to developmental characteristics brought from childhood and these characteristics are seen as transference in therapeutic relationship. Psychodynamic therapy focuses on how present losses and stressors revive past losses and traumas (Gabbard & Bennett, 2009). Although there are many opinions and theories in this field, these will not be discussed here. Psychodynamic psychotherapy can be applied short or long term. Long-term psychodynamic psychotherapy aims to provide an insight and reevaluation of experiences related to the depressive pathology. In the first and one study in this field, changes caused by psychodynamic psychotherapy on left anterior hippocampus, amygdala, subgenual cingulate and medial prefrontal cortex regions are investigated using fMRI in MDD. Depression patients were applied psychodynamic therapy for 15 months and activation values obtained before the therapy was compared with those obtained after the therapy. After psycho- therapy, it was found that depressive symptoms decreased, accompanied by a decline of activity in the cingulate and medial prefrontal cortex, which was high before the therapy (Buchheim et al., 2012). Findings obtained through psycho- dynamic psychotherapy show that abnormal activities in the prefrontal cortex and cingulate cortex in the depression are regulated, which supports results of the CBT and IPT.

Discussion

In just over two decades, developments in a branch of biology—the discipline of Neuroscience—promise to enhance our understanding of mental illness and mental health, and expand our treatment strategies and tools. They are leading to a major shift in how we understand the relationships between brain biology and effects of experience.

Three major developments were Neuroimaging technologies, concept of neuroplasticity and Self directed Neuroplasticity were helpful in understanding the effects of psychotherapy on the Brain.

Learning experiences can modify how genes will express. The altering of early life temperament tendencies involved biological changes in the brain. Whether the source is genetics or experience, observed behavior is always an expression of how the brain is working now. We don't inherit a behavior control gene. We inherit a brain structure with certain propensities; at least some of these tendencies can be altered with neuroplastic changes in that structure.

Mental disorders should then be regarded as nothing other than chemical imbalances, and psychiatrists should not treat individuals, but their brains. brain may only be properly understood as a social and historical organ, along the lines of a ‘social neuroscience’ or ‘neurophenomenology’.

Neuro-plasticity is a prerequisite for any enduring change in behavior cognition, and emotion, which is the focus of psychotherapy. In order to produce lasting effects, psychotherapy should arrive at restructuring neural networks, particularly in the sub cortical-limbic system which is responsible for unconscious emotional motivations and dispositions.

The established role of procedural memory and emotional learning, the implicit nature of early acquired relational patterns, the crucial importance of attachment, intercorporality and empathy, in contrast to a decreasing role of repression and declarative memory, have shifted the emphasis from insight-oriented, interpretative or cognitive techniques towards procedural and emotional learning. Psychotherapy may thus be regarded as a new attachment relationship which is able to regulate affective homeostasis and restructure attachment-related implicit memory. Psychotherapy seems to be mainly based on cortical ‘top-down’ mechanisms, and pharmacotherapy on subcortical ‘bottom-up’ mechanisms.

Psychotherapy is a form of attachment relationship. A physiological process capable of regulating neurophysiology and altering underlying neural structure. When patients participate in psychotherapy, they first of all activate the implicit memory system and then engage the mechanism whereby implicitly stored material can be modified

The clear implication for humans is that learning literally changes the structure and function of the brain. Now it may seem a big leap from a snail to a human. But if psychotherapy is thought of as a form of learning, then when therapists talk to patients, they cause them to learn, perhaps changing their brain function and, perhaps, for the long run.

New science of "information transduction" that explores how information experienced as human cognitive behavior (thoughts, words, images, emotions, meaning etc.) is transformed into other forms of information expressed as the physical structure of our genes and proteins and visa versa pioneered by Thomas.

Conclusion

Following a long period of mutual neglect, contemporary neuroscience and psychotherapy have entered a new stage of their relationship. With growing sophistication in its methods, neuroscience has started to identify neural correlates not only of mental disorders but also of therapeutic changes. The traditional dualism of psychological and somatic psychiatry seems no longer tenable, since even fleeting emotions or thoughts have been shown to leave their traces in the brain. This offers two diverging paths for psychiatry. On the one hand, neurobiology will claim hegemony over psychological approaches, implying some kind of reductionism which holds that all psychological states are 'really' brain states.

Psychotherapy alone has so far been largely ineffective for diseases like schizophrenia, where there is strong evidence of structural, as well as functional, brain abnormalities. So it seems that if the brain is severely disordered, then talk therapy cannot alter it. But it is clear that talk therapy can alter brain function.

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UNIT- XIII

CONTEMPORARY ISSUES IN BEHAVIOUR MEDICINE

Class Presentation

Subject: Behavioural Medicine

Presented by: Barsleeeby Alex Daniel

IInd Year M.Phil Clinical Psychology

POST GRADUATE INSTITUE OF BEHAVIORAL AND MEDICIAL SCIENCES, RAIPUR, C.G.

Syllabus

- Research and development in health psychology
- Psychobiology
- Psychophysiology
- Psychoneuroimmunology
- Sociobiology
- Effects of psychotherapy on the biology of brain

Development In Health Psychology

A number of trends within medicine, psychology and the health care system have combined to make the emergence of health psychology inevitable. It is safe so say that health psychology is one of the most important developments in the filed of psychology in the past 50 years. Following factors have led to the development of health psychology.

Changing Patterns of Illness

Until 20th century, the major cause of illness and death in the United States were acute disorder- especially tuberculosis, pneumonia, and other infectious diseases. Acute disorders are short term medical illnesses, often the result of a viral or bactireal invader and usually amenable to cure. But now chronic illness- especially heart disease, cancer and diabetes-are the main contributors to disability and death, especially in industrialized countries.Chronic illness are slowly developing with which people for a long time. Often, chronic illness cannot be cured but, rather, only managed by the patient and provider working together.

Chronic diseases are in which psychological and social factors are implicated as causes . For example personal health habit such as diet and smoking are implicated in development of heart disease and cancer.other than that, people may live with chronic disease for many years, psychological issues arise in connection with them. Health psychologist help the chronically ill adjust psychologically and socially to their changing health state. They help those with chronic illness develop treatment regimens, many of which involve self-care.

1. Advances in Technology

   The field of health psychology is changing almost daily because new issues arise that require the input of psychologist. For example, new technologies now make make it possible to identify the genes that contribute to many disorders.

2. Impact of Epidemiology

In establishing the goal and concern of health psychology and health care endeavor more broadly,morbidity and mortality statistics are essential. Health psychologist need to know the major causes of disease in the country, particularly those diseases that lead to early death , so as reduce their occurrence. For example knowing that cardiac disease is major cause of premature death has led to a national wide effort to reduce risk among those most vulnerable , including smoking reduction effort to reduce risk factors among those most vulnerable, including smoking reduction efforts, implementation of dietary changes, cholesterol reduction techniques, increased exercise, and weight loss.

3. Expanded Health Care Services

Health psychology represents an important perspective on these issues for several reasons : Because containing health care cost is so important, health psychology`s main emphasis on prevention- namely, modifying people`s risky health behaviours before they ever become ill -has potential to reduce the number of dollars devoted to the management of illness. Health psychologist have done substantial research on what makes people satisfied or dissatisfied with their health care. Thus, they can help in the design of user-friendly health care systems.

The health care industry employs many millions of individuals in a variety of jobs. Nearly ever individual in the country has direct contact with the health care system as recipient of services. Thus, its impact on people is enormous

4. Increased Medical Acceptance

Although Health psychologist have been employed in health setting for many years, their value isincreasingly recognized by physician and other health care professionals.

At one time, the role of psychologist in health care was largely confined to the task of administrating tests and interpreting the test result of individuals who were suspected of being psychologically disturbed. In health settings,psychologist saw only the “ problem patients” who were difficult for medical staff to manage or whose physical complaints that could be readily attributed to medical problems and were therefore thought to be outside the psychologist`s province of expertise.

Demonstrated Contributions to Health

Few brief examples,

1. Health psychologist have <u>developed a variety of short-term interventions</u> to address a wide variety of health-related problems, including managing pain; modifying bad health habits such as smoking and managing side effects or treatment effects associated with a range of chronic disease.
   - Psychologist inform patients about the procedures and sensation involved in unpleasant medical procedures such as surgery improves there adjustment to those procedure
   - Metholodological
     Contribution
   - Experiments

In an experiment researcher creates two or more conditions that differ from each other in exact and predetermined ways. People are randomly assigned to experience these different conditions, randomly assigned to experience these different conditions, and their reactions are measured. Experiments conducted by health care practitioners to evaluate treatments or intervention and their reactions are measured. Experiments conducted by health care practitioners to evaluate treatments or interventions and their effectiveness over time are also called randomized clinical trials.

Experiments have been the mainstay of science, because they often provide more definitive answer to problems than other research methods. When we manipulate a variable and see its effect, we can establish a cause-effect relationship definitively. For this reason, experiments and randomized clinical trials have been the mainstay of health psychology research. However, sometimes it is impractical to study issues experimentally. People cannot, for example, be randomly assigned to diseases.

1. Correlation studies

In Congregational studies health psychologist measures whether a change in one variable corresponds with changes in another variable. A Congregational study might identify, for example, that people who are higher in hostility have a higher risk for cardiovascular disease.

2. Prospective design

<u>Prospective research</u> looks forward in time to see how a group of individual change, or how a relationship between two variables change, over time. For example, if we were to find that hostility develops relatively in early life but other risk factors for heart disease and recognize that the reverse direction of causality- namely, that heart disease hostility - is unlikely. Health psychologists conduct many prospective studies in order to understand the risk that relate to certain health conditions.

A particular type of longitudinal research, in which researcher observe the same people over a long period of time.

- Retrospective Research

Retrospective Research looks backward in time, and attempt to reconstruct the conditions that led to a current situation. Retrospective methods, for example were critical in identifying the risk factors that lead to the development of AIDS.

Psychophysiology: Theory and Methods

How does mental and emotional life tie in with the workings of the body? Psychophysiology is the branch of psychology that studies the behavior of the individual in a biological context. It is an attempt to chart the mutual interactions between psychological processes and the workings of the body, giving equal emphasis to both.

A fundamental principle of psychophysiology is that thoughts and feelings cannot exist apart from the body. It follows that a full understanding of psychological processes depends on understanding the biological context from which they proceed. Due to its emphasis on integrating our understanding of mental and physiological processes,
psychophysiology has contributed to research methods and theory building in behavioral medicine and to the neurosciences of cognition and emotion. Psychophysiology does this by providing a theoretical basis and a set of measurement methods that help to disentangle relationships between psychology and biology and between our thoughts and emotional experience in relation to good and poor health. From this perspective, psychophysiologists bring a physiological emphasis to the study of behavior and mental processes, and one expression of this emphasis is the contributions psychophysiology has made is to the understanding of emotions and their impact on good and poor health. Although it is accepted in psychophysiology that thoughts and feelings do not exist without the brain and the body, it is necessary to emphasize that the thoughts and feelings of interest are not equivalent to or directly reducible to these physiological processes.

The emphasis of psychophysiology, like that of behavioral medicine itself, is primarily on the whole person. However, it is necessary to measure the functions of specific systems, such as the cardiovascular system, endocrine system, or immune system in the course of psychophysiological investigations. This calls for a methodology that allows emotional experience to be studied simultaneously with physiological functioning in ways that are unobtrusive and minimally invasive. This ensures that the person being studied is behaving in a normal manner, as in everyday life, and is not reacting unduly to the apparatus or laboratory setting. Psychophysiological principles have been used to study responses to stress in the laboratory, responses to stressors in daily life, and individual differences in such responses.

Behavioral medicine is both a science and an approach to clinical practice. These two parts are concerned with the influence of behavioral factors on health and disease. Behavioral medicine holds that states of health can be influenced by overt behaviors, such as dietary habits, and by covert behaviors, such as emotional states and stress responses. This perspective leads behavioral medicine researchers to ask questions about the ways that emotional states and stress responses can affect health through their influence on physiology. The goal is to bring to light how our behaviors and our ways of perceiving and reacting to the world may affect our wellbeing for better or worse. Such research addresses questions in several major areas, including (1) how the body responds during positive and negative emotion states, (2) how a given person may differ from one time to the next in stress reactivity, (3) the ways in which persons differ from one another in their stress responses, and (4) on the positive side, to establish the effects of behavior on good health and longevity. To carry out such research, behavioral medicine draws in part on the theory and methods developed in the field of psychophysiology.

In laboratory studies, persons are often exposed to stressors to determine how they react to such challenges both emotionally and physiologically. The results are thought to indicate how emotionally relevant events and behavioral stressors can affect physiology in daily life and therefore whether they may contribute to disease. As one example, a commonly used stressor is public speaking. This challenges the subject to make up a short speech and deliver it without notes and to do so in a fluent and convincing manner. Public speaking is stressful because most persons wish to avoid the embarrassment of doing poorly and to be seen as masterful and competent by observers in the laboratory. Using this method, the social world can be modeled in a small way in the laboratory, and the participant's disposition is invoked to produce a stress response. During public speaking, this process of social evaluation, along with the resulting fear and anxiety, produces substantial increases in heart rate and blood pressure and stress hormones, including catecholamines and cortisol. The person's mood states usually are assessed at rest before the task begins and again at the end using paper-and-pencil measures or brief interviews. Similarly, autonomic reactions are often measured at rest and during stress using automated blood pressure monitors and impedance cardiographs, and endocrine responses may be observed using saliva or blood sampling. In this manner, the person's psychological, cardiovascular, and endocrine reactions may be measured to provide a picture of how physiological reactions are set off by psychologically meaningful events.

This research strategy can then be extended to compare different kinds of people for potential differences in their physiological reactivity to stress. One common example is for the researcher to identify young, healthy individuals who have a family history of high blood pressure and also to find those with no such history.

These family history groups can then be compared in the laboratory for differences their stress responses, perhaps using the public speaking stressor or some other method. This allows potential differences in stress reactivity to be assessed in relation to a family history of this prevalent cardiovascular disease. It is then possible to follow such persons for a period of years to establish which persons become hypertensive and which retain a normal blood pressure. Do persons from the family history group have a greater likelihood of becoming hypertensive in middle age? Are persons with greater reactions to stress more likely to become hypertensive, regardless of family history? Such studies therefore allow potential interactions between family history and stress reactivity to be studied. If persons with a family history of hypertension who are also highly reactive to social stress are much more likely to become hypertensive, then we would conclude that the family history created a biologically based risk factor that was enhanced by an elevated level of stress responsivity. In contrast, should risk of hypertension be increased equally by high reactivity in persons with and without a family history of hypertension, we would conclude that family history and reactivity tendencies contribute to hypertension risk in an additive manner.

Although the laboratory provides a well-controlled environment with an extensive range of measurement techniques, ambulatory methods have been used with increasing frequency outside the laboratory to document how challenges in persons' daily lives can affect cardiovascular, endocrine, or immune systems. Such methods measure the person's responses to naturalistic stressors, such as work stress, or challenges in the home, such as family conflict or the stress of caring for a chronically ill spouse. Such studies rely on small, lightweight monitors that can be worn comfortably as persons go about their daily routines. These monitors can make reliable measurements in a wide range of circumstances. Such systems are able to track heart rate, blood pressure, and physical activity. In addition, the person usually reports on their subjective state using brief paper diaries or personal digital assistants.

As in laboratory studies, this ambulatory method may be used to estimate the interaction of stress responses and disease risk. Persons with and without a family history of hypertension may be compared as they go about their daily lives. As in the laboratory, persons with the largest or most prolonged reactions to stress at home or at work are suspected of having greater risk of future disease, and again, they may be followed up for actual occurrence of hypertension in future years. Ambulatory systems currently in use include traditional Holter electrocardiographs, blood pressure monitors, and impedance cardiographs. The success of these systems has led several commercial companies to develop reliable products for research and clinical use.

Although some research focuses on family history, other work seeks to connect psychological dispositions, such as hopelessness, depression, or hostile style to disease risk. Studies using this strategy may compare highly hostile persons with nonhostile individuals with a specific hostility provoking interaction, such as harassing comments during work on a difficult task. By measuring physiological reactions to such specific challenges in persons with different psychological characteristics, a clearer picture may be developed of the psychological and physiological interplay that is suspected of contributing to disease.

While much of this research focuses on negative emotion states, stress responses, and risk of disease, there is a growing interest in positive emotional states and in studying persons who tend frequently to experience the positive emotions of joy and happiness. As in the above examples such persons can be selected for their emotion traits using a combination of self-report techniques and in laboratory tests of brain function. Persons high in typical positive affect can then be compared to those with less positive affective states in their resistance to the effects of stress and in their long-term states of health.

The research examples listed above all depend on testing persons while they are relaxed and resting, as well as when they are under stress or perhaps in a pleasurable mood. For these reasons, it is desirable to use measurement methods that do not cause discomfort or distress. Behavioral medicine research has therefore relied on methods of psychophysiological measurement that are noninvasive or minimally invasive and cause the volunteer no discomfort. The examples above focused on the cardiovascular system which can be studied using methods such as the electrocardiogram, blood pressure monitoring, and impedance cardiography to measure pumping action of the heart and constriction of the blood vessels, and, occasionally, fluid output to assess kidney function. Stress research often uses additional methods to track responses of the endocrine system, involving collection of urine, blood, or saliva for measurement of stress hormones and other substances associated with stress and pain responses. Still, other studies examine the immune system here using minimally invasive techniques in the collection of blood for later measurement of the numbers of immune system cells and their biological activity. Closely related to these physiological measurements is the need to classify persons as to personality and temperament characteristics to establish relationships between acute stress responses or chronic allostatic responses in the lab or in daily life. These considerations call for use of interviews or paper-and-pencil measures of personality and mood states. Finally, the application of such psychophysiological techniques calls for appropriate selection of tasks and ways to analyze the data.

Psychobiology

Biological psychologists study the animal roots of behavior, relating actions and experiences to genetics and physiology. Biological psychology is the study of the physiological, evolutionary, and developmental mechanisms of behavior and experience. It is approximately synonymous with the terms biopsychology, psychobiology, physiological psychology, and behavioral neuroscience.

The term biological psychology emphasizes that the goal is to relate biology to issues of psychology. Neuroscience includes much that is relevant to behavior but also includes more detail about anatomy and chemistry. Biological psychology is more than a field of study. It is also a point of view. It holds that we think and act as we do of certain brain mechanisms, which we evolved because animals with these mechanisms survived and reproduced better than animals with other mechanisms. Biological explanations of behavior fall into four categories: physiological, ontogenetic, evolutionary, and functional (Tinbergen, 1951). A physiological explanation relates a behavior to the activity of the brain and other organs. It deals with the machinery of the body—for example, the chemical reactions that enable hormones to influence brain activity and the routes by which brain activity controls muscle contractions. The term ontogenetic comes from Greek roots meaning the origin (or genesis) of being. An ontogenetic explanation describes how a structure or behavior develops, including the influences of genes, nutrition, experiences, and their interactions. For example, the ability to inhibit impulses develops gradually from infancy through the teenage years, reflecting gradual maturation of the frontal parts of the brain. An evolutionary explanation reconstructs the evolutionary history of a structure or behavior. The characteristic features of an animal are almost always modifications of something found in ancestral species (Shubin, Tabin, & Carroll, 2009). For example, monkeys use tools occasionally, and humans evolved elaborations on those abilities that enable us to use tools even better (Peeters et al., 2009). Evolutionary explanations also call attention to features left over from ancestors that serve little or no function in the descendants. For example, frightened people get "goose bumps"—erections of the hairs— especially on their arms and shoulders. Goose bumps are useless to humans because our shoulder and arm hairs are so short and usually covered by clothing. In most other mammals, how- ever, hair erection makes a frightened animal look larger and more intimidating . An evolutionary explanation of human goose bumps is that the behavior evolved in our remote ancestors and we inherited the mechanism.

A functional explanation describes why a structure or behavior evolved as it did. Within a small, isolated population, a gene can spread by accident through a process called genetic drift. For example, a dominant male with many offspring spreads all his genes, including some that helped him become dominant and other genes that were neutral or possibly disadvantageous. However, a gene that is prevalent in a large population presumably provided some advantage—at least in the past, though not necessarily today. A functional explanation identifies that advantage. For example, many species have an appearance that matches their background . A functional explanation is that camouflaged appearance makes the animal inconspicuous to predators. Some species use their behavior as part of the camouflage. For example, zone-tailed hawks, native to Mexico and the southwestern United States, fly among vultures and hold their wings in the same posture as vultures. Small mammals and birds run for cover when they see a hawk, but they learn to ignore vultures, which pose no threat to a healthy animal. Because the zone-tailed hawks resemble vultures in both appearance and flight behavior, their prey disregard them, enabling the hawks to pick up easy meals (W. S. Clark, 2004).

Psychoneuroimmunology

Psychoneuroimmunology (PNI) is the study of the functional relationships between central nervous system, behavior, and immune system. These relationships have been documented to be multidirectional. For example, while behavior can influence immune processes through changes in nervous system signals, immune signals have been shown to alter the function of the central nervous system, thereby influencing behavior.

Further, all systems exert regulatory control over each other, forming a complex communication network. PNI research aims at describing this network and thus to contribute to the understanding of the behavioral and biological mechanisms underlying the links between psychosocial factors and health as well as disease development and progression. Psychosocial factors studied in PNI thereby range from negative psychological states such as depression and anxiety, to social support, interpersonal relationships, and personality factors. Disease-related processes investigated include cancer, susceptibility to infection, wound healing, HIV/AIDS, autoimmune diseases, and cardiovascular diseases. One important PNI branch focuses on how stress and stress-related neuroendocrine processes affect health aswell as disease development and progression. As such, PNI is truly interdisciplinary, integrating not only knowledge from immunology, neuroscience, and psychology, but also from areas such as psychiatry, endocrinology, physiology, and pharmacology.

Sociobiology

Sociobiology, the systematic study of the biological basis of social behaviour. The term sociobiology was popularized by the American biologist Edward O. Wilson in his book Sociobiology: The New Synthesis (1975). Sociobiology attempts to understand and explain animal (and human) social behaviour in the light of natural selection and other biological processes. One of its central tenets is that genes (and their transmission through successful reproduction) are the central motivators in animals' struggle for survival, and that animals will behave in ways that maximize their chances of transmitting copies of their genes to succeeding generations. Since behaviour patterns are to some extent inherited, the evolutionary process of natural selection can be said to foster those behavioural (as well as physical) traits that increase an individual's chances of reproducing.

Sociobiology has contributed several insights to the understanding of animal social behaviour. It explains apparently altruistic behaviour in some animal species as actually being genetically selfish, since such behaviours usually benefit closely related individuals whose genes resemble those of the altruistic individual. This insight helps explain why soldier ants sacrifice their lives in order to defend their colony, or why worker honeybees in a hive forego reproduction in order to help their queen reproduce. Sociobiology can in some cases explain the differences between male and female behaviour in certain animal species as resulting from the different strategies the sexes must resort to in order to transmit their genes to posterity.

Sociobiology is more controversial, however, when it attempts to explain various human social behaviours in terms of their adaptive value for reproduction. Many of these behaviours, according to one objection, are more plausibly viewed as cultural constructs or as evolutionary by-products, without any direct adaptive purpose of their own. Some sociobiologists—Wilson in particular—have been accused of attributing adaptive value to various widespread but morally objectionable behaviours (such as sexism and racism), thereby justifying them as natural or inevitable. Defenders of sociobiology reply that at least some aspects of human behaviour must be biologically influenced (because competition with other species would select for this trait); that evolutionary explanations of human behaviour are not defective in principle but should be evaluated in the same way as other scientific hypotheses; and that sociobiology does not imply strict biological determinism.

Reference

http://www.britannica.com/EBchecked/topic/551863/sociobiology